ACX30585.1
GQ351269.1:20492-33422
uncultured Candidatus Thioglobus sp.

PmbA_TldD_1st PmbA_TldD_2nd PmbA_TldD_3rd FAD_binding_3 5-FTHF_cyc-lig Thioredoxin_2 Thioredoxin_2 DDE_Tnp_IS1595 ? DEAD Helicase_C DbpA DUF4258 CopG_antitoxin ? Aminotran_1_2 ? AICARFT_IMPCHas MGS HTH_8 Dus

AEF83759.1
CP001843.1:1365528-1380290
Treponema primitia ZAS-2

RNase_HII ? DUF3786 ? GalKase_gal_bdg GHMP_kinases_N GHMP_kinases_C SurE ARM_TT21_C TPRs BrnT_toxin CopG_antitoxin ThiI THUMP ThiI_fer Aminotran_5 HTH_5 Glycos_transf_4 AA_kinase ACT_9 ACT_9 NAD_binding_3 Homoserine_dh ?

AFM13307.1
CP002959.1:2736670-2755275
Turneriella parva DSM 21527

? ? DDE_Tnp_IS1595 Phage_integrase ? ? HKD ResIII Helicase_C BrnT_toxin CopG_antitoxin DDE_Tnp_IS1595 HsdM_N N6_Mtase Methylase_S Methylase_S DUF6161 ? HTH_29 rve rve_3

AFM13954.1
CP002959.1:3473206-3485726
Turneriella parva DSM 21527

? ? DUF4184 Flagellin_N ? Lcl_C ? BrnT_toxin BrnA_antitoxin ? ? DUF7273 ? ? ?

AFM14065.1
CP002959.1:3578021-3590120
Turneriella parva DSM 21527

? BrnT_toxin ParE_toxin ? PIN MazE_antitoxin ? BrnT_toxin BrnA_antitoxin CoiA_N SH3_3 ? SH3_3 ? ?

AFZ48047.1
CP003940.1:2319275-2331311
Cyanobacterium stanieri PCC 7202

Pyr_redox_2 Pep_deformylase AMP-binding AMP-binding_C DUF3153 TPRs DnaJ ? Slr1339 BrnT_toxin ? ACT_9 ACT_9 ACT_9 ACT AA_kinase GCS2 DUF305 ? AhpC-TSA CarboxypepD_reg

ALO46702.1
CP013189.1:2236175-2258098
Pseudohongiella spirulinae

Na_Pi_cotrans Na_Pi_cotrans PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran ? Peptidase_S13 BrnT_toxin CopG_antitoxin UvrA_inter AAA_29 UvrA_DNA-bind UvrA_inter ABC_tran COesterase COesterase HD DUF502 Radical_SAM TauE

AOP33522.1
CP015217.1:1341351-1357600
Leptospira tipperaryensis

HHH_5 KH_NusA_2nd KH_NusA_1st NusA_N RimP_N ? YjeF_N Carb_kinase ? RibD_C Glyoxalase BrnT_toxin CopG_antitoxin ? NamZ_C NamZ_N cNMPBD ? ? ?

ATU05928.1
CP017714.1:1127220-1150974
Candidatus Gracilibacteria bacterium HOT-871

LysM LysM CHAP ? ? T2SSE ? ? ChlI ATPase ATPase Zn_ribbon_LapB BrnT_toxin CopG_antitoxin N_methyl LRRs N_methyl HYR Bact_surface_Ig-like LRRs LRRs LRRs ? LRRs LRRs ? CopG_antitoxin OB_RNB OB_RNB CSD2 RNB S1 TPR_MLP1_2 ODF2_C Helical_needle ? GreA_GreB

BAF89639.1
AP009384.1:4172256-4191812
Azorhizobium caulinodans ORS 571

Gln-synt_C Autotransporter SBP_bac_5 ? ? Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 DnaB_bind Toprim_4 DNAG_N Zn_ribbon_DnaG BrnT_toxin BrnA_antitoxin YqeY CPSase_sm_chain GATase FCD GntR CBS CBS CoA_transf_3 MFS_1 MFS_1

BAP56488.1
AP014633.1:2630311-2642377

Thioploca ingrica

HTH_Tnp_IS630 DDE_3 DDE_5 DDE_Tnp_1 PP-binding ? ? BrnT_toxin ? CheW 4HB_MCP_1 HAMP HAMP_2 HAMP_2 MCPsignal MS_channel_2nd cNMPBD PDDEXK_2 Yae1_N HTH_Rpn_C S4 HSP15_C COQ7

BAY08185.1
AP018172.1:1574648-1586951
Calothrix sp. NIES-2098

AIPR CoiA_N ? ? ? ? N6_Mtase BrnT_toxin ? RPA CRISPR_Cas2 ? DUF6444 ? ?

BAZ36853.1
AP018278.1:13050-27446
Calothrix sp. NIES-4101

Rep3_C Rep3_N ? PhdYeFM_antitox TacA1 Acetyltransf_7 DUF6279 ? RHH_1 BrnT_toxin BrnA_antitoxin ? PIN DDE_Tnp_1 GIIM RVT_1 adh_short_C2 ?

BBB57518.1
AP017965.1:36506-46182

Candidatus Megaera polyxenophila

ParB_N ? ? ? N6_Mtase ? CopG_antitoxin BrnT_toxin ? DDE_Tnp_1 IstB_IS21 Mu-transpos_C_2 rve HTH_29 ? ? ParA-Soj-NTPase

BCG64206.1
AP023240.1:2202066-2216927
Methyloprofundus sp.

DUF2959 AA_kinase Chorismate_bind Anth_synt_I_N DDE_3 Upx-like PIN ? BrnT_toxin CopG_antitoxin UCP028175 MliC ? Pan_kinase BPL_C BPL_LplA_LipB HTH_11 DUF6714 PilZ

CAA6810898.1
CACVAW010000043.1:833-12441
uncultured Campylobacterales bacterium

CbiA ? ? ? TrwB_AAD_bind Rep3_C Rep3_N ? BrnT_toxin CopG_antitoxin DUF2316 ParE_toxin ? DUF7201 PdxA PdxJ

CAB1057319.1
CACVKW010000335.1:400-4297
Olavius sp. associated proteobacterium Delta 1

? BrnA_antitoxin BrnT_toxin BrnT_toxin CopG_antitoxin ParE_toxin PhdYeFM_antitox Formyl_trans_N Formyl_trans_C_2 PIG-L

CAB1061575.1
CACVKW010000888.1:60581-71959
Olavius sp. associated proteobacterium Delta 1

PqiA PqiA ? HKD HKD PLDc_N STAS Sulfate_transp DUF1269 BrnT_toxin CopG_antitoxin RFamide_26RFa WHD_MalT AAA_16 Aldedh CT_A_B CT_C_D ?

CRY95784.1
LN853377.1:1-2279
uncultured prokaryote

BrnT_toxin BrnT_toxin BrnA_antitoxin ? REP_ORF2-G2P ? ?

CUS04705.2
LN890655.2:3079488-3092412
Candidatus Promineifilum breve

DotU ? ? ? ? PIN AAA AAA_assoc_2 MgsA_C BrnT_toxin CopG_antitoxin SH3_2 Peptidase_M20 Chromate_transp Chromate_transp META CofD Abhydrolase_1 GerE

DAE29540.1
BK059095.1:7269-11508
virus sp. ctkyY8

? ? TMP_3 ? ? ? ? ? CopG_antitoxin ? ? ? ? ? ?

EAQ98049.2
CM002299.1:742514-756154

Congregibacter litoralis KT71

HTH_18 Arabinose_bd GST_C_2 GST_N_2 DoxX Y2318_C Isochorismatase Pirin_C Pirin HTH_1 LysR_substrate BrnT_toxin CopG_antitoxin Formyl_trans_N ? HTH_29 HTH_21 rve TonB_dep_Rec_b-barrel Plug Zn_ribbon_ISL3 HTH_28 DDE_Tnp_ISL3

EJF06241.1
AKKQ01000062.1:4063-10149
Thiovulum sp. ES

CcdA PemK_toxin BrnT_toxin CopG_antitoxin ? ? Citrate_synt Ligase_CoA ? DUF805 Citrate_bind Citrate_synth_N

EKD26636.1
AMFJ01035557.1:147-713
uncultured bacterium

BrnT_toxin CopG_antitoxin

EKD27097.1
AMFJ01035347.1:221-729
uncultured bacterium

? CopG_antitoxin

EKD28870.1
AMFJ01034651.1:5258-8192
uncultured bacterium

MazE_antitoxin PIN ? ParE_toxin ? HTH_3 RelB BrnT_toxin CopG_antitoxin

EKD30205.1
AMFJ01034121.1:1-7195
uncultured bacterium (gcode 4)

GatB_N GatB_Yqey ? CopG_antitoxin Methyltransf_12 UPF0004 Radical_SAM zf_Tbcl_2 ? ?

EKD45346.1
AMFJ01028478.1:1-5111
uncultured bacterium

Ribosomal_S20p GTP_EFTU GTP_EFTU_D2 EFG_C BipA_C Y1_Tnp ? CopG_antitoxin MurJ

EKD51959.1
AMFJ01025953.1:158-4811
uncultured bacterium

BrnT_toxin CopG_antitoxin ? PilZ Receiver OmpA Laminin_II YfiO Radical_SAM

EKD66019.1
AMFJ01021653.1:1624-17363
uncultured bacterium (gcode 4)

? T2SSE MatE Polysacc_synt ? ? PSD13_N ? ? ? HTH_40 RQC RecQ_Zn_bind Helicase_C DEAD SYY_C-terminal tRNA-synt_1b CSD

EKD67647.1
AMFJ01021180.1:44-2553
uncultured bacterium

DUF4153 BrnT_toxin CopG_antitoxin tRNA-synt_2c

EKD71890.1
AMFJ01019493.1:27-3491
uncultured bacterium

Cytidylate_kin2 ? ? Lactamase_B_2 ? CopG_antitoxin

EKD72348.1
AMFJ01019366.1:53-1115
uncultured bacterium

BrnT_toxin CopG_antitoxin ? GerE

EKD86208.1
AMFJ01013925.1:1106-1837
uncultured bacterium

? ? CopG_antitoxin

EKD86754.1
AMFJ01013669.1:1-1024
uncultured bacterium

BrnT_toxin CopG_antitoxin ? ?

EKD91278.1
AMFJ01011928.1:20-5917
uncultured bacterium

? Ribosomal_L10 RL10_C Ribosomal_L12_N Ribosomal_L12 BrnT_toxin CopG_antitoxin HTH_17 Glyco_transf_4 Glycos_transf_1 Glyco_tran_WecG 23S_rRNA_IVP Epimerase

EKE00252.1
AMFJ01008322.1:1738-4202
uncultured bacterium

? TraC-like NOMO1-like_2nd PrgI BrnT_toxin CopG_antitoxin

EKE15404.1
AMFJ01004427.1:3626-8647
uncultured bacterium

BrnT_toxin CopG_antitoxin ACT_7 ? Glutaredoxin ? Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 Glyco_transf_4

EMS80477.1
APJX01000002.1:175267-186321
Desulfotignum phosphitoxidans DSM 13687

? ? OEP OEP SapC ? ? Y1_Tnp BrnT_toxin HTH_3 HicB HigB-like_toxin HTH_3 HicB_lk_antitox RHH_1 ParE_toxin

ETR69402.1
ATBP01000654.1:1227-4239
Candidatus Magnetoglobus multicellularis str. Araruama

BrnT_toxin KilA-N DDE_Tnp_ISAZ013

ETR69528.1
ATBP01000623.1:160-4529

Candidatus Magnetoglobus multicellularis str. Araruama

DUF5615 BrnT_toxin ? DUF2335 ?

ETR70109.1
ATBP01000488.1:576-5162

Candidatus Magnetoglobus multicellularis str. Araruama

TetR_N TetR_C_13 adh_short SnoaL_2 SnoaL_2 Oxidored_FMN BrnT_toxin

ETW98483.1
KI931842.1:13527-25237
Candidatus Entotheonella factor

? ? ? FA_desaturase Rieske LigXa_C Bac_luciferase ? BrnT_toxin ? MFS_1 DUF3071 Abhydrolase_1 ? LamininG TSP_3 Acetylase RHH_1

GAK56308.1
DF820464.1:699419-715342

Candidatus Vecturithrix granuli

CcmE DsbD_TM K319L-like_PKD ? CMAS ADH_N ADH_zinc_N Aldo_ket_red BrnT_toxin CopG_antitoxin Amidohydro_2 HSDR_N_2 ResIII Helicase_C EcoEI_R_C HsdM_N N6_Mtase ? Methylase_S Methylase_S UPF0175

GAK58796.1
DF820468.1:283218-303678

Candidatus Vecturithrix granuli

DUF362 Fer4_5 Fer4_5 Fer4_21 Fer4_8 Fer4_7 Fer4_16 Fer4_7 Transketolase_N Transket_pyr Transketolase_C SBP_bac_1 BPD_transp_1 BPD_transp_1 BrnT_toxin NikA-like Caspase AP-ATPase Fer4_16 Fer4_21 Fer2_4 Fer4_10 NADH_4Fe-4S SLBB Complex1_51K 2Fe-2S_thioredx 2Fe-2S_thioredx AFOR_C AFOR_N

GBD95275.1
BDSY01000013.1:12971-18629
bacterium BMS3Abin06

FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 ? ? CopG_antitoxin HigB-like_toxin HTH_3 PhdYeFM_antitox Ribosomal_L17 RNA_pol_A_CTD RNA_pol_A_bac S4 Ribosomal_S4

GBE05954.1
BDTC01000044.1:57751-72333
bacterium BMS3Abin10

MgsA_C AAA_assoc_2 AAA DHDPS TatA_B_E TatC ? Zn_ribbon_8 ? BrnT_toxin CopG_antitoxin Se-cys_synth_N SelA GTP_EFTU GTP_EFTU_D2 Beta-barrel_SelB WHD_2nd_SelB WHD_3rd_SelB GDP_Man_Dehyd Glycos_transf_4 CoA_binding_3 Wzy_C Glyco_transf_9

GBF80889.1
BDQK01000013.1:242515-255185
Aphanothece sacrum FPU1

SurA_N_3 Rotamase PEP-CTERM ? ? ? Beta-lactamase2 Uma2 BrnT_toxin BrnA_antitoxin Acyl_transf_1 ACP_syn_III_C ACP_syn_III FA_synthesis Peptidase_S13 Peptidase_S13 ? KARI_C KARI_N

GBF82669.1
BDQK01000017.1:189975-204233
Aphanothece sacrum FPU1

? MFS_1 MFS_1 RecO_C RecO_N Peptidase_M50 HhH-GPD EndIII_4Fe-2S ? HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB BrnT_toxin BrnA_antitoxin Pps STY-kinase Lipase_GDSL_2 RNA_pol_Rpb6 ? DUF1818

GBR74075.1
BGZN01000028.1:10392-21480
Candidatus Termititenax aidoneus

Anticodon_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 HMGL-like HCS_D2 LeuA_dimer Orn_Arg_deC_N CdaA_N DAC Pantoate_transf Receiver Methyltransf_21 BrnT_toxin CopG_antitoxin Cpn60_TCP1

GBR75001.1
BGZN01000127.1:1-2519
Candidatus Termititenax aidoneus

? BrnT_toxin CopG_antitoxin MatE MatE

GHT25848.1
BOAN01000006.1:19024-28312
Planctomycetales bacterium

YoeB_toxin Glycos_transf_2 Glu_cyclase_2 Lactamase_B Flavodoxin_2 ? ? ? ? N_methyl SBP_bac_10 Y1_Tnp ApbE ? DDE_3 HTH_Tnp_IS630

GHT58465.1
BOAU01000042.1:8976-16815
Spirochaetia bacterium

BrnT_toxin BrnA_antitoxin IPPT RNA_pol_Rpb6 PTA_PTB TPRs TPRs TPRs TPRs SBP_bac_8 BPD_transp_1

GHU23020.1
BOBG01000019.1:2821-22155
Spirochaetia bacterium

RhaA ? AA_kinase ACT_9 ACT_9 NAD_binding_3 Homoserine_dh ? PALP HU-DNA_bdg DUF4469 MmeI_N MmeI_hel MmeI_Mtase MmeI_TRD MmeI_C BrnT_toxin BrnA_antitoxin Oxidored_nitro Oxidored_nitro Bac_export_1 TPRs TPRs TPRs TPR_BSK1_C TPRs TPRs TPRs TPRs TPRs Bac_export_2 ?

GHU65850.1
BOBN01000011.1:5348-16642
Spirochaetia bacterium

ARG Glyco_hydro_4C Glyco_hydro_4 GntR UTRA DUF2281 YafQ_toxin Lactamase_B_2 Fer4_9 Aldo_ket_red BrnT_toxin CopG_antitoxin AAA_14 DUF4143 Hydrolase ? ? Mannitol_dh_C Mannitol_dh APH

GHV03382.1
BNUO01000003.1:52251-66813
Campylobacterota bacterium

Guanylate_cyc DinB ABC_tran ABC_membrane T2SSF T2SSF tRNA-synt_1b LysR_substrate HTH_1 Cache_3-Cache_2 HAMP DUF948 MCPsignal BrnT_toxin CopG_antitoxin Glyco_transf_9 ? ? Pterin_bind DNA_pol3_delta2 HobA

GHV03522.1
BOBW01000014.1:185-12601
Spirochaetia bacterium

SBP_bac_1 DUF3502 BPD_transp_1 BPD_transp_1 MatE MatE Epimerase ? ? BrnT_toxin ? Acetyltransf_4 Cupin_2 GFO_IDH_MocA_C3 GFO_IDH_MocA DinB NTP_transferase Thy1

GHV05037.1
BOBW01000027.1:44-6779
Spirochaetia bacterium

BrnT_toxin BrnA_antitoxin AbiEi_4 AbiEii MatE MatE YitT_membrane DUF2179 RsaX20 Lactamase_B_2 Metalloenzyme

GHV36968.1
BNUS01000004.1:122-15323
Spirochaetia bacterium

HTH_18 BPD_transp_1 BPD_transp_1 SBP_bac_8 Glyco_hydro_88 HTH_19 BrnT_toxin CopG_antitoxin Glyco_hydro_28 DHDPS AP_endonuc_2 Hydrolase Glyco_hydro_3 Alginate_lyase

GHV70976.1
BNUV01000058.1:94-8235
Spirochaetia bacterium

APH Acetylase Rce1-like Acetylase Pox_D5 ? Peptidase_S24 BrnT_toxin ? HATPase_c_3 HSDR_N_2 N6_N4_Mtase

GHV84950.1
BNVB01000011.1:24299-40584
Spirochaetia bacterium

? SpoIIM DUF8413 RDD ? HTH_19 CfpA_C ChlI Mg_chelatase Mg_chelatase_C BrnT_toxin BrnA_antitoxin PHP Amidohydro_1 cNMPBD TPRs TPRs cNMPBD ? dCache_1 HAMP His-Kinase His-Kinase Receiver

GHV94873.1
BNVF01000017.1:553-8868
Spirochaetia bacterium

PBECR2 BrnT_toxin CopG_antitoxin PseudoU_synth_2 LRRs AMP-binding_C_2 AMP-binding POR TPP_enzyme_C POR_N SSF

GHW02336.1
BNUE01000003.1:17718-37034
candidate division SR1 bacterium

REC1 RuvC_1 Methyltransf_1N DNA_binding_1 Fic RuvC_1 RuvC_1 Cas_Cas4 Cas_Cas1 CRISPR_Cas2 PBECR3 CopG_antitoxin PI3K_P85_iSH2 Lactamase_B_3 Xan_ur_permease ? ? Impact_N

GJE16691.1
BPQT01000014.1:20387-36827
Methylobacterium marchantiae

MarR_2 Receiver Aminotran_3 ? Aminotran_1_2 AMP-binding Acyltransferase MFS_1 TetR_C_33 BrnT_toxin HTH_3 PNPOx_N DUF3108 LbH_EIF2B LbH_EIF2B MoaE AI-2E_transport Phage_integrase Phage_int_SAM_1

GJL49054.1
BQIU01000003.1:16236-36827
Nitrospirales bacterium

TonB_dep_Rec_b-barrel Plug STN FecR_C FecR DUF4880 Sigma70_r4_2 Sigma70_r2 Sigma70_r4_2 Sigma70_r2 CzcB_C Beta-barrel_RND_2 BSH_RND FtsX MacB_PCD Sulfotransfer_3 TPRs TPRs TPRs TPRs TPRs BrnT_toxin CopG_antitoxin TonB_dep_Rec_b-barrel Plug STN FecR Sigma70_r4_2 Sigma70_r2 ? ParE_toxin TonB_dep_Rec_b-barrel Plug STN

GJL71002.1
BQJA01000003.1:7633-29222
Nitrosomonas sp.

Sbt_1 Receiver Trans_reg_C HAMP His-Kinase His-Kinase Usp Sulfate_transp RNase_T ExoI_SH3 ExoI_C ? BrnT_toxin CopG_antitoxin PALP Thr_dehydrat_C Thr_dehydrat_C PilJ Terpene_synth_C MCPsignal FimL_C FimL_2nd FimL_C Hpt Hpt Hpt His-Kinase CheW Receiver MutL_C DNA_mis_repair HATPase_c_3 DeoRC Rib_5-P_isom_A PhoU PhoU

GJL87583.1
BQJF01000017.1:3427-15369
Minwuia thermotolerans

CBS Voltage_CLC 4HBT_2 ? Lactamase_B CoA_transf_3 CMD Pantoate_transf DUF4258 CopG_antitoxin DAO Hist_deacetyl ? DSBA GFA ROS_MUCR

GJM07068.1
BQJM01000013.1:22203-35646
marine bacterium B5-7

adh_short_C2 STY-kinase ? DAHP_synth_1 Trp_Tyr_perm AAA_14 DUF4143 ? BrnT_toxin CopG_antitoxin dnstrm_HI1420 RecR_C Toprim_4 RecR_ZnF RecR_HhH YbaB_DNA_bd DNA_pol3_tau_5 DNA_pol3_gamma3 DNAX_ATPase_lid DNA_pol3_delta2 DNA_binding_1 Methyltransf_1N MepB-like

GJM40788.1
BQJW01000148.1:1505-10549
Ardenticatenaceae bacterium

PFK PFK PFK PUA_4 Methyltrans_RNA TP_methylase ? CopG_antitoxin N6_Mtase Methylase_S Methylase_S

GJQ22203.1
BQML01000101.1:431-1740
Bacteroidia bacterium

? BrnT_toxin

GJQ60960.1
BQMY01000001.1:4655-20367
Melioribacteraceae bacterium

HNH_2 ? Pel9A-like_beta_helix Por_Secre_tail PRTase-CE NADAR nSTAND3 ? BrnT_toxin CopG_antitoxin Beta_helix Por_Secre_tail

GMO57094.1
BTIB01000012.1:890-14358
Rickettsiales bacterium

? tRNA_m1G_MT ABC_tran MacB_PCD FtsX DUF3299 TrbL BrnT_toxin CopG_antitoxin HicB_lk_antitox HicA_toxin ?

GMO62129.1
BTIB01000097.1:264-10845
Rickettsiales bacterium

? Hydrolase_like Hydrolase_6 LptF_LptG ? ? BrnT_toxin ? ketoacyl-synt Ketoacyl-synt_C TrmE_N MnmE_helical SSO2105_N ? ?

GMO65363.1
BTHY01000226.1:76-3670
Treponemataceae bacterium

PIN ? ? BrnT_toxin ?

GMR22159.1
BTRH01000047.1:1-7049
Acidobacteriota bacterium

Usp Usp BrnT_toxin CopG_antitoxin Cys_Met_Meta_PP Cys_Met_Meta_PP Guanylate_cyc ? AbiEii PALP

HAC64093.1
DLXW01000182.1:3241-15478
Cyanothece sp. UBA12306

CN_hydrolase NAD_synthase FHA PAS GGDEF EAL Aminotran_1_2 ? ? ? ? BrnT_toxin DUF6364 Phosphorylase

HAJ56176.1
DMEU01000002.1:2324-14898
Candidatus Omnitrophota bacterium

MshEN T2SSF_N T2SSF T2SSF PMSR SelR PK ? PseudoU_synth_2 DUF3656 Peptidase_U32 ? CopG_antitoxin ? ? Lipoprotein_18 Zn_ribbon_LysW ? DUF6504

HAL03060.1
DMLF01000183.1:3123-19780
Lachnospiraceae bacterium

PHP Hydrolase_3 Zn_ribbon_3 Epiregulin_C PP_kinase_C PP_kinase_C_1 PP_kinase PP_kinase_N Ppx-GppA_III Ppx-GppA DUF815 Zn_Ribbon_1 HTH_3 BrnT_toxin ? SBP_bac_8 GGDEF PyrI_C PyrI OTCace OTCace_N Nitroreductase Nitroreductase AAA_lid_6 AAA

HAM36445.1
DMMY01000025.1:117526-133617
Elusimicrobia bacterium

? AAA_32 LonB_AAA-LID Lon_C Zn_Ribbon_1 CDP-OH_P_transf Receiver DAO Y2_Tnp BrnT_toxin CopG_antitoxin BrnT_toxin ? RF-1 Fer4_8 Fer2_3 Succ_DH_flav_C FAD_binding_2 Sdh_cyt

HAO21923.1
DMPH01000345.1:1-5704
Desulfobacteraceae bacterium

BrnT_toxin CopG_antitoxin Aminotran_1_2 Cytidylate_kin Peptidase_M20 Ni_insertion ?

HAO22593.1
DMPH01000427.1:130-3986
Desulfobacteraceae bacterium

Polbeta HEPN ? BrnT_toxin CopG_antitoxin ParE_toxin LRRs

HAP67534.1
DMSO01000174.1:813-11621
Nitrospinae bacterium

PDDEXK_1 PDDEXK_1 PDDEXK_1 UvrD_C UvrD-helicase BrnT_toxin CopG_antitoxin Peptidase_M16 Mug135_C Peptidase_M16_C TsaD ILVD_EDD_N

HAR49835.1
DMVQ01000182.1:7402-18139
Smithella sp.

Thiolase_C FAD_syn Flavokinase Semialdhyde_dh Semialdhyde_dhC PMT_2 TPRs Dala_Dala_lig_C Macro BrnT_toxin CopG_antitoxin DUF2442 DUF4160 Pur_DNA_glyco AI-2E_transport ? TPRs TPRs TPRs TPRs TPRs

HAZ00419.1
DNHP01000240.1:1-2853
Opitutia bacterium

Mem_trans Mem_trans Thr_dehydrat_C Thr_dehydrat_C PALP ? CopG_antitoxin

HAZ07792.1
DNHR01000030.1:20977-29248
Elusimicrobia bacterium

? CopG_antitoxin RHH_1 ParE_toxin BrnT_toxin BrnA_antitoxin BrnT_toxin CopG_antitoxin Peptidase_M48 ? FIST_C FIST NadA

HBA61838.1
DNKL01000150.1:1292-9455
Elusimicrobia bacterium

Hydrolase Tricorn_N ? Aminotran_1_2 Multi_Drug_Res ? ? ? CopG_antitoxin ?

HBC44888.1
DNMI01000014.1:11950-27533
Candidatus Collierbacteria bacterium

FtsK_alpha FtsK_SpoIIIE FtsK_gamma HTH_25 ? ? tRNA-synt_2c tRNA_SAD Baseplate_J ? BrnT_toxin CopG_antitoxin RuvX ? ? YceG ? RNA_pol_Rpb2_2 RNA_pol_Rpb2_3 RNA_pol_Rpb2_45 RNA_pol_Rpb2_6 RNA_pol_Rpb2_7

HBC44941.1
DNMI01000014.1:53567-61777
Candidatus Collierbacteria bacterium

? ? SecY KilA-N ADK Peptidase_M24 TIR ? CopG_antitoxin eIF-1a Ribosomal_L36 Ribosomal_S13 Ribosomal_S11 Ribosomal_S4 S4 RNA_pol_A_bac RNA_pol_A_CTD

HBE01767.1
DNNS01000180.1:219-4097
Spirochaetia bacterium

Peptidase_M20 EpsG BrnT_toxin CopG_antitoxin

HBF12526.1
DNQF01000066.1:1-4862
Deltaproteobacteria bacterium

Casc1_N SurE PCMT Peptidase_M23 Pribosyltran Ribosomal_S21 YqeY BrnT_toxin

HBF13842.1
DNQF01000160.1:30-13222
Deltaproteobacteria bacterium

DUF4258 MqsA_antitoxin CopG_antitoxin BrnT_toxin BrnA_antitoxin ? BrnT_toxin BrnT_toxin CopG_antitoxin ? ? PutA_N Pro_dh Aldedh Methyltransf_25

HBO96792.1
DOFQ01000046.1:1-4585
Candidatus Omnitrophota bacterium

Glycos_transf_1 Phosphodiest Synaptojanin ? Synaptojanin CDP-OH_P_transf ? ? CopG_antitoxin

HBS25667.1
DOIT01000030.1:4748-8470
Gammaproteobacteria bacterium

DUF4258 ? CcdB CcdA Thioredox_DsbH ? DsrC RS_preATP-grasp-like ATP-grasp PurT_C

HCB93571.1
DOYB01000179.1:1-2843
Selenomonas sp.

Beta-Roll Beta-Roll Beta-Roll Beta-Roll BrnT_toxin ? TGS tRNA_SAD

HCC37012.1
DOZS01000136.1:259-4895
Treponema sp.

DEAD Helicase_C_2 BrnT_toxin BrnA_antitoxin ? ? Rrf2

HCC84758.1
DOZC01000067.1:1-11520
Candidatus Paceibacterota bacterium

DNA_processg_A Toprim Topoisom_bac Zn_ribbon_Top1 Zn_ribbon_Top1 CTP_transf_like ? NAD_synthase Hydrolase_3 F420_ligase ? CopG_antitoxin Thymidylat_synt HIT CTP_transf_like dCMP_cyt_deam_1 RibD_C dUTPase

HCH33292.1
DPHD01000069.1:41358-51923
Oceanospirillaceae bacterium

EamA EamA ? DUF1289 ThiF Impact_N DUF1949 TetR_C_7 Lactamase_B_2 ? CopG_antitoxin YjdM Zn_Ribbon_YjdM ? DbpA Helicase_C DEAD Glyoxalase Rbx_binding Pyr_redox_2 Rubredoxin Aldedh

HCM43272.1
DPYK01000216.1:23-4616
Candidatus Omnitrophota bacterium

? CopG_antitoxin ? Tmemb_14 VTT_dom HbpS-like Oxidored_FMN YjdM

HCT84689.1
DPKH01000058.1:727-6952
Candidatus Margulisiibacteriota bacterium

BrnT_toxin CopG_antitoxin DUF1016_N ? Methyltransf_25 ? Methyltransf_11 Peptidase_U32 DUF3656

HDH01021.1
DRCJ01000326.1:707-1214
Nitrospirae bacterium

BrnT_toxin CopG_antitoxin

HDM79392.1
DRBG01001132.1:143-3125
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin Guanylate_kin RNA_pol_Rpb6 MMR_HSR1 YchF-GTPase_C

HDP34981.1
DSBB01000310.1:29-5171

Candidatus Hydrogenedentota bacterium

? CopG_antitoxin BPD_transp_1 BPD_transp_1 SBP_bac_1

HDP81075.1
DSAT01000420.1:267-4406
Spirochaetes bacterium

2-Hacid_dh_C NAD_binding_2 NAD_binding_11 BrnT_toxin CopG_antitoxin Ala_racemase_C Ala_racemase_N

HDQ45675.1
DSAP01000145.1:207-7408
bacterium

? ? PemK_toxin Cupin_2 Lactamase_B Phage_int_SAM_1 Phage_integrase Peptidase_S8 Peptidase_S8 FlgD_ig BrnT_toxin CopG_antitoxin DUF4160

HDY68366.1
DRGJ01000280.1:512-6715
Candidatus Scalindua sp.

Tcf25 ABC_tran ABC-ATPase ABC2_membrane_3 ABC2_membrane_3 MazE_antitoxin PIN PK_C BrnT_toxin CopG_antitoxin

HEC29562.1
DRJK01000122.1:3588-11125
Gammaproteobacteria bacterium

BrnT_toxin ? MTS RLMG_N Radical_SAM EAL cNMPBD ? tRNA-synt_2e tRNA_synt_2f DALR_1

HED13547.1
DRJH01000241.1:368-2729
Gammaproteobacteria bacterium

BrnT_toxin CopG_antitoxin Zeta_toxin HepT-like Polbeta

HES76106.1
DSED01000186.1:1-1531
bacterium

ACR_tran BrnT_toxin CopG_antitoxin Receiver

HET6893561.1
DASSDR010000994.1:521-4810
Pyrinomonadaceae bacterium

ACT ACT ? ? ? MazE_antitoxin PIN MR_MLE_N MR_MLE_C BrnT_toxin CopG_antitoxin

HET7312788.1
DASSKM010000005.1:3226-20674
Salinisphaera sp.

Glycos_transf_2 Glyco_trans_2_3 Acyl-CoA_dh_1 PIG-L Methyltransf_25 AA_permease Peptidase_S58 HTH_6 BrnT_toxin CopG_antitoxin PRODH Pro_dh-DNA_bdg Pro_dh Aldedh DUF4142 Receiver GerE ? CheB_methylest CEP89 PAS His-Kinase His-Kinase Pribosyltran

HET9098683.1
DASTAD010000028.1:42595-53827
Candidatus Saccharimonadales bacterium

? GmrSD_C GmrSD_N ? PIN PhdYeFM_antitox ? CopG_antitoxin ? Abhydrolase_1 Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 AI-2E_transport CysS_C tRNA-synt_1e

HEU0021009.1
DASTLM010000172.1:1-4737
Dehalococcoidia bacterium

Hydant_A_N Hydantoinase_A Hydant_A_C BrnT_toxin CopG_antitoxin PIG-L Hydantoinase_B

HEU5281975.1
DASTZQ010000064.1:18598-29951
Gammaproteobacteria bacterium

Ribosomal_L34 Ribonuclease_P YidD YidC_periplas 60KD_IMP TrmE_N MnmE_helical ANKs ANKs ANKs BrnT_toxin CopG_antitoxin Pribosyltran Indigoidine_A PBP Pyr_redox_2 DUF234 ATPase_2 DUF2177

HEV7235739.1
DASYFX010000484.1:1-1733
Ktedonobacteraceae bacterium

? DUF4258 CopG_antitoxin ?

HEV7453642.1
DASYJQ010000004.1:792-7495
Candidatus Saccharimonadales bacterium

RNase_T ? Hydrolase YjbR ? Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 ?

HEV7504632.1
DASYJD010000028.1:14645-27529
Thermoanaerobaculia bacterium

TP_0789-like AA_permease_2 DUF6855 ? RHH_1 YjbR RuvB_N AAA_lid_4 WHD_RuvB BrnT_toxin CopG_antitoxin DAGK_cat YegS_C DoxX Methyltransf_11 Pribosyltran AMMECR1 AMMECR1 ? DUF6938 DUF6938

HEX2788549.1
DATAJP010000003.1:1038306-1052159
Ignavibacteria bacterium

Por_Secre_tail Reprolysin_5 Lipocalin_4 Por_Secre_tail InhA-like_MAM Peptidase_M14 RF-1 PCRF ? MamF_MmsF ? BrnT_toxin CopG_antitoxin BcrAD_BadFG GATase NAD_synthase GMP_synt_C CAP DUF5798 DUF3450 PhoH

HEX5139744.1
DASUGT010000094.1:149-3442
Dehalococcoidia bacterium

Anth_synt_I_N Chorismate_bind Acetylase NUDIX BrnT_toxin CopG_antitoxin GATase

HEX5641470.1
DASUMX010000081.1:62-5101
Thermoleophilia bacterium

GIY_YIG_2 ATP-grasp_4 ? SLT ? CopG_antitoxin ?

HFD32927.1
DRLV01000268.1:1-2713
Gammaproteobacteria bacterium

HTH_3 ? Peptidase_S9 BrnT_toxin ? ?

HFD87699.1
DRMC01000432.1:1-3587
Gammaproteobacteria bacterium

DUF1415 ? CopG_antitoxin ? OsmC

HFE37242.1
DROU01000130.1:168-2637
Gammaproteobacteria bacterium

? Thioredoxin_3 ArsP_1 ? ?

HFQ79914.1
DRPV01000029.1:80-10075
Desulfobacterales bacterium

MFS_1 MFS_1 MerR_1 ? BrnT_toxin CopG_antitoxin Receiver NtrC-AAA AAA_lid_14 HTH_8 AAA_2 ClpB_D2-small ALAD ARMT1-like_dom ATP_bind_3 ?

HFS85359.1
DRRH01000121.1:1-2387
Epsilonproteobacteria bacterium

? Gp49 dnstrm_HI1420 ? CopG_antitoxin BSH_RND HH_YBHG

HGG59987.1
DRPO01000370.1:727-5290
Gammaproteobacteria bacterium

SelO ? CopG_antitoxin DUF7939 BatD BatD BatD

HGJ66983.1
DSUF01000639.1:1-8189
bacterium

RMMBL Beta-Casp Lactamase_B_6 ? PmbA_TldD_1st PmbA_TldD_2nd PmbA_TldD_3rd ADH_N ADH_zinc_N Alpha_L_fucos Discoidin BrnT_toxin ? Peptidase_C1

HHB13111.1
DRQY01000293.1:1-3112
Chromatiales bacterium

APH BrnT_toxin CopG_antitoxin TOP_N Peptidase_M3

HHI96954.1
DROK01000112.1:76-3106
Thermodesulfatator atlanticus

? His-Kinase His-Kinase Phage_integrase DUF4258 ?

HHY53802.1
DUWG01000001.1:6060-12475
Chloroflexota bacterium

BMC XisH XisI Aldedh BrnT_toxin RHH_1 BMC EutQ Cob_adeno_trans MreB_Mbl PilM_2 ? ParE_toxin

HIC08744.1
DTUX01000063.1:679-2638
Aquificales bacterium

SNase ? ? ?

HID66004.1
DTXP01000293.1:2-2120
Aquificaceae bacterium

BrnT_toxin CopG_antitoxin ? ? ? ?

HIE30574.1
DTYX01001141.1:1-12733
Candidatus Poribacteria bacterium

? zf-HC2 Sigma70_r4_2 Sigma70_r2 TPRs LysM YkuD ECH_1 Peptidase_M20 BrnT_toxin Fumerase Fumerase_C Receiver NtrC-AAA AAA_lid_14 HTH_8 Pyr_redox_2 Fer4_21 Fer2_BFD Zn_ribbon_InsA DAO

HIP19968.1
DQSP01000067.1:1-3374
Sulfurimonas sp.

bAvd Lcl_C RVT_1 Lcl_C BrnT_toxin CopG_antitoxin GTP_cyclohydroI

HIP59425.1
DQTU01000097.1:5127-15545
Campylobacterales bacterium

VitK2_biosynth VitK2_biosynth Sirtuin ThiS MoaE MoeA_N MoCF_biosynth MoeA_C ? AhpC-TSA BrnT_toxin CopG_antitoxin ABC_tran ABC-ATPase PIN PhdYeFM_antitox Aldedh ? DUF7897

HIP61127.1
DQTR01000052.1:180-8318
Sulfurovum sp.

YkuD ABM Radical_SAM SPASM BrnT_toxin CopG_antitoxin Aldedh Aldedh POTRA POTRA Omp85 ?

HIP73210.1
DQTY01001074.1:114-7841
Anaerolineae bacterium

? TGT BrnT_toxin CopG_antitoxin GHMP_kinases_N GHMP_kinases_C TPP_enzyme_C Fer4_9 POR PFOR_II POR_N

HIU78461.1
DVNQ01000050.1:7458-20441
Candidatus Avilachnospira avicola

ABC_tran ABC_tran BD-FAE CitMHS HTH_1 LysR_substrate HTH_1 LysR_substrate tRNA_edit Acylphosphatase BrnT_toxin ? TP_methylase P5CR_dimer F420_oxidored Aldose_epim MTS PSP1_3rd PSP1 DNApol3-delta_C DNA_pol3_delta2

HIV47467.1
DXIW01000043.1:1-7011

Candidatus Acutalibacter stercorigallinarum

Peptidase_Prp Ribosomal_L21p ? PyrI_C PyrI OTCace OTCace_N Transcrip_reg TACO1_YebC_N Acetylase FMN_red BrnT_toxin BrnA_antitoxin Mntp Phosphorylase

HJD59774.1
DYDV01000044.1:22855-33963
Rickettsia endosymbiont of Omalisus fontisbellaquei

? EPSP_synthase RNA_pol_Rpb6 ACPS Rotamase_3 SecA_DEAD P-loop_SecA SecA_SW SEC-C NTP_transf_2 HEPN BrnT_toxin CopG_antitoxin ? ANKs ? ANKs ? ?

HJP18262.1
DASCNB010000122.1:3772-20975
Nitrospinota bacterium

STY-kinase cNMPBD cNMPBD HEATs HEATs HEATs HEATs HEATs Zn_ribbon_4 Zn_ribbon_Brz FHA Guanylate_cyc FHA FHA STY-kinase STY-kinase AMP-binding Peptidase_S24 CopG_antitoxin Ferritin ? 3Beta_HSD ketoacyl-synt Ketoacyl-synt_C PP-binding Amino_oxidase

HJQ07948.1
DATFHU010000004.1:5139-21737
Candidatus Saccharimonadales bacterium

? Glyco_hydro_15 TPRs ? Spermine_synth Peptidase_M28 ? FAD_binding_4 FAD-oxidase_C Flavodoxin_2 BrnT_toxin CopG_antitoxin DUF5753 Beta-lactamase ? Acetylase DUF5753 SLT_3

HJT23312.1
DATFDD010000158.1:1-2381
bacterium

Plug TonB_dep_Rec_b-barrel BrnT_toxin

HJY62582.1
DATEVQ010000013.1:1-4761
Ignavibacteria bacterium

Aldedh BrnT_toxin CopG_antitoxin HicA_toxin HicB_lk_antitox TPRs TPRs vWA vWA DUF6600

HJY62823.1
DATEVQ010000067.1:37-7240
Ignavibacteria bacterium

Big tRNA-synt_2b Seryl_tRNA_N ScdA_N Hemerythrin Cytochrom_C nos_propeller BrnT_toxin

HKJ40983.1
DATFMO010000096.1:148-2474
Sunxiuqinia sp.

Biotin_lipoyl ? BrnT_toxin CopG_antitoxin GHMP_kinases_C GHMP_kinases_N

HKJ68296.1
DATFLS010000405.1:131-4658
bacterium

Glyco_hydro_67M DUF4258 CopG_antitoxin APH ?

HKJ98432.1
DATFKQ010000060.1:52876-66990
Desulfotignum sp.

? ABC_tran ABC-ATPase oligo_HPY ABC_tran oligo_HPY Aminotran_5 MFS_1 BrnT_toxin CopG_antitoxin zf-dskA_traR PAS GAF_2 His-Kinase His-Kinase Zn_ribbon_NOB1 Biotin_lipoyl PYC_OADA Biotin_carb_C CPSase_L_D2 Biotin_carb_N BPL_C BPL_LplA_LipB FMN_red MarR

HKU18601.1
DATEJI010000108.1:203218-217033
Candidatus Saccharimonadales bacterium

PG_binding_1 HlyIII Glyoxalase Endonuc-PvuII ? GuanidinoAc_N-MeTrfase GuanidinoAc_N-MeTrfase TrmB ? CopG_antitoxin ? Hydrolase_3 MethyltransfD12 DUF7687 DUF7690 ? ?

HKV99299.1
DATEGZ010000070.1:28308-37244
Vicinamibacterales bacterium

EamA EamA DUF5916 ADH_N ADH_zinc_N ? Yip1 CBS OMP_b-brl BrnT_toxin CopG_antitoxin DUF3024

HKY87048.1
DATEDM010000365.1:1-9582
Pseudorhodoplanes sp.

? RelA_SpoT Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 Toprim_4 DNAG_N Zn_ribbon_DnaG BrnT_toxin BrnA_antitoxin LprI Pyrophosphatase YqeY CPSase_sm_chain

HLA49435.1
DATESY010000087.1:219-3875
Candidatus Saccharimonadales bacterium

? DUF4618 CopG_antitoxin NTP_transf_2 HTH_27 Fic MazE_antitoxin PIN PhdYeFM_antitox PIN

HLB34139.1
DATEBV010000275.1:139-9355
Chthoniobacterales bacterium

Glyco_transf_9 DUF993 ? Fer4_13 FolB DciA Pyr_redox_dim Pyr_redox_2 BrnT_toxin CopG_antitoxin

HLB60766.1
DATEBW010000102.1:8168-17088
Patescibacteria group bacterium

FIST FIST_C FliJ His-Kinase His-Kinase Receiver LigD_N UDG VIT1 ? BrnT_toxin CopG_antitoxin DUF8173 MazG

HLB60887.1
DATEBW010000121.1:1-10997
Patescibacteria group bacterium

Polysacc_synt BrnT_toxin BrnA_antitoxin PD40 Peptidase_S9_N ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N ? Apolipoprotein Spermine_synth Peptidase_M28

HLF47796.1
DATDPN010000010.1:336-3767
Candidatus Methylomirabilota bacterium

DUF4258 CopG_antitoxin Bac_luciferase FAD_binding_6 NAD_binding_1 TPP_enzyme_N

HLG18584.1
DATDPG010000023.1:5786-15735
Bdellovibrionota bacterium

RNase_H_2 ? Aconitase Aconitase_C ? Archease RtcB DUF6036 ? CopG_antitoxin 2H ? ?

HLO42856.1
DATGXK010000006.1:52691-68617
Leadbetterella sp.

Dioxygenase_C Lyase_1 ADSL_C Abhydrolase_1 CMD FAD_binding_3 CoA_trans CoA_trans DUF983 Thiolase_N Thiolase_C BrnT_toxin CopG_antitoxin Amidohydro_2 Acyl-CoA_dh_C Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N Meth_synt_2 Abhydrolase_6 adh_short AMP-binding_C_3 AMP-binding

HLP48272.1
DATGWR010001130.1:422-8370
Candidatus Kapaibacterium sp.

VWA BrnT_toxin CopG_antitoxin ? NAD_synthase CN_hydrolase ? ? ?

HLP48584.1
DATGWR010001211.1:171-2161
Candidatus Kapaibacterium sp.

DUF2442 DUF4258 CopG_antitoxin ParE_toxin Unstab_antitox

HLP58266.1
DATGXF010000270.1:1-9599
Candidatus Deferrimicrobium sp.

? ? ? DUF4143 AAA_14 BrnT_toxin ? zf-CDGSH Fer4_19 zf-CDGSH Peptidase_M48 Peptidase_M48_N PepSY_TM_like_2 DUF4920 Peptidase_M20 Condensation AMP-binding

HLX12459.1
DATGFY010000302.1:1-11426
Bacteroidota bacterium

? Glyoxalase Por_Secre_tail BrnT_toxin CopG_antitoxin Glycos_transf_2 Glyco_transf_9 Glyco_transf_4 Glycos_transf_1 Methyltransf_23 Epimerase_2 Methyltransf_11

HMK55918.1
DATHDF010000036.1:1926-12462
Dissulfurispiraceae bacterium

FdhD-NarQ PhosphMutase DUF6345 Fascin DUF7910 ? ? DUF6056 TPRs BrnT_toxin CopG_antitoxin

HMK56479.1
DATHDF010000068.1:8999-21313
Dissulfurispiraceae bacterium

PAS NtrC-AAA AAA_lid_14 HTH_8 ? Molydop_binding Molybdopterin Molybdop_Fe4S4 Fer4_7 Fer4_7 NADH-G_4Fe-4S_3 Fer2_4 HTH_1 MoaC Rep_fac-A_C ? BrnT_toxin CopG_antitoxin ? ? ? ? AMMECR1 polyprenyl_synt

HMK60788.1
DATHDG010000022.1:23485-34933
Dissulfurispiraceae bacterium

TonB_2 ExbD MotA_ExbB PAS His-Kinase His-Kinase PfkB ThiI NFACT-R_2 ? BrnT_toxin CopG_antitoxin DNA_alkylation ? ? Apolipoprotein Transgly_assoc ?

HMK61921.1
DATHDG010000046.1:1-5494
Dissulfurispiraceae bacterium

PemK_toxin BrnT_toxin ? ? Fer4 Fer4_5 Fer4_5 Fer4_21 Fer4_7 NapD Molydop_binding Molybdopterin Molybdop_Fe4S4 TAT_signal

HMN71203.1
DAOLAM010000017.1:5780-17288
Rhodoblastus sp.

GyrI-like DAP_epimerase DAP_epimerase UPF0004 Radical_SAM AICARFT_IMPCHas MGS Ribosomal_L19 ? tRNA_m1G_MT BrnT_toxin CopG_antitoxin PRC_RimM RimM Ribosomal_S16 SRP_SPB SRP54 SRP54_N HTH_37 HicB-like_2 HicA_toxin Aminotran_1_2

HMN76976.1
DAOLAT010000026.1:116-6471
Burkholderiaceae bacterium

MFS_1 AAA_28 NMN_transporter DivIC Phage_integrase Phage_integrase Phage_int_SAM_4 RelE HTH_3 BrnT_toxin CopG_antitoxin Acetylase

HMO78312.1
DAOLBU010000021.1:8320-19840
Candidatus Paceibacterota bacterium

? RecA_C RecA_N Glucodextran_B FtsK_gamma FtsK_SpoIIIE FtsK_alpha FtsK_4TM DUF7342 ? ? DUF4258 CopG_antitoxin RelB YafQ_toxin Y1_Tnp ?

HMO78898.1
DAOLBR010000005.1:13240-25220
Candidatus Paceibacterota bacterium

N_methyl ? Nlig-Ia DNA_ligase_aden OB_DNA_ligase HHH_5 HHH_2 BRCT GatC Amidase DUF4145 ? BrnT_toxin CopG_antitoxin ? ? ? ? ? Lipoprotein_21

HMP74065.1
DAOLCW010000121.1:198-6536
Kiritimatiellia bacterium

BrnT_toxin CopG_antitoxin PhdYeFM_antitox Zeta_toxin LacI Peripla_BP_3 AASDHPPT_N ACPS Shikimate_dh_N Ldh_1_N SDH_C A24_N_bact Peptidase_A24

HMV42851.1
DAOLJY010000132.1:74-8668
Leptospiraceae bacterium

MBOAT Lipase_GDSL BrnT_toxin CopG_antitoxin Peptidase_A24 A24_N_bact Aminotran_1_2 ? adh_short_C2 23S_rRNA_IVP 4HBT_2 ?

HMY80757.1
DAOQQF010000019.1:114-20827
Candidatus Absconditabacterales bacterium

? DUF937 ? TPRs SMC_N SLH ? RNase_H MMR_HSR1 RsgA_GTPase MMR_HSR1 KH_dom-like S1 TraG-D_C HerA-FtsK dCache_1 ? ?

HNN00362.1
DAORDF010000133.1:121-7179
Turneriella sp.

? ? BrnT_toxin BrnA_antitoxin ROK MFS_1 Glyco_hydro_1 Tll0287-like DUF4139

HNT97925.1
DAOQUF010000139.1:33-4633
Elusimicrobiales bacterium

? BrnT_toxin CopG_antitoxin ? ? Gpr1_Fun34_YaaH Peptidase_C1 Pps Pps ?

HNW81985.1
DAOQAY010000026.1:1-14481
bacterium

? AdoHcyase SRP54_N SRP54 PilZ PUA_4 Methyltrans_RNA Porin_5 Porin_5 Porin_5 Porin_5 sCache_like HAMP PAS His-Kinase His-Kinase BrnT_toxin CopG_antitoxin PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran PhoU PhoU Collagen Beta_helix_4

HNW93708.1
DAOPZR010000411.1:1-2952
bacterium

? BrnT_toxin CopG_antitoxin ? ? MatE

HNX24737.1
DAOPZT010000122.1:71-6436
Spirochaetota bacterium

BrnT_toxin ? ? ? Guanylate_cyc HAMP PAS Receiver PAS PAS HisKA_2 His-Kinase

HNZ35616.1
DAOPWF010000058.1:1-9629
Syntrophales bacterium

DUF6627 Peptidase_C39_2 TPRs TPRs Lipase_GDSL_2 ? BrnT_toxin CopG_antitoxin Lipase_GDSL_2 Band_7_1 ZnR Zn_Ribbon_1 ? GerE PAS PAS ?

HOD12129.1
DAOPPE010000003.1:19025-29731
Candidatus Omnitrophota bacterium

? 23S_rRNA_IVP ? ? HTH_26 Cupin_2 23S_rRNA_IVP Receiver BrnT_toxin CopG_antitoxin HisKA_7TM His-Kinase His-Kinase 23S_rRNA_IVP HD_5 GAF_2 Receiver 23S_rRNA_IVP His-Kinase GAF_2

HOI83716.1
DAOSDK010000330.1:1-1612
Campylobacterales bacterium

Asn_synthase BrnT_toxin CopG_antitoxin Polbeta ?

HOJ29886.1
DAOSDB010000076.1:3-8413
Spirochaetota bacterium

BPD_transp_1 BPD_transp_1 BrnT_toxin CopG_antitoxin His-Kinase His-Kinase PAS HAMP sCache_like Trans_reg_C Receiver PhoU PhoU ABC_tran PBP_like_2 PBP_like_2

HOU41285.1
DAOSMQ010000221.1:1-4741
Promineifilum sp.

? RNase_PH RNase_PH_C BrnT_toxin CopG_antitoxin Peptidase_M20 ? PhdYeFM_antitox

HOV20432.1
DAOOWE010000235.1:1-3579
Ottowia sp.

BrnT_toxin CopG_antitoxin ? PhdYeFM_antitox PIN T2SSE

HOW74989.1
DAOOTQ010000011.1:6084-19960
Candidatus Competibacteraceae bacterium

MazG MazG NA-iREase1 N6_N4_Mtase DUF3362 Radical_SAM Radical_SAM_N ThiW ABC_tran ABC_tran CbiQ Phage_integrase NADAR BrnT_toxin CopG_antitoxin RHH_1 PIN YchF-GTPase_C MMR_HSR1 Pept_tRNA_hydro Ribosomal_TL5_C Ribosomal_L25p Pribosyl_synth Pribosyltran_N

HOW81294.1
DAOOTA010000022.1:1-12195
Spirochaetota bacterium

DUF4230 DUF456 sCache_2 HAMP MCPsignal DUF2442 DUF4160 ? Glug Glug Glug Glug Glug Glug Glug BrnT_toxin CopG_antitoxin RimP_N DUF150_C NusA_N S1 KH_NusA_1st KH_NusA_2nd HHH_5 IF2_N GTP_EFTU EF-G_D2 IF-2 GTP_EFTU_D2 RBFA TruB_N

HOX17471.1
DAOOSZ010000073.1:47-4721
Spirochaetales bacterium

? CopG_antitoxin MazE_antitoxin PIN PhdYeFM_antitox ParE_toxin Gal_mutarotas_2 Glyco_hydro_31_2nd Glyco_hydro_31_3rd RsmI_C

HOZ02987.1
DAOOPX010000005.1:4262-16632
Candidatus Woesebacteria bacterium

PIN Trp_repressor PALP tRNA-synt_1b tRNA_bind FtsH_ext AAA AAA_lid_3 Peptidase_M41 ? BrnT_toxin CopG_antitoxin 5_3_exonuc_N 5_3_exonuc Ribonuclease_P Glyco_transf_4 Glycos_transf_1 Glyco_hydro_39 WHD_HrcA HrcA NUDIX GrpE

HPA49902.1
DAOONQ010000052.1:140-2575
Methanofastidiosum sp.

BrnT_toxin CopG_antitoxin PRiA4_ORF3 Methyltransf_25

HPA56969.1
DAOONB010000150.1:1-5062
bacterium

? BrnT_toxin CopG_antitoxin C2 CxxCxxCC Glycos_transf_4 TPR_DNAAF5 Glyco_tran_28_C

HPB83544.1
DAOOLF010000597.1:108-2357
Spirochaetota bacterium

? BrnT_toxin CopG_antitoxin HEPN Polbeta

HPE70098.1
DAOPKN010000091.1:1-7159
Thermotogota bacterium

Cohesin Peptidase_C1 Peptidase_C1 EF-hand_7 BrnT_toxin ? MerR_1 Methyltransf_25 PmbA_TldD_1st PmbA_TldD_3rd PmbA_TldD_3rd ?

HPF16192.1
DAOPKA010000005.1:1-13520

Thermotogota bacterium

? Peptidase_M20 DUF454 ? HD PolyA_pol_RNAbd PolyA_pol FGS BrnT_toxin CopG_antitoxin DUF2202 ? ? DUF6144 DUF2624 ?

HPH03655.1
DAOPHP010000153.1:1-6663
Spirochaetota bacterium

RecC_4th Exonuc_V_gamma ? SBP_bac_3 GGDEF Receiver Hpt BrnT_toxin CopG_antitoxin

HPI38817.1
DAOPFC010000213.1:1-4429
Ignavibacteriaceae bacterium

CTP_synth_N Aldedh BrnT_toxin BrnA_antitoxin DUF5009 Peptidase_M17

HPJ40401.1
DAOPDZ010000559.1:15-2184
Spirochaetota bacterium

? ? BrnT_toxin CopG_antitoxin

HPK56674.1
DAOPBR010000029.1:1-6014
Spirochaetota bacterium

HTH_29 ? DUF6364 ? BrnT_toxin CopG_antitoxin RHH_1 PIN Beta-lactamase2 ? ? NUDIX

HPK56858.1
DAOPBR010000035.1:1-10234
Spirochaetota bacterium

EamA ? CopG_antitoxin PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I ATP_bind_3 ? ? RecG_dom3_C Helicase_C DEAD RecG_wedge ?

HPO16799.1
DAONNH010000168.1:5700-13299
Candidatus Hydrogenedentota bacterium

BrnT_toxin CopG_antitoxin PhoPQ_related RsgA_GTPase Methyltransf_25 HGTP_anticodon tRNA-synt_2b His-Kinase His-Kinase Iso_dh

HQF30859.1
DAOOFO010000253.1:1-2592
Hyphomicrobiales bacterium

Aconitase_C BrnT_toxin CopG_antitoxin DUF1223 HSP90

HQH28499.1
DAOODG010001027.1:1-1405
Oligoflexia bacterium

BrnT_toxin CopG_antitoxin Hemerythrin ?

HQJ59831.1
DAOLSJ010000112.1:1-3121
bacterium

vWF_A VWA YfbK_C ? BrnT_toxin ?

HQK63498.1
DAOLQI010000068.1:1-2004
Methanofastidiosum sp.

NAGPA BrnT_toxin

HQK76544.1
DAOLRB010000519.1:397-2771
Candidatus Hydrogenedentota bacterium

Oxidored_FMN BrnT_toxin CopG_antitoxin

HQM83314.1
DAOLNE010000006.1:14772-24252
bacterium

PMT_2 DNA_binding_1 OMP_b-brl HipA_C Couple_hipA HTH_3 CGGC DUF4258 CopG_antitoxin ? GrpB HAD_2 MscL Metal_resist ? Sigma70_r4_2 Sigma70_r2

HQS84922.1
DAOMDB010000124.1:1143-3948
Alphaproteobacteria bacterium

BrnT_toxin CopG_antitoxin DDE_Tnp_1

HQU48647.1
DAOMBL010000089.1:1-6876
Casimicrobiaceae bacterium

Ammonium_transp BrnT_toxin CopG_antitoxin HicB_lk_antitox HicA_toxin Epimerase Glycos_transf_4 Epimerase Methyltransf_21

HRD97872.1
DAORNK010000231.1:1-7376
Rubrivivax sp.

? ? Aminotran_1_2 MLTR_LBD HTH_31 ? BrnT_toxin CopG_antitoxin DUF4124

HRI48039.1
DAOMYS010000499.1:1-2300

Ignavibacteriaceae bacterium

BrnT_toxin CopG_antitoxin DEAD Helicase_C

HRI57837.1
DAOMYU010000770.1:203-1686
Anaerolineae bacterium

Methyltransf_25 BrnT_toxin CopG_antitoxin

HRK60921.1
DAOMWF010000001.1:1-9338
Candidatus Omnitrophota bacterium

Ribonuc_red_lgC Ribonuc_red_sm ? ? ? ? BrnT_toxin CopG_antitoxin TPMT ? Zn_ribbon_TFIIB MS_channel_1st MS_channel_2nd MS_channel_3rd ? ?

HRN68272.1
DAOMSB010000050.1:140-11137
Promineifilum sp.

? MRM3-like_sub_bind SpoU_methylase ? DUF6847 Rv0078B AAA AAA_assoc_2 MgsA_C SH3_2 SH3_2 BrnT_toxin CopG_antitoxin Peptidase_M20 Chromate_transp Chromate_transp ? ?

HRN70848.1
DAOMRU010000157.1:91-3887
Candidatus Woesebacteria bacterium

FTSW_RODA_SPOVE BrnT_toxin CopG_antitoxin Beta_helix fn3

HRN95873.1
DAOMRL010000002.1:3923-19605
Candidatus Levyibacteriota bacterium

P-loop_TraG ? ? ? ? DNA_pol3_beta_3 DNA_pol3_beta_2 DNA_pol3_beta Bac_DnaA_C Bac_DnaA DnaA_N BrnT_toxin CopG_antitoxin ? FOL3_C Mur_ligase_M AI-2E_transport YibE_F Sod_Fe_N Sod_Fe_C

HRP76963.1
DAOMPH010000126.1:1-8835
Rhodocyclaceae bacterium

ABC_tran ABC_membrane NAD_binding_1 FAD_binding_6 BrnT_toxin CopG_antitoxin CitMHS ? ParE_toxin HVO_A0114 TumE ?

HRT20910.1
DAOTNI010000024.1:1-16856

Candidatus Hydrogenedentota bacterium

AF_1845 Aldo_ket_red TAT_signal LeuD_C LeuD Aconitase_C GntR Glyco_hydro_106 ? GFO_IDH_MocA_C2 GFO_IDH_MocA BrnT_toxin CopG_antitoxin BNR_2 IMP_cyclohyd Lyase_1

HRW78115.1
DAOTGU010000358.1:18-1741
Candidatus Sabulitectum sp.

HTH_34 ? BrnT_toxin CopG_antitoxin

HRX15663.1
DAOTGZ010000051.1:9-8142
Spirochaetota bacterium

? Transglutaminase BrnT_toxin CopG_antitoxin Peripla_BP_3 LacI Cellulase Glyco_hydro_12 GIY-YIG UDPGT NYN

HRX49558.1
DAOTGD010000271.1:320-2362
Spirochaetota bacterium

HicB_lk_antitox HicA_toxin PIN ? BrnT_toxin CopG_antitoxin

HSF32017.1
DATIBH010000307.1:1-9417
Candidatus Tectomicrobia bacterium

Glyoxalase DUF4258 ? DUF1326 ? ? 3keto-disac_hyd BrnT_toxin CopG_antitoxin RVT_1 RVT_1 ? RVT_1

HSM81528.1
DATIVG010000198.1:734-5402
Nodosilinea sp.

CobN-Mg_chel BrnT_toxin FAS_N SpoIIE Peripla_BP_5 DUF4278

HSN55420.1
DATIUZ010000551.1:505-3484
Candidatus Sulfomarinibacteraceae bacterium

Phage_int_SAM_4 Phage_integrase BrnT_toxin CopG_antitoxin Methyltransf_25 Peptidase_M23 ?

HSO71367.1
DATITK010000155.1:121-1612
Thermodesulfobacteriota bacterium

Aminotran_5 BrnT_toxin

HSR95746.1
DATIPJ010000048.1:1-11953
Kofleriaceae bacterium

? Peptidase_S8 BrnT_toxin HTH_3 NMT1 BPD_transp_1 ABC_tran K_trans K_trans_C STY-kinase TPRs CsgG

HSV61257.1
DATILH010000208.1:29770-44747
Variovorax sp.

ADH_N ADH_zinc_N MFS_1 AMP-binding AMP-binding_C Cons_hypoth698 TctC NIPSNAP OMPdecase BrnT_toxin CopG_antitoxin Acetyltransf_7 Aminotran_3 Aminotran_1_2 AAA_26 Radical_SAM BATS p450

HSW48289.1
DATIJS010000091.1:37-2043
Candidatus Saccharimonadales bacterium

Peptidase_M23 LysM LysM BrnT_toxin CopG_antitoxin

HSX00334.1
DATIIQ010000041.1:12478-25033
Patescibacteria group bacterium

HTH_MmpR5 MarR_2 FtsX FtsX ABC_tran PadR ? ? ? ? CopG_antitoxin ? BrnT_toxin BrnA_antitoxin tRNA_edit Glycos_transf_2 CarboxypepD_reg CarboxypepD_reg SdrD_B

HSX02631.1
DATIIV010000020.1:108912-122320
Candidatus Saccharimonadia bacterium

FAD_binding_4 ? ABC_membrane ABC_tran PAS His-Kinase His-Kinase ? ? AI-2E_transport BrnT_toxin CopG_antitoxin ? Glyoxalase MFS_1 HxlR MFS_1 ABC2_membrane

HTA76389.1
DATHNZ010000148.1:1-4854
bacterium

GCV_T DUF6036 Flg_bbr_C HTH_24 Fic Fic_N BrnT_toxin CopG_antitoxin Glyco_hydro_12

HTB32412.1
DATHNK010000375.1:1-8950
Bacteroidia bacterium

MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 S1-P1_nuclease 2-ph_phosp GCV_T_C GCV_T PIG-L BrnT_toxin ? DUF1801 ? OprB

HTE21628.1
DATHJH010000005.1:29116-43592
Candidatus Limnocylindria bacterium

Peptidase_S24 IMS IMS_HHH IMS_C DUF4113 HEPN ? ? Pox_VP8_L4R SduA_C BrnT_toxin NikA-like Fic_N Fic adh_short His-Kinase PAS PAS His-Kinase His-Kinase Receiver

HTK03382.1
DATJGY010000019.1:1-10905
Alphaproteobacteria bacterium

Peptidase_S74 DUF559 Epiregulin_C LytR_C TPRs TPRs Peptidase_M32 Prenyltransf BrnT_toxin CopG_antitoxin Glyco_tran_WecG MreB_Mbl PEGA

HTK03784.1
DATJGY010000049.1:1-5641
Alphaproteobacteria bacterium

? BrnT_toxin CopG_antitoxin 5_3_exonuc_N 5_3_exonuc Glyco_transf_4 Glycos_transf_1 Fapy_DNA_glyco H2TH zf-FPG_IleRS VTT_dom SMC_N SMC_N ?

HUC87531.1
DATKHL010000002.1:221-6408
Candidatus Binatia bacterium

LPG_synthase_TM ? ? ? MazE_antitoxin PemK_toxin PsbP_2 ? CopG_antitoxin

HUD03209.1
DATKGW010000004.1:5973-19378
Patescibacteria group bacterium

T4SS_pilin HSP20 PDDEXK_5 Glycos_transf_2 ? ? ? BrnT_toxin CopG_antitoxin Glycos_transf_2 ? HIT ? HTH_3 Hydrolase Val_tRNA-synt_C Anticodon_1 tRNA-synt_1

HUD07412.1
DATKHC010000023.1:8226-20519
Candidatus Saccharimonadales bacterium

Methyltransf_31 MATCAP MATCAP DUF4325 His-Kinase FitA-like_RHH PIN ? Peptidase_M15 BrnT_toxin CopG_antitoxin ? ? ? Glyco_trans_2_3 Zn_ribbon_NrdR ?

HUI70290.1
DATJAF010000226.1:1-3400
Spirochaetia bacterium

BrnT_toxin Antitox_RHH ParA-Soj-NTPase ParB_N HerA-FtsK HerA_C

HUT45866.1
DATLAH010000219.1:1-6231
Sedimentisphaerales bacterium

? Peptidase_M55 ? BrnT_toxin CopG_antitoxin ? GH_C SSO3021_2nd GH_cat DUF5010

HUT65459.1
DATKZJ010000088.1:228-5489
Spirochaetota bacterium

BrnT_toxin CopG_antitoxin MatE MatE Cupin_2 ketoacyl-synt Ketoacyl-synt_C ABC_tran ?

HUV07621.1
DATKXM010000347.1:6-2645
Spirochaetia bacterium

UvrD-helicase UvrD_C UvrD_C BrnT_toxin

HUV62257.1
DATKWR010000005.1:996-19078
Sedimentisphaerales bacterium

Wzy_C Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 Glycos_transf_1 Beta_helix Glyco_transf_4 Glycos_transf_1 Glycos_transf_2 DegT_DnrJ_EryC1 DegT_DnrJ_EryC1 BrnT_toxin CopG_antitoxin Glycos_transf_4 Bac_transf Radical_SAM VanZ DUF5060 Beta-lactamase Nitroreductase DJ-1_PfpI

HUW21065.1
DATKWK010000010.1:258-8948
Candidatus Bathyarchaeia archaeon

MreD CBM_4_9 TAT_signal DUF4258 CopG_antitoxin ? TAT_signal

HVA10857.1
DATLTZ010000051.1:333-5414
Candidatus Dormiibacterota bacterium

? Phage_integrase Phage_int_SAM_1 NDK bPH_2 SurA_N_3 Rotamase ? CopG_antitoxin ?

HVA11094.1
DATLTZ010000092.1:1-5721
Candidatus Dormiibacterota bacterium

FAD-oxidase_C HTH_23 BrnT_toxin CopG_antitoxin His_Phos_1 Peptidase_S11 LemA Peptidase_M48

HVA66712.1
DATLUA010000046.1:82281-95858
Elusimicrobiota bacterium

Thiolase_N Thiolase_C GTP_EFTU EF-G_D2 RF3_C zf-dskA_traR ? ? Methyltransf_23 ? BrnT_toxin CopG_antitoxin Lyase_aromatic Urocanase_N Urocanase Urocanase_C PAS His-Kinase His-Kinase Hemerythrin Hemerythrin CBS CBS TauE

HVE44121.1
DATLOT010000028.1:376-10780
Gammaproteobacteria bacterium

? N_methyl N_methyl Aminotran_5 FumaraseC_C Lyase_1 DUF3458_C DUF3458 Peptidase_M1 Peptidase_M1_N ? BrnT_toxin CopG_antitoxin Methyltransf_23 HAD_2 ?

HVT05447.1
DASZZV010000430.1:5351-13467
Thermoanaerobaculia bacterium

NTP_transf_2 ? PhdYeFM_antitox HicB_lk_antitox DUF1664 ? ? BrnT_toxin CopG_antitoxin Dynamitin ? ? ? BrnT_toxin ParE_toxin

HVW22908.1
DASZVP010000001.1:29987-44573
Candidatus Saccharimonadales bacterium

Sulfate_transp PilO Pox_P4B PilM_2 ? ? WZC_N BrnT_toxin CopG_antitoxin MFS_1 MMR_HSR1 YchF-GTPase_C bPH_3 cNMPBD HTH_Crp_2 Pribosyltran FtsJ S4

HWO17669.1
DATMCI010000022.1:54038-69901
Kofleriaceae bacterium

ABC-ATPase ? PglZ BrxL_N BrxL_ATPase Lon_C ? ? DUF433 BrnT_toxin MqsA_antitoxin ? Bac_luciferase ? ? FTP Peptidase_M4 Peptidase_M4_C PPC AAA_33

HXC39734.1
DATNCF010000365.1:1876-15811
Burkholderiales bacterium

Arg_tRNA_synt_N tRNA-synt_1d DALR_1 SPOR DSBA DSBA Rhodanese DSBA adh_short BrnT_toxin CopG_antitoxin LysE Abhydrolase_1 PhaC_N B_lactamase-like_C Lactamase_B MerR_1 MerR-DNA-bind Thiolase_N Thiolase_C Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1

HXE31305.1
DATNAC010000103.1:1-7665
Terriglobales bacterium

CPSase_L_D1 CPSase_L_D2 MGS PIN RHH_5 ? Glyoxalase ? CopG_antitoxin Tricorn_N Tricorn_N PD40 Tricorn_N Peptidase_S9 Na_H_Exchanger ? ?

HXK31807.1
DATETC010000299.1:1-999
Candidatus Paceibacterota bacterium

fn3 BrnT_toxin BrnA_antitoxin

HXK50298.1
DAOSWU010000159.1:1-5064
Clostridiales bacterium

? BrnT_toxin CopG_antitoxin YgbB DUF748

HXS99886.1
DATNGC010000032.1:3921-13545
Elusimicrobiota bacterium

? PDDEXK_1 FGGY_N FGGY_C ? BrnT_toxin CopG_antitoxin ? Hydrolase ABC_tran ABC_membrane Aminotran_1_2 4HBT ?

HXT83768.1
DATNDA010000263.1:374-2316
Verrucomicrobiia bacterium

BrnT_toxin CopG_antitoxin DDE_Tnp_1

HXV60458.1
DASPGQ010000184.1:4887-19616
Vicinamibacteria bacterium

ParA-Soj-NTPase GDP_Man_Dehyd DUF2730 GAF_2 TPRs TPRs TPRs MacB_PCD FtsX MacB_PCD FtsX ? BrnT_toxin CopG_antitoxin Esterase Esterase Peptidase_S51 CPSase_L_D2 Glycos_transf_1 Glyco_transf_5 adh_short_C2

HXV78963.1
DASPFZ010000113.1:26-4113
Candidatus Binatia bacterium

Rv0959_3rd VWA_2 LexA_DNA_bind Peptidase_S24 23S_rRNA_IVP Phage_int_SAM_4 Phage_integrase BrnT_toxin

HYA93327.1
DASPPQ010000201.1:1-11880
Thermodesulfobacteriota bacterium

DLH DUF1059 HAMP Cache_3-Cache_2 DctP DUF8082 PatA_N Lambda_Bor DJ-1_PfpI BrnT_toxin CopG_antitoxin RHH_1 EamA EamA ADH_N ? ?

HYN07351.1
DASQFI010000030.1:74886-84763
Vicinamibacterales bacterium

ResIII AEP-TOTE BrnT_toxin CopG_antitoxin SEC-C PIN RHH_1 STY-kinase TPRs TPRs NYN OST-HTH TIR DUF2846

HZL69605.1
DASRKS010001047.1:1-3077
Candidatus Limnocylindrales bacterium

Peptidase_M56 Hydrolase BrnT_toxin CopG_antitoxin ?

HZQ29608.1
DASRFS010000015.1:7158-16328
Patescibacteria group bacterium

DUF5666 FMN_bind ApbE ApbE NQR2_RnfD_RnfE FAD_binding_6 NAD_binding_1 DUF1861 ? Replic_Relax BrnT_toxin CopG_antitoxin PhdYeFM_antitox ParE_toxin DUF4258 DUF2283 MMR_HSR1 YchF-GTPase_C ?

KAA3604705.1
QQUF01000016.1:170634-184622
Calditrichaeota bacterium

HTH_8 AAA_lid_14 NtrC-AAA Receiver DUF2203 Carboxyl_trans adh_short_C2 AtuA ? ? BrnT_toxin BrnA_antitoxin MM_CoA_mutase TPRs TAtT TPRs TPRs FKBP_C FKBP_C ParE_toxin RelB-like RelB_like_antitoxin SpoU_methylase

KAA3620228.1
QQUE01000002.1:52304-70041
Calditrichaeota bacterium

DUF3891 DUF3891 OmdA DoxX DUF2141 YitT_membrane DUF2179 DUF3124 ? ? SpoIID HTH_40 HRDC RQC RecQ_Zn_bind Helicase_C DEAD Peptidase_M14 TfoX_N DHDPS G_glu_transpept G_glu_transpept His-Kinase His-Kinase HAMP

KAB2875309.1
WBVC01000068.1:1566-6287
Bauldia sp.

HpcH_HpaI ATPaseInh_sub_z SAICAR_synt PurS DUF16 GATase_5 ? BrnT_toxin BrnA_antitoxin FGAR-AT_linker

KAB2891610.1
WBUQ01000007.1:56348-73600
Desulfobulbaceae bacterium

? DUF2523 Zot type_II_gspD_N0 Secretin_N Secretin_N Secretin ASCH ? Phage_integrase BrnT_toxin CopG_antitoxin HNH Lon_C Lon_lid AAA AAA_PrkA LON_substr_bdg ClpB_D2-small AAA_2 zf-C4_ClpX CLP_protease Trigger_C FKBP_C Trigger_N Thioredoxin_4

KAB2922966.1
WBUG01000038.1:22391-30925
Bacteroidetes bacterium

? CopG_antitoxin Ferritin Por_Secre_tail DUF5009 CtpB_N-like PDZ_6 Peptidase_S41 CHD_PDZ-protease Receiver Trans_reg_C HAMP His-Kinase His-Kinase

KAB8318790.1
JXCB02000002.1:93091-115190
Tolypothrix campylonemoides VB511288

TPRs TPRs RNaseH-CATRA+Caspase STY-kinase CHASE2 Fer2 NAD_binding_1 FAD_binding_6 FHA cNMPBD FHA Fer4_5 DUF928 GTP_cyclohydro2 DHBP_synthase Semialdhyde_dh Semialdhyde_dhC_1 BrnT_toxin CopG_antitoxin SurA_N_2 PIN Enolase_C Enolase_N Fer4_12 Glyoxalase SLH OprB

KAF3982286.1
QPIM01000053.1:6863-21894
Methylococcales symbiont of Hymedesmia sp. n. MRB-2018

? HsdM_N N6_Mtase Virulence_RhuM Methylase_S Methylase_S REase SWI2_SNF2 SWI2_SNF2 T1R_D3-like ParE_toxin PhdYeFM_antitox BrnT_toxin CopG_antitoxin HSP15_C S4 TP_methylase Cons_hypoth95 TP_methylase CbiD CbiC BPD_transp_1 BPD_transp_1_N

KHD07399.1
JSZA02000098.1:4564-22768

Candidatus Thiomargarita nelsonii

Pribosyltran_N Pribosyl_synth Ribosomal_L25p Ribosomal_TL5_C Pept_tRNA_hydro NTP_transf_5 ? Peptidase_S24 SocA_Panacea BrnT_toxin ? ? DUF2085 E2-CBASS ? ? Laminin_I HTH_3

KHE91635.1
JRYO01000188.1:3705-14083
Candidatus Scalindua brodae

Protoglobin DUF4438_C DUF4438 ? CarboxypepD_reg Cytochrome_B Fer4_20 Pyr_redox_2 Fer4_7 Cytochrom_B_C BrnT_toxin CopG_antitoxin ? DUF3584

KIM03135.1
JQIQ01000099.1:46-3539
Sulfurovum sp. PC08-66

? ATP-grasp_2 Ligase_CoA CoA_binding Ligase_CoA ? CopG_antitoxin ParE-like_toxin

KIM10129.1
JQIT01000015.1:276-13335
Sulfuricurvum sp. PC08-66

? PhdYeFM_antitox YoeB_toxin BrnT_toxin CopG_antitoxin Zn_ribbon_DnaG DNAG_N Toprim_4 DnaB_bind DnaGprimase_HBD STY-kinase AAA_16 GAF PAS PAS His-Kinase His-Kinase ? Receiver CheW ?

KKP46105.1
LBOZ01000014.1:2330-10468

Candidatus Woesebacteria bacterium GW2011_GWA2_33_28

LAGLIDADG_2 LAGLIDADG_3 Ribonuc_red_lgC Ribonuc_red_lgC TSCPD 23S_rRNA_IVP Cob_adeno_trans ? CopG_antitoxin dCMP_cyt_deam_1 ? PIN Cupin_3 ? DUF8128 TrwB_AAD_bind

KKP47517.1
LBOZ01000004.1:33449-44490

Candidatus Woesebacteria bacterium GW2011_GWA2_33_28

PIN ? PhdYeFM_antitox PIN tRNA_bind FtsH_ext AAA Intein_splicing AAA AAA_lid_3 Peptidase_M41 VTT_dom BrnT_toxin CopG_antitoxin 5_3_exonuc_N 5_3_exonuc Glyco_transf_4 Glycos_transf_1 ? Fapy_DNA_glyco H2TH zf-FPG_IleRS ? Glycos_transf_1 Glyco_transf_4

KKP60155.1
LBPR01000029.1:3791-11193

Candidatus Roizmanbacteria bacterium GW2011_GWC2_34_23

Amidase GatB_N GatB_Yqey ? HEPN Polbeta 23S_rRNA_IVP BrnT_toxin CopG_antitoxin ? ? PT-HINT

KKP66466.1
LBPX01000032.1:1-9478

Candidatus Roizmanbacteria bacterium GW2011_GWC2_35_12

DCD Thymidylat_synt DHFR_1 ? ? Queuosine_synth TGT TGT BrnT_toxin Metal_resist CopG_antitoxin DUF3271 Formyl_trans_C

KKQ37065.1
LBTI01000029.1:3779-16088

Candidatus Woesebacteria bacterium GW2011_GWA1_37_7

Glycos_transf_4 ? ? ? PMT_2 ? DUF4342 DUF4258 CopG_antitoxin HTH_38 Ferric_reduct Zn_ribbon_recom Recombinase DUF218 HAD MFS_1

KKQ38939.1
LBTJ01000001.1:48660-56585

Candidatus Roizmanbacteria bacterium GW2011_GWA2_37_7

Wzy_C GIY-YIG BrnT_toxin BrnA_antitoxin Hydrolase_2 ? ? Glyco_transf_4 Glycos_transf_1 Alpha-amylase SusG_C OB_DEPS-1_6th

KKQ41162.1
LBTP01000046.1:2560-7023

Microgenomates group bacterium GW2011_GWC1_37_8

NUDIX Ribosomal_S3Ae CopG_antitoxin NUDIX ? SET TEBP_beta HTH_3 DUF5678

KKQ53456.1
LBUC01000017.1:5745-14951

Candidatus Falkowbacteria bacterium GW2011_GWF2_38_1205

N_methyl PspC BrnT_toxin CopG_antitoxin PIN PhdYeFM_antitox GalP_UDP_transf GalP_UDP_tr_C Transgly Transpeptidase Big DNA_pol3_beta_3 DNA_pol3_beta_2 DNA_pol3_beta Bac_DnaA_C Bac_DnaA DnaA_N

KKQ74352.1
LBUZ01000034.1:2882-10584

Candidatus Woesebacteria bacterium GW2011_GWB1_38_5b

Glyoxalase ? ? DUF5674 ? HicA_toxin HicB_lk_antitox BrnT_toxin BrnA_antitoxin ? Big_13 CHB_HEX_C_1 Pur_ac_phosph_N fn3 ?

KKQ74526.1
LBUZ01000030.1:1-8773

Candidatus Woesebacteria bacterium GW2011_GWB1_38_5b

DUF5680 RibD_C ? ? DUF1801 ? ? ? CopG_antitoxin Gp49 HTH_3 His-Kinase DUF4325

KKR03387.1
LBWD01000002.1:40227-50809

Microgenomates group bacterium GW2011_GWC1_39_12

Wzy_C ? ? YfhO ? NPCBM GT87 ? DUF4258 CopG_antitoxin ?

KKR31951.1
LBXN01000056.1:3819-10845

Candidatus Gottesmanbacteria bacterium GW2011_GWC2_39_8

SLBB HHH_3 Competence MBL_MAP1B GDPD Glyco_transf_4 Glycos_transf_1 Abhydrolase_1 VTT_dom BrnT_toxin CopG_antitoxin

KKR47145.1
LBYE01000010.1:1-8629

Parcubacteria group bacterium GW2011_GWC1_40_13

? Cu-oxidase_3 BrnT_toxin CopG_antitoxin ABC_tran_CTD PCRF RF-1 Synaptojanin ABC_tran FtsX_ECD FtsX CTP_synth_N GATase LAGLIDADG_3

KKR78623.1
LBZW01000030.1:1-14723

Candidatus Nomurabacteria bacterium GW2011_GWA2_40_9

GTP_EFTU GTP_EFTU_D2 EFG_C BipA_C Sigma70_r4 HARE-HTH ? YceG YceG tRNA_Me_trans tRNA_Me_trans_M tRNA_Me_trans_C MgtC tRNA-synt_2c tRNA_SAD BrnT_toxin CopG_antitoxin Ribosomal_L21p RecR_C Toprim RecR_HhH DnaB_C DnaB tRNA-synt_1b ?

KKR80767.1
LBZY01000010.1:1-9158

Candidatus Daviesbacteria bacterium GW2011_GWC1_40_9

Polbeta TMCO5 SpoIID bPH_2 DUF5657 ? SmpB DZR_2 Pribosyltran BrnT_toxin CopG_antitoxin ? PhdYeFM_antitox ? DUF5652 ?

KKS79872.1
LCEW01000027.1:982-7186

Candidatus Beckwithbacteria bacterium GW2011_GWA2_43_10

? ? CopG_antitoxin Pribosyltran_N Pribosyl_synth Pribosyltran dNK GATase NAD_synthase GMP_synt_C dCMP_cyt_deam_1 PNPase

KKT47291.1
LCIC01000011.1:16382-33241

Parcubacteria group bacterium GW2011_GWC2_44_17

PrgI TraC-like P-loop_TraG ? ? NYN T4SS_pilin BrnT_toxin CopG_antitoxin Peptidase_M23 LysM LysM tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 tRNA-synt_1 Anticodon_1 Sua5_yciO_yrdC ABC_tran UvrA_DNA-bind ABC_tran ?

KKT64681.1
LCIY01000053.1:116-6542

Candidatus Woesebacteria bacterium GW2011_GWA2_44_33

BrnT_toxin CopG_antitoxin ? DUF5667 TPR_28 DUF5667 DUF7768

KKU03496.1
LCKS01000001.1:105054-122025

Candidatus Amesbacteria bacterium GW2011_GWC2_45_19

PMT_2 PTPS_related ? PMT_2 PMT_2 Glycos_transf_2 Glyco_transf_4 Glycos_transf_1 BrnT_toxin CopG_antitoxin Mannosyl_trans2 Glycos_transf_2 DUF3863 Glycos_transf_2 Glyco_transf_4 Glycos_transf_1 Glyco_transf_4 Glycos_transf_1

KKU81046.1
LCOR01000047.1:5575-13749

Parcubacteria group bacterium GW2011_GWA2_47_8

AI-2E_transport TP_methylase TatD_DNase BrnT_toxin CopG_antitoxin tRNA-synt_1g tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 ? ? CARDB Kre9_KNH

KKU87956.1
LCOZ01000005.1:1257-13109

Candidatus Beckwithbacteria bacterium GW2011_GWC2_47_9

DUF6922 Glyco_tran_WecG Aldedh Aldedh Glycos_transf_1 Glyco_transf_4 MreB_Mbl MerR HTH_17 BrnT_toxin ? ? PEGA Sua5_yciO_yrdC TsaE HIT NYN

KKU93826.1
LCPE01000016.1:1942-10211

Candidatus Amesbacteria bacterium GW2011_GWC1_48_10

ATP-cone DUF5652 MgtE DUF7282 DUF3566 ? Phage_holin_4_2 BrnT_toxin CopG_antitoxin Beta-Roll Beta-Roll Beta-Roll LTD Peptidase_S26 Peptidase_M73 ? PMT_2

KKW10734.1
LCQF01000027.1:15488-29727
Parcubacteria group bacterium GW2011_GWA2_49_9

HTH_3 ? SNF2-rel_dom Helicase_C GmrSD_N DUF6364 YafQ_toxin HhH-GPD BrnT_toxin CopG_antitoxin ? DUF2339 DUF2339 DUF3999 HhH-GPD RNHCP Trypsin

KKW46142.1
LCSD01000032.1:8328-15564

Candidatus Kaiserbacteria bacterium GW2011_GWA2_58_9

? BrnT_toxin BrnA_antitoxin NTP_transferase MS_channel_1st_1 MS_channel_1st_1 HTH_20 Ribosomal_L12 Ribosomal_L12_N RL10_C Ribosomal_L10

KPA14344.1
JPDT01001541.1:546-1101
Candidatus Magnetomorum sp. HK-1

BrnT_toxin ?

KPA17665.1
JPDT01000564.1:1-16113
Candidatus Magnetomorum sp. HK-1

CitMHS NUDIX Receiver His-Kinase His-Kinase Receiver Receiver His-Kinase His-Kinase PAS Receiver His-Kinase ? AAA ? ? ? ArsA_ATPase ArsA_ATPase ArsA_ATPase Lactamase_B Beta-Casp RMMBL Oxidored_FMN Pyr_redox_2

KQC12376.1
LKUE01000363.1:233-7380
Desulfuromonas sp. SDB

CopG_antitoxin MazE_antitoxin PemK_toxin Rce1-like ? BrnT_toxin CopG_antitoxin APH NYN CutA1 PEMT PEMT ?

KXK48627.1
LLZO01000230.1:7196-22870
Chlorobi bacterium OLB5

Phenol_MetA_deg ? Thiolase_N Thiolase_C CDP-OH_P_transf PgpA MoCF_biosynth CinA_KH CinA ABC_tran BrnT_toxin CopG_antitoxin UbiA ? SBBP SBBP SBBP Por_Secre_tail PSII_BNR PSII_BNR PSII_BNR ? Por_Secre_tail

KYC52829.1
LNJB01000036.1:130-5794
Candidatus Methanofastidiosum methylothiophilum

LamininG ? ? T2SSE BrnT_toxin CopG_antitoxin

KYJ86428.1
LNKT01000034.1:196893-206012
Sulfurovum riftiae

? Rhomboid ? Metallothio_Pro GidB PqqD Thioredoxin_2 BrnT_toxin CopG_antitoxin Asn_synthase GATase_7 ? Hpr_kinase_C

MAE59488.1
NZAF01000176.1:1-9776
Halobacteriovorax sp.

? ? SSB HTH_17 Phage_int_SAM_3 Phage_integrase BrnT_toxin CopG_antitoxin HTH_3 HTH_3 ThiF ? TPRs Radical_SAM

MAT57706.1
PAAX01000014.1:129373-143286
Ignavibacteriae bacterium

? ADI MFS_1 MFS_1 ? ? Cytochrom_C HGTP_anticodon tRNA-synt_2b GlyRS_ins_1 BrnT_toxin CopG_antitoxin ? ? ABC2_membrane_3 GIY-YIG ABC2_membrane_3 DrrA1-3_C ABC_tran

MAY27093.1
PAKU01000036.1:56869-69162
Polycyclovorans sp.

FMN_bind Rnf-Nqr HhH-GPD EndIII_4Fe-2S ? Ribosomal_L25p Ribosomal_TL5_C Pept_tRNA_hydro MMR_HSR1 YchF-GTPase_C BrnT_toxin CopG_antitoxin Dus Pro_isomerase Calcineurin Acetyltransf_10 MFS_1_like DbpA DeaD_dimer Helicase_C DEAD

MAZ77673.1
PANR01000030.1:2345-16592

Legionellaceae bacterium

? AstE_AspA_cat MFS_1 ? ? MOV-10_beta-barrel PQ-loop ? CopG_antitoxin ? ? Aminotran_1_2 TMP_3 ResIII UvrB_inter UvrB_3rd Helicase_C UvrB UVR RebB

MBA1421574.1
VCAW01000252.1:202-3160
Epsilonproteobacteria bacterium

BrnT_toxin CopG_antitoxin ? Cons_hypoth95 NLPC_P60 SH3_7 SH3_6 N_NLPC_P60

MBA1432535.1
VCAX01000015.1:5659-19271
Epsilonproteobacteria bacterium

OAD_beta Receiver CheW Methyltransf_36 Molybdopterin Molydop_binding Aminotran_3 OEP Methyltransf_28 BrnT_toxin CopG_antitoxin T2SSE ArsC GatC ? Beta-prop_RSE1_2nd Phage_int_SAM_1 Phage_integrase

MBA2494323.1
JACCYR010000235.1:193-4021
Acidobacteria bacterium

DHO_dh Pribosyltran ? ? OTCace_N OTCace BrnT_toxin BrnA_antitoxin

MBA3047808.1
VMTK01000062.1:22800-34281
Candidatus Falkowbacteria bacterium

Ribosomal_L5 Ribosomal_L5_C Ribosomal_S14 Ribosomal_S8 Ribosomal_L6 Ribosomal_L6 Ribosomal_L18p Ribosomal_L18p Ribosomal_S5 Ribosomal_S5_C Ribosomal_L27A BrnT_toxin CopG_antitoxin SecY ABC_tran MacB_PCD FtsX ? Na_Ca_ex Na_Ca_ex ADK

MBA3533002.1
JACCSL010000609.1:84-7838
Ardenticatenales bacterium

? FeeM Aminotran_5 DoxX_2 Abhydrolase_1 UDPGT Methyltransf_25 BrnT_toxin ? ? DUF2981

MBA3537615.1
JACCSX010000291.1:318-7078
Tatlockia sp.

? CopG_antitoxin PFIN ? Phage_integrase DDE_Tnp_Tn3 DUF4158 YhcG_C DUF1016_N

MBA3582895.1
JACDDA010000013.1:329-5147
Gammaproteobacteria bacterium

? ? CopG_antitoxin ? DDE_Tnp_IS66 ? DDE_Tnp_Tn3

MBA3632707.1
JACDDO010000046.1:1-13320
Acidobacteria bacterium

CDO_I Lipase_GDSL E1_dh Transket_pyr Transketolase_C BrnT_toxin ? Biotin_lipoyl E3_binding 2-oxoacid_dh TonB_C FGS STY-kinase EamA EamA DUF2231

MBA3724345.1
JACDET010000028.1:3636-15200
Candidatus Levyibacteriota bacterium

Peptidase_C39_2 TPRs YkuD TruB_N TruB_C_2 HAD_2 ? DnaJ NUDIX BrnT_toxin CopG_antitoxin NTPase_I-T SelR Glyoxalase Dockerin_1 GO-like_E_set Glyoxal_oxid_N Pur_ac_phosph_N Big ? Peptidase_U49

MBA3956911.1
JACDHD010000002.1:57691-69869
Parachlamydiaceae bacterium

ParG DAHP_synth_1 ? MepB-like DUF2992 zf-RING_2 ? BrnT_toxin CopG_antitoxin His_Phos_1 zf-RING_2 Sel1 Sel1 Sel1 Sel1 TPR_27 Abhydrolase_1 OsmC NodZ

MBA4389783.1
PNOF01000005.1:67442-73069
Syntrophus sp. (in: bacteria)

BrnT_toxin CopG_antitoxin sCache_2 DUF3467 Inhibitor_I42 EF-hand_5 EF-hand_7 EF-hand_6 tRNA-synt_2b HGTP_anticodon Creatinase_N Peptidase_M24

MBA5248555.1
VCHC01000122.1:593-11698
Gammaproteobacteria bacterium

Cons_hypoth95 Pyrophosphatase HRDC DNA_pol_A_exo1 HAMP His-Kinase His-Kinase Receiver HTH_8 TrkA_N TrkA_C TrkA_N TrkA_C TrkH BrnT_toxin CTD10 MC6 ?

MBA5249336.1
VCHC01000433.1:1-1405
Gammaproteobacteria bacterium

ANTAR Receiver ? ?

MBC2713731.1
JABZFM010000001.1:66851-79718
Desulfobacteraceae bacterium

DRTGG FeS Fe_hyd_lg_C Fe_hyd_lg_C ? DRTGG ? 2Fe-2S_thioredx 2Fe-2S_thioredx Complex1_51K SLBB_2 NADH_4Fe-4S Fer4_10 BrnT_toxin CopG_antitoxin Thiolase_C Thiolase_N ECH_1 3HCDH 3HCDH 3HCDH_N LolA NadA MazE_antitoxin PIN

MBC6721015.1
JACSET010000011.1:24174-41789
Treponema sp. Marseille-Q4130

ABC_tran NIL BPD_transp_1 DUF5312 ? Big Big LRRs LRRs LRRs LRRs SBP_bac_5 TP_0584_C BrnT_toxin BrnA_antitoxin DUF8086 ? Cpn60_TCP1 Pallilysin ? ?

MBC7376475.1
JACMRR010000103.1:1-8891
Burkholderiaceae bacterium

p450 BrnT_toxin CopG_antitoxin TonB_dep_Rec_b-barrel Plug ? ABC_tran BCA_ABC_TP_C ABC_tran BPD_transp_2 BPD_transp_2

MBC8127156.1
JACMIY010000256.1:1-6388
Verrucomicrobiales bacterium

GATase_5 BrnT_toxin CopG_antitoxin Trigger_N FKBP_C Trigger_C CLP_protease zf-C4_ClpX AAA_2 ClpB_D2-small POR

MBC8245929.1
JACNJE010000181.1:1-10134
Verrucomicrobia bacterium

Pec_lyase ? AstE_AspA_cat BATS Radical_SAM DEAD Helicase_C RecQ_Zn_bind RQC HRDC SGSH_C Sulfatase RNase_H_2 BrnT_toxin CopG_antitoxin STY-kinase ?

MBC8386992.1
JACNIX010000103.1:8829-17820
Actinomycetota bacterium

? CorA AAA_14 DUF4143 HEPN NTP_transf_2 HTH_17 ? Tricorn_N BrnT_toxin CopG_antitoxin PRiA4_ORF3

MBC8549495.1
JACNFJ010000338.1:391-6118
Candidatus Brocadiales bacterium

HTH_3 Peptidase_S24 BrnT_toxin CopG_antitoxin SKI Lipase_GDSL_2 Receiver His-Kinase His-Kinase PAS PAS HAMP Tll0287-like

MBC8555405.1
JACNFJ010001578.1:151-2821
Candidatus Brocadiales bacterium

TIR ? TIR BrnT_toxin BrnA_antitoxin DrHyd

MBD3245165.1
WJJL01000060.1:1-9275
Candidatus Moraniibacteriota bacterium

? N6_Mtase HSDR_N_2 ? ? BrnT_toxin CopG_antitoxin ArsP_1 DUF1576 Zn_ribbon_6 DUF553 Zn_ribbon_RanBP YidD HrcA

MBD3339002.1
WJKD01000264.1:59-1896
Promethearchaeota archaeon

HTH_3 BrnT_toxin CopG_antitoxin ?

MBD3816771.1
JACXUA010000410.1:1-1947
Halothiobacillus sp.

DUF808 BrnT_toxin CopG_antitoxin Chromate_transp Chromate_transp

MBD3878732.1
JACXWG010000017.1:1177-20008
Quinella sp. 1Q5

RNA_pol_A_bac RNA_pol_A_CTD Ribosomal_L17 FhaA_N FHA Anticodon_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Pribosyltran DZR_2 CPSase_sm_chain GATase CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS BrnT_toxin ? Lactamase_B Competence DUF4131 Mrr_cat Helicase_C SNF2-rel_dom EH_Signature ? OmpA RmuC YpjP

MBD5403851.1
JAAVBJ010000014.1:1-16988
Treponema sp.

FlgD_ig PEGA RadC UPF0758_N ATPase_2 Alpha-amylase Y_phosphatase3 ParA GmrSD_N BrnT_toxin BrnA_antitoxin PEPCK_GTP PEPCK_N ? HTH_3 HTH_3 ?

MBD5408642.1
JAAVBI010000030.1:4072-15874
Treponema sp.

Pur_ac_phosph_N ? PIN BrnT_toxin ? ? ? BrnT_toxin BrnA_antitoxin ? ? PIN DUF8402 fn3 Ig_RIMBP2 ? ?

MBD5426599.1
JAAVAS010000027.1:5737-17467
Treponema sp.

? Zeta_toxin PIN ? BrnT_toxin BrnA_antitoxin Cyclophil_like2 Transgly Transpeptidase BiPBP_C ? DUF6261 LRRs LRRs LRRs Pept_tRNA_hydro

MBD5485090.1
JAAUZY010000031.1:60333-68978
Lachnospiraceae bacterium

SBP_bac_8 ? ? Sigma70_r2 Sigma70_r4_2 ? Sigma70_r2 Sigma70_r4_2 DUF4179 BrnT_toxin ?

MBE0496277.1
JACUTV010000039.1:1-7364
Campylobacterales bacterium

PAS HTH_3 HipA_C BrnT_toxin CopG_antitoxin ? YjbR DUF4143 AAA_14 ? MbnB_TglH_ChrH DUF2063

MBE2234497.1
JACSRL010000456.1:1-4649
Anaerolinea sp.

MoCoBD_1 MoCoBD_2 MoCoBD_2 BrnT_toxin CopG_antitoxin Aminotran_1_2 2-Hacid_dh_C

MBE3065151.1
JADBLZ010000496.1:1-3973
Spirochaetes bacterium

BrnT_toxin CopG_antitoxin nSTAND2 Guanylate_cyc nSTAND2 AAA_16 ?

MBE3118265.1
JADBLG010000267.1:1-2601
Candidatus Atribacteria bacterium

? AAA_14 DUF4143 BrnT_toxin CopG_antitoxin CHASE4

MBE6136618.1
SVBE01000014.1:139-16562
Erysipelotrichaceae bacterium

BPD_transp_1 RCC1 WD40_RLD OppC_N BPD_transp_1 BPD_transp_1_N SBP_bac_5 Pribosyltran Virulence_RhuM Fic DUF1016_N YhcG_C YhcG_C BrnT_toxin CopG_antitoxin AAA_13 ABC_tran BPD_transp_1 SBP_bac_3 PNPase HD_5 AA_permease_2 HD_5

MBE7469584.1
JABTUK010000002.1:275887-297749
Anaerolineales bacterium

PRAI PRAI NAD_binding_10 DUF2867 PALP tRNA-synt_1b Trp_syntA HTH_29 TPRs TPRs TPRs YfiO SLT BrnT_toxin CopG_antitoxin GAF_2 PAS His-Kinase His-Kinase LysM Calx-beta Calx-beta His-Kinase His-Kinase HAMP dCache_3 TrmE_N MnmE_helical ?

MBE7533491.1
JABTTU010000002.1:1568636-1586521
Ardenticatenaceae bacterium

? CHAP His_Phos_1 STY-kinase ? HEPN_RiboL-PSP SH3_3 Ub-Mut7C Mut7-C BrnT_toxin CopG_antitoxin GAF_2 SpoIIE STY-kinase AAA_16 TPRs TPR_MalT TPRs DUF433 Receiver Trans_reg_C Nitroreductase Abhydrolase_1 cNMPBD

MBE8189663.1
JAAOIF010000030.1:1-9132

Candidatus Thioglobus sp.

His-Kinase HAMP DNA_pol_A_exo1 SIS GATase_6 PHP BrnT_toxin CopG_antitoxin Sua5_yciO_yrdC MazE_antitoxin PemK_toxin tRNA-synt_2c tRNA_SAD Snf7 DHHA1

MBE8221312.1
JAAOIH010000002.1:38861-55050
Bdellovibrionales bacterium

? AAA_14 DUF4143 ? ResB ResB Cytochrom_C_asm Cpn10 Cpn60_TCP1 BrnT_toxin CopG_antitoxin FAD-oxidase_C FAD_binding_4 Hpt His-Kinase His-Kinase Receiver HNH HATPase_c_3 Topo-VIb_trans

MBE9018512.1
JADEXX010000285.1:1-7943
Chroococcidiopsidales cyanobacterium LEGE 13417

SNF2-rel_dom Intein_splicing SNF2-rel_dom DUF3670 ? BrnT_toxin ? ? NTP_transf_2 HepT-like ? Methyltransf_31 N_methyl Pilin_GH

MBE9525719.1
JADFVK010000046.1:1-6221
Proteobacteria bacterium

DUF4124 PhdYeFM_antitox UPF0175 DUF3368 ? CopG_antitoxin BrnT_toxin PIN BrnT_toxin CopG_antitoxin MazE_antitoxin PemK_toxin ? ?

MBF0101382.1
JADGCZ010000070.1:167-12796
Desulfobacterales bacterium

PAS His-Kinase His-Kinase PAS His-Kinase His-Kinase Receiver DUF4160 DUF2442 Receiver His-Kinase His-Kinase ATPase ATPase KaiB BrnT_toxin ? Hpt H-kinase_dim His-Kinase CheW HEPN Polbeta PAS His-Kinase His-Kinase Receiver

MBF0103479.1
JADGCZ010000388.1:156-3396
Desulfobacterales bacterium

SpoIIE His-Kinase ? CopG_antitoxin NTP_transf_2 HEPN CheW

MBF0103805.1
JADGCS010000002.1:159-11331
Deltaproteobacteria bacterium

Receiver His-Kinase His-Kinase PAS PAS Receiver DUF4379 DUF4379 DUF4379 Fer4_10 FAD_binding_4 FAD-oxidase_C ? Fer4_8 CCG CCG DAGK_cat YegS_C BrnT_toxin CopG_antitoxin AbiEii AF_1314_C ?

MBF0106891.1
JADGCS010000114.1:246-13989
Deltaproteobacteria bacterium

DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 DUF4215 TEN_NHL DUF4215 DUF4215 AbiEii BrnT_toxin CopG_antitoxin ? DUF4143 AAA_14 ParE_toxin ?

MBF0135443.1
JADGCO010000009.1:30342-47217
Magnetococcales bacterium

LD_cluster2 ? ? HATPase_c_3 HSP90 ? ParA-Soj-NTPase ? BrnT_toxin CCDC90-like Fer4_2 Fer4_16 Fer4_7 Fer4 Fer4_5 Fer4_5 DUF362 DUF6599 Fer4_10 Aldo_ket_red CheW Receiver HTH_8 AAA_lid_14 NtrC-AAA PAS_10 Receiver

MBF0200628.1
JADGBO010000047.1:123-17442
Desulfamplus sp.

Hpt Receiver His-Kinase His-Kinase His-Kinase Receiver His-Kinase His-Kinase Receiver Receiver Receiver NMT1_2 CheW ? BrnT_toxin KH_I_RBM43 Receiver Hpt H-kinase_dim His-Kinase CheW Acyltransferase Arginosuc_synth Arginosuc_syn_C Lyase_1 ASL_C2 ?

MBF0201898.1
JADGBO010000185.1:56-6838
Desulfamplus sp.

RepB_primase Resolvase HTH_7 DUF4160 DUF2442 BrnT_toxin CopG_antitoxin ? Phage_integrase ? PemK_toxin RHH_1

MBF0217450.1
JADGBR010000035.1:14062-33333
Candidatus Omnitrophota bacterium

BrnT_toxin BrnA_antitoxin Rrf2 PBP_like_2 BPD_transp_1 BPD_transp_1 ABC_tran BrnT_toxin CopG_antitoxin ? CHASE DUF2730 His-Kinase His-Kinase Receiver Receiver Hpt CHASE PAS PAS HAMP PAS His-Kinase His-Kinase Receiver Receiver Hpt Receiver His-Kinase FGGY_N FGGY_C

MBF0217882.1
JADGBR010000086.1:196-2278
Candidatus Omnitrophota bacterium

RmuC ? NikA-like

MBF0238656.1
JADGBI010000066.1:12254-25978
SAR324 cluster bacterium

Receiver Oxidored_molyb MamF_MmsF ? SHMT PilZ CTP_transf_like BrnT_toxin ? TPRs TPRs TPRs TPRs TPRs TPR_NPHP3 Hexapep Hexapep Hexapep Acetyltransf_11 Glyco_hydro_20b Glyco_hydro_20 N_methyl T2SSG N_methyl T2SSG DUF433

MBF0239951.1
JADGBI010000158.1:1-10910
SAR324 cluster bacterium

SpoIIE GAF AAA_16 STY-kinase ? BrnT_toxin BrnA_antitoxin Phage_integrase Phage_int_SAM_5 DUF6788 ? HTH_17 BT4734-like_N

MBF0240706.1
JADGBI010000296.1:330-3274
SAR324 cluster bacterium

BrnT_toxin ? YdfE_C PhdYeFM_antitox ? PIN PhdYeFM_antitox

MBF0245325.1
JADGBK010000240.1:221-3686
Planctomycetota bacterium

DUF4258 BrnA_antitoxin DUF4143 Y1_Tnp DUF488 DUF488

MBF0254132.1
JADGBA010000042.1:578-9110
Candidatus Omnitrophota bacterium

ATPgrasp_ST 7TM_transglut Transglutaminase Transglut_i_TM ATP-synt_ab CopG_antitoxin HI0933_like HI0933_like_1st Calcineurin BrnT_toxin CopG_antitoxin Fic RHH_3 Y1_Tnp

MBF0260110.1
JADGBC010000248.1:349-2627
Desulfamplus sp.

Gp49 dnstrm_HI1420 BrnT_toxin CopG_antitoxin HicA_toxin HicB

MBF0274720.1
JADGAW010000079.1:1-8787
Nitrospinota bacterium

DUF4143 BrnT_toxin CopG_antitoxin HGTP_anticodon tRNA-synt_2b Radical_SAM SPASM PCP_red ? His-Kinase His-Kinase

MBF0275063.1
JADGAW010000128.1:1-6306
Nitrospinota bacterium

CCDC39 SMC_hinge KIF21A ? CopG_antitoxin UVR UvrB Helicase_C UvrB_3rd UvrB_inter ResIII

MBF0275583.1
JADGAW010000230.1:1-3462
Nitrospinota bacterium

BrnT_toxin CopG_antitoxin Aminotran_1_2 WHD_RuvB AAA_lid_4 RuvB_N Metal_resist

MBF0280310.1
JADGAV010000040.1:18-13091
SAR324 cluster bacterium

DUF4423 ? GTP_EFTU GTP_EFTU_D2 EFG_III EFG_IV EFG_C STAS Aminotran_1_2 ? ANG-1 Peptidase_S74 BrnT_toxin CopG_antitoxin LamininG Big

MBF0286860.1
JADGAT010000006.1:48181-69868
SAR324 cluster bacterium

? ? Abhydrolase_1 A_deaminase adh_short Y1_Tnp Peptidase_S74 BrnT_toxin CopG_antitoxin LamininG Big LamininG LamininG K319L-like_PKD LamininG Kelch_KLHDC2_KLHL20_DRC7 Kelch_KLHDC2_KLHL20_DRC7 K319L-like_PKD LamininG ADH_N ADH_zinc_N Uma2 Histidinol_dh DUF2797 Abhydrolase_1

MBF0361861.1
JADFZW010000016.1:50369-65035
Oligoflexia bacterium

NTase_sub_bind ? ? PriA_3primeBD ResIII Zn_ribbon_PriA Helicase_C PriA_C RNase_PH RNase_PH_C Ham1p_like ANKs ANKs ANKs BrnT_toxin CopG_antitoxin ? ? DUF445 Calcineurin Zn_dep_PLPC ?

MBF0377330.1
JADFZQ010000007.1:22040-38555
Desulfamplus sp.

ABC_tran ABC-ATPase oligo_HPY ABC_tran oligo_HPY MazG MazG Colicin_V PhoH 7TMR-HDED 7TM-7TMR_HD HD YbeY PdxJ BrnT_toxin ? DHH DHHA2 Aminotran_1_2 adh_short_C2 FAD_oxidored Fer4 Fer4_6 Pyr_redox_2 NAD_binding_8 Fer4_7 FlpD FlpD

MBF0413451.1
JADFZJ010000047.1:25588-29830
Desulfamplus sp.

BrnT_toxin CopG_antitoxin ? Sec10_N RelB YafQ_toxin DUF3782 DUF8196 PIN

MBF0441806.1
JADFZC010000185.1:22-5563

Oligoflexales bacterium

7TMR-DISMED2 7TMR-DISM_7TM PAS Hpt His-Kinase BrnT_toxin CopG_antitoxin TPRs TPRs

MBF0484638.1
JADFYE010000024.1:8161-20529
Candidatus Omnitrophota bacterium

Asp_Glu_race PTPS QueC MotA_ExbB ExbD Radical_SAM QueF ? CopG_antitoxin RNase_PH RNase_PH_C Ham1p_like DUF748 ? SpvB

MBF0485833.1
JADFYI010000008.1:39578-50029
Candidatus Omnitrophota bacterium

T2SSF T2SSF N_methyl PilS ? Collar TraU ? ? ? ? TatA_B_E TatC Radical_SAM HemN_C ? ? DUF1318

MBF0486115.1
JADFYI010000017.1:14344-28346
Candidatus Omnitrophota bacterium

PilZ DEAD Helicase_C Ada_Zn_binding Ada_Zn_binding ODP Flavodoxin_1 Cation_efflux ZT_dimer ? Methyltransf_31 BrnT_toxin CopG_antitoxin GATase_2 Glu_syn_central Glu_synthase GXGXG

MBF0489817.1
JADFYF010000017.1:1-18366
Candidatus Omnitrophota bacterium

N_methyl TPRs TPRs Acyl_transf_3 ? TPRs TPRs HigB-like_toxin HTH_3 BrnT_toxin CopG_antitoxin Fic ? Phosphorylase CBM_48 Alpha-amylase Alpha-amylase Isoamylase_C Glyco_transf_4 Glycos_transf_1 DUF4912

MBF0493356.1
JADFYH010000044.1:6133-16662
Deltaproteobacteria bacterium

? Cytochrom_c3_2 K319L-like_PKD Cytochrome_C7 HTH_3 NYN ? HTH_3 BrnT_toxin ? Phage_integrase Phage_int_SAM_1

MBF0493995.1
JADFYD010000017.1:1-18410
Candidatus Omnitrophota bacterium

? ? HEATs HEATs HEATs DUF3492 Glycos_transf_1 PelG MPI_cupin_dom Cellulase DUF5060 ? CopG_antitoxin HD_3 DAHP_synth_1 Glyco_hydro_77

MBF0504851.1
JADFXX010000012.1:49826-64149
Candidatus Omnitrophota bacterium

DDE_Tnp_IS66 ? LamininG BrnT_toxin CopG_antitoxin ? Enolase_N Enolase_C DivIC JAKMIP_CC3 HTH_3 RVT_1

MBF0505192.1
JADFXX010000038.1:1-2115
Candidatus Omnitrophota bacterium

? BrnT_toxin CopG_antitoxin ? PMT_2

MBF0506234.1
JADFXU010000012.1:28068-40902
Nitrospirae bacterium

UPF0150 HsdM_N N6_Mtase Methylase_S Methylase_S Methylase_S REase SWI2_SNF2 T1R_D3-like YgjP-like ? BrnT_toxin CopG_antitoxin Sod_Cu Methyltransf_21 LptF_LptG LptF_LptG Fe-S_biosyn ALAD

MBF0511462.1
JADFXW010000037.1:1-7478
Candidatus Omnitrophota bacterium

? BrnT_toxin CopG_antitoxin ISOA1-3_C Alpha-amylase CBM_48 ? Exonuc_VII_S Exonuc_VII_L tRNA_anti_2 ? ?

MBF0532384.1
JADFXQ010000018.1:1996-27023
Candidatus Omnitrophota bacterium

MGS CPSase_L_D2 CPSase_L_D1 CPSase_L_D3 CPSase_L_D2 CPSase_L_D1 GATase CPSase_sm_chain ? TPRs LptM_cons ? AbiEii ? BrnT_toxin CopG_antitoxin Pyr_redox_2 Fer4_20 GXGXG Glu_synthase Glu_syn_central GATase_2 NAD_synthase CN_hydrolase GATase Gln-synt_C-ter Gln-synt_C GSIII_N P-II

MBF0619669.1
JADFWS010000058.1:2070-11488
Candidatus Omnitrophota bacterium

? BrnT_toxin CopG_antitoxin DUF1015 Zn_ribbon_ACC Carboxyl_trans ABC_tran OB_MalK TOBE GAF_2 GGDEF GGDEF ROK

MBF1215415.1
JABZOL010000005.1:3047-16644
Haemophilus parainfluenzae

YafQ_toxin RelB Transgly_assoc HTH_26 Methylase_S Methylase_S ? ? BrnT_toxin ? Rep3_C Rep3_N KfrB PsbQ MobA_MobL ? BrnT_toxin

MBF1264026.1
JABZPG010000894.1:49-1070
Lautropia mirabilis

ResIII BrnT_toxin

MBF2054090.1
JACYMC010000155.1:10153-27411
Candidatus Sericytochromatia bacterium

Methyltransf_23 CNNM CBS CBS CorC_HlyC ? Glyco_trans_1_2 TPRs ? Zn_ribbon_TFIIB ZnR Zn_ribbon_TFIIB DUF3375 BrnT_toxin CopG_antitoxin DUF4194 AAA_29 SbcC_Walker_B Cupin_8 DDE_Tnp_1_4

MBF2057551.1
JACYMF010000069.1:365-16694

Cyanobacterium sp. T60_A2020_053

ParE_toxin DDE_Tnp_1 HEATs HEATs HEATs PepSY_TM PNP_phzG_C PNPOx_N LeuA_dimer HCS_D2 HMGL-like ? BrnT_toxin ? Trigger_C FKBP_C Trigger_N BT1 FMN_dh FMN_dh Hexapep Hexapep FGAR-AT_linker AIRS AIRS_C AIRS AIRS_C Receiver GerE

MBF2057939.1
JACYMF010000088.1:85-11013

Cyanobacterium sp. T60_A2020_053

? CopG_antitoxin BrnT_toxin S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 N6_Mtase ? Cons_hypoth95 BrnT_toxin BrnA_antitoxin BrnT_toxin BrnA_antitoxin HicA_toxin ? BrnT_toxin

MBF2097215.1
JACYLR010000036.1:1-3007

Gloeomargaritaceae cyanobacterium C42_A2020_066

ATP-grasp_IQCH ATP-grasp_IQCH Arm-DNA-bind_2 Phage_integrase BrnT_toxin

MBF8250027.1
JACVYF010000021.1:23-12713
Candidatus Levyibacteriota bacterium

? ? RelB YafQ_toxin Thioredoxin Pyr_redox_2 ? BrnT_toxin CopG_antitoxin ? LAGLIDADG_1 DUF4258 DUF2283

MBF8277024.1
JACVXV010000090.1:574-13804
Candidatus Brocadiaceae bacterium

tRNA-synt_1g tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 SPASM Radical_SAM ? HicB_lk_antitox Wzt_C ABC_tran ABC2_membrane ? CopG_antitoxin ? His-Kinase His-Kinase PAS HAMP ? HEPN Polbeta ALAD

MBF8285518.1
JACSDE010000543.1:1-3445
Anaerolineales bacterium

? Phage_integrase Phage_int_SAM_4 BrnT_toxin RelE HTH_3 DUF1015

MBH0204921.1
JABMDM010000021.1:350-5618
Nitrospira sp.

? CopG_antitoxin DUF6714 ? HKD FAD_binding_3 ? TPM_phosphatase

MBI1320849.1
WGMK01000026.1:54600-66315

Candidatus Hydrogenedens sp.

Glycos_transf_2 ? ? BrnT_toxin CopG_antitoxin Peptidase_M19 Abhydrolase_6 PAF-AH_p_II GFO_IDH_MocA_C3 GFO_IDH_MocA PfkB AMP-binding_C AMP-binding STAS

MBI1390920.1
WGLV01000069.1:44195-59291
bacterium

Rrf2 ? ? ? ? GH_cat GH_C Glyco_hydro_59_C BrnT_toxin BrnA_antitoxin PQQ_2 GFO_IDH_MocA GFO_IDH_MocA_C2 HicB_lk_antitox ? MZB WHD_HRQ1 Helicase_C DEAD NAD_binding_2 NAD_binding_11

MBI1727835.1
JACORX010000307.1:153-6026
Candidatus Rokuibacteriota bacterium

PIG-L AMP-binding AMP-binding_C Bac_luciferase Oxygenase-NA Ada_Zn_binding DNA_binding_1 Sigma70_r4_2 Sigma70_r2 BrnT_toxin

MBI1733286.1
JACORW010000102.1:1-9235
Gammaproteobacteria bacterium

DUF488 HTH_3 HicB_lk_antitox HicA_toxin BrnT_toxin CopG_antitoxin HTH_3 STY-kinase His-Kinase His-Kinase Trans_reg_C Receiver Plug TonB_dep_Rec_b-barrel cpYpsA

MBI1768877.1
JACOSO010000045.1:348-3854
Bacteroidetes bacterium

BrnT_toxin CopG_antitoxin 23S_rRNA_IVP ? HVO_A0114 RVT_1

MBI1857478.1
JACOTW010000014.1:142-14584
Candidatus Saccharibacteria bacterium

His_Phos_1 ATP_bind_3 FtsH_ext AAA AAA_lid_3 Peptidase_M41 MurJ GTP_EFTU GTP_EFTU_D2 RF3_C EFG_C LepA_C ? BrnT_toxin CopG_antitoxin ? ADK WHD_HrcA HrcA GrpE HSP70 DnaJ DnaJ_C

MBI1861337.1
JACOTR010000116.1:13490-32898
Deltaproteobacteria bacterium

ABC_tran ? CCDC39 SLT SLT_superhelical Spermine_synth ? ? BrnT_toxin BrnA_antitoxin ? ? Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans TauD AMP-binding AnkG_N

MBI1871694.1
JACOUC010000077.1:4111-18613
Chlamydiae bacterium

PMT_2 TPRs TPRs Polysacc_deac_1 Lipase_GDSL_2 GATase_7 Asn_synthase DUF6290 DUF4258 MqsA_antitoxin BrnT_toxin CopG_antitoxin Hexapep CopG_antitoxin NUDIX_4 HhH-GPD NTP_transferase APH NTP_transf_5 ?

MBI1939029.1
JACOUS010000030.1:206160-221243
Ignavibacteriales bacterium

PorZ_N_b_propeller Por_Secre_tail ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_Xtn ABC_tran_CTD Methyltransf_23 Glyoxalase ? ROK Rossmann-like_BshC CC_BshC BrnT_toxin CopG_antitoxin EPSP_synthase SpoU_methylas_C SpoU_methylase ? DeoC THF_DHG_CYH_C THF_DHG_CYH YmdB

MBI1961200.1
JACOVK010000020.1:1-5973
Parcubacteria group bacterium

? ? Fic DrHyd BrnT_toxin CopG_antitoxin AAA ? ?

MBI2008735.1
JACOWU010000084.1:1-4236
Candidatus Amesbacteria bacterium

Fervidolysin_NPro-like DUF378 DUF4956 Peptidase_C39_2 NAD_binding_1 DUF4258 ?

MBI2009800.1
JACOWY010000010.1:1-6684
Candidatus Chisholmbacteria bacterium

? HTH_Crp_2 ? PhdYeFM_antitox PIN BrnT_toxin CopG_antitoxin ? ?

MBI2009869.1
JACOWY010000017.1:5083-13022
Candidatus Chisholmbacteria bacterium

ChlI AAA_25 AAA_25 Zn_ribbon_LapB RHH_1 PIN ? ? Pro_CA_2 ? CopG_antitoxin ? S1 S1 S1 S1 ? ? ? ?

MBI2020669.1
JACOXG010000023.1:80-9078
Candidatus Daviesbacteria bacterium

? ? ? ? ? tRNA_anti-like FKBP15 BrnT_toxin ? His-Kinase His-Kinase

MBI2039647.1
JACOYE010000007.1:99452-110872
Candidatus Microgenomates bacterium

MASE11 His-Kinase TED_complement Beta-lactamase2 GIY-YIG HNH_2 ? bPH_2 DUF4258 CopG_antitoxin Amidase_3 Penicillinase_R Fic Fic ? ? PMSR

MBI2050235.1
JACOYR010000001.1:58984-69159
Parcubacteria group bacterium

? ? ? ? ? ? ? BrnT_toxin CopG_antitoxin ? Sigma70_r2 Sigma70_r4_2 ? ? SdpI DUF1648 Lactamase_B_6 Beta-Casp RMMBL

MBI2059498.1
JACPAV010000008.1:54297-75038
Nitrospirae bacterium

NTP_transferase Hexapep_GlmU Glyco_hydro_57 GalP_UDP_transf GalP_UDP_tr_C Glyco_transf_5 Glyco_trans_1_4 PFK ? ? BrnT_toxin CopG_antitoxin MGH1-like_GH Glyco_hydro_63 GDE_N GDE_C Transketolase_N Transket_pyr Transketolase_C_1 LacAB_rpiB TAL_FSA PGI NAD_binding_2 6PGD

MBI2066269.1
JACOZU010000033.1:1-6970
Candidatus Woesebacteria bacterium

Fuseless Cas_Cas4 Hydrolase_4 Gp49 HTH_3 BrnT_toxin CopG_antitoxin ? Ribosomal_L33 ? PGA_cap

MBI2098043.1
JACOZJ010000002.1:722-6915
Candidatus Wildermuthbacteria bacterium

ABC_tran FtsX_ECD FtsX ? ? ? ? ? BrnT_toxin CopG_antitoxin ?

MBI2104648.1
JACPAY010000007.1:54675-69911
Candidatus Omnitrophota bacterium

QRPTase_N QRPTase_C ResIII UvrB_inter UvrB_3rd Helicase_C UvrB UVR BPL_LplA_LipB BPL_LplA_LipB_2 BPL_C Pan_kinase Cpn10 Cpn60_TCP1 Cpn10 ? CopG_antitoxin GATase NAD_synthase GMP_synt_C MS_channel_1st MS_channel_2nd MS_channel_3rd DsbD_N PK Trns_repr_metal OEP OEP

MBI2119888.1
JACPBF010000023.1:48698-57110
Elusimicrobia bacterium

? ? CopG_antitoxin tRNA-synt_2b tRNA_edit tRNA-synt_2b HGTP_anticodon His_Phos_1 CobU DBI_PRT CobS Aminotran_1_2

MBI2167583.1
JACPCP010000019.1:16768-35370
Candidatus Omnitrophota bacterium

? PBP_dimer Transpeptidase FTSW_RODA_SPOVE TatA_B_E Glyco_hydro_77 ? AbiEii BrnT_toxin CopG_antitoxin Gp49 HTH_3 DUF1016_N SNase N6_N4_Mtase ResIII Tn7_TnsA-like_N UvrD-helicase UvrD_C

MBI2326957.1
JACPEV010000003.1:134696-146363
Candidatus Curtissbacteria bacterium

SmpB CTP_synth_N GATase YafQ_toxin ? 3-dmu-9_3-mt 3-dmu-9_3-mt 3-dmu-9_3-mt 3-dmu-9_3-mt AHSA1 HTH_5 BrnT_toxin CopG_antitoxin ? DZR_2 Pribosyltran PMT_2 HTH_3 LAGLIDADG_3 PIN PhdYeFM_antitox

MBI2337391.1
JACPFA010000046.1:6079-25299
Deltaproteobacteria bacterium

? Thioredox_DsbH ? Acyltransferase Ribul_P_3_epim Formyl_trans_C Formyl_trans_N Pep_deformylase BrnT_toxin CopG_antitoxin PriA_C Helicase_C Zn_ribbon_PriA DEAD PriA_3primeBD ? ? DUF3047 Band-7 NTP_transferase

MBI2411509.1
JACPGJ010000008.1:514825-527701
Candidatus Kerfeldbacteria bacterium

DUF7387 HicA_toxin Cupin_2 Peptidase_M24 ? ? Ppnp BrnT_toxin CopG_antitoxin RHH_1 PIN NuiA Cu-binding_MopE Beta_helix DUF378 Peptidase_M20

MBI2436770.1
JACPGR010000053.1:200-5973
Candidatus Magasanikbacteria bacterium

CoA_binding Succ_CoA_lig BrnT_toxin CopG_antitoxin Citrate_synt Citrate_synt PDDEXK_3 HMBD MJ0043_C HD DUF559

MBI2441219.1
JACPGM010000095.1:29429-34049
Lentisphaerae bacterium

BrnT_toxin CopG_antitoxin DUF6036 ? Zip Creatininase PP-binding ?

MBI2442760.1
JACPGQ010000025.1:50-6000
Candidatus Levyibacteriota bacterium

Glyco_tran_WecG Glyco_tran_WecG MreB_Mbl PEGA Sortase Peripla_BP_2 BrnT_toxin

MBI2476385.1
JACPHR010000021.1:47955-56550
Candidatus Taylorbacteria bacterium

Acylphosphatase DoxX SpoU_methylase ? zf-FPG_IleRS H2TH Fapy_DNA_glyco BrnT_toxin CopG_antitoxin Mur_ligase_C Mur_ligase_M ? RuvA_C TrmB DUF397 RuvA_C HHH_5 RuvA_N

MBI2476455.1
JACPHR010000026.1:181-14713
Candidatus Taylorbacteria bacterium

? FTSW_RODA_SPOVE MethyltransfD12 ? Glycos_transf_1 Glyco_transf_4 Glyco_transf_4 CoA_binding_3 Bac_transf Epimerase BrnT_toxin CopG_antitoxin SecD_1st SecDF_P1_head SecD_SecF_C Sec_GG SecD_SecF_C HMGL-like IPMS_D2 LeuA_dimer ?

MBI2596432.1
JACPKP010000011.1:78942-96037
Candidatus Daviesbacteria bacterium

SBP_bac_1 PG_binding_4 PG_binding_4 VanW G5 LigT_PEase LigT_PEase GDE_C LPD3 TsaD TsaD tRNA-synt_1 Anticodon_1 BrnT_toxin CopG_antitoxin ? ? Ribosomal_TL5_C Ribosomal_L25p DUF4012 FIMAH Epimerase RVT_3 MatE Polysacc_synt

MBI2605552.1
JACPKS010000044.1:63162-74486
Deltaproteobacteria bacterium

Peptidase_Mx HKD HKD Leu_Phe_trans HKD HKD ? CopG_antitoxin ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N NTase_sub_bind Polbeta ? ClpS Ferritin

MBI2613931.1
JACPLF010000007.1:210812-222709
Candidatus Levyibacteriota bacterium

RuvC Ribosomal_L25p Ribosomal_TL5_C TPRs Anticodon_1 tRNA-synt_1 ? PIN ? BrnT_toxin CopG_antitoxin Trans_reg_C DUF4012 ? ? AbiEii ?

MBI2621176.1
JACPLG010000003.1:106580-116083
Candidatus Levyibacteriota bacterium

PhdYeFM_antitox adh_short ? Redoxin Band-7 PIN RHH_1 ? CopG_antitoxin HicA_toxin HicB_lk_antitox ? ? ? Fervidolysin_NPro-like Peptidase_S8

MBI2641695.1
JACPMD010000120.1:1-3298
Candidatus Roizmanbacteria bacterium

? Glutaredoxin ? UbiA ?

MBI2817540.1
JACPPJ010000056.1:5853-16642
Acidobacteria bacterium

? ? CopG_antitoxin CarboxypepD_reg oligo_HPY ABC_tran oligo_HPY ABC_tran BPD_transp_1 OppC_N BPD_transp_1 BPD_transp_1_N SBP_bac_5

MBI2890699.1
JACPQY010000050.1:11582-24820
Nitrospirae bacterium

? MarR HGD-D HGD-D BcrAD_BadFG 4HBT ? BrnT_toxin CopG_antitoxin DctP Rotamase_2 SurA_N_3 ? Abhydrolase_1 TAL_FSA OMP_b-brl

MBI2891250.1
JACPQY010000071.1:10106-27005
Nitrospirae bacterium

HTH_17 Trypsin PDZ_2 PDZ_6 ABC_tran DrrA1-3_C ABC2_membrane_2 DUF7088 ABC_transp_aux FtsJ S4 ? BrnT_toxin CopG_antitoxin ? DUF1844 Trypsin PDZ_2 PDZ_2 Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small cNMPBD HTH_Crp_2 HATPase_c_3

MBI2906193.1
JACPQD010000006.1:4702-19139
Chloroflexi bacterium

PTP-NADK Big CopD CopC Peptidase_M20 ? TPRs BrnT_toxin CopG_antitoxin ? TPRs TPRs TPRs HYR HYR HYR HYR PKD_4 PKD_4 DJ-1_PfpI ? Glyoxalase

MBI2980836.1
JACPSD010000018.1:139-3906
Deltaproteobacteria bacterium

SLT DUF4438_C DUF4438 DUF5128 CopG_antitoxin Methyltransf_3 FumaraseC_C Lyase_1

MBI3242812.1
JACPWC010000061.1:707-15216
Chloroflexi bacterium

Peptidase_A8 Methyltransf_11 Methyltransf_31 OsmC TPR_MalT TPRs TPRs AAA_16 BTAD DUF5915 BrnT_toxin ? DUF5915 Anticodon_1 tRNA-synt_1 HYDIN_VesB_CFA65-like_Ig PCYCGC Cys_Met_Meta_PP ? ICMT

MBI3282789.1
JACPWX010000047.1:1-11656
Candidatus Curtissbacteria bacterium

YfhO YfhO PMSR ? RRM_1 Thioredoxin DsbD_TM ? BrnT_toxin CopG_antitoxin GEVED Peptidase_S51 DDE_Tnp_ISL3 HTH_Tnp_ISL3 GTP_EFTU GTP_EFTU_D2 GTP_EFTU_D3

MBI3319880.1
JACPXQ010000178.1:1-6712
Candidatus Omnitrophota bacterium

Glycos_transf_N ? DUF5678 BrnT_toxin CopG_antitoxin Fic RHH_3 DUF6010 Zn_ribbon_Top1 Zn_ribbon_Top1 Topoisom_bac Toprim

MBI3335860.1
JACPYC010000103.1:1-2506
Candidatus Portnoybacteria bacterium

? BrnT_toxin CopG_antitoxin Transketolase_N

MBI3372956.1
JACPYS010000025.1:164-7447
Betaproteobacteria bacterium

PEMT BrnT_toxin CopG_antitoxin Cupin_2 DUF3024 ? AcetylCoA_hydro AcetylCoA_hyd_C PhaC_N Abhydrolase_1 DUF4405

MBI3540852.1
JACQAV010000061.1:1-2915
Deltaproteobacteria bacterium

PCRF RF-1 BrnT_toxin BrnA_antitoxin DUF7107

MBI3559384.1
JACQBH010000078.1:65-4851
Candidatus Gottesmanbacteria bacterium

DsbB ? Transgly Transpeptidase ? CopG_antitoxin Ribosomal_L34 23S_rRNA_IVP Ribonuclease_P ?

MBI3575960.1
JACQBF010000310.1:1-7333
Gammaproteobacteria bacterium

? BrnT_toxin CopG_antitoxin ? DHO_dh ACT ACT Lactamase_B ILVD_EDD_N ILV_EDD_C FAD_binding_4

MBI3601504.1
JACQCF010000004.1:33158-53693
Candidatus Omnitrophota bacterium

? ? BACON_2 TEN_YD-shell TEN_YD-shell ? AgrB TEN_YD-shell DUF6531 RHS_repeat RHS_repeat RHS_repeat RHS_repeat HYDIN_VesB_CFA65-like_Ig Glucodextran_B Glucodextran_B ? BrnT_toxin CopG_antitoxin DUF6933 ? ? ? TrbC TraC_F_IV P-loop_TraG

MBI3615611.1
JACQCH010000019.1:10811-17391
Candidatus Omnitrophota bacterium

HKD ? YjeF_N ? DoxX Gp49 HTH_3 BrnT_toxin CopG_antitoxin Zn_ribbon_21 4HBT Nitroreductase ? DUF5674 ?

MBI3620440.1
JACQCK010000048.1:1-8034
Candidatus Roizmanbacteria bacterium

DUF359 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 tRNA-synt_1 Anticodon_1 Dus Dus CutA1 ? CopG_antitoxin HTH_5 DUF2703 ?

MBI3631690.1
JACQCP010000078.1:668-2411
Candidatus Staskawiczbacteria bacterium

BrnT_toxin CopG_antitoxin MS_channel_1st_1

MBI3801295.1
JACQEL010000909.1:1-4454
Deltaproteobacteria bacterium

Glyoxalase Meth_synt_2 BrnT_toxin CopG_antitoxin ? GFO_IDH_MocA Gal80p_C-like

MBI3917203.1
JACQGN010000138.1:89-12394
Betaproteobacteria bacterium

SGL SGL TctC TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N BrnT_toxin CopG_antitoxin MqsA_antitoxin HicA_toxin ? ? ? ANKs Viral_helicase1 AAA_19

MBI3985066.1
JACQHJ010000007.1:1-30735
Candidatus Levyibacteriota bacterium

? Rho_RNA_bind ATP-synt_ab LysM LysM Peptidase_M23 MazE_antitoxin ? DUF4143 AAA_14 zf_Tbcl zf_Tbcl_2 BrnT_toxin CopG_antitoxin TPRs TPRs LamininG LamininG LamininG LamininG LamininG LamininG LamininG Gly_rich_2 Gly_rich_2 DUF2341 ? Peptidase_S74 Phage_int_SAM_1 Phage_int_SAM_1 AbiEii

MBI4050514.1
JACQJJ010000004.1:2274-14425
Candidatus Doudnabacteria bacterium

GKRP_SIS_N SIS cNMPBD HTH_Crp_2 Glutaredoxin ABC_membrane PAP2_3 Pyr_redox_2 ? BrnT_toxin CopG_antitoxin PGA_cap FtsH_ext AAA AAA_lid_3 Peptidase_M41 ? Ribonucleas_3_3 dsrm SMC_N VTT_dom

MBI4061818.1
JACQJQ010000188.1:1-2809
Elusimicrobia bacterium

Carb_kinase HNH_5 BrnT_toxin CopG_antitoxin ACPS ? SIS

MBI4065215.1
JACQJW010000017.1:74179-88691
Candidatus Gottesmanbacteria bacterium

RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 Ribosom_S12_S23 Ribosomal_S7 TAT_signal HhH-GPD TrAA12 ? BrnT_toxin CopG_antitoxin Thioredoxin_10 Redoxin DsbD_TM Thioredoxin Colicin_V CAP DNA_binding_1 adh_short_C2 ?

MBI4089303.1
JACQKL010000016.1:6678-18977
Candidatus Levyibacteriota bacterium

AI-2E_transport ? RuvC Ribosomal_L25p Ribosomal_TL5_C TPRs Anticodon_1 tRNA-synt_1 tRNA-synt_1 BrnT_toxin CopG_antitoxin Trans_reg_C DUF4012 RVT_3 R3H KH_KhpA-B 60KD_IMP YidD

MBI4091647.1
JACQKO010000005.1:1-2813
Candidatus Levyibacteriota bacterium

? Cupin_2 PNPOx_N ? CopG_antitoxin

MBI4159649.1
JACQNN010000008.1:259-2645
Candidatus Wolfebacteria bacterium

Metalloenzyme BrnT_toxin CopG_antitoxin SGSH_C

MBI4208183.1
JACQOV010000009.1:1642-18541
Deltaproteobacteria bacterium

Thioredox_DsbH Glyco_hydro_88 Glyco_hydro_76 NTP_transf_2 HEPN Peptidase_S8 PAP2 ? SNF SNF BrnT_toxin CopG_antitoxin S4 PseudoU_synth_2 DUF5679 ABC_tran ABC_membrane ABC_tran ABC_membrane Methyltrans_RNA PUA_4 Peptidase_M16_C Peptidase_M16 Peptidase_M16_C Peptidase_M16

MBI4226974.1
JACQRF010000050.1:1-3107
Candidatus Omnitrophota bacterium

ParE_toxin BrnT_toxin CopG_antitoxin T2SSF T2SSF N_methyl T2SSG ?

MBI4247979.1
JACQRQ010000021.1:222-6944
Elusimicrobia bacterium

? CopG_antitoxin RelB3_C ParE_toxin ? ? SSF Sigma70_r2 Sigma70_r4_2

MBI4352413.1
JACQQY010000008.1:5249-12967
Candidatus Omnitrophota bacterium

? ? CopG_antitoxin MreB_Mbl MreC ? PBP_dimer Transpeptidase FTSW_RODA_SPOVE HAMP GGDEF

MBI4403867.1
JACQOU010000156.1:1883-8569
Deltaproteobacteria bacterium

SBP_bac_5 HTH_26 ? FliL ? CBAH DUF4143 ? BrnA_antitoxin

MBI4412252.1
JACQOY010000190.1:113-2234
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin HicA_toxin UPF0150 Arc ?

MBI4430072.1
JACQPT010000252.1:342-5875
Ignavibacteriales bacterium

BrnT_toxin CopG_antitoxin Por_Secre_tail TfoX_N Por_Secre_tail Beta-prop_ACSF4

MBI4432845.1
JACQQX010000054.1:46585-57659
Candidatus Omnitrophota bacterium

MlaE ABC_tran MlaD YCF81_N PIN IspD YgbB SATase_N Hexapep BrnT_toxin CopG_antitoxin tRNA-synt_1e DALR_2 Thymidylate_kin DNA_pol3_delta2 tRNA-synt_1g Anticodon_1 Antitox_RHH ParE_toxin

MBI4518808.1
JACQVA010000253.1:1448-6764
Deltaproteobacteria bacterium

WD40_RLD BrnT_toxin ? DUF4258 DUF2283 ? PIN PIN VapB_antitoxin

MBI4558584.1
JACQVH010000178.1:1-5913
Candidatus Hydrogenedentota bacterium

GTP_EFTU BrnT_toxin CopG_antitoxin ? ? FGS STY-kinase DUF433 PhyH ?

MBI4587880.1
JACQWA010000064.1:1-5611
Candidatus Rokuibacteriota bacterium

SNase Lactamase_B PemK_toxin DctP ? AbiEii SNase BrnT_toxin CopG_antitoxin DUF2283

MBI4598440.1
JACQWM010000060.1:1-6826
Candidatus Omnitrophota bacterium

PMT_2 Glycos_transf_2 PMT_2 PIN RHH_5 Lambda_Bor Lambda_Bor BrnT_toxin CopG_antitoxin ? ?

MBI4619727.1
JACQWO010000029.1:1-6305
Desulfobacterales bacterium

? DUF7479 Methyltransf_31 Zn_ribbon_HMPTM Radical_SAM AMP-binding AMP-binding_C_2 Oxidored_FMN PseudoU_synth_1 PseudoU_synth_1 BrnT_toxin

MBI4632843.1
JACQWR010000007.1:111723-127116
Deltaproteobacteria bacterium

ABC_sub_bind Fer4_16 VTT_dom DUF2339 DUF2339 ? ParD_like ParE-like_toxin BrnT_toxin CopG_antitoxin Ldh_1_C Ldh_1_N PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV Nitroreductase ? ? FGAR-AT_linker AIRS AIRS_C AIRS AIRS_C

MBI4648997.1
JACQWN010000587.1:291-8000
Bacteroidia bacterium

BrnT_toxin ? SufBD_N SUFBD_core ABC_tran SufBD_N SUFBD_core MNS-STAND DUF8371 PseudoU_synth_2 Pantoate_transf

MBI4720453.1
JACQXT010000103.1:1392-5771
Chitinivibrionia bacterium

? ? BrnT_toxin CopG_antitoxin RHH_1 PemK_toxin Flavodoxin_2 RHH_3 ? PhdYeFM_antitox

MBI4801527.1
JACQYT010000029.1:246-7409
Elusimicrobia bacterium

FGS ? MqsA_antitoxin ? FGS ? DUF4384 ? CopG_antitoxin

MBI4973010.1
JACRPE010000001.1:16936-28146
Candidatus Roizmanbacteria bacterium

? EamA EamA Flavokinase BrnT_toxin CopG_antitoxin Dockerin_1 Cohesin MctB ? CopG_antitoxin Peptidase_M16 Peptidase_M16_C A24_N_bact Peptidase_A24 ABC2_membrane_6 ABC2_membrane_6 ABC_tran AAA_18

MBI4982976.1
JACROS010000103.1:14693-19037
Candidatus Omnitrophota bacterium

ORF6N ? CopG_antitoxin ? ? ? Smr YajQ Receiver

MBI4991516.1
JACRNZ010000144.1:1015-8894
Candidatus Gottesmanbacteria bacterium

? PPS_PS GHMP_kinases_N ? ? Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 BrnT_toxin CopG_antitoxin AhpC-TSA 1-cysPrx_C ArfC SNase

MBI5057397.1
JACRMS010000033.1:99420-114096
Nitrospirae bacterium

NAD_binding_8 ORF6N ? Usp Usp HAMP His-Kinase His-Kinase Receiver ? BrnT_toxin CopG_antitoxin MauJ Unstab_antitox ? HTH_8 AAA_lid_14 NtrC-AAA Receiver Lon_C Lon_lid AAA Receiver HAMP His-Kinase His-Kinase

MBI5122429.1
JACRJK010000016.1:1951-12870
Candidatus Roizmanbacteria bacterium

? Phos_pyr_kin NUDIX NAD_synthase CTP_transf_like CTP_transf_like DUF359 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 ? ? CopG_antitoxin ? Ribonuc_2-5A TM7S3_TM198 BrnT_toxin CopG_antitoxin Peptidase_M16 Peptidase_M16_C

MBI5139415.1
JACRHN010000047.1:326-5718
Candidatus Nomurabacteria bacterium

? N_methyl N_methyl SPM ? CopG_antitoxin

MBI5147737.1
JACRHY010000021.1:4211-20083
Parcubacteria group bacterium

? ? WHD_NrtR NUDIX tRNA_NucTran2_2 HD PolyA_pol_RNAbd PolyA_pol ? NDK tRNA-synt_2 N-OB BrnT_toxin CopG_antitoxin Lactamase_B Competence DUF4131 CobA_CobO_BtuR RNase_HII ? PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 N-OB

MBI5149650.1
JACRHP010000028.1:17166-32797
Candidatus Omnitrophota bacterium

DUF4131 Competence ? BrnT_toxin CopG_antitoxin TPRs YfiO LptE DNA_pol3_delta_C Ribosomal_S20p MurJ GIY-YIG UvrC_M UVR UvrC_RNaseH_dom

MBI5182317.1
JACRGT010000122.1:243-7016
Nitrospirae bacterium

? ? GcvP2_C GDC-P GDC-P GCV_T_C GCV_T DUF433 BrnT_toxin ?

MBI5192827.1
JACRGV010000056.1:8551-19780
Nitrospirae bacterium

Cytochrom_C_2 RlmL_1st THUMP UPF0020 SnoaL_4 ? AAA_33 ? ? BrnT_toxin CopG_antitoxin FAD-depend_C Xre_MbcA_ParS_C RES CorA Aminotran_4 CopG_antitoxin

MBI5201835.1
JACRDR010000287.1:1-10985
Elusimicrobia bacterium

OMP_b-brl CutA1 FAD_binding_6 NAD_binding_1 Maf ? His-Kinase Flavin_Reduct BrnT_toxin CopG_antitoxin ? LIAS_N Radical_SAM E1_dh Transket_pyr Transketolase_C Biotin_lipoyl

MBI5206371.1
JACRDZ010000027.1:1-2574
Candidatus Firestonebacteria bacterium

ACR_tran P-II NKAIN ? BrnT_toxin ?

MBI5211917.1
JACREW010000023.1:1-14657
Nitrospirae bacterium

DEAD Helicase_C DUF4912 Rho_N VWA ? CbbQ_C GATase_2 Glu_syn_central Glu_synthase GXGXG HhH-GPD BrnT_toxin CopG_antitoxin AA_kinase MobB Aminotran_3 Aldedh

MBI5215338.1
JACREZ010000018.1:3080-14803
Ignavibacteriae bacterium

Amino_oxidase SQS_PSY SQS_PSY DUF1573 TMEM131_like_N DUF1573 HYDIN_VesB_CFA65-like_Ig Haem_bd ? BrnT_toxin ? Sdh_cyt FAD_binding_2 Succ_DH_flav_C Fer2_3 Fer4_8 DUF4160 DUF2442 ?

MBI5215966.1
JACREZ010000027.1:21865-35721
Ignavibacteriae bacterium

Ribosomal_S30AE Hpr_kinase_N Hpr_kinase_C HOOK Peptidase_M23 DUF2795 POTRA Omp85 DNA_pol3_delta2 DNApol3-delta_C tRNA-synt_1g Anticodon_3 tRNA_bind BrnT_toxin CopG_antitoxin DinB Y1_Tnp ? ? CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel ?

MBI5235211.1
JACRCE010000030.1:13866-23546
Deltaproteobacteria bacterium

MreC MreB_Mbl PUA_2 ATP-sulfurylase APS_kinase M20_dimer TPRs Thioredoxin_2 Receiver MMR_HSR1_Xtn MMR_HSR1 GTP-bdg_M GTP-bdg_N BrnT_toxin Metal_resist CopG_antitoxin DUF4258 MqsA_antitoxin

MBI5236102.1
JACRCE010000079.1:1-12798
Deltaproteobacteria bacterium

Glyco_hydro_57 MutS_I MutS_II MutS_III MutS_V MutS_C AMIN Amidase_3 GlnE GlnD_UR_UTase GlnD_3rd HD ACT_ACR9_3rd Phage_int_SAM_1 Phage_integrase PilZ Receiver ? BrnT_toxin CopG_antitoxin

MBI5238086.1
JACRCF010000127.1:14300-27207
Deltaproteobacteria bacterium

FGS Fer4_10 Fer4_7 Pyr_redox_3 ? Cytochrom_c3_2 Cytochrome_C7 Cytochrome_C7 Cytochrome_C7 Cytochrome_C7 Cytochrome_C7 Cytochrome_C7 Cytochrome_C7 Mtrc-MtrF_II-IV_dom DUF2238 HD GAF Helicase_C DEAD BrnT_toxin CopG_antitoxin GreA_GreB_N

MBI5251053.1
JACRDE010000424.1:1568-9542
Desulfomonile tiedjei

PIN BrnT_toxin CopG_antitoxin NTP_transf_2 HEPN KAP_NTPase ParE_toxin Unstab_antitox BPD_transp_2

MBI5356338.1
JACRNH010000007.1:10595-22249
Candidatus Collierbacteria bacterium

PGK PGK PGK Peptidase_M23 Cytidylate_kin Cpn60_TCP1 Cpn10 PIN PhdYeFM_antitox BrnT_toxin CopG_antitoxin Band-7 60KD_IMP CTP_transf_like ? SBP_bac_8 PG_binding_4 VanW

MBI5400704.1
JACRKV010000028.1:4048-10683
Candidatus Yonathbacteria bacterium

BrnT_toxin CopG_antitoxin ? Cupredoxin_1 Peptidase_M1_N Peptidase_M1 ERAP1_C CDP-OH_P_transf Pribosyltran DZR_2 PilM_2 FtsA

MBI5402357.1
JACREY010000008.1:68812-79219
Ignavibacteriae bacterium

Bac_DNA_binding tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c Methyltransf_11 DUF2238 PhdYeFM_antitox YoeB_toxin Anticodon_2 BrnT_toxin CopG_antitoxin Anticodon_2 tRNA-synt_1c TP_methylase SpoU_sub_bind SpoU_methylase ? Abhydrolase_1 Aminotran_4

MBI5417903.1
JACRIQ010000136.1:354-2367
Candidatus Poribacteria bacterium

Memo BrnT_toxin CopG_antitoxin YokU

MBI5418366.1
JACRIQ010000175.1:51128-63544
Candidatus Poribacteria bacterium

Peptidase_M16 Peptidase_M16_C Radical_SAM B12-binding Radical_SAM B12-binding PhdYeFM_antitox ParE_toxin HicB_lk_antitox ? BrnT_toxin CopG_antitoxin MazE_antitoxin ParE_toxin Uma2 Uma2 Uma2 Spermine_synth

MBI5470078.1
JACRFH010000027.1:261-8080
Candidatus Kaiserbacteria bacterium

Ribosomal_L2_C Ribosomal_L2_N Ribosomal_L23 Ribosomal_L4 Ribosomal_L3 ? ? DUF4258 CopG_antitoxin TPRs TPRs ?

MBI5470634.1
JACRFH010000063.1:19705-28524
Candidatus Kaiserbacteria bacterium

ParE_toxin PhdYeFM_antitox ? HD NusA_N S1 KH_NusA_1st KH_NusA_2nd Methyltransf_11 DHHA2 DHH BrnT_toxin BrnA_antitoxin ? Trigger_N Trigger_C AbiEii DUF6922 DUF6629 CopG_antitoxin

MBI5558775.1
JACRCD010000075.1:1-11229
Deltaproteobacteria bacterium

PhzC-PhzF ? Unstab_antitox DEDD_Tnp_IS110 ? DUF4268 AAA_14 DUF4143 BrnT_toxin CopG_antitoxin RNase_II_C_S1 RNB HTH_RNase_II MMR_HSR1 KH_2 tRNA-synt_1c Anticodon_2

MBI5593108.1
JACRBZ010000081.1:1-4895
Deltaproteobacteria bacterium

ABC_tran BCA_ABC_TP_C BrnT_toxin CopG_antitoxin Lactamase_B_2 RMMBL ? ? ? ABC-ATPase

MBI5625379.1
JACRDQ010000044.1:148934-165037
Elusimicrobia bacterium

Pro_isomerase HEATs HEATs HEATs HEATs HEATs ? ? ? Trypsin PRK ? BrnT_toxin CopG_antitoxin DUF4440 Trypsin ? FecR Radical_SAM PALP

MBI5641824.1
JACRHE010000220.1:408-7096
Nitrospirae bacterium

? CopG_antitoxin OmpA PAS His-Kinase His-Kinase BacA FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma LolA ?

MBI5642642.1
JACRET010000013.1:1-11246
Deltaproteobacteria bacterium

Prok-E2_D DUF2016 Prok-JAB ThiF ? ? ? ? BrnT_toxin CopG_antitoxin PRiA4_ORF3 PIN MazE_antitoxin Gp3-like Zn_ribbon_DnaG Toprim_2 DUF6094

MBI5647246.1
JACRFB010000040.1:43425-58479
Ignavibacteriae bacterium

HPPK FolB ? ? ? ? tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2 BrnT_toxin CopG_antitoxin Anticodon_2 tRNA-synt_1c Peptidase_M50B Lactamase_B Competence DUF4131 Metalloenzyme Tyr_Deacylase HD_4

MBI5656680.1
JACRPV010000110.1:8890-25929
Geobacter sp.

PmbA_TldD_3rd PmbA_TldD_2nd PmbA_TldD_1st RsgA_GTPase ? HicB_lk_antitox HicA_toxin FRG BrnT_toxin CopG_antitoxin Acetylase SMC_N NAD_kinase_C NAD_kinase AAA AAA_assoc_2 MgsA_C PFK S-methyl_trans

MBI5729569.1
JACRFC010000008.1:1-2942
Ignavibacteriales bacterium

ABC_tran_2 adh_short Ribonuc_L-PSP BrnT_toxin CopG_antitoxin

MBI5748066.1
JACRGN010000072.1:1-3860
Nitrospinota bacterium

Pur_ac_phosph_N BrnT_toxin CopG_antitoxin DDE_Tnp_1

MBI5788625.1
JACRJN010000100.1:8992-29978
Candidatus Schekmanbacteria bacterium

TraB_PrgY_gumN HHH_8 HHH_5 DNA_pol_B_thumb PHP TMEM260-like ? LYTB S1 S1 S1 S1 S1 S1 CLP_protease Peptidase_S49 BrnT_toxin CopG_antitoxin MethyltransfD12 EPSP_synthase Phosphodiest MurJ Beta_helix Beta_helix ThiC_Rad_SAM

MBI5873161.1
JACRLT010000015.1:2632-13168
Candidatus Omnitrophota bacterium

ParE_toxin NTP_transf_2 HepT-like ? DNA_pol3_delta2 DNAX_ATPase_lid DNA_pol3_gamma3 DnaX_C RecR_HhH RecR_ZnF Toprim_4 RecR_C PilZ ? CopG_antitoxin ? DUF5674 PatG_C Polysacc_synt Glyco_hydro_1 Glyco_hydro_1 ThiS

MBI9055943.1
JAEINU010000015.1:4838-14926
Bacteroidales bacterium

DUF6364 PIN PIN PhdYeFM_antitox TerS_N AAA_14 DUF4143 rve_3 rve HTH_21 HTH_Tnp_1 DUF4258 CopG_antitoxin MazE_antitoxin PIN AAA_14 DUF4143 Fic DIP2311-like_C DUF6364 PIN

MBI9070706.1
JAEINS010000003.1:182727-204180
Melioribacteraceae bacterium

HD AcetylCoA_hydro AcetylCoA_hyd_C HRDC DNA_pol_A_exo1 ? HATPase_c_4 PMSR SNase SNase BrnT_toxin CopG_antitoxin GATase_5 AIRS_C FGAR-AT_linker HAD_2 ABC_sub_bind PAS GAF_2 PAS His-Kinase Receiver HDOD Receiver HD_5

MBI9097520.1
JAEINM010000035.1:186-10106
Spirochaetaceae bacterium

? PemK_toxin AAA_14 DUF4143 DDE_Tnp_IS66_C ? BrnT_toxin CopG_antitoxin AAA_14 DUF4143 TetR_N TetR_C_29 BSH_CzcB ABC_tran ABC_tran ABC2_membrane_3

MBK6433542.1
JADIZS010000057.1:5289-15874
Candidatus Amarolinea dominans

? ? N6_N4_Mtase DUF2442 BrnT_toxin DUF4160 DUF2442 ? LysM LysM GIDA Bac_luciferase ?

MBK6580364.1
JADJAW010000071.1:1205-14270
Sandaracinaceae bacterium

NAD_synthase GATase Peptidase_M20 CLP_protease His_Phos_1 Presenilin Yop-YscD_cpl Sigma54_activ_2 AAA_lid_14 HTH_8 SspB BrnT_toxin MqsA_antitoxin AbiEi_4 ? HipA_C HTH_Tnp_1 TnpB_IS66 LZ_Tnp_IS66

MBK6618511.1
JADJBK010000006.1:1702048-1715394
Nitrosomonas sp.

DUF7387 HicA_toxin ? ? Glyco_trans_1_4 GFO_IDH_MocA ? ? CopG_antitoxin HigB-like_toxin HTH_3 ADH_N ADH_zinc_N GFO_IDH_MocA GFO_IDH_MocA_C3 Hepar_II_III_N Hepar_II_III_C Glyco_trans_4_4 Glycos_transf_1 PIN

MBK6623130.1
JADJBL010000045.1:234329-245651
Saprospirales bacterium

Lipocalin_4 CHU_C Lectin_L-type_dom ArsC MFS_1 Receiver EAD11 Glyoxalase ? BrnT_toxin ? ? 2-oxoacid_dh 2EXR 2-oxoacid_dh BrnT_toxin ? NOV_C ?

MBK7219584.1
JADJHJ010000026.1:35797-49942
Candidatus Promineofilum glycogenicum

? MRM3-like_sub_bind SpoU_methylase DotU ? ? DUF6847 ? BrnT_toxin CopG_antitoxin DLH SH3_2 Peptidase_M20 TPRs META ?

MBK7259187.1
JADJHP010000009.1:224456-237562

Ignavibacteriae bacterium

Cass2 ? Sde182_NH-like RibD_C ? PhdYeFM_antitox ParE_toxin BrnT_toxin CopG_antitoxin ? Gp49 HTH_3 DUF4160 IstB_IS21 ?

MBK7662727.1
JADJMB010000001.1:892401-906469
Sterolibacteriaceae bacterium

PK_C PK HbpS-like MOFRL DUF4147 NAD_binding_11 NAD_binding_2 AP_endonuc_2 TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N EamA EamA BrnT_toxin CopG_antitoxin Ureidogly_lyase Allantoicase Allantoicase Polysacc_deac_1 OHCU_decarbox Urate_ox_N Transthyretin

MBK7894999.1
JADJOJ010000006.1:154914-168363
Candidatus Leptofilum gracile

ADH_N ADH_zinc_N MUG113 ? SNIPE Beta-prop_THOC3 MerR_1 Bac_luciferase BrnT_toxin ? HTH_18 3-dmu-9_3-mt Radical_SAM Radical_SAM_C Acetylase SBP_bac_3 SBP_bac_5 G3P_acyltransf Prok-JAB

MBK8430949.1
JADJSZ010000011.1:18379-35329
Candidatus Hadersleviella danica

Pyr_redox_2 NDH2_C Phage_holin_9 LON_substr_bdg AAA Lon_lid Lon_C BrnA_antitoxin ? ? TetR_N TetR_C_24 BrnT_toxin CopG_antitoxin Calx-beta Calx-beta Calx-beta Calx-beta Calx-beta Calx-beta LTD Ig_Rha78A_N CotH LTD ? Flg_new_2 Flg_new_2 Flg_new_2

MBK8568493.1
JADKBX010000001.1:15197-34263
Nitrosomonadales bacterium

ACR_tran ACR_tran OEP OEP FapA_N HDOD ? AAA_14 DUF4143 BrnT_toxin CopG_antitoxin ? PDDEXK_1 Transposase_20 HRDC RQC RecQ_Zn_bind Helicase_C DEAD Rotamase SurA_N_2 BolA

MBK8652938.1
JADJZS010000010.1:3637-25384
Haliscomenobacter sp.

RAMPs Cas_APE2256 ? CARF_Card1 Card1_endonuc Csm6_6H Cas_Cas02710 MJ1673 ? BrnT_toxin ? AAA_30 UvrD_C_2 DUF7017 S1CSD-TOTE-2 Cas_Cas1 BT4734-like_N CRISPR_Cas2 Apiosidase Collagen_bind_2

MBK8704309.1
JADKAE010000009.1:172392-189927
Saprospiraceae bacterium

? Plug CarbopepD_reg_2 CBM_4_9 PKD PKD DapB_N DapB_C Synaptojanin EamA EamA Peptidase_S46 BrnT_toxin ? Sortilin-Vps10 Sortilin-Vps10 PKD_4 LamininG Por_Secre_tail DUF4340 Ham1p_like ?

MBK8871817.1
JADJZR010000024.1:17201-27224
Elusimicrobia bacterium

Peripla_BP_2 ? PIN Aminotran_4 Chorismate_bind ? NifU Nfu_N BrnT_toxin CopG_antitoxin GBBH-like_N ParA RHH_1 ? HepT-like NTP_transf_2

MBK8899625.1
JADJXK010000001.1:106010-122595
Candidatus Leptofilum proximum

Receiver CBS CBS His-Kinase HisKA_3 GAF_2 GAF_2 GerE Receiver Aldedh STY-kinase Fer2_2 Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 BrnT_toxin NikA-like DUF2283 DnaJ_C HSP70 TIR EAD7 ? GerE Receiver His-Kinase HisKA_3 ?

MBK8988103.1
JADJXA010000007.1:226218-239020

Candidatus Leptovillus gracilis

ATG22 VTT_dom VTT_dom HKD HKD Glyco_transf_4 Glycos_transf_1 Cu-oxidase_4 Cu-oxidase_4 BrnT_toxin CopG_antitoxin FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma Ubie_methyltran ArabinogalactanP Thioredoxin_4 AAA_18 adh_short

MBK9008760.1
JADJWS010000005.1:94805-109317

Candidatus Defluviilinea gracilis

Radical_SAM SPASM DUF2180 Glyco_transf_4 Glyco_trans_1_4 ? ? AAA VWA_CoxE BrnT_toxin ? ARG GAF_2 GAF HD_5 Receiver 4HBT_2 CBM9_1 Glyco_hydro_18 Glyco_trans_1_4 Glyco_transf_4

MBK9115196.1
JADJWJ010000005.1:24798-39821
Betaproteobacteria bacterium

Cytochrome_C7 Paired_CXXCH_1 Paired_CXXCH_1 Mtrc-MtrF_II-IV_dom Cytochrome_C7 Cytochrom_c3_2 GFO_IDH_MocA_C3 GFO_IDH_MocA Phasin_2 Receiver His-Kinase His-Kinase 7TMR-DISM_7TM Receiver GerE ? ? BrnT_toxin CopG_antitoxin Zn_ribbon_FGT1_2 adh_short_C2 Rrf2 Acetylase Molybdopterin Molydop_binding Fer4_7 Fer4_11 Fer4_11

MBK9172897.1
JADJWP010000002.1:1284726-1302897
Chloracidobacterium sp.

DUF1569 DUF7575 TPRs TPRs TPRs TPRs DUF7575 TPRs TPRs TPRs DUF7575 TolB_N TPRs TPRs TPRs RMD1-3 DUF7575 TolB_N TPRs TPRs TPRs DUF7575 Tricorn_2nd Tricorn_N Beta-prop_THOC3 Tricorn_N Tricorn_N Tricorn_N DUF7575 BrnT_toxin CopG_antitoxin DUF7575 TPRs TPRs TPRs TPR-S Beta-prop_THOC3 PD40 TPRs TPRs ? Transp_cyt_pur ?

MBK9216304.1
JADJXP010000002.1:1530707-1552413
Chloracidobacterium sp.

Wzy_C CBM_4_9 TPRs DegT_DnrJ_EryC1 HhH-GPD EndIII_4Fe-2S Peptidase_M28 TPRs TPRs TPRs TPRs TPRs CbiX BrnT_toxin CopG_antitoxin ? Hydrolase_3 TPRs Peptidase_M1 Ivy ? DAHP_synth_1

MBK9238588.1
JADKDJ010000041.1:831-11668
Candidatus Rhodoferax odensensis

? ? ? ? ? IstB_IS21 IstB_IS21 BrnT_toxin CopG_antitoxin BrnT_toxin BrnA_antitoxin GerE Receiver His-Kinase HisKA_3 GAF_2 Molybdop_Fe4S4 Molybdopterin Molydop_binding Fer4_7 Fer4_11

MBK9925601.1
JADKIW010000001.1:2131676-2145881
Candidatus Defluviilinea proxima

? Glycos_transf_2 ? Methyltransf_12 ? Sulfotransfer_1 Acyltransferase BrnT_toxin ? TIR Melibiase_2 Melibiase_C DUF2207_C DUF2207 MerR_1 Aldo_ket_red bact-PGI_C SIS

MBL0058999.1
JADKHR010000001.1:1809286-1827438
Elusimicrobia bacterium

? Cytochrome_C7 Cytochrom_c3_2 ? TMEM175 Methyltransf_23 DUF2141 MipA Glyoxalase BrnT_toxin CopG_antitoxin COX1 NorB_cytochrome_c-like HTH_Crp_2 cNMPBD Zn_ribbon_recom Recombinase Resolvase ? DUF1016_N SNase ?

MBL0177371.1
JADKJW010000031.1:212603-229826

Ignavibacteria bacterium

Ribosomal_L36 Ribosomal_S13 Ribosomal_S11 Ribosomal_S4 S4 RNA_pol_A_bac RNA_pol_A_CTD Ribosomal_L17 MMR_HSR1 BrnT_toxin CopG_antitoxin Sortilin-Vps10 Sortilin-Vps10 WD40_WDHD1_1st Por_Secre_tail Synaptojanin ? DNA_pol3_gamma3 DNAX_ATPase_lid DNA_pol3_delta2 TPR_CcmH_CycH Glyco_transf_9

MBL0350198.1
JADKIY010000002.1:840148-850484
Elusimicrobia bacterium

GIY-YIG ? HTH_Tnp_1 RDD ? Rce1-like ? BrnT_toxin CopG_antitoxin DUF5684 ? ? ? ? ?

MBL1128618.1
RBMZ01000021.1:1-9584
Chloroflexota bacterium

CHAP ? His_Phos_1 STY-kinase ? HEPN_RiboL-PSP SH3_3 Ub-Mut7C Mut7-C BrnT_toxin CopG_antitoxin GAF_2 SpoIIE STY-kinase AAA_16

MBL1130428.1
RBMZ01000065.1:1452-15965
Chloroflexota bacterium

Usp ABC_tran BPD_transp_1 BPD_transp_1 PBP_like_2 Usp MshEN Trans_reg_C BrnT_toxin ? Trans_reg_C Receiver His-Kinase His-Kinase PAS HAMP CHASE4 Hydantoinase_B GmrSD_N CHRD ?

MBL1213925.1
RBMR01000012.1:49637-60739
Ignavibacteriae bacterium

TonB_C DUF3379 Sigma70_r4_2 Sigma70_r2 TauE ? ? ? BrnT_toxin CopG_antitoxin Inositol_P Guanylate_cyc Sua5_yciO_yrdC Sua5_C TauE Lipocalin_4 GATase

MBL1275159.1
NVUB02000045.1:1274-3919
Ectothiorhodospiraceae bacterium

BrnT_toxin CopG_antitoxin Hydrolase EamA EamA

MBL3589334.1
JAESPF010000021.1:1077-15364

gamma proteobacterium endosymbiont of Lamellibrachia anaximandri

Ub-Mut7C Mut7-C Rhodanese ? DUF4114 PEP-CTERM Y1_Tnp PEP-CTERM BrnT_toxin CopG_antitoxin HTH_24 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase PAS DUF2157 YceG Thymidylate_kin DNA_pol3_delta2 DNApol3-delta_C

MBL3591843.1
JAESPF010000225.1:125-677
gamma proteobacterium endosymbiont of Lamellibrachia anaximandri

BrnT_toxin CopG_antitoxin

MBL4818056.1
JAESQZ010000065.1:39-3657
Deltaproteobacteria bacterium

Acyltransferase GPAT_C BrnT_toxin CopG_antitoxin Acetyltransf_7

MBL4818822.1
JAESQZ010000175.1:17274-20635
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin ? Bac_export_1 Bac_export_3 FliP FliO FliMN_C

MBL4942423.1
JAESPR010000117.1:3230-10773
Colwellia sp.

tRNA-synt_2 BrnT_toxin CopG_antitoxin Nitroreductase HTH_3 ? DoxX Peptidase_S49 Peptidase_S49 Asparaginase Asparaginase_C

MBL68345.1
PBLV01000026.1:15272-26814
Verrucomicrobiales bacterium

Sulfatase SGSH_C BrnT_toxin CopG_antitoxin FGS AXE1 DLH Sulfatase SGSH_C TAT_signal HXXSHH PSCyt1 PSD3 PSD5 PA14 PSD5 PSD4 PSCyt3 PSD2 Glyco_hydro_16

MBL6972986.1
JADHTS010000009.1:11225-20114
Sulfurimonas sp.

SMC_N Phage_integrase Phage_int_SAM_4 ? ? ? ? ? BrnT_toxin CopG_antitoxin ? ? DUF2158 ? ParE_toxin ?

MBL6990275.1
JADHTQ010000025.1:109-12266
Bacteriovoracaceae bacterium

? ? BrnT_toxin CopG_antitoxin ? Hydrolase_4 FBPase_2 Alpha-amylase ? ABC_membrane ABC_tran

MBL7005321.1
JADHTU010000272.1:1-5128
Gammaproteobacteria bacterium

Histidinol_dh DUF6172 DUF1456 DUF1456 CcdB CcdA DDE_Tnp_1_6 BrnT_toxin CopG_antitoxin

MBL7049784.1
JADHUV010000032.1:3809-20643
Nitrospira sp.

T2SSM_b T2SSC HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase PAS HAMP DUF4390 PRA-CH UPF0102 BrnT_toxin CopG_antitoxin C_GCAxxG_C_C SPOR SPOR TPRs TPRs TPRs PgaA_barrel Polysacc_deac_1 GHL13 Glycos_transf_2 PgaD Usp Usp

MBL7072990.1
JADHWG010000024.1:13492-25217
Candidatus Omnitrophota bacterium

? AAA_16 BrnT_toxin CopG_antitoxin DUF5666 DUF5666 Radical_SAM Bac_DNA_binding LpxD LbH_EIF2B Hexapep Hexapep Epimerase GGDEF

MBL7073307.1
JADHWG010000043.1:2194-11548
Candidatus Omnitrophota bacterium

? ? DUF4258 HTH_17 DUF8208 HTH_31 RepA_C Phage_integrase BrnT_toxin CopG_antitoxin HTH_3 Xre_MbcA_ParS_C ? ? BrnA_antitoxin ? ?

MBL7078128.1
JADHWP010000002.1:5078-16710
Candidatus Shapirobacteria bacterium

Pyr_redox_2 Thioredoxin Hydrolase FtsX MacB_PCD ABC_tran RND-MFP_C Beta-barrel_YknX BSH_RND ? BrnT_toxin CopG_antitoxin TPRs Anticodon_1 tRNA-synt_1 Asp_protease_2 DUF5678 DUF5667 ?

MBL7158191.1
JADHYM010000006.1:11866-29743
Candidatus Omnitrophota bacterium

TPP_enzyme_C POR tRNA-synt_1c tRNA-synt_1c Anticodon_2 tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2 GIY-YIG GIY-YIG GIY-YIG BrnT_toxin CopG_antitoxin ROK ? OEP OEP BSH_RND Beta-barrel_RND_2 RND-MFP_C ACR_tran ACR_tran ACR_tran

MBL7159199.1
JADHYG010000009.1:8839-21022

Candidatus Microgenomates bacterium

? ? DUF3996 ? ? ? Peptidase_S26 RDD BrnT_toxin CopG_antitoxin ? NTP_transf_2 HEPN ? SpoU_methylase SpoU_methylase PAP2

MBL7960354.1
JAEUSJ010000078.1:1429-13915
bacterium

GreA_GreB_N GreA_GreB ? Glyco_transf_4 Glycos_transf_1 GDPD CPSase_sm_chain GATase PseudoU_synth_1 BrnT_toxin CopG_antitoxin PseudoU_synth_1 PseudoU_synth_1 Bac_DNA_binding Peptidase_S49 CLP_protease SBP_bac_5 SBP_bac_5 SBP_bac_5

MBL8109573.1
JAEURC010000044.1:1-6940
Anaerolineales bacterium

? NTP_transf_2 HEPN Spore_III_AA PTPase R3H BrnT_toxin ? Thymidylate_kin Peptidase_S51 DUF1706 ?

MBL8429090.1
JAEUOG010000023.1:21213-26226
Dechloromonas sp.

DUF1801 BrnT_toxin CopG_antitoxin YoeB_toxin PhdYeFM_antitox DUF302 Cytochrome_CBB3 AhpC-TSA ?

MBL9128158.1
JAEUHF010000751.1:1-1887
Verrucomicrobiales bacterium

PhdYeFM_antitox ? ? ? BrnT_toxin ?

MBM2814689.1
JAERMQ010000017.1:41377-56831
Ignavibacteria bacterium

NusB TIG TIG TIG TIG TIG TIG TIG FGS HEPN Polbeta Por_Secre_tail ? PGI DUF4258 ? TGT DUF5618 gag-asp_proteas ? ? ?

MBM2814835.1
JAERMQ010000019.1:32367-44926
Ignavibacteria bacterium

? NTP_transf_2 HEPN ? DUF4276 ABC-ATPase PDZ_2 Peptidase_M28 PA BrnT_toxin CopG_antitoxin ? dCMP_cyt_deam_1 RibD_C TonB_C TonB_C B12-binding Beta-prop_THOC3 Por_Secre_tail

MBM2815778.1
JAERMQ010000039.1:8498-23995
Ignavibacteria bacterium

VWA BatA ? Cytochrom_C PFK Peptidase_S8 Por_Secre_tail TMEM260-like TMEM260_2 ? BrnT_toxin BrnA_antitoxin ? DUF4258 DUF2283 ? ? FIST FIST_C

MBM3124134.1
VGNF01000027.1:158-12327
Chloroflexi bacterium

SIMPL NurA HAS-barrel GT87 Ion_trans_2 HerA-FtsK HerA_C BrnT_toxin ? MazE_antitoxin PIN ? ? OsmC YhhN

MBM3131101.1
VGOB01000230.1:460-5563
Chloroflexi bacterium

Kelch_KLHDC2_KLHL20_DRC7 ? UbiA Cyt_bd_oxida_I Cyt_bd_oxida_II BrnT_toxin

MBM3261761.1
VGJQ01000038.1:1-5250
Candidatus Kaiserbacteria bacterium

? HepT-like Polbeta BrnT_toxin BrnA_antitoxin ? RmuC LemA TPM_phosphatase

MBM3265963.1
VGJU01000413.1:89-2173
Candidatus Zixiibacteriota bacterium

NTP_transf_2 WCX WYL WYL HTH_11 BrnT_toxin ?

MBM3283214.1
VGJN01000049.1:425-8716
Candidatus Gottesmanbacteria bacterium

AAA_14 DUF4143 ? DUF11 InhA-like_MAM CarboxypepD_reg CarboxypepD_reg Peptidase_S8 BrnT_toxin CopG_antitoxin LamininG

MBM3377453.1
VGHN01000008.1:1-7673
Betaproteobacteria bacterium

DUF2779 BrnT_toxin CopG_antitoxin YafQ_toxin RelB TspO_MBR Lycopene_cycl FAD_binding_3 PUCC UbiA

MBM3406815.1
VGHA01000022.1:17068-30562
Betaproteobacteria bacterium

? Ureidogly_lyase DIOX_N 2OG-FeII_Oxy DctM DctQ DctP ? CopG_antitoxin SBP_bac_11 TctC Asp_Glu_race AMP-binding AMP-binding_C_3 ECH_1 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_C

MBM3705198.1
VGAX01000035.1:241-7223
Actinomycetota bacterium

BrnT_toxin CopG_antitoxin ParA-Soj-NTPase SAF CbiA T2SSE T2SSF T2SSF

MBM3708046.1
VGBA01000106.1:287-6423
Actinomycetota bacterium

AbiEii DUF6922 BrnT_toxin CopG_antitoxin CobA_CobO_BtuR DBI_PRT DUF4872 BtrH_N ARM_TT21_2nd ?

MBM3709082.1
VGAS01000034.1:418-6943
Actinomycetota bacterium

CorA BrnT_toxin CopG_antitoxin ? Serpin HicA_toxin HicB_lk_antitox Germane HGD-D

MBM4050442.1
VGYK01002439.1:1-1752
Planctomycetota bacterium

BrnT_toxin ? DUF4962

MBM4146104.1
VGWX01000038.1:213-9410
Nitrospira sp.

PAS GGDEF EAL BrnT_toxin BrnA_antitoxin AsnC_trans_reg Slp Peptidase_M1 Cofac_haem_bdg PDZ_2 Thioredoxin_3 HTH_5 SBF

MBM4153981.1
VGWF01000011.1:1-11817
Lentisphaerae bacterium

Peripla_BP_4 Bact_hydrolase ? LamininG ? PQQ_2 PQQ_2 PQQ_2 DUF4874 DUF4832 BrnT_toxin CopG_antitoxin MgtC DUF4010 NPCBM

MBM4162604.1
VGWB01000292.1:358-3790
Ignavibacteria bacterium

Tricorn_C1 PhdYeFM_antitox ParE_toxin HigB-like_toxin HTH_3 Gp49 HTH_3 BrnT_toxin CopG_antitoxin

MBM4167510.1
VGWC01000145.1:91-2028
Ignavibacteria bacterium

UvrB_3rd Helicase_C UvrB UVR BrnT_toxin HTH_3

MBM4251766.1
VGTI01000010.1:71923-84636
Deltaproteobacteria bacterium

? DUF4336 ? AAA_14 DUF4143 Cu-oxidase_2 Cytochrom_C SCO1-SenC DUF285 DUF285 BrnT_toxin CopG_antitoxin Rax2 Rax2 Rax2 Rax2 Rax2 Rax2

MBM4310282.1
VGSH01000024.1:107-10760
Deltaproteobacteria bacterium

? ? ? ? BrnT_toxin CopG_antitoxin ORF6N Chorismate_synt SKI Pribosyltran OTCace_N OTCace DHOase

MBM5791825.1
VGPT01000056.1:270-5266

Cyanobacteria bacterium M_surface_10_m1_298

SNase ? Peptidase_M15 BrnT_toxin CopG_antitoxin ? ? ? NTP_transf_2 HepT-like ?

MBN1101778.1
JAFGPH010000002.1:1-7603
Deltaproteobacteria bacterium

? BrnT_toxin CopG_antitoxin B12-binding Radical_SAM B12-binding Radical_SAM ? PIN DUF5618 FG-GAP_3 CsgG

MBN1224298.1
JAFGNQ010000347.1:1-5581
Candidatus Aminicenantes bacterium

? BrnT_toxin CopG_antitoxin Zn_Ribbon_1 STY-kinase TPRs Beta-lactamase ? DUF2442 DUF4160 ?

MBN1363334.1
JAFGLS010000016.1:9499-26990
Syntrophaceae bacterium

? His-Kinase NOV_C Helicase_C SNF2-rel_dom DUF4143 AAA_14 PhdYeFM_antitox ParE_toxin LysR_substrate BrnT_toxin CopG_antitoxin RNA_pol_A_CTD WCX WYL HTH_11 ARPP-1 ? ? ?

MBN1363430.1
JAFGLS010000027.1:193-12658
Syntrophaceae bacterium

ACR_tran YknX_C Beta-barrel_RND_2 HH_MFP_RND BSH_CusB OEP OEP OEP OEP TetR_C_4 TetR_N ? Beta-prop_THOC3 Beta-prop_IFT122_1st BrnT_toxin CopG_antitoxin Polyketide_cyc2

MBN1433654.1
JAFGKQ010000076.1:1-4611
Candidatus Fermentibacterales bacterium

? FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 Por_Secre_tail BrnT_toxin CopG_antitoxin AAA_14 DUF4143

MBN1452403.1
JAFGKO010000241.1:58998-67664
Anaerolineales bacterium

Aminotran_1_2 Pribosyl_synth Pribosyltran_N Acetylase 2-ph_phosp Pribosyltran OMPdecase OTCace OTCace_N BrnT_toxin CopG_antitoxin Amidohydro_1

MBN1525621.1
JAFGJO010000061.1:193-5066
Spirochaetales bacterium

BrnT_toxin BrnA_antitoxin DUF6290 DUF4277 ? ? AAA_23 ABC-ATPase DUF3419

MBN1586269.1
JAFGIU010000026.1:3352-11611
Candidatus Omnitrophota bacterium

BrnA_antitoxin ArsC Receiver Receiver His-Kinase GAF HisKA_7TM ACP_syn_III_C ACP_syn_III His-Kinase His-Kinase HisKA_7TM BrnT_toxin CopG_antitoxin

MBN1586970.1
JAFGIU010000067.1:1-5521
Candidatus Omnitrophota bacterium

RibD_C Lum_binding Lum_binding BrnT_toxin CopG_antitoxin ? MgtC RHS_repeat

MBN1760540.1
JAFGHA010000243.1:10948-16404
Chitinispirillaceae bacterium

Por_Secre_tail BrnT_toxin CopG_antitoxin BcrAD_BadFG Methyltransf_21 ? Y2_Tnp Y2_Tnp BrnT_toxin

MBN1781348.1
JAFGGN010000062.1:37412-57040
bacterium

GATase Acetylase Receiver NHL NHL ? POTRA Omp85 ? BrnT_toxin CopG_antitoxin ? TPM_phosphatase TPM_phosphatase Alpha_L_fucos PA14 Arylsulfotran_2 CBM_4_9 Por_Secre_tail TEN_YD-shell

MBN1782741.1
JAFGGN010000107.1:16286-31287
bacterium

TonB_dep_Rec_b-barrel Plug CarbopepD_reg_2 SurA_N_2 PhdYeFM_antitox PIN Macro DarT ? BrnT_toxin CopG_antitoxin MFS_1 ? SocA_Panacea ? RMMBL Lactamase_B_2 Big Fib_succ_major

MBN1839670.1
JAFGFU010000017.1:256-4121
Campylobacterales bacterium

BrnT_toxin CopG_antitoxin NERD HTH_Tnp_1_2 ? Phage_Coat_B ? ?

MBN1866359.1
JAFGFN010000048.1:30331-44689
Candidatus Sumerlaeota bacterium

FHA FHA Metallophos_2 ? N_methyl ? GHMP_kinases_N Synaptojanin BrnT_toxin BrnA_antitoxin RuvB_N AAA_lid_4 WHD_RuvB HigB_toxin ? DUF8426 NOV_C Unstab_antitox ParE_toxin

MBN1915783.1
JAFGFJ010000006.1:132583-148215
Candidatus Dojkabacteria bacterium

? ? HMGL-like ACT_9 Acyl_transf_3 ? ? BrnT_toxin CopG_antitoxin ? His-Kinase PAS PspC LiaI-LiaF-like_TM1 LiaI-LiaF-like_TM1 DUF4097 LiaF-like_C ? Transgly_assoc

MBN1997219.1
JAFGEC010000157.1:5416-19861
candidate division KSB1 bacterium

Peptidase_S41 TrbL_3 SpoIIE ? ? DUF6602 ? BrnT_toxin CopG_antitoxin GAF_2 SpoIIE SpoIIAA-like Peptidase_S8 Lysine_decarbox PAS GAF_2 PAS PAS PAS HisKA_2 His-Kinase Receiver

MBN2001485.1
JAFGEC010000676.1:1111-16139
candidate division KSB1 bacterium

Hydrolase_like Hydrolase_6 Glyco_hydro127C Glyco_hydro127M Beta-AFase-like_GH127_cat SSF Peptidase_M24 23S_rRNA_IVP AAA_14 ? BrnT_toxin CopG_antitoxin ? Big Big DUF2141 ? Pribosyltran OTCace_N OTCace DHOase

MBN2016843.1
JAFGDR010000011.1:8673-23710
Candidatus Cloacimonadota bacterium

? TraB_PrgY_gumN Regulator_TrmB TrmB DUF2062 TP_0481_C Radical_SAM Spermine_synth BrnT_toxin CopG_antitoxin Fer4_9 Aldo_ket_red DUF123 SfsA SfsA_N AhpC-TSA HTH_34 ? TPRs

MBN2018458.1
JAFGDR010000061.1:163-2996
Candidatus Cloacimonadota bacterium

? ? ? BrnT_toxin CopG_antitoxin ? ?

MBN2062797.1
JAFGDF010000330.1:1-10885
Deltaproteobacteria bacterium

? Fe-ADH ADH_Fe_C ADH_N ADH_zinc_N Beta-lactamase2 Beta-lactamase2 Transpeptidase Transgly Glyco_transf_9 Phage_integrase Phage_int_SAM_4 BrnT_toxin CopG_antitoxin

MBN2065182.1
JAFGDG010000002.1:1-3085
Thermoplasmatota archaeon

AP_endonuc_2 BrnT_toxin CopG_antitoxin DUF123 TrkA_C CBS CBS Voltage_CLC

MBN2072762.1
JAFGDA010000061.1:1-6955
Actinomycetota bacterium

BrnT_toxin CopG_antitoxin PIN MazE_antitoxin Flavodoxin_5 PPDK_N ? MFS_1

MBN2103221.1
JAFGCO010000019.1:1-6562
bacterium

RHS_repeat RHS_repeat RHS_repeat BrnT_toxin BrnA_antitoxin ? PSII_BNR PSII_BNR PSII_BNR PSII_BNR TEN_YD-shell RHS_repeat RHS_repeat TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat

MBN2266916.1
JAFFZZ010000054.1:223-779
Candidatus Babeliaceae bacterium

BrnT_toxin CopG_antitoxin

MBN2271532.1
JAFGTU010000396.1:3639-13776
Sedimentisphaerales bacterium

PDDEXK_3 Aconitase_C DUF7074 TAT_signal Sulfatase SGSH_C Iso_dh DUF1015 Aminotran_5 BrnT_toxin CopG_antitoxin Sigma70_r2 Sigma70_r4_2 ? PDZ_2 ?

MBN2279326.1
JAFGTP010000008.1:130-11395
Candidatus Neomarinimicrobiota bacterium

? Por_Secre_tail Por_Secre_tail GHL10 Golvesin_C Por_Secre_tail HTH_24 ROK DUF2130 ? BrnT_toxin CopG_antitoxin

MBN2313942.1
JAFGTG010000181.1:1-5169
Sedimentisphaerales bacterium

2-Hacid_dh DUF72 ? DUF2442 BrnT_toxin CopG_antitoxin DUF4113 IMS_C IMS

MBN2362944.1
JAFGSQ010000011.1:37947-43241
candidate division WOR-3 bacterium

BrnT_toxin CopG_antitoxin Rho_N Por_Secre_tail ? PAS GGDEF DeoC SRAP

MBN2414143.1
JAFGSB010000175.1:17577-26227
candidate division KSB1 bacterium

MFS_1 MFS_1 BrnT_toxin CopG_antitoxin Pect-lyase_RHGA_epim Beta_helix DUF188 Radical_SAM_N Radical_SAM DUF3362 Por_Secre_tail Lactamase_B 4HBT

MBN2435789.1
JAFGRW010000096.1:9532-25154
Spirochaetes bacterium

? ? H_PPase HD_3 DUF4143 AAA_14 Transglutaminase HTH_18 ? Acetyltransf_7 Peripla_BP_3 LacI Cellulase Glyco_hydro_12 ? FGS AAA_14 DUF4143

MBN2542171.1
JAFGQG010000057.1:49693-62780
bacterium

? Methyltransf_11 YjeF_N Carb_kinase ADH_N ADH_zinc_N FAD_binding_4 Fer4_10 Fer4_16 CGGC BrnT_toxin CopG_antitoxin MM_CoA_mutase Glyoxalase_4 Cys_Met_Meta_PP CobA_CobO_BtuR AlaDh_PNT_N AlaDh_PNT_C AcetylCoA_hydro AcetylCoA_hyd_C

MBN2561675.1
JAFGQA010000262.1:1-8533
Phycisphaerae bacterium

? ParE_toxin HEATs HEATs HEATs ? ? ? BrnT_toxin CopG_antitoxin RHH_1 PIN ? ? ? ?

MBN2769198.1
JAFGVW010000010.1:291-3368
Spirochaetes bacterium

Gp49 HTH_3 BrnT_toxin CopG_antitoxin DDE_Tnp_1_6 DUF772

MBN2772386.1
JAFGVW010000197.1:9532-25154
Spirochaetes bacterium

? ? H_PPase HD_3 DUF4143 AAA_14 Transglutaminase HTH_18 ? Acetyltransf_7 Peripla_BP_3 LacI Cellulase Glyco_hydro_12 ? FGS AAA_14 DUF4143

MBN2824287.1
JAFGVH010000104.1:1-5654
Campylobacterales bacterium

? ? BrnT_toxin CopG_antitoxin Molybdopterin Molybdop_Fe4S4 Fer4_Nqo3 NADH-G_4Fe-4S_3 Fer2_4 DUF389 PDDEXK_9

MBN2824305.1
JAFGVH010000107.1:1-6770
Campylobacterales bacterium

? ? BrnT_toxin CopG_antitoxin ? Methyltransf_25 GmrSD_C GmrSD_N DXPR_C DXP_redisom_C DXP_reductoisom CTP_transf_1

MBN2824395.1
JAFGVH010000124.1:585-4643
Campylobacterales bacterium

UPF0175 BrnT_toxin ? ? ParE_toxin Radical_SAM EamA EamA DNA_alkylation

MBN2824815.1
JAFGVH010000194.1:81-2323
Campylobacterales bacterium

YMR265C_C CopG_antitoxin ABC_tran ?

MBN2825842.1
JAFGVH010000384.1:1-5729
Campylobacterales bacterium

? Fic HTH_24 Transgly Transpeptidase BiPBP_C BrnT_toxin CopG_antitoxin QueH

MBN2839230.1
JAFGUR010000288.1:86-958
Fusobacteriaceae bacterium

? BrnT_toxin CopG_antitoxin

MBN2852051.1
JAFGUQ010000091.1:1-1236
Clostridia bacterium

? BrnT_toxin BrnA_antitoxin ?

MBN2872073.1
JAFGUH010000105.1:17930-28898
Halothiobacillaceae bacterium

AA_kinase Acetylase SAM_MT ? ATP-grasp_4 Patatin DUF2058 DEAD Helicase_C BrnT_toxin CopG_antitoxin CbbQ_C

MBN2886698.1
JAFGTZ010000309.1:368-6464
Chromatiaceae bacterium

BrnT_toxin ? ParE_toxin RelB-like RuBisCO_small RuBisCO_large_N RuBisCO_large RuBisCO_small CsoS2_M CsoS2_M CsoS2_M

MBN37456.1
PBMI01000017.1:27513-45969
Opitutia bacterium

TSP3_bac TSP3_bac Cadherin ? PAD_porph FGS FGS TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N BrnT_toxin CopG_antitoxin Helicase_C SNF2-rel_dom SNF2_assoc DHHA1 tRNA_SAD tRNA-synt_2c Iso_dh NusA_N KH_NusA_1st KH_NusA_2nd IF2_N IF2_N GTP_EFTU EF-G_D2 IF-2 GTP_EFTU_D2 RBFA

MBN4058284.1
JAFIUL010000010.1:1-8517
Mariprofundus ferrooxydans

? ? BrnT_toxin CopG_antitoxin HflX_C MMR_HSR1 GTP-bdg_M GTP-bdg_N 6TM_PilS His-Kinase His-Kinase Receiver HTH_8 ? ?

MBN4063648.1
JAFITY010000030.1:2541-7602
Cardiobacterium sp. AH-315-I02

PhdYeFM_antitox ? ? ParE_toxin RHH_1 Gp49 HTH_37 BrnT_toxin CopG_antitoxin VapB_antitoxin Poly_export SLBB ?

MBN8440319.1
JAFLDS010000001.1:257563-278334
Thauera sp.

Receiver NtrC-AAA AAA_lid_14 HTH_50 CopD ABC_tran ABC2_membrane NYN OST-HTH HNH_2 CTD11 AAA_23 DUF2326 BrnT_toxin CopG_antitoxin Pps Pps Pps Pps Pps Pps Pps Pps Glyoxalase_5 Glyoxalase TPP_enzyme_C POR DUF6537 TP_methylase Cytochrome_CBB3 Cytochrom_D1 Cytochrome_CBB3

MBN8504792.1
JAFLDE010000032.1:24818-38416
Burkholderiales bacterium

TPRs TPRs TPRs GAF_2 His-Kinase His-Kinase MM_CoA_mutase MM_CoA_mutase MeaB B12-binding BrnT_toxin CopG_antitoxin ? Abhydrolase_6 DUF6597 HTH_18 DUF6265 Peptidase_S9_N Peptidase_S9_N Peptidase_S9

MBN8682100.1
JAFLBP010000007.1:3271-20829
Chitinophagales bacterium

LbH_EIF2B NTP_transferase ? ? 3HCDH_N 3HCDH ECH_1 DUF779 Aldedh AraC_N HTH_18 BrnT_toxin ? SHOCT DUF697 RtcB Calcineurin SDH_alpha SDH_beta ATP-grasp_4 ?

MBN9070959.1
JAFKJW010000067.1:27932-39517
Hyphomicrobiales bacterium

PrmA SCO1-SenC ? CreA Hexapep Rce1-like adh_short BrnT_toxin CopG_antitoxin PcrA_UvrD_tudor UvrD_C UvrD-helicase UvrD-helicase ? Glyoxalase_7 OmpA YkuD EamA EamA

MBO4266431.1
JAGAFR010000036.1:1-5208
Lachnospiraceae bacterium

? MgtC Phage_integrase ALP_N ? BrnT_toxin

MBO4320450.1
JAGADS010000227.1:20250-25697
Treponema sp.

? TEN_YD-shell TEN_YD-shell PIN ? TEN_YD-shell ? TEN_YD-shell BrnT_toxin CopG_antitoxin PAF-AH_p_II

MBO4461498.1
JAGAAH010000044.1:158-4109
Lachnospiraceae bacterium

VTC DUF4956 CotH BrnT_toxin

MBO4888059.1
JAFZQZ010000075.1:418-6342
Bacillota bacterium

BrnT_toxin BrnA_antitoxin Phage_integrase ParA-Soj-NTPase Fer4_NifH ParB_N NUDIX ? Big_14

MBO4956498.1
JAGAUJ010000008.1:2246-19205
Rickettsiales bacterium

Mtase_EDM2 Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 Toprim_2 DNAG_N Zn_ribbon_DnaG DivIC SPOR ? Bro-N ? CopG_antitoxin DUF401 ? SPOR Gate FeoB_C Gate FeoB_Cyto FeoB_N FeoA ?

MBO5731026.1
JAGBXW010000058.1:6139-14373
Treponema sp.

NTase_sub_bind Formyl_trans_N AIRS_C AIRS ? BrnA_antitoxin ? ? HTH_3 ? CBS CBS Na_H_Exchanger YitT_membrane DUF2179

MBO6235899.1
JAGBLV010000225.1:349-2205
Schwartzia sp. (in: firmicutes)

SLH Porin_O_P BrnT_toxin ?

MBO6303574.1
JAGBKP010000012.1:374-14002
Selenomonadaceae bacterium

GtrA_DPMS_TM FBPase_2 BrnT_toxin ? Receiver GGDEF EAL MatE GARS_A Flagellin_C ? CBM_48 Alpha-amylase Alpha-amylase_C

MBO7105594.1
JAGCOE010000104.1:1-14031
Fibrobacter sp.

CBM9_2 VWA Flg_new_2 Flg_new_2 Flg_new_2 ? ? PseudoU_synth_2 ? ? BrnT_toxin CopG_antitoxin ? FloA ABC_tran ABC_tran_Xtn

MBP1716431.1
JAGFWZ010000930.1:1-4546
Deltaproteobacteria bacterium

TEN_YD-shell AFOR_C AFOR_N AFOR_N BrnT_toxin CopG_antitoxin SPASM

MBP3222107.1
JAGBKK010000037.1:47412-56527
Neisseriaceae bacterium

TbpB_B_D LeuA_dimer HCS_D2 HMGL-like Porin_1 BrnT_toxin BrnA_antitoxin CydP Cyt_bd_oxida_I Cyt_bd_oxida_II Nitro_FeMo-Co YbgT_YccB Cyd_oper_YbgE

MBP3365459.1
JAGBGX010000002.1:59682-73931
Treponema sp.

His_biosynth NYN NYN OST-HTH ? TEN_YD-shell NADase_transloc TEN_YD-shell TEN_YD-shell TEN_YD-shell BrnT_toxin BrnA_antitoxin Imm43 Barstar ? ? ? Phage_integrase

MBP3417042.1
JAGBFT010000099.1:196-4307
Spirochaetaceae bacterium

PhdYeFM_antitox YoeB_toxin TdpA_C Glycos_transf_2 ? ? BrnA_antitoxin

MBP3760667.1
JAGAYM010000056.1:4482-13509
Ruminococcus sp.

Bro-N Phage_pRha Phage_pRha MobA_MobL ? BrnT_toxin BrnA_antitoxin ? ? Dockerin_1 BrnT_toxin BrnA_antitoxin Receiver LytTR

MBP5162310.1
JAGDDI010000156.1:1504-12896
Spirochaetales bacterium

AAA LON_substr_bdg BrnT_toxin BrnA_antitoxin Vut_1 PHP PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV PGM_PMM_III PGM_PMM_I Glyco_hydro_94 GH94_b-supersand SBP_bac_1

MBP5469569.1
JAGCWO010000633.1:375-3594
Candidatus Riflebacteria bacterium

Y1_Tnp ? CopG_antitoxin ATPase_2 DUF234

MBP5788747.1
JAGCPQ010000030.1:353-4732
Neisseriaceae bacterium

TusA MotA_ExbB ExbD BrnT_toxin BrnA_antitoxin Hydrolase E1-E2_ATPase HMA

MBP5789790.1
JAGCPQ010000271.1:788-10101
Neisseriaceae bacterium

YceD Ribosomal_L32p FA_synthesis BrnT_toxin BrnA_antitoxin Beta-barrel_CAF17_C GCV_T ABC_membrane ABC_tran HMGL-like HCS_D2 LeuA_dimer HTH_31 HigB-like_toxin Ldh_1_N Ldh_1_C

MBP6103676.1
JAGNYL010000005.1:21318-31542
Gammaproteobacteria bacterium

GidB SAM_GIDA_C GIDA_C_1st GIDA MnmE_helical TrmE_N 60KD_IMP YidC_periplas YidD Ribonuclease_P Ribosomal_L34 BrnT_toxin CopG_antitoxin RHH_3 RelB3_C ParE_toxin DUF1653 Peptidase_M15 HTH_25 Bro-N

MBP6277002.1
JAGNVG010000047.1:1-8823
Limnohabitans sp.

RlmL_1st THUMP UPF0020 PIN BrnT_toxin CopG_antitoxin UPF0149 SEC-C Abhydrolase_1 TctC AMP-binding_C AMP-binding Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N Abhydrolase_1

MBP6533654.1
JAGNQI010000018.1:127-8327
Arenimonas sp.

? VanZ Methyltransf_28 adh_short_C2 LpqE-like DUF533 DUF697 ? BrnT_toxin CopG_antitoxin HPPK FolB TsaD Ribosomal_S21 YqeY

MBP6781695.1
JAGNMV010000467.1:1-2878
Ottowia sp.

adh_short_C2 SBP_bac_5 BrnT_toxin CopG_antitoxin

MBP6881257.1
JAGOPW010000004.1:51749-63197
Candidatus Paceibacterota bacterium

? NicO ABC_tran NifU_N Aminotran_5 SUFBD_core SufBD_N ? BrnT_toxin CopG_antitoxin ? Metalloenzyme Enolase_N Enolase_C Glyco_hydro_68 Glyco_hydro_32N GST_N_3 Ribosomal_S30AE

MBP6919980.1
JAGOOP010000006.1:61640-74154
Candidatus Omnitrophota bacterium

CbiA CbiA Lactamase_B DsrE FUR Catalase Catalase-rel B12-binding Radical_SAM ? CopG_antitoxin DUF378 PNPOx_N MipA PHP_C APH ORF6N B12-binding Radical_SAM

MBP7095708.1
JAGOKR010000023.1:21897-37149
Spirochaetia bacterium

GGDEF Tll0287-like ? FAD-oxidase_C FAD_binding_4 DUF1667 Pyr_redox_2 Fer2_BFD DAO FCD GntR BrnT_toxin CopG_antitoxin cNMPBD LUD_dom Aminotran_1_2 HisKA_7TM HisKA_2 His-Kinase Receiver MFS_1

MBP7552795.1
JAGOCL010000087.1:1-8649
Spirochaetes bacterium

MFS_1 ABC2_membrane ? ? DUF4382 ? AbiEii BrnT_toxin CopG_antitoxin ABC_tran

MBP7584242.1
JAGODF010000104.1:1-11752
Spirochaetes bacterium

FtsK_SpoIIIE FtsK_alpha FtsK_4TM AA_permease_2 OMP_b-brl RNase_J_C RMMBL RNase_J_b_CASP Lactamase_B dUTPase BrnT_toxin CopG_antitoxin ? Helicase_C_3 ResIII ERCC3_RAD25_C Polbeta HepT-like DUF2019 ?

MBP7654845.1
JAGOBX010000569.1:298-2351
Candidatus Dependentiae bacterium

Cnd1 BrnT_toxin CopG_antitoxin

MBP7709660.1
JAGNLL010000001.1:219072-232085
Rickettsiales bacterium

Mur_ligase_C Mur_ligase_M Mur_ligase HhH-GPD EndIII_4Fe-2S Fic MerR_1 ? S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C CRCB S4 PseudoU_synth_2 BrnT_toxin CopG_antitoxin Ala_racemase_C Ala_racemase_N PRA-CH GCS2 PTPS 4HBT_2 SRP54_N SRP54 SRP_SPB

MBP7781660.1
JAGNJQ010000007.1:7422-19466
Burkholderiales bacterium

TP_methylase NAD_binding_7 CbiK Ribosomal_L27 Ribosomal_L21p ? DUF455 BrnT_toxin CopG_antitoxin DUF455 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 ? PriA_C Helicase_C Zn_ribbon_PriA DEAD PriA_3primeBD Calcineurin Glyco_transf_22

MBP8082985.1
JAGNAP010000055.1:1-12633
Spirochaetes bacterium

NadA Rrf2 PALP Cys_Met_Meta_PP N6_Mtase Methylase_S MazE_antitoxin PIN DUF4258 CopG_antitoxin PhdYeFM_antitox PIN HTH_18 Acetylase Isochorismatase GH_cat

MBP9022399.1
JAGPET010000028.1:365-9993
Spirochaetes bacterium

DUF6364 ParE_toxin BrnT_toxin CopG_antitoxin RHH_1 PIN Beta-lactamase2 NUDIX ANKs ANKs CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg bMG3 MG2 A2M_BRD A2M TED_complement bMG10

MBP9097024.1
JAGPDN010000451.1:1-2615
Ignavibacteria bacterium

? ? BrnT_toxin Oxidored_q4

MBP9145980.1
JAGPDF010000060.1:14882-27723
Thermoanaerobaculia bacterium

TPRs TPRs TPRs TPRs TPRs UnbV_ASPIC FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 Y1_Tnp LiaI-LiaF-like_TM1 zf-HC2 zf-HC2 Sigma70_r4_2 Sigma70_r2 Rce1-like BrnT_toxin CopG_antitoxin ? Peptidase_M64 M64_N YoeB_toxin PhdYeFM_antitox ?

MBP9501647.1
JAGOZU010000021.1:32357-40778
Promineifilum sp.

AAA_assoc_2 MgsA_C SH3_2 BrnT_toxin CopG_antitoxin Esterase ? ? NEDD4_Bsd2 Ribosomal_L9_N Ribosomal_L9_C HTH_25 RodZ_C

MBP9670533.1
JAGOVT010000038.1:494-7418
Candidatus Woesebacteria bacterium

BrnT_toxin CopG_antitoxin ? TPRs N_methyl DEC-1_N PilW_C N_methyl ?

MBP9700528.1
JAGOVC010000047.1:1-7094
Candidatus Woesebacteria bacterium

Anticodon_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 ? BrnT_toxin CopG_antitoxin DUF8128 TrwB_AAD_bind ?

MBP9702176.1
JAGOVG010000001.1:120665-133281
Candidatus Woesebacteria bacterium

Polysacc_synt RVT_3 DUF4012 Ribosomal_L25p Ribosomal_TL5_C TPRs TrmB MFS_1 BrnT_toxin CopG_antitoxin TrmB ? EamA EamA PIG-L ? ?

MBP9703093.1
JAGOVG010000014.1:11654-19209
Candidatus Woesebacteria bacterium

DDAH_eukar GerE ? ? ? DALR_1 tRNA-synt_1d Arg_tRNA_synt_N tRNA_bind ? NikA-like Receiver His-Kinase His-Kinase PAS TctA

MBP9726054.1
JAGOUN010000004.1:13450-25413
Gammaproteobacteria bacterium

? ? ? LysE dUTPase PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV TauE BrnT_toxin CopG_antitoxin Ala_racemase_N F420_oxidored P5CR_dimer CCB3_YggT CCB3_YggT 5-FTHF_cyc-lig RuvX ?

MBP9726387.1
JAGOUN010000012.1:23317-33390
Gammaproteobacteria bacterium

Pro_dh Pro_dh-DNA_bdg PRODH TerC RNase_T CopG_antitoxin BrnT_toxin ? 1-cysPrx_C AhpC-TSA Glutaredoxin PMT Arnt_C EamA DegT_DnrJ_EryC1 Epimerase

MBP9727299.1
JAGOUN010000069.1:215-8772
Gammaproteobacteria bacterium

SNF2-rel_dom Helicase_C BrnT_toxin CopG_antitoxin NTP_transferase Epimerase HepT-like Polbeta PGI

MBP9782013.1
JAGOSJ010000103.1:1-3584
Candidatus Woesebacteria bacterium

AbiEii ? ? CopG_antitoxin Acetylase Glycos_transf_2 Mannosyl_trans2

MBP9836594.1
JAGOSI010000012.1:990-12419
Candidatus Paceibacterota bacterium

? Acetylase Peptidase_S51 ? ? Fic WHD_DprA DNA_processg_A BrnT_toxin CopG_antitoxin ? TfoX_N Adenine_glyco DUF6263 DUF218 Seryl_tRNA_N tRNA-synt_2b

MBP9854201.1
JAGOSC010000018.1:22812-32267
Candidatus Omnitrophota bacterium

HEPN GIY-YIG YhdP MazE_antitoxin PIN AAA_33 ? ? CopG_antitoxin DUF202 Pyr_redox_2 Pyr_redox_dim Rhodanese Sulf_transp DUF6691 Cation_efflux ZT_dimer ?

MBP9866127.1
JAGOQZ010000215.1:1-3756
Candidatus Omnitrophota bacterium

? BrnT_toxin CopG_antitoxin ? tRNA_edit ?

MBQ0163371.1
JAGHSB010000248.1:122-6405
Candidatus Treponema scatequi

BrnT_toxin BrnA_antitoxin Put_DNA-bind_N CoA_binding Aminotran_5 2-Hacid_dh_C Transp_cyt_pur RsgA_GTPase eIF-1a

MBQ1335785.1
JAFOIJ010000160.1:116-1832
Selenomonadaceae bacterium

BrnT_toxin ? DUF6444 DDE_Tnp_IS66

MBQ1510663.1
JAFOMW010000302.1:1-5820
Selenomonadaceae bacterium

Hexapep ? Macro Peptidase_C26 TrmO_N tRNA-synt_2 N-OB GreA_GreB GreA_GreB_N BrnT_toxin BrnA_antitoxin

MBQ1914274.1
JAFOVM010000161.1:100-2180
Selenomonadaceae bacterium

BrnT_toxin ? ? ? Hexapep

MBQ1914408.1
JAFOVM010000195.1:1-9300
Selenomonadaceae bacterium

? Peptidase_C26 TrmO_N tRNA-synt_2 N-OB GreA_GreB GreA_GreB_N BrnT_toxin BrnA_antitoxin ? GmrSD_N T1RH-like_C T1R_D3-like SWI2_SNF2 REase ? PIN

MBQ2204403.1
JAFPEL010000042.1:9-10346
Lachnospiraceae bacterium

BrnT_toxin BrnA_antitoxin Peripla_BP_2 FecCD ABC_tran ? DUF2229 DUF2229 BcrAD_BadFG BcrAD_BadFG ChlI Mg_chelatase Mg_chelatase_C

MBQ2465912.1
JAFPID010000689.1:259-3098
Treponema sp.

? ? ? ? Peptidase_M24 Creatinase_N MurJ

MBQ2678235.1
JAFQXQ010000072.1:13232-25767
Bacillota bacterium

His_biosynth IGPD Aminotran_1_2 HisG tRNA-synt_His BrnT_toxin BrnA_antitoxin ApbE NusG_II Hpre_diP_synt_I CHB_HEX_C_1 CHB_HEX_C_1 fn3

MBQ2997097.1
JAFQRI010000128.1:373-11432
Oscillibacter sp.

Peptidase_M20 EamA EamA SIS HTH_6 Glt_symporter OCD_Mu_crystall OCD_Mu_crystall Enolase_N Enolase_C BrnT_toxin BrnA_antitoxin

MBQ3347919.1
JAFRBG010000001.1:15005-28597
Candidatus Saccharibacteria bacterium

? ? ? FemAB FemAB FemAB E1-E2_ATPase Hydrolase Cation_ATPase_C ? BrnT_toxin BrnA_antitoxin TsaD TsaD Fib_alpha HTH_28 SPOUT_MTase NUDIX ?

MBQ3829461.1
JAFPTX010000018.1:233-7473
Spirochaetales bacterium

ChlI Mg_chelatase Mg_chelatase_C BrnT_toxin ? H_PPase HD ?

MBQ3836879.1
JAFPTY010000368.1:1-6730
Treponema sp.

LB_356 NTP_transf_8 DUF6788 SBP_bac_3 BrnA_antitoxin HTH_3 Hyphal_reg_CWP BrnT_toxin BrnA_antitoxin ?

MBQ3922702.1
JAFPPV010000304.1:193-3855
Spirochaetales bacterium

FGS ? BrnT_toxin BrnA_antitoxin PFK

MBQ4403776.1
JAFPVP010000083.1:13828-24908
Selenomonadaceae bacterium

AAA AAA_lid_3 ? ? ? Sel1 Sel1 Sel1 Sel1 Sel1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 CTB BrnT_toxin ? FHA FhaA_N Ribosomal_L17 RNA_pol_A_CTD RNA_pol_A_bac S4 Ribosomal_S4 Ribosomal_S11 Ribosomal_S13

MBQ6360710.1
JAFRLH010000042.1:54476-75122
Lachnospiraceae bacterium

FA_synthesis PP-binding Ribonucleas_3_3 dsrm SMC_N Tyr_Deacylase LtrA tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C BrnT_toxin ? Cation_efflux ZT_dimer ? Lyase_aromatic Lyase_aromatic BTAD AAA_16 Na_H_antiporter Na_H_antiporter

MBQ6407982.1
JAFRMU010000124.1:106-5360
Butyrivibrio sp.

HTH_28 PhoU ? DUF3847 MobA_MobL BrnT_toxin BrnA_antitoxin ?

MBQ6480932.1
JAFRIF010000187.1:1-2724
Anaerolineaceae bacterium

AhpC-TSA MazE_antitoxin PIN BrnT_toxin BrnA_antitoxin VanZ ?

MBQ6566119.1
JAFRVF010000151.1:1-3204
Treponema sp.

? BrnT_toxin BrnA_antitoxin

MBQ6623655.1
JAFREV010000118.1:1-7051

Mogibacterium sp.

Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B Cna_B BrnT_toxin BrnA_antitoxin DFP Flavoprotein

MBQ6976461.1
JAFSLZ010000304.1:1-9046
Selenomonadaceae bacterium

tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C BrnT_toxin ? Mur_ligase_M Mur_ligase_C His-Kinase His-Kinase Receiver Hpt TPP_enzyme_N

MBQ7199327.1
JAFSUE010000270.1:131-1755
Selenomonadaceae bacterium

DUF445 BrnT_toxin

MBQ7454225.1
JAFSVS010000221.1:214-2801
Selenomonadaceae bacterium

BrnT_toxin ? Zn_ribbon_LapB AAA_25 ChlI Cation_ATPase_C

MBQ7477617.1
JAFSWY010000048.1:81-4405
Selenomonadaceae bacterium

RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 BrnT_toxin ? ? AAA-ATPase_like

MBQ7500962.1
JAFTAK010000257.1:499-5302
Clostridia bacterium

BrnT_toxin BrnA_antitoxin Rubrerythrin Rubrerythrin RUBY_RBDX Ni_insertion AIRC SHOCT-like_DUF2089-C Asn_synthase ? ?

MBQ7644935.1
JAFSWH010000252.1:249-9172
Spirochaetales bacterium

? SDF Alpha-amylase Pept_tRNA_hydro DUF4416 ? ORF6N BrnT_toxin BrnA_antitoxin

MBQ8199563.1
JAFTWX010000077.1:117753-143690
Lachnospiraceae bacterium

YfhO YfhO YfhO YfhO GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C Radical_SAM HemN_C Sigma70_r2 Sigma70_r4 T2SSE ChlI Mg_chelatase Mg_chelatase_C DNA_processg_A BrnT_toxin ? LRRs LRRs LRRs LRRs LRRs LRRs LRRs LRRs Ig_Eae LRRs MshEN T2SSE T2SSF_N T2SSF T2SSF N_methyl A24_N_bact Peptidase_A24 PilM_2 PilN

MBQ8348545.1
JAFTQW010000013.1:1-4367
Ruminococcus sp.

BrnA_antitoxin ? Cortex-I_coil ? MPS2 HTH_3 ? BrnT_toxin ?

MBQ8560123.1
JAFUOU010000031.1:142-2481
Spirochaetaceae bacterium

CHB_HEX_C_1 Big DUF6273 ? CopG_antitoxin

MBQ9017384.1
JAFUBU010000084.1:9668-15461
Candidatus Saccharibacteria bacterium

BrnT_toxin BrnA_antitoxin ? GtrA_DPMS_TM Epimerase Competence ? TsaE

MBQ9334232.1
JAFSTN010000036.1:177-12700
Lachnospiraceae bacterium

HGSNAT_cat ? FGAR-AT_linker AIRS_C GATase_5 QueH QueH 2-Hacid_dh_C Arg_tRNA_synt_N tRNA-synt_1d DALR_1 AI-2E_transport BrnT_toxin UPF0154

MBQ9348481.1
JAFTAN010000817.1:1-2969
Oscillibacter sp.

Cyc-maltodext_C Alpha-amylase BrnT_toxin ? ? ?

MBQ9360483.1
JAFTAP010000058.1:31515-46801
Lachnospiraceae bacterium

S4 FtsJ MFS_1 Peripla_BP_3 LacI MatE MatE NAD_kinase NAD_kinase_C ? SMC_N BrnT_toxin BrnA_antitoxin Glyco_transf_5 Glycos_transf_1 NFACT_N NFACT_N NFACT-R_1 SEC-C HisKA_7TM GGDEF EAL IF3_N IF3_C Ribosomal_L35p

MBQ9480500.1
JAFTDM010000734.1:1-3401
Selenomonadaceae bacterium

? BrnT_toxin ? HEPN_Swt1 DUF499

MBQ9615326.1
JAFSQR010000158.1:1-8444
Selenomonadaceae bacterium

DUF445 DUF445 BrnT_toxin ? PilZ HD_5 MCPsignal CCDC39 HAMP 4HB_MCP_1 Rubrerythrin RUBY_RBDX Gln-synt_C

MBQ9620129.1
JAFSQX010000385.1:159-2228
Neisseriaceae bacterium

? ? PFIN BrnT_toxin BrnA_antitoxin

MBQ9630249.1
JAFSRC010000097.1:1-3695
Treponema sp.

S-AdoMet_synt_C BrnT_toxin CopG_antitoxin HhH-GPD EndIII_4Fe-2S Flavodoxin_4 MS_channel_1st MS_channel_2nd MS_channel_3rd HTS

MBQ9683478.1
JAFTFY010000030.1:33956-46473
Neisseriaceae bacterium

Isochorismatase URO-D ? ? SPMIP5 DUF465 ACT Formyl_trans_N BrnT_toxin BrnA_antitoxin GlnE GlnD_UR_UTase GlnE GlnD_UR_UTase Fer2 Ribosomal_S20p DUF4435 ABC-ATPase Glyco_hydro_3 SpoU_methylase MRM3-like_sub_bind

MBR0118760.1
JAFRWA010000073.1:50-2558
Eubacterium sp.

BrnT_toxin ? DMRL_synthase UPF0547 MscL ?

MBR0420015.1
JAFRJA010000155.1:62-9221
Erysipelotrichaceae bacterium

CBM_X2 LRRs BrnT_toxin BrnA_antitoxin Fic HTH_24 Phage_int_SAM_1 Phage_integrase Phage_integrase Phage_int_SAM_1 Phage_integrase Phage_integrase

MBR1373960.1
JAFUTF010000376.1:1-2472
bacterium

BrnT_toxin BrnA_antitoxin ? DnaB_C Toprim_2

MBR1615645.1
JAFUXS010000433.1:9265-27888
Treponema sp.

LRRs LRRs LRRs LRRs LRRs LRRs VWA_CoxE AAA_lid_8 bpMoxR LRRs LRRs Pectate_lyase_4 HTH_3 BrnT_toxin BrnA_antitoxin RelB YoeB_toxin HTH_18 HTH_5 HMA E1-E2_ATPase Hydrolase Toprim_2 DnaB_C

MBR1730358.1
JAFUZV010000189.1:10440-22908
Selenomonadaceae bacterium

ABC_tran Methyltransf_11 Serpin FlK RmlD_sub_bind PARG_cat_microb Pribosyltran_N Pribosyl_synth BrnT_toxin ? HATPase_c_3 HSP90 HSP90 HAD_2 Ribosomal_S20p ? Rhodanese CstA CstA CstA

MBR1743001.1
JAFVAA010000634.1:1-4008
Lachnospiraceae bacterium

? ? Arm-DNA-bind_4 Phage_int_SAM_3 Phage_integrase BrnT_toxin CopG_antitoxin

MBR1796597.1
JAFVAY010000002.1:187447-201845
Clostridiales bacterium

CBS NTP_transferase DegT_DnrJ_EryC1 DegT_DnrJ_EryC1 Acetyltransf_6 Epimerase NeuB SAF Epimerase_2 CTP_transf_3 BrnT_toxin Guanylate_cyc ? PycTM HTH_26 SLFN_AlbA_2 HATPase_c_4 Sigma70_r4 ? UPF0175

MBR1804874.1
JAFVBP010000018.1:1-12523
Selenomonadaceae bacterium

adh_short adh_short_C2 Flavodoxin_4 NUMOD4 HNH_3 Mac Hexapep Transgly ? Arginosuc_synth Arginosuc_syn_C BrnT_toxin ? Cupin_2 DUF3280 FlgT_N PASTA Transpeptidase PBP_dimer ? ? TelA

MBR1852085.1
JAFVCJ010000027.1:54-6863
Lachnospiraceae bacterium

BrnT_toxin BrnA_antitoxin Radical_SAM MatE MatE Transposase_31 Sigma70_r2 Sigma70_r4_2 DUF4179 HTH_3

MBR1874069.1
JAFVCW010000072.1:16249-30439
Eubacterium sp.

ketoacyl-synt Ketoacyl-synt_C LRRs Flg_new HTH_5 LRRs Aldolase_II DUF5662 Big Big Big Big BrnT_toxin ? Thiamine_BP BPD_transp_1 NMT1 ABC_tran Cthe_2159 PAPS_reduct

MBR3622198.1
JAFXOC010000113.1:95-1616
Selenomonadaceae bacterium

? FliS FliT BrnT_toxin ?

MBR3654882.1
JAFXOV010000040.1:57-4498
Elusimicrobia bacterium

BrnT_toxin ? ? ? NikA-like MobC Relaxase ? RepL

MBR3721404.1
JAFXMX010000035.1:204-12777
Selenomonadaceae bacterium

TatD_DNase Anticodon_1 tRNA-synt_1g tRNA-synt_1g TP_methylase PSP1_3rd PSP1 DNA_pol3_delta2 DUF327 ? BrnT_toxin ? Glyco_hydro_28 Pect-lyase_RHGA_epim fn3 ? PDDEXK_2

MBR4569863.1
JAFXRA010000124.1:1-6646
Candidatus Riflebacteria bacterium

AAA_14 DUF4143 MarR HTH_3 BrnT_toxin CopG_antitoxin ? Peptidase_M50 CTP_transf_1 Prenyltransf

MBR4629846.1
JAFXSM010000020.1:165419-174677
Treponema sp.

? ? Peripla_BP_2 FecCD ABC_tran AAA_18 ? BrnT_toxin BrnA_antitoxin Flg_new Flg_new Flg_new

MBR4737415.1
JAFZMK010000179.1:1-8115
Rhodocyclaceae bacterium

DUF1318 ? ? ? ? ? Flagellin_N Flagellin_IN Flagellin_C BrnT_toxin BrnA_antitoxin ? BrnA_antitoxin 4HBT BrnT_toxin

MBR5407839.1
JAFZOS010000083.1:468-25799
Lachnospiraceae bacterium

MacB_PCD FtsX ? PDDEXK_2 DivIC ? RUBY_RBDX Rubrerythrin Zn_Ribbon_1 YARHG BrnT_toxin ? HTH_3 Y1_Tnp Glyco_hydro_94 GH94_b-supersand LacI Peripla_BP_3 Glyco_hydro_2_N2 Glyco_hydro_2 Mannosidase_ig Ig_mannosidase Glyco_hydro_36N Melibiase Glyco_hydro_36C

MBR5412241.1
JAFZOV010000052.1:1-12415
Fibrobacter sp.

? RF-1 PCRF Rotamase SurA_N_3 BrnT_toxin CopG_antitoxin Nitroreductase Nitroreductase ? SMC_N HisKA_7TM GGDEF EAL GGDEF EAL ?

MBR5412459.1
JAFZOV010000068.1:320-10888
Fibrobacter sp.

Fib_succ_major Fib_succ_major Fib_succ_major Acetyltransf_7 Multi_Drug_Res Fib_succ_major ? BrnT_toxin BrnA_antitoxin Fib_succ_major ? DUF4160 DUF2442 LPAM_1 Fib_succ_major Fib_succ_major

MBR5413304.1
JAFZOV010000132.1:63-5453
Fibrobacter sp.

? RF3_C EF-G_D2 GTP_EFTU SPOUT_MTase Metallophos_2 DUF4258 BrnA_antitoxin

MBR5913577.1
JAFZIV010000024.1:1-14111
Selenomonadaceae bacterium

Sigma70_r4_2 SDF Amidase_2 BrnT_toxin ? TRCF Helicase_C DEAD CarD_TRCF_RID UvrB_inter CTB PAS GGDEF EAL Beta-Roll Beta-Roll

MBR5969915.1
JAFZLB010000066.1:1-6340
Lachnospiraceae bacterium

BPD_transp_1 Alpha-amylase Alpha-amylase HTH_31 Couple_hipA HipA_C ParE_toxin RelB Polbeta BrnT_toxin ? ?

MBR5998476.1
JAFZLM010000218.1:134-3343
Deltaproteobacteria bacterium

DUF4186 BrnT_toxin BrnA_antitoxin tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1

MBR6165774.1
JAFXZS010000017.1:61431-74855
Clostridia bacterium

MazE_antitoxin DUF6076 Cytidylate_kin Acyltransferase LYTB S1 S1 S1 S1 F420_ligase SH3_3 BrnT_toxin BrnA_antitoxin DNA_alkylation Acetylase ? DcuC 5-FTHF_cyc-lig NUDIX

MBR6934588.1
JAFWQX010000073.1:14606-25276
Clostridia bacterium

? NFACT_N NFACT-R_1 BrnT_toxin BrnA_antitoxin URO-D Peptidase_M3 Peptidase_M3_N GDPD MFS_1 MFS_1 MFS_2 His_Phos_1 YjbQ

MBR7063830.1
JAFWUJ010000082.1:430-1776
Treponema sp.

Rep3_N Rep3_C BrnT_toxin

MBS0154217.1
JAFEBE010000005.1:278293-291777
Nitrospira sp.

PIN Mrr_N Mrr_cat ? pP_pnuc_1 ? ? ? BrnT_toxin CopG_antitoxin Imm_superinfect TIR Aldo_ket_red ? ? HTH_8 AAA_lid_14 NtrC-AAA Receiver

MBS0165654.1
JAFEBH010000008.1:597-11965
Nitrospira sp.

B12-binding Radical_SAM PmbA_TldD_1st PmbA_TldD_3rd PmbA_TldD_1st PmbA_TldD_2nd PmbA_TldD_3rd ? TPRs LDcluster4 PMSR BrnT_toxin CopG_antitoxin DrHyd

MBS0184555.1
JAFEBM010000066.1:36781-47974
Nitrospira sp.

Sigma70_r2 Sigma70_r4_2 An_peroxidase HI0933_like TPR_MLP1_2 ? ACCA N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP BrnT_toxin CopG_antitoxin DUF4265

MBS0236782.1
JAFECK010000035.1:1-5998
Proteobacteria bacterium

Transpeptidase BufA1 New_glue BufA1 DUF4258 CopG_antitoxin Acetyltransf_8 Ribosomal_L19 tRNA_m1G_MT ? DUF2610

MBS0438122.1
JAFEEK010000059.1:1-4136
Proteobacteria bacterium

TPRs TPRs TPRs PIN MazE_antitoxin BrnT_toxin CopG_antitoxin DUF6090 ?

MBS0619449.1
JAFEGD010000218.1:1-11740
Spirochaetes bacterium

ClpS Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small TMC Calcineurin ? AhpC-TSA 1-cysPrx_C Orn_Arg_deC_N Orn_DAP_Arg_deC ? BrnT_toxin BrnA_antitoxin HTH_Tnp_1 HTH_21 rve DNA_methylase DNA_methylase RE_Eco29kI ? TMEM175

MBS1162926.1
JAGSYY010000123.1:1-3812
Burkholderiaceae bacterium

BrnT_toxin CopG_antitoxin GMC_oxred_C GMC_oxred_N Rrf2 Acetylase Fer4_2 Fer4_11

MBS1259514.1
JAANXD010000096.1:10998-15628
Candidatus Scalindua arabica

? BrnT_toxin CopG_antitoxin ? ? ? ? Nitroreductase Nitroreductase APS_kinase GTP-eEF1A_C GTP_EFTU_D2 GTP_EFTU

MBS1768736.1
JAFDVK010000012.1:494102-503736
Acidobacteria bacterium

RibD_C Methyltransf_11 ? SnoaL_2 Glyoxalase MgtE_N CBS CBS MgtE MgtC BrnT_toxin CopG_antitoxin ? TMEM175 DinB ? ? PEMT

MBS1816942.1
JAFDVO010000005.1:42401-60307
Acidobacteria bacterium

BSH_RND Beta-barrel_RND_2 YknX_C PadR Methyltransf_31 HTH_5 Gp_dh_N Gp_dh_C MFS_1 MFS_1 Rhodanese Rhodanese BrnT_toxin rve_3 rve HTH_21 HTH_Tnp_1 CopG_antitoxin ? N6_N4_Mtase ResIII Fic Fic_N ?

MBS3756951.1
JAGXKN010000081.1:1-4894
Desulfobacterales bacterium

GARS_A GARS_C AIRC Sua5_yciO_yrdC NAPRTase_N NAPRTase_C UPF0150 BrnT_toxin CopG_antitoxin ?

MBS6699563.1
JAGZWU010000017.1:14987-32280
Faecalibacterium prausnitzii

BD-FAE Peptidase_S9 ? HD FtsX ABC_tran ? Receiver His-Kinase BrnT_toxin ? BcrAD_BadFG BcrAD_BadFG DUF2229 ? PFK PFK Lipase_GDSL_2 Nitroreductase

MBS9782440.1
JAGYHK010000008.1:1-10361
Arcobacter sp.

TPRs TPRs TPRs TPRs TPRs TPRs WHD_NWD1 AAA_16 DUF4062 UPF0175 DUF3368 BrnT_toxin CopG_antitoxin RelB-like HTH_29 rve rve_3 YhcG_C DUF1016_N ? T4SS-DNA_transf Sec6

MBT3011668.1
JAHHGJ010000007.1:71-7192
Candidatus Thiodiazotropha sp. (ex Lucina pensylvanica)

? BrnT_toxin CopG_antitoxin Glyoxalase GGDEF ? OsmC Pyr_redox_dim Pyr_redox_2 ?

MBT3191023.1
JABGOW010000033.1:1-5520
Verrucomicrobia bacterium

Phage_CRI HTH_17 ? Phage_integrase ? BrnT_toxin HEPN HTH_18

MBT3199063.1
JABGPF010000029.1:871-20501
Phycisphaerales bacterium

Sulfatase SGSH_C GFO_IDH_MocA_C3 GFO_IDH_MocA AP_endonuc_2 ? 3keto-disac_hyd HEATs HEATs HEATs HEATs GFO_IDH_MocA_C2 GFO_IDH_MocA PfkB BrnT_toxin CopG_antitoxin CAP SASA Sulfatase ? ANKs ANKs ANKs ?

MBT3214512.1
JABGPG010000165.1:2309-10292
Deltaproteobacteria bacterium

BPD_transp_1 BPD_transp_1 EF_0837-like_N SelA PfkB Aldolase LacI Peripla_BP_3 BrnT_toxin CopG_antitoxin

MBT3235764.1
JABGPI010000020.1:57673-69952
Bdellovibrionales bacterium

? DUF2249 Hemerythrin CheD Receiver ? ODP ? BrnT_toxin BrnA_antitoxin FG-GAP_3 FG-GAP_3 FG-GAP_3 DUF374 EFG_C EFG_IV EFG_III EF-G_D2 GTP_EFTU ? AIRS AIRS_C Rhodanese AAA_SelU

MBT3249509.1
JABGPU010000014.1:36812-40920
Candidatus Paceibacterota bacterium

? ? CopG_antitoxin ? ? ? YafQ_toxin TPRs Methyltransf_23

MBT3294246.1
JABGQM010000004.1:142912-161369
Verrucomicrobia bacterium

VWA VIT ARPP-1 zf-HC2 Sigma70_r4_2 Sigma70_r2 A2M_recep TED_complement A2M A2M_BRD MG2 ? PFK BrnT_toxin CopG_antitoxin 4HBT_2 DUF6874 AMP-binding_C AMP-binding MR_MLE_C MenC_N UbiA ECH_1

MBT3297104.1
JABGQM010000091.1:126-12440
Verrucomicrobia bacterium

FGS Lipase_GDSL_2 BrnT_toxin CopG_antitoxin ? Peripla_BP_3 HTH_18 Matrix_Paramyxo_N HATPase_c_4 DUF4062 LamininG GMPPB_C DUF6055

MBT3321690.1
JABGQV010000090.1:1-14977
Anaerolineae bacterium

? Sirtuin GAF_2 His-Kinase His-Kinase ADH_zinc_N ADH_N G_glu_transpept MscL UvrD-helicase UvrD_C PcrA_UvrD_tudor BrnT_toxin CopG_antitoxin ? FAD_binding_3 RecO_C RecO_N dCMP_cyt_deam_1 CorC_HlyC CBS CBS CNNM Hexapep Hexapep LbH_EIF2B NTP_transf_3

MBT3341663.1
JABGRH010000041.1:1-9530
Gemmatimonadetes bacterium

Glyco_hydro_95_cat Glyco_hyd_65N_2 IPT IPPT BrnT_toxin CopG_antitoxin Glyco_transf_4 Glyco_trans_1_4 ? NAD_binding_10 SdpI FAD_binding_2 Succ_DH_flav_C NUDIX

MBT3347526.1
JABGRB010000029.1:11281-29026
Thiotrichales bacterium

AAA_14 DUF4143 HTH_3 OMP_b-brl Autotransporter DUF4172 Fic His-Kinase PAS HAMP Phosphonate-bd BrnT_toxin CopG_antitoxin TrkA_N Na_H_Exchanger TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N TetR_N Pyr_redox_dim Pyr_redox_2 DUF4143 AAA_14 Sigma70_r2 Sigma70_r4_2

MBT3347794.1
JABGRB010000033.1:1714-18405
Thiotrichales bacterium

COX4_pro ? PpnN_C Lysine_decarbox DUF4478 Aminotran_3 EAL Receiver Receiver Hpt Receiver His-Kinase His-Kinase GAF PAS Receiver PAS BrnT_toxin ? ParE_toxin DUF6364 RES Xre_MbcA_ParS_C Xre-like-HTH HTH_8 SoFic-like_C Fic Fic_N

MBT3382713.1
JABGRN010000006.1:64000-79981
Prolixibacteraceae bacterium

Receiver Trans_reg_C adh_short CN_hydrolase ? TAT_signal AP_endonuc_2 ? Ldh_2 BrnT_toxin CopG_antitoxin Lactamase_B_2 OEP OEP ACR_tran YknX_C BSH_RND Cupin_2 FAD_binding_2 Succ_DH_flav_C

MBT3473972.1
JABGTF010000275.1:8203-18557
Gammaproteobacteria bacterium

Receiver Hpt BrnT_toxin CopG_antitoxin NIT His-Kinase His-Kinase Receiver Hpt DUF6901 ? Chorismate_synt MFS_1_like GGDEF

MBT3712172.1
JABILV010000102.1:221-3509
Anaerolineae bacterium

BrnT_toxin CopG_antitoxin ? Radical_SAM Fer4_7 Gly_radical PFL-like

MBT3749860.1
JABIMO010000506.1:1-1890
Bacteroidetes bacterium

YoeB_toxin BrnT_toxin CopG_antitoxin DDE_Tnp_1

MBT4087896.1
JABIES010000249.1:55-8845
Deltaproteobacteria bacterium

NMT1_3 DctM GCV_H BrnT_toxin CopG_antitoxin ? MFS_1 Phage_integrase Molybdopterin COesterase COesterase

MBT4262767.1
JABICZ010000018.1:21998-34712
Deltaproteobacteria bacterium

PALP NIR_SIR_ferr NIR_SIR NIR_SIR_ferr NIR_SIR Fer4_20 Pyr_redox_2 Fer4 PALP Methyltransf_11 Cupin_2 Bac_globin BrnT_toxin CopG_antitoxin DUF6326 HTH_3 HigB_toxin HTH_3 Lactamase_B DUF2971

MBT4288426.1
JABIDG010000333.1:9360-16671
Deltaproteobacteria bacterium

BPD_transp_1 Hydrolase_3 Regulator_TrmB TrmB Thr_dehydrat_C Thr_dehydrat_C PALP PNPase HTH_5 ? BrnT_toxin CopG_antitoxin Rubredoxin Glu_synthase

MBT4289299.1
JABIDG010000434.1:1-4725
Deltaproteobacteria bacterium

? DUF4143 AAA_14 BrnA_antitoxin BrnT_toxin BrnT_toxin CopG_antitoxin His-Kinase

MBT4290879.1
JABIDG010000619.1:312-5263
Deltaproteobacteria bacterium

MazE_antitoxin PIN ? Unstab_antitox BrnT_toxin CopG_antitoxin PIN PhdYeFM_antitox PIN PhdYeFM_antitox HTH_3 HigB-like_toxin

MBT4668580.1
JABHWN010000128.1:1-8340
Candidatus Ruthturnera sp.

ParA-Soj-NTPase Peptidase_S41 PDZ_6 CtpB_N-like MlaC HTH_5 LGT Thymidylat_synt Zn_Ribbon_YjdM YjdM BrnT_toxin CopG_antitoxin LamininG Big

MBT4837030.1
JABJPU010000230.1:7265-14577
Methylococcales bacterium

ACR_tran BrnT_toxin ? ? Polyketide_cyc2 MFS_1 ? SRAP Zn_ribbon_21

MBT4936780.1
JABJNA010000115.1:371-8911
Candidatus Peregrinibacteria bacterium

T2SSE LDcluster4 DCD Gp49 ? BrnT_toxin CopG_antitoxin ? ? DUF3179 Adenylsucc_synt DDE_Tnp_IS66

MBT4937115.1
JABJNA010000185.1:1-11436
Candidatus Peregrinibacteria bacterium

PycTM ? ? DMRL_synthase Glycos_transf_4 BrnT_toxin CopG_antitoxin ? Peptidase_M23 Glyco_hydro2_C5 SLH SLH SLH PKD_4 PKD_4 PKD_4 PKD_4 PKD_4 ? ParE_toxin PhdYeFM_antitox

MBT4971898.1
JABJLO010000060.1:30055-47800
Thiotrichales bacterium

AAA_14 DUF4143 HTH_3 OMP_b-brl Autotransporter DUF4172 Fic His-Kinase PAS HAMP Phosphonate-bd BrnT_toxin CopG_antitoxin TrkA_N Na_H_Exchanger TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N TetR_N Pyr_redox_dim Pyr_redox_2 DUF4143 AAA_14 Sigma70_r2 Sigma70_r4_2

MBT4972501.1
JABJLO010000104.1:1-2394
Thiotrichales bacterium

HTH_3 OMP_b-brl BrnT_toxin ? Bax1-I

MBT5759207.1
JABIVB010000100.1:1-2796
Candidatus Neomarinimicrobiota bacterium

Big CADHERIN Por_Secre_tail Gp49 HTH_3 BrnT_toxin ?

MBT63564.1
PBYS01000075.1:176-10151
Puniceicoccaceae bacterium

HTH_8 AAA_lid_14 NtrC-AAA Receiver G_glu_transpept G_glu_transpept DUF1573 MauE Rhodanese FumaraseC_C Lyase_1 BrnT_toxin CopG_antitoxin YoeB_toxin PhdYeFM_antitox MazE_antitoxin PemK_toxin ?

MBT7116876.1
JABHKJ010000012.1:8800-19517
Campylobacteraceae bacterium

? Band-7 Aminotran_4 Aldedh Helicase_C DEAD S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 ? BrnT_toxin ? Rhodanese

MBT8351558.1
JAHHLT010000537.1:800-3806
Deltaproteobacteria bacterium

AAA_14 DUF4143 BrnT_toxin CopG_antitoxin ?

MBT8367681.1
JAHHLY010001106.1:1-1644
Deltaproteobacteria bacterium

TIR BrnT_toxin CopG_antitoxin Pribosyltran

MBT8420282.1
JAHHNN010000233.1:1-3491
Gammaproteobacteria bacterium

? ? ? PhdYeFM_antitox BrnT_toxin ? ? ?

MBU0480904.1
JAHJDD010000029.1:3445-16243
Proteobacteria bacterium

? His_Phos_1 CobU DBI_PRT CobS Aminotran_1_2 CbiA GATase_3 CobD_Cbib BrnT_toxin Toast-rack_PspC-Cterm ThiC_Rad_SAM MazE_antitoxin PemK_toxin Peptidase_M20 PIN PhdYeFM_antitox

MBU0482545.1
JAHJDD010000070.1:686-9996
Proteobacteria bacterium

ABC_tran BPD_transp_1 SBP_bac_11 Nitro_FeMo-Co Nitro_FeMo-Co Radical_SAM Oxidored_nitro Oxidored_nitro BrnT_toxin CopG_antitoxin AbiEii HTH_31

MBU0506596.1
JAHJDR010000350.1:1-7854
bacterium

B12-binding Radical_SAM ? BrnT_toxin ? OMPdecase Pribosyltran 4HBT_2 Ala_racemase_N F420_oxidored P5CR_dimer CCB3_YggT

MBU0576176.1
JAHJBS010000019.1:1-9316
Patescibacteria group bacterium

DNA_ligase_A_N HTH_DeoR Fic DUF5654 Carb_kinase SMC_N DUF4258 CopG_antitoxin SpoIID T4SS_pilin T4SS_pilin Prenyltransf ?

MBU0578384.1
JAHJBU010000002.1:18465-29639
Patescibacteria group bacterium

DUF5654 AbiEi_4 AbiEii Carb_kinase SMC_N NTP_transf_2 HEPN DUF4258 CopG_antitoxin DUF4325 His-Kinase SpoIID ? PIN T4SS_pilin Defensin_beta T4SS_pilin

MBU0582029.1
JAHJBX010000005.1:8574-21370
Alphaproteobacteria bacterium

? CreA Hexapep YetF_N DUF421 adh_short ? Hydrolase_4 OsmC BrnT_toxin CopG_antitoxin YjdM Tlde1_dom PcrA_UvrD_tudor UvrD_C UvrD-helicase UvrD-helicase OmpA YkuD YqhI DLH

MBU0710795.1
JAHIZH010000062.1:978-16207
bacterium

? SRP54_N SRP54 UPF0004 Radical_SAM TRAM_2 BrnT_toxin BrnA_antitoxin DUF3800 ? BrnT_toxin CopG_antitoxin NTP_transferase LbH_EIF2B UPF0004 Radical_SAM dNK Formyl_trans_N MGS AICARFT_IMPCHas MreB_Mbl

MBU0951956.1
JAHIVV010000036.1:5887-23933
Elusimicrobia bacterium

HKD ResIII SNF2-rel_dom Helicase_C MmeI_Mtase Eco57I M_Eco57I_C ? ResIII AEP-TOTE ? PIN Transposase_mut BrnT_toxin CopG_antitoxin ? DDE_Tnp_1_4

MBU0975139.1
JAHIVL010000101.1:920-14412
Patescibacteria group bacterium

Glycos_transf_1 Glyco_transf_4 DNA_ligase_A_C DNA_ligase_A_M DNA_ligase_A_M DNA_ligase_A_N Fic YafQ_toxin DUF6364 Carb_kinase SMC_N DUF4258 CopG_antitoxin SpoIID Pox_VERT_large NET2A_C T4SS_pilin T4SS_pilin Prenyltransf NUDIX MDD_C MVD-like_N

MBU1016790.1
JAHIUT010000021.1:1117-8616

Patescibacteria group bacterium

BrnT_toxin CopG_antitoxin AAA_14 DUF4143 Fic DUF4143 AAA_14 TPRs TACO1_YebC_N Transcrip_reg Transcrip_reg RuvC

MBU1043393.1
JAHITY010000039.1:644-5442
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin BrnT_toxin CopG_antitoxin PhdYeFM_antitox YoeB_toxin Thy1 HD_5 GGDEF GAF PAS

MBU1087522.1
JAHITE010000087.1:171-7325
Candidatus Omnitrophota bacterium

? ? CopG_antitoxin Rubrerythrin RsgA_GTPase ? Rhodanese MCPsignal HAMP

MBU1088820.1
JAHITA010000112.1:1-4192

Patescibacteria group bacterium

ABC_tran MacB_PCD FtsX ? ? ? DUF4258 CopG_antitoxin

MBU1104409.1
JAHISR010000015.1:203-12637
Patescibacteria group bacterium

TsaD SBP_bac_1 Methyltransf_11 RecA_N RecA_C Cupin_2 BrnT_toxin CopG_antitoxin RNase_J_C RNase_J_b_CASP Lactamase_B_2 UDG Lon_lid AAA ? tRNA-synt_1g tRNA-synt_1g 3D G5

MBU1122140.1
JAHISK010000092.1:296-4638
Candidatus Omnitrophota bacterium

? N6_N4_Mtase ? BrnT_toxin CopG_antitoxin DUF4160 DUF2442

MBU1181689.1
JAHJQU010000193.1:1-3113
Proteobacteria bacterium

Y1_Tnp BrnT_toxin RHH_1 DUF4258 DUF2283 GntR

MBU1188458.1
JAHJQV010000027.1:83224-99893
Gammaproteobacteria bacterium

PhdYeFM_antitox FAD_binding_7 AAA_23 ABC-ATPase OLD-like_TOPRIM UvrD-helicase DUF2711 NAPRTase NAPRTase_N Isochorismatase DUF4258 CopG_antitoxin LysR_substrate HTH_1 MS_N MS_TIM-barrel MS_C ICL ICL ? Peptidase_S9_N BNR_2 DUF8300 S6PP_C

MBU1207567.1
JAHJQK010000117.1:289-11502
Proteobacteria bacterium

Lactamase_B NTP_transf_5 EarA AAA_14 CopG_antitoxin TraG-D_C ? BrnT_toxin CopG_antitoxin ? AAA_lid_3 Peptidase_M78 His-Kinase TsaD Peptidase_M50

MBU1326801.1
JAHJOJ010000010.1:1-9929
Patescibacteria group bacterium

? Nramp CBS CBS MgtE_N GTP_EFTU GTP_EFTU EFG_C LepA_C RHH_1 HigB-like_toxin BrnT_toxin CopG_antitoxin TrmB ? ? ?

MBU1726065.1
JAHJJA010000006.1:15424-23085
Candidatus Omnitrophota bacterium

? ParE_toxin UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C APC1_C MazE_antitoxin PIN ShlB POTRA_2 ? NikA-like RHH_1 ParE_toxin TPS

MBU1863887.1
JAHJGZ010000151.1:468-2532
Candidatus Omnitrophota bacterium

ORF6N ORF6N ? CopG_antitoxin

MBU2444034.1
JAHJVF010000010.1:211-3920
Bacteroidetes bacterium

Gp49 HTH_3 CNNM CBS CorC_HlyC BrnT_toxin CopG_antitoxin YafQ_toxin

MBU2514687.1
JAHJTV010000081.1:171-6325
bacterium

? BrnT_toxin CopG_antitoxin DUF7120 PemK_toxin ? DUF996 BNR_2 HicB_lk_antitox HicA_toxin

MBU2515937.1
JAHJTV010000122.1:432-4467
bacterium

? ? DUF772 Y2_Tnp Y2_Tnp

MBU2574779.1
JAHJSU010000281.1:10508-22314
Elusimicrobia bacterium

N6_Mtase ? BrnT_toxin CopG_antitoxin ? ? ? ACR_tran SBP_bac_5 BPD_transp_1

MBU2609372.1
JAHJRW010000006.1:1-5264
Chloroflexota bacterium

BPD_transp_1 OppC_N BPD_transp_1 BPD_transp_1_N SBP_bac_5 DHOase ?

MBU2631950.1
JAHJRT010000635.1:1-10404
Proteobacteria bacterium

BCA_ABC_TP_C ABC_tran Fer4_11 AFOR_N AFOR_C TetR_N GMC_oxred_N GMC_oxred_C ABM HGD-D BrnT_toxin CopG_antitoxin DPBB_1 ? Phospholipase-like EamA

MBU2632098.1
JAHJRC010000002.1:1-2657
Patescibacteria group bacterium

Ribosomal_L33 ? Ser_hydrolase BrnT_toxin ?

MBU2644894.1
JAHJRI010000150.1:12848-25573
bacterium

ABC_tran ABC_tran PhdYeFM_antitox PIN TrkA_C TrkA_N Na_H_Exchanger DUF3124 PIN FitA-like_RHH BrnT_toxin CopG_antitoxin PIN PhdYeFM_antitox PIN VapB_antitoxin Fe-ADH AAA_14 DUF4143

MBU3910959.1
JAHIEN010000002.1:1-4428
Candidatus Omnitrophota bacterium

UPF0020 DUF6199 ELYS ? CopG_antitoxin RelE MqsA_antitoxin ORF6N HTH_19 DUF6037 DUF6171

MBU3917587.1
JAHIEK010001029.1:846-10114
bacterium

? DDE_Tnp_IS240 BrnT_toxin CopG_antitoxin ? Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 B12-binding Radical_SAM

MBU3950290.1
JAHIFE010000230.1:1-5643
Proteobacteria bacterium

? BrnT_toxin LAMB4 ? PIN CBM_4_9 ? YafQ_toxin Epimerase

MBU4016679.1
JAHIGH010000125.1:187-9205

Patescibacteria group bacterium

? TypeIII_RM_meth N6_N4_Mtase ? ResIII HI_1054_N HI_1054 ? Bro-N ? CopG_antitoxin RuvA_N HHH_5 RuvA_C RuvB_N AAA_lid_4 WHD_RuvB

MBU4098200.1
JAHIHU010000030.1:181-7243
Patescibacteria group bacterium

? TypeIII_RM_meth N6_N4_Mtase ? ResIII HI_1054_N HI_1054 ? ? ? CopG_antitoxin

MBU4098592.1
JAHIHU010000096.1:293-6393
Patescibacteria group bacterium

LeuA_dimer HCS_D2 HMGL-like ? ? CLC_0170 DUF7076 ? PhdYeFM_antitox ? BrnT_toxin CopG_antitoxin PsbP_2

MBU4200775.1
JAHIJO010000175.1:1-6039
Verrucomicrobia bacterium

DUF4874 DUF4832 BrnT_toxin CopG_antitoxin TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs

MBU4229472.1
JAHIKD010000095.1:322-13138
Proteobacteria bacterium

FG-GAP_3 FG-GAP_3 ParE_toxin RHH_1 ? MazE_antitoxin PIN BrnT_toxin CopG_antitoxin tRNA-synt_His tRNA-synt_His HGTP_anticodon N-OB tRNA-synt_2 HEATs Hydrolase PapZ_C RapZ-like_N Sigma54_DBD Sigma54_CBD Sigma54_AID

MBU4312265.1
JAHILR010000135.1:24-4906
Candidatus Omnitrophota bacterium

? gag-asp_proteas ? ? Macro ? ? CopG_antitoxin PhdYeFM_antitox

MBU4313153.1
JAHILR010000207.1:1-3254
Candidatus Omnitrophota bacterium

Xre_MbcA_ParS_C RES RHH_5 ParE_toxin BrnT_toxin CopG_antitoxin DUF3368 DUF6290 Xre_MbcA_ParS_C

MBU4375874.1
JAHINC010000002.1:19919-31237
Candidatus Omnitrophota bacterium

Sec_GG SecD_1st SecDF_P1_head SecD_SecF_C Sec_GG SecD_SecF_C Cation_efflux ZT_dimer ABhydrolase AA_permease_2 PTS_EIIA_2 Receiver NYN HTH_Tnp_ISL3 ? Peripla_BP_5 CopG_antitoxin ? ? ?

MBU4445358.1
JAHIOG010000377.1:336-5308
bacterium

Flavodoxin_1 FmdE DsbD_2 Unstab_antitox ? DUF6290 ? BrnT_toxin CopG_antitoxin

MBU4487033.1
JAHIPF010000407.1:1-7692

Candidatus Delongbacteria bacterium

ResIII Mrr_cat_2 AbiEii BrnT_toxin CopG_antitoxin zf-HC2 Sigma70_r4_2 Sigma70_r2 PGK TIM ?

MBU6170187.1
JAGWXY010000006.1:12687-28762
Verrucomicrobia bacterium

UDPGP PRA-CH Ribosomal_L19 YdjY SQHop_cyclase_N Cys_Met_Meta_PP Helicase_C SNF2-rel_dom SNF2_assoc SWIM BrnT_toxin CopG_antitoxin Lactamase_B Rotamase SurA_N_3 AI-2E_transport Beta-lactamase tRNA-synt_1b DJ-1_PfpI

MBU6338391.1
JAGWYT010000001.1:42912-66126
Rickettsiales bacterium

? His-Kinase His-Kinase Receiver His-Kinase Methyltransf_12 MASE8 His-Kinase His-Kinase Receiver His-Kinase His-Kinase Gp49 dnstrm_HI1420 Receiver BrnT_toxin CopG_antitoxin HlyC ? Phospholipase-like Beta-Roll Beta-Roll HCBP_related HCBP_related Beta-Roll Beta-Roll Beta-Roll Beta-Roll He_PIG Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll Toluene_X Phage_integrase Phage_int_SAM_1 Peptidase_M48 TPRs

MBU8911115.1
JAHOYS010000222.1:106-5351
Desulfobacterales bacterium

? BrnT_toxin CopG_antitoxin Pyridoxal_deC POR TPP_enzyme_C PFOR_II POR_N

MBU8913656.1
JAHOYX010000144.1:1-5831
Spirochaetales bacterium

? RHH_1 PIN BrnT_toxin CopG_antitoxin ? oligo_HPY ABC_tran oligo_HPY ABC_tran ?

MBV1914505.1
JAHRWD010000032.1:63424-78112
Pseudomonadales bacterium

TPRs ChAPs Receiver Glu_synthase GCV_T GCV_T_C GCV_H GDC-P GDC-P GcvP2_C PUA_3 Methyltrans_SAM BrnT_toxin CopG_antitoxin ? HAD NUDIX TANGO2 TauE LGT

MBV5330188.1
JAFLKX010000339.1:9681-21925
Chlorobium sp.

? DDE_Tnp_1_2 DUF772 HEATs ? BrnA_antitoxin PHP DNA_pol3_alpha DNA_pol3_finger BPD_transp_2 BPD_transp_2 ABC_tran ABC_tran Peripla_BP_4 DeoC

MBV9073269.1
JAFAUE010000176.1:1-19176
Acidobacteriota bacterium

Vut_1 DMFA2_C Big Big Big LamininG Big Big Big Big Big Big LamininG AhpC-TSA ? RecG_dom3_C Helicase_C Aldose_epim Peptidase_S9 Tricorn_N Tricorn_N Tricorn_2nd BrnT_toxin CopG_antitoxin HhH-GPD Pur_ac_phosph_N Pur_ac_phosph_N Gly-zipper_YMGG FtsX MacB_PCD FtsX MacB_PCD

MBV9388443.1
JAFASY010000233.1:144-3608

Chroococcidiopsidaceae cyanobacterium CP_BM_ER_R8_30

? BrnT_toxin CopG_antitoxin PMT_2 TPR_CcmH_CycH

MBV9390020.1
JAFASY010000314.1:13415-25615

Chroococcidiopsidaceae cyanobacterium CP_BM_ER_R8_30

Beta-prop_WDR5 Beta-prop_WDR3_1st HMA_2 DUF5132 Hydrolase E1-E2_ATPase HMA_2 HMA_2 HMA_2 UDG ADH_zinc_N ADH_N BrnT_toxin CopG_antitoxin Cas12f1-like_TNB ? ? SnoaL_2 ? Lcl_C

MBV9576340.1
JAFAWK010000282.1:1-7094
Gammaproteobacteria bacterium

TrbL VirB8 CagX TrbI T4SS-DNA_transf DUF4156 ? BrnT_toxin CopG_antitoxin ?

MBW1650611.1
JAFDMF010000023.1:127-11873
Deltaproteobacteria bacterium

RelB-like PK_C PK BrnT_toxin ? SPL Anti-Pycsar_Apyc1 N6_Mtase PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I ? Secretin Secretin_N type_II_gspD_N0

MBW1676364.1
JAFDLX010000090.1:33-9698
Deltaproteobacteria bacterium

Fer4_8 Fer4_4 2Fe-2S_thioredx Complex1_51K NADH_4Fe-4S Fer4_20 Pyr_redox_2 PIN RHH_1 ZnR ? BrnT_toxin ? ?

MBW1695123.1
JAFDLF010000429.1:1-1080
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin ?

MBW1698679.1
JAFDLK010000053.1:11693-29336
Deltaproteobacteria bacterium

BBP2_2 ? Wzy_C DUF4838 Glyco_hydro_67N Receiver DUF218 Polysacc_synt_2 CoA_binding_3 BrnT_toxin ? AbiEii AbiEi_4 HigB-like_toxin HTH_3 ? rve rve_2

MBW1717033.1
JAFDLW010000221.1:41-7218
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin Glyoxalase_4 ATPase_2 Amidohydro_2 23S_rRNA_IVP PriA_C Helicase_C RUBY_RBDX Zn_ribbon_PriA DEAD PriA_3primeBD ?

MBW1717450.1
JAFDLW010000315.1:3-3456
Deltaproteobacteria bacterium

HTH_23 Y1_Tnp PEP-CTERM ? PIN DUF433 DUF5615 BrnT_toxin RHH_1

MBW1723919.1
JAFDLB010000015.1:1-7795
Deltaproteobacteria bacterium

HK Radical_SAM ? VapB_antitoxin PIN C_GCAxxG_C_C BrnT_toxin CopG_antitoxin Peptidase_M90 ? Radical_SAM Radical_SAM

MBW1768661.1
JAFDKO010000623.1:1-4246
Deltaproteobacteria bacterium

MFS_1 MFS_1 BrnT_toxin CopG_antitoxin Slp Slp Abhydrolase_1 ?

MBW1784409.1
JAFDKT010000331.1:1-4746
Deltaproteobacteria bacterium

VapB_antitoxin PIN CN_hydrolase BrnT_toxin CopG_antitoxin SBP_bac_3

MBW1788169.1
JAFDKU010000308.1:284-5766
Deltaproteobacteria bacterium

Calcineurin DUF2196 Spermine_synth DUF188 BrnT_toxin BrnA_antitoxin ? Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N ?

MBW1854355.1
JAFDJX010000255.1:1-2676
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin UbiD_N UbiD UbiD_C

MBW1896436.1
JAFDJD010000228.1:170-3024
Deltaproteobacteria bacterium

BrnT_toxin RHH_10 Glycos_transf_2 ? Cytochrome_C7 His-Kinase

MBW1929131.1
JAFDIV010000010.1:273-10745
Deltaproteobacteria bacterium

HTH_23 Y1_Tnp PEP-CTERM ? PIN DUF433 DUF5615 BrnT_toxin RHH_1 NTP_transf_2 HEPN PIN DUF6364 Polysacc_synt ?

MBW1931254.1
JAFDIV010000050.1:9873-14416
Deltaproteobacteria bacterium

Y1_Tnp BrnT_toxin CopG_antitoxin Rv0078B PIN MazE_antitoxin ? NTP_transf_2 PIN

MBW1999633.1
JAFDIC010000293.1:155-4819
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin Abhydrolase_1 DUF2283 Fer4_7 4HBT

MBW2003814.1
JAFDHR010000043.1:6148-12325
Deltaproteobacteria bacterium

FdhD-NarQ SurE BrnT_toxin CopG_antitoxin ? Asp_protease_2 ? N6_N4_Mtase N6_N4_Mtase HTH_11 ParB_N Peptidase_S24 HIT

MBW2097917.1
JAFDGK010000030.1:4977-14437
Deltaproteobacteria bacterium

Pyr_redox_2 Pyr_redox_2 Fer4_7 FlpD Fer4_8 Fer4_17 Fer4_22 FAD_binding_6 NAD_binding_1 DHODB_Fe-S_bind ? ? BrnT_toxin CopG_antitoxin HTH_3 HigB-like_toxin

MBW2098807.1
JAFDGK010000102.1:1-6339
Deltaproteobacteria bacterium

BrnT_toxin ? AbiEii AbiEi_4 HigB-like_toxin HTH_3 ? rve rve_3

MBW2196069.1
JAFDFK010000566.1:106-1911
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin HTH_11 ?

MBW2237771.1
JAFDEU010000031.1:1-10673
Deltaproteobacteria bacterium

UPF0182 HK Radical_SAM ? VapB_antitoxin PIN C_GCAxxG_C_C BrnT_toxin CopG_antitoxin Peptidase_M90 ? ? Radical_SAM Radical_SAM ? HTH_5

MBW2238911.1
JAFDEU010000203.1:499-5799
Deltaproteobacteria bacterium

CopG_antitoxin HTH_3 HigB-like_toxin BrnT_toxin CopG_antitoxin AAA_14 DUF4143 RelB-like ? HicA_toxin UPF0150 HicB

MBW2319176.1
JAFDDZ010000302.1:317-3247
Deltaproteobacteria bacterium

PhdYeFM_antitox YoeB_toxin Polbeta HEPN MazE_antitoxin PIN BrnT_toxin CopG_antitoxin

MBW2364642.1
JAFDDM010000176.1:153-3891
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin Pyridox_ox_2 DUF2817 Lactamase_B_3 DUF4386

MBW2369188.1
JAFDDL010000320.1:1-5139
Deltaproteobacteria bacterium

? BrnT_toxin CopG_antitoxin Polbeta HepT-like AMP-binding AMP-binding_C_3 zf-ChsH2 OB_ChsH2_C Thiolase_N

MBW2576331.1
JAFDBA010000260.1:1-3046
Deltaproteobacteria bacterium

? ? ? BrnT_toxin CopG_antitoxin PhdYeFM_antitox

MBW4651667.1
JAHHHI010000006.1:23512-39309
Kaiparowitsia implicata GSE-PSE-MK54-09C

ParB_N HTH_ParB ? Pilin_GH ? DUF6464 DUF4278 HsdM_N N6_Mtase BrnT_toxin ? HicB_lk_antitox ? DDE_3 HTH_32 HTH_23 ResT-TelK_cat ?

MBW5290165.1
JACRUS010000005.1:23899-36497

Candidatus Ruthia sp. Asou_11_S2

? ? HI0933_like Redoxin rve HTH_38 ? Phage_int_SAM_4 Phage_integrase BrnT_toxin CopG_antitoxin TauE AsnC_trans_reg HTH_24 NIR_SIR_ferr PAPS_reduct Rnf-Nqr FeS Fer4_7 Fer4_21

MBW7858336.1
JACFNH010000231.1:1-4001
Leptonema sp. (in: Bacteria)

Exonuc_V_gamma BrnT_toxin BrnA_antitoxin ?

MBW7889337.1
JACFMZ010000155.1:1-4014
Bacteroidetes bacterium

? CTP_transf_like Aminotran_1_2 Zn_ribbon_8 BrnT_toxin CopG_antitoxin ?

MBX2809751.1
JAHQVN010000162.1:208-10634
Cellvibrionaceae bacterium

CbiX CbiX BrnT_toxin CopG_antitoxin cobW CobW_C CobN-Mg_chel ? Gp49 dnstrm_HI1420 CbiA GATase_3

MBX3011684.1
JAHBVH010000017.1:1-12560
Caldilineaceae bacterium

Tricorn_N Tricorn_N PD40 PD40 STY-kinase FHA Trans_reg_C ABC_tran_CTD ABC_tran ABC_tran_Xtn ABC_tran Peptidase_S58 Lyase_1 ADSL_C FtsJ Adenyl_transf BrnT_toxin ? MFS_1

MBX3179433.1
JAHBVV010000020.1:95843-108694
Candidatus Hydrogenedentota bacterium

TPRs Wzy_C TPRs Glyco_transf_4 Glycos_transf_1 Epimerase ? ParE_toxin Sde182_NH-like BrnT_toxin CopG_antitoxin DUF2809 Acyltransferase NUDIX ? ABC_tran ABC_tran_Xtn ABC_tran ?

MBX3292480.1
JAHBSJ010000005.1:164-13348
Acidobacteria bacterium

OMPdecase ? ? Acetylase Gln-synt_C Gln-synt_N AAA TRAM_2 Radical_SAM UPF0004 BrnT_toxin CopG_antitoxin Beta-prop_IFT122_1st Beta-prop_EML_2 Beta-prop_THOC3

MBX3650645.1
JAHCAG010000005.1:138523-145914
Burkholderiales bacterium

Peptidase_M48 BrnT_toxin CopG_antitoxin MTS DUF2189 TctC TauD DUF3644 ?

MBX7148659.1
JAIBAG010000010.1:140-14794
bacterium

? ? ? Peptidase_S8 ? BrnT_toxin CopG_antitoxin ? ATC_hydrolase Peptidase_C25 Cu-binding_MopE FG-GAP_3 Trypan_PARP ? ? TetR_N

MBX7149878.1
JAIBAG010000044.1:8115-12186
bacterium

? CopG_antitoxin GreA_GreB ? GreA_GreB Thx DUF5684 DUF3187

MBX7235193.1
JAIBAP010000027.1:21371-31586
Caldilineales bacterium

PRAI 23S_rRNA_IVP DUF433 DUF5615 23S_rRNA_IVP Phage_integrase Phage_int_SAM_4 ABC1 BrnT_toxin CopG_antitoxin Rce1-like Gp49 dnstrm_HI1420 MazE_antitoxin PemK_toxin ?

MBX9578018.1
JAIEQF010000025.1:33459-44119
Chthoniobacterales bacterium

MlaE ABC_tran MlaD Helical_needle FtsK_4TM FtsK_alpha FtsK_SpoIIIE FtsK_gamma HTH_25 ? UPF0004 Radical_SAM TRAM_2 BrnT_toxin CopG_antitoxin MlaE

MBX9598654.1
JAIEPY010000087.1:5550-25563
Burkholderiales bacterium

SBP_bac_5 PTA_PTB Malic_M malic ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_CTD Adenine_glyco tRNA-synt_1b MMR_HSR1 ATP-cone ATP-cone Ribonuc_red_lgN Ribonuc_red_lgC ? CopG_antitoxin GDP_Man_Dehyd PTR2 ABC2_membrane AAA-ATPase_like PDDEXK_9 PDDEXK_2 PDDEXK_2

MBX9599212.1
JAIEPY010000126.1:180-12386
Burkholderiales bacterium

PRK PRK UVR UvrB Helicase_C UvrB_3rd UvrB_inter ResIII Bax1-I ? CopG_antitoxin DUF3427 DUF8090 Helicase_C ResIII HKD NUDIX Zn_ribbon_NUD Hexapep_2 Hexapep THDPS_N_2 TMP-TENI HK Phos_pyr_kin

MBX9867580.1
JAIETT010000258.1:259-5807
Burkholderiales bacterium

? CopG_antitoxin SNaCT7 TrfA MobA_MobL ? Hydro_N_hd Trns_repr_metal

MBY0309631.1
JAIEOU010000002.1:1974-18082
Patescibacteria group bacterium

? RecA_C RecA_N Glucodextran_B FtsK_gamma FtsK_SpoIIIE FtsK_alpha FtsK_4TM TrmB PG_binding_1 Y1_Tnp BrnT_toxin CopG_antitoxin LCAT ? ? DUF1360 CDP-OH_P_transf Ribosomal_L10

MCA0237542.1
JAIUPL010000016.1:67911-86015
Bacteroidetes bacterium

? TonB_C AI-2E_transport Na_H_Exchanger PMT_2 Acetylase DEAD Helicase_C BrnT_toxin ? FtsK_4TM ? ? ? ABC2_membrane_3 Peptidase_M1 LytTR Receiver

MCA0388209.1
JAIUOE010000003.1:363504-380676
Bacteroidetes bacterium

Phage_int_SAM_1 Phage_integrase ? ABC_membrane ABC_tran ? ? ? STY-kinase BrnT_toxin CopG_antitoxin PIN RH3_dom SufBD_N SUFBD_core ABC_tran SufBD_N SUFBD_core Aminotran_5

MCA0433659.1
JAIUNJ010000004.1:320213-331400
Proteobacteria bacterium

Exonuc_VII_S S4 FtsJ CPT Chorismate_synt His_Phos_1 adh_short_C2 PNPOx_N PNP_phzG_C BrnT_toxin CopG_antitoxin adh_short_C2 Cys_Met_Meta_PP EF-hand_5 Trypsin PDZ_2 PNPOx_N Aminotran_3

MCA1622928.1
JAIVJJ010000028.1:46294-62916
Acidobacteria bacterium

Sigma70_ECF STY-kinase TolB_N TPRs TPRs TPRs Transp_cyt_pur Anthrone_oxy Polysacc_deac_1 CN_hydrolase Fer4_20 Pyr_redox_2 BrnT_toxin BrnA_antitoxin ? DHO_dh Fer4_21 Lactamase_B_2 Peptidase_M14 Peptidase_S9 Tricorn_2nd Tricorn_2nd Tricorn_N Pyrid_ox_like

MCA3176509.1
JADCHG010000387.1:171-4538
Burkholderiales bacterium

RloB ABC-ATPase BrnT_toxin CopG_antitoxin GmrSD_N HEPN_RiboL-PSP

MCA3585994.1
JADCCX010000126.1:257-11894
Methylocystis sp.

ALAD BrnT_toxin BrnA_antitoxin Semialdhyde_dh Semialdhyde_dhC_1 AcnX_3rd AcnX_2nd AcnX AcnX_swivel_put ACT PGDH_inter 2-Hacid_dh_C Aminotran_5 DndB ?

MCA9329247.1
JAGQNL010000053.1:316-2356
Candidatus Saccharibacteria bacterium

? ? CopG_antitoxin

MCA9343046.1
JAGQMZ010000021.1:1-5428
Candidatus Saccharibacteria bacterium

KH_KhpA-B R3H ParE_toxin PhdYeFM_antitox Fic ? PIN HicB_lk_antitox ? CopG_antitoxin Glycos_transf_1 Acyl_transf_3

MCA9369968.1
JAGQLT010000029.1:1995-9858
Candidatus Woesebacteria bacterium

Mur_ligase_M Mur_ligase_C ? BrnT_toxin CopG_antitoxin Prenyltransf AA_kinase polyprenyl_synt SQS_PSY Lycopene_cyc Lycopene_cyc Amino_oxidase

MCA9379246.1
JAGQLI010000119.1:229-7186
Candidatus Dojkabacteria bacterium

CobW_C cobW MscL ? BrnT_toxin CopG_antitoxin ? Sigma70_r2 Sigma70_r4_2 DUF2202 ? New_glue

MCA9407810.1
JAGQKR010000129.1:196-13266
Candidatus Omnitrophota bacterium

PKD_4 He_PIG He_PIG CADHERIN FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 He_PIG PASTA Cu-binding_MopE Big DUF6531 TEN_YD-shell TEN_YD-shell TEN_YD-shell RHS_repeat TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell Aspzincin_M35 ? ? ? BrnT_toxin CopG_antitoxin ? Glyco_transf_4

MCA9731110.1
JAGQII010000029.1:616-12079
candidate division KSB1 bacterium

ASD1_dom Alpha-L-AF_C Glyco_hydro_39 BrnT_toxin CopG_antitoxin GH43_C2 Glyco_hydro_43 MutS_V MutS_III MoCF_biosynth AdoMet_dc Ham1p_like ADH_zinc_N_2 ADH_N

MCA9939186.1
JAGQFX010000189.1:1-7148
Anaerolineales bacterium

Radical_SAM TRAM_2 AhpC-TSA LGT ? tRNA-synt_1g Anticodon_3 DUF2142 PMT_2 ? BrnT_toxin Metal_resist CopG_antitoxin

MCB0086776.1
JAGRCN010000125.1:4887-12989
Caldilineaceae bacterium

tRNA-synt_2 N-OB ? LTD LysM ? Aminotran_4 RraA-like BrnT_toxin CopG_antitoxin MoCoBD_2 MoCoBD_1

MCB0169381.1
JAGRCA010000035.1:10992-21397
Anaerolineae bacterium

Ala_racemase_N D-ser_dehydrat BrnT_toxin CopG_antitoxin DNA_pol_A DNA_pol_A_exo1 5_3_exonuc 5_3_exonuc_N CTP_transf_1 DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_topoisoIV ? DAHP_snth_FXD DAHP_synth_1 PDH_N PDH_C

MCB0227969.1
JAGRBU010000091.1:76-1053
Anaerolineae bacterium

? ? ?

MCB0415087.1
JAGRAA010000343.1:1-2515
Bdellovibrionales bacterium

BrnT_toxin BrnA_antitoxin ? ?

MCB0627459.1
JAGQXF010000767.1:192-2153
Saprospiraceae bacterium

BrnT_toxin ? HxlR DJ-1_PfpI ?

MCB1057949.1
JAGQSE010000692.1:252-3441
Acidobacteria bacterium

Pps Pps Pps DinB_2 BrnT_toxin CopG_antitoxin Lipase_GDSL_2

MCB1460778.1
JAGRLN010000599.1:1-2974
Nitratireductor sp.

? Methyltransf_30 DAO BrnT_toxin CopG_antitoxin

MCB1614712.1
JAGRJA010000065.1:3362-10154
Pseudomonadales bacterium

? LHH SUKH_5 HTH_3 RelE BrnT_toxin CopG_antitoxin TEN_YD-shell ? Ubiquitin_2 RH3_dom ParE_toxin Methyltransf_11

MCB1980876.1
JAGRFE010000086.1:140-8266
Rhodoferax sp.

DUF2730 BrnT_toxin CopG_antitoxin MFS_1 MFS_1 ? ArgJ ArgJ PEP-CTERM Peptidase_S8 fn3_5

MCB1986332.1
JAGRFL010000194.1:414-8238
Nitrosomonas sp.

HicA_toxin UPF0150 ? Tox-PAAR-like Tox-GHH2 ? CCDC90-like ? BrnT_toxin CopG_antitoxin ? PhdYeFM_antitox PIN

MCB2052304.1
JAGREQ010000390.1:1-1914
Novosphingobium sp.

GDP_Man_Dehyd BrnT_toxin CopG_antitoxin

MCB2147221.1
JAHLLR010000029.1:17408-31535
Deltaproteobacteria bacterium

? TauE Pyr_redox_2 Rubredoxin_C Fer4 Fer4_11 Prismane ? Pyr_redox_3 BrnT_toxin CopG_antitoxin Cupin_2 Cys_Met_Meta_PP PEMT Acetylase Aldo_ket_red ?

MCB2181104.1
JAHLLM010000005.1:22355-35404
Desulfobulbaceae bacterium

Receiver His-Kinase His-Kinase Receiver Receiver His-Kinase His-Kinase dCache_3 PNPase Ysc84 His-Kinase cNMPBD Flagellin_C Flagellin_N BrnT_toxin HicB_lk_antitox Phage_integrase ? Bac_DNA_binding ATP_bind_3 HcyBio CBS CBS

MCB4791843.1
JAJAPY010000008.1:69620-79524
Candidatus Liberimonas magnetica

DUF8412 OmdA ? ? Abi_alpha BrnT_toxin CopG_antitoxin DUF2975 ? CopG_antitoxin DUF8412 OmdA DUF4160 DUF2442 DUF4160 DUF2442 N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP

MCB4792050.1
JAJAPY010000011.1:11244-26775
Candidatus Liberimonas magnetica

Tricorn_2nd Tricorn_N TolB_N TonB_2 TPRs ATG16 OmpA ABC_tran ABC_tran UvrA_DNA-bind UvrA_inter ABC_tran ? TIR IBR BrnT_toxin CopG_antitoxin CBS CBS Succ_DH_flav_C FAD_binding_2 FlgO Pantoate_ligase Cellulase Pantoate_transf Pantoate_transf

MCB8959121.1
JACKBM010000004.1:217344-241962
Ardenticatenales bacterium

HTH_77 AAA_16 GerE AMP-binding AMP-binding_C TPR_MalT TPR_MalT AAA_16 BTAD Glyoxalase Glyoxalase YceI DUF2254 PMT BrnT_toxin CopG_antitoxin Thiolase_C_1 Thiolase_N ACP_syn_III_C HMG_CoA_synt_N HMG-CoA_red ? Big DUF8230 STY-kinase Big DUF8230

MCB8979523.1
JACKBS010000009.1:197340-212664
Ardenticatenaceae bacterium

BPD_transp_1 Inositol_P HAD Aminotran_5 Hydrolase Peptidase_C39_2 SH3_3 SH3_3 CAP BrnT_toxin CopG_antitoxin MTHFR S-methyl_trans LytR_cpsA_psr Radical_SAM Mob_synth_C Glyoxalase Glycos_transf_2 PALP

MCB9006158.1
JACKBR010000004.1:1-7333
Ardenticatenaceae bacterium

? Glutaredoxin BrnT_toxin CopG_antitoxin EamA EamA Glyoxalase DeoC ? GRDB Neur_chan_memb Neur_chan_LBD ?

MCB9060627.1
JACKAK010000002.1:309718-323970
Halobacteriovoraceae bacterium

AI-2E_transport cNMPBD TPRs Aminotran_3 AstA Aldedh ? BrnT_toxin ? YgaP-like_TM ? ? ? ? ?

MCB9081522.1
JACJXV010000018.1:56373-71632
Lewinellaceae bacterium

Polbeta ? GAF_2 His_kinase 7TMR-DISM_7TM His_kinase His-Kinase Receiver LytTR DUF1330 DUF6090 TP_0481_C BrnT_toxin ? ? EAD11 SKA1 HTH_29 HTH_33 DDE_3 ? Hydrolase_like Hydrolase_6

MCB9433381.1
JACKCD010000030.1:37787-51554
Ardenticatenaceae bacterium

Mrr_cat MethyltransfD12 Pept_tRNA_hydro ? UvrB_inter CarD_TRCF_RID DEAD Helicase_C RecG_dom3_C TRCF ? ? BrnT_toxin CopG_antitoxin ? ? HTH_24 ROK NamZ_N NamZ_C BPD_transp_1 BPD_transp_1

MCB9814278.1
JACKGF010000001.1:539764-553331
Candidatus Nomurabacteria bacterium

? Inositol_P ? ? ? LCAT ? DUF4258 CopG_antitoxin CofD ? Glyco_transf_4 Glycos_transf_1 Glyco_trans_4_4 Glycos_transf_1 Glycos_transf_1 Methyltransf_11

MCC0042914.1
JACKJY010000001.1:105194-116436
Brucellaceae bacterium

GrpE GIY-YIG Gly_transporter Gly_transporter PfkB DUF2214 EndIII_4Fe-2S HhH-GPD DUF2244 BrnT_toxin BrnA_antitoxin HTH_18 DNA_binding_1 ? ? ? IU_nuc_hydro HSP20

MCC2624650.1
JAJFBE010000001.1:542463-555987
Burkholderiales bacterium

LptF_LptG LptF_LptG Peptidase_M17 Calcineurin 5_nucleotid_C DUF498 ParE_toxin DUF6290 BrnT_toxin ? Lyase_1 FumaraseC_C GARS_N GARS_A GARS_C MotA_ExbB Methyltransf_23 Polyketide_cyc Ub-RnfH

MCC2666616.1
JAJFBR010000055.1:1-6525
Gammaproteobacteria bacterium

ANKs ANKs ANKs ANKs DUF4143 AAA_14 ? ? CopG_antitoxin OEP OEP

MCC5790918.1
JAFIGF010000259.1:132-3925
Opitutales bacterium

? DUF433 ? BrnT_toxin CopG_antitoxin ? HicB_lk_antitox HicA_toxin

MCC5835494.1
JAFIGG010000067.1:1-10284
Opitutales bacterium

DUF463 ? HRDC RQC RecQ_Zn_bind Helicase_C DEAD FUR TPRs ABhydrolase-like_N NadA MFS_1 BrnT_toxin CopG_antitoxin GlcNAc_2-epim

MCC6275125.1
JADLCJ010000197.1:622-8820
Leptospiraceae bacterium

7TMR-DISM_7TM His-Kinase His-Kinase Beta-lactamase ? DUF4258 CopG_antitoxin ? IspD ? YjgF_endoribonc MgtC Methyltransf_23

MCC6472629.1
JADLDQ010000071.1:23-11197
Burkholderiales bacterium

AMP-binding AMP-binding_C FAD_binding_5 CO_deh_flav_C Fer2 Molybdop_Fe4S4 Molybdopterin Molydop_binding ? DUF3418 OB_NTP_bind HA2_C WHD_HA2 Helicase_C AAA_22 Phage_integrase Phage_int_SAM_4 BrnT_toxin

MCC6598591.1
JADLAT010000017.1:251466-266629
Alphaproteobacteria bacterium

? N_methyl cNMPBD adh_short Inhibitor_I78 GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C PEMT BrnT_toxin CopG_antitoxin ? DUF892 UVR UvrB Helicase_C UvrB_3rd UvrB_inter ResIII ? Sdh_cyt Sdh_cyt

MCC6868406.1
JADZAQ010000090.1:1347-16386
Burkholderiales bacterium

? SLAC1 ? ? adh_short Thiolase_C Thiolase_N MFS_1 BrnT_toxin CopG_antitoxin FGGY_C FGGY_N DUF2889 ATP-grasp_2 Ligase_CoA CoA_binding Ligase_CoA Fic HTH_31

MCC7183361.1
JADZCO010000063.1:10258-26570
Rhodocyclaceae bacterium

MAPEG Biotin_carb_N CPSase_L_D2 Biotin_carb_C PCC_BT Biotin_lipoyl MM_CoA_mutase B12-binding MeaB Carboxyl_trans Biotin_carb_N CPSase_L_D2 Biotin_carb_C PCC_BT Biotin_lipoyl ? BrnT_toxin CopG_antitoxin CTP_transf_3 Trm112p LpxK ExbD MotA_ExbB tRNA_anti_2 Exonuc_VII_L

MCC8110265.1
JAJBUI010000011.1:31561-39657
Ruminococcus sp.

HTH_24 ? ? Asp_aminotransf PDZ_2 Trypsin Dockerin_1 ? BrnT_toxin BrnA_antitoxin

MCC8377842.1
JAJIYI010000028.1:68555-85911
Rickettsia endosymbiont of Graphium doson

Ribosomal_S15 TruB_C_2 TruB_N TLC 120_Rick_ant 120_Rick_ant ? ? Exonuc_VII_S BrnT_toxin CopG_antitoxin Bestrophin-like ? ? DUF4172 Fic ? SSF

MCD4655485.1
JAIOTS010000664.1:1-8601
bacterium

Transposase_29 HTH_28 ? WHD_4th_Lhr DEAD ? FGS TfoX_N BrnT_toxin RHH_10 SIMPL

MCD4667049.1
JAIOSY010000015.1:1-2830
Sulfurimonas sp.

? ? BrnT_toxin CopG_antitoxin TAT_signal DctP

MCD4706203.1
JAIOSV010000007.1:16067-25874
Candidatus Sabulitectum sp.

Redoxin DUF2196 BrnT_toxin CopG_antitoxin ADK ADK ? SYY_C-terminal tRNA-synt_1b UPRTase Peptidase_M23 UvrB_inter CarD_TRCF_RID DEAD Helicase_C TRCF

MCD4708981.1
JAIOSV010000112.1:43143-51344
Candidatus Sabulitectum sp.

GcpE GcpE_C Gln-synt_N Gln-synt_C DUF5128 ? Phage_integrase Phage_integrase Phage_int_SAM_4 Methyltransf_31 BrnT_toxin CopG_antitoxin HTH_3 Gp49

MCD4747155.1
JAIOSH010000501.1:183-2858
Bacteroidales bacterium

BrnT_toxin CopG_antitoxin Rce1-like CHB_HEX_C_1 Glyco_hydro_92

MCD4750379.1
JAIOSJ010000322.1:6347-18770
Thermoanaerobaculales bacterium

ParE-like_toxin DUF6364 CarboxypepD_reg CarboxypepD_reg HicA_toxin ? BrnT_toxin CopG_antitoxin TIG TIG Ephrin_rec_like Ephrin_rec_like Ephrin_rec_like Nucleos_tra2_N Nucleos_tra2_N Gate Nucleos_tra2_C ? Chorismate_bind Aminotran_4 HTH_5

MCD4789202.1
JAIORT010000276.1:7405-14494
Bacteroidales bacterium

DUF5723 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 MORN_2 TonB_C GCV_H Ketoacyl-synt_2 Polysacc_deac_1 LolA BrnT_toxin ?

MCD5390897.1
JAJOKU010000116.1:67-6147
candidate division NPL-UPA2 bacterium

? BrnT_toxin CopG_antitoxin ? ? V-ATPase_G_2 V_ATPase_I V_ATPase_I ATP-synt_C ATP-synt_C vATP-synt_E

MCD6116943.1
JAGGVB010000151.1:1981-17181
bacterium

OEP OEP BSH_RND Beta-barrel_YknX YknX_C ABC_tran MacB_PCD FtsX STAS Pps Pps Pps Pps Pps MacB_PCD FtsX BrnT_toxin ? ? Aminoglyc_resit HigB-like_toxin Peptidase_M78 FMN_dh FMN_dh polyprenyl_synt

MCD6116998.1
JAGGVB010000157.1:1-9983
bacterium

Por_Secre_tail RGL11_C fn3 ? BrnT_toxin CopG_antitoxin Phospholip_B HTH_5 CutC ? Peptidase_C26 GWxTD_dom

MCD6212763.1
JAGGRU010000297.1:172-11757
Sulfurovum sp.

? GATase BrnT_toxin CopG_antitoxin CofH_C Radical_SAM GATase_6 SIS SIS Choline_kinase APH NMN_transporter Thiamine_BP TonB_dep_Rec_b-barrel Plug

MCD6214389.1
JAGGRT010000308.1:98-2885
Candidatus Desulfofervidus sp.

DUF2281 PIN ? PIN DUF4258 CopG_antitoxin

MCD6329659.1
JAGHCS010000139.1:134-6407
Candidatus Cloacimonadota bacterium

DUF6263 DUF6263 Por_Secre_tail Radical_SAM HVO_2753_ZBP Spermine_synth BrnT_toxin CopG_antitoxin

MCD6533374.1
JAGGWR010000110.1:1-8049
Candidatus Tharpella aukensis

Cytochrome_C7 TPRs VWA_2 VWA ? iHEPN-DUF4381 vWA BrnT_toxin CopG_antitoxin ? HTH_18 ? Gly_kinase

MCD6534213.1
JAGGWR010000214.1:1-8252
Candidatus Tharpella aukensis

PIN ? CopG_antitoxin Cas_Cas1 CRISPR_Cas2 CRISPR_Cas2 SLFN_AlbA_2 HATPase_c_4 Fic-like_C Cas_NE0113 A_deaminase Ketoacyl-synt_C ketoacyl-synt ketoacyl-synt

MCD7772694.1
JAJQHL010000036.1:1-3842
Ruminococcus sp.

SecG ? ? Zn_Ribbon_1 BrnT_toxin BrnA_antitoxin ?

MCD7815085.1
JAJQGP010000176.1:13462-22854
Lachnospiraceae bacterium

Receiver Trans_reg_C HAMP His-Kinase His-Kinase ? BrnT_toxin ? DUF6034 DUF6034 ? ? ABC_tran ? Trypan_PARP CARDB

MCD8040182.1
JAJQDC010000011.1:27367-37787
Clostridia bacterium

Ribosomal_L2_C Ribosomal_L2_N Ribosomal_L23 Ribosomal_L4 Ribosomal_L3 Ribosomal_S10 DUF1980_C cobW CobW_C cobW BrnT_toxin CopG_antitoxin DUF2188 PFF1_TM EFG_C EFG_IV EFG_III EF-G_D2 GTP_EFTU

MCD8132020.1
JAJQBK010000059.1:416-8488
Lachnospiraceae bacterium

BrnT_toxin ? HEATs Glyco_hydro_3 ACT_9 AA_kinase Homoserine_dh NAD_binding_3 ACT_4 HTH_23 NUDIX_4 HhH-GPD SfsA SfsA_N

MCD8220584.1
JAJQAC010000001.1:1-9069
Clostridiales bacterium

Radical_SAM ThiG ThiF ThiS-like ThiS ? Phos_pyr_kin Aldo_ket_red BrnT_toxin ? FlK Neopullulanase-like_C Alpha-amylase Alpha-amylase Alpha-amylase_N

MCD8482044.1
JAJOKB010000044.1:192-10675
Verrucomicrobia bacterium

DRTGG DHH DHHA2 ? BrnT_toxin CopG_antitoxin Acetyltransf_10 HTH_18 DUF2149 MotA_ExbB CobN-Mg_chel TonB_dep_Rec_b-barrel

MCE2523819.1
JAGWAS010000022.1:12501-30342
Rhodobacteraceae bacterium

Transket_pyr Transketolase_C Biotin_lipoyl E3_binding E3_binding 2-oxoacid_dh Redoxin ? Pyr_redox_2 Reductase_C Cons_hypoth95 CopG_antitoxin NAPRTase NAPRTase_N Isochorismatase Acetylase His-Kinase DNA_gyraseB Toprim GyrB_hook DNA_gyraseB_C Molybdop_Fe4S4 Molybdopterin Molydop_binding IGPS Fer4_7 Fer4_11

MCE2559977.1
JAGWAG010000142.1:111-10464
Acidobacteria bacterium

? AhpC-TSA DsbD_N ? ? BrnT_toxin CopG_antitoxin ANKs ANKs ? ARG Cupin_3 PQQ PQQ

MCE2927791.1
JAJTHB010000027.1:17413-28435
Rickettsiales bacterium

E1_dh ? Transket_pyr Transketolase_C ? Biotin_lipoyl E3_binding 2-oxoacid_dh CaMKII_AD SnoaL_3 BrnT_toxin CopG_antitoxin ? Pyr_redox_2 Pyr_redox_dim N_methyl ? COQ9 His-Kinase PAS

MCE2991908.1
JAJTHY010000081.1:11100-19460
Candidatus Jidaibacter sp.

TrmB Fic Fic_N HPIH TENA_THI-4 HTH_31 Haem_oxygenas_2 ? ? CopG_antitoxin PBECR3 Peptidase_S24 IMS IMS_HHH IMS_C DUF4113 ?

MCE3232520.1
JAJONM010000020.1:9716-24344
Rickettsiaceae bacterium

N_methyl FtsZ_C Tubulin N_methyl HTH_8 AAA_lid_14 NtrC-AAA Receiver ? MutS_I MutS_II MutS_III MutS_V MutS_C ? Inositol_P BrnT_toxin CopG_antitoxin GIDA SQS_PSY PAS Sod_Fe_N Sod_Fe_C PIN PhdYeFM_antitox

MCE3237335.1
JAJONP010000008.1:3851-22185
Gammaproteobacteria bacterium

HD_assoc_2 HD ParB_N HTH_ParB ParB_C ? AAA_14 DUF4143 ANKs ANKs ANKs ANKs DUF4143 AAA_14 ? ? CopG_antitoxin OEP OEP BSH_RND Beta-barrel_RND_2 YknX_C ACR_tran ANKs ANKs ANKs ? AAA_14

MCE3239064.1
JAJONP010000152.1:1-4608
Gammaproteobacteria bacterium

DHDPS ? BrnT_toxin CopG_antitoxin ? EI24 ? CN_hydrolase AbiEii ?

MCE5293775.1
JAJFUY010000123.1:114-10153
Chlamydiales bacterium

NMT1 Peptidase_M24 Trp_Tyr_perm SufE ? ? ? CopG_antitoxin ? ? GKRP-like_C GKRP_SIS_N

MCE7918891.1
QWHE01000021.1:16696-29296
Chloroflexi bacterium CFX1

MTTB Peptidase_M20 Pterin_bind S-methyl_trans ? ? Peptidase_C69 BrnT_toxin ? Pirin Pirin_C NUDIX ? Methyltransf_25 tRNA_edit TMEM164

MCE7987491.1
QWII01000079.1:97-16900
Caldilinea sp. CFX5

? YbjQ_1 ? ? ? GATase_2 Glu_syn_central Glu_synthase GXGXG BrnT_toxin BrnA_antitoxin Aconitase RRXRR HNH_5 RHH_1 Aconitase_C LeuD DUF433 NTP_transf_2

MCE9588840.1
JAIOPL010000032.1:576-14950
Verrucomicrobia bacterium

Zn_ribbon_9 CT398_CC ? SIS_2 A24_N_bact Peptidase_A24 Peptidase_A24 ABC_membrane ABC_tran MGS AICARFT_IMPCHas PdxJ BrnT_toxin CopG_antitoxin Metallophos_2 ? Sigma70_r2 Sigma70_r4 Thioredoxin_2 STY-kinase ?

MCF6198103.1
JAKIUO010000011.1:16780-28515
Hyphomicrobiaceae bacterium

TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N SelO BrnT_toxin CopG_antitoxin Glyoxalase Exosortase_EpsH VIT VWA_3 ? Receiver Trans_reg_C HAMP His-Kinase His-Kinase

MCF6209126.1
JAKIUN010000001.1:47119-64022
Gammaproteobacteria bacterium

CCDC39 Peptidase_M23 CtpB_N-like PDZ_2 Peptidase_S41 Polysacc_deac_2 CSD DUF480 SET Methyltransf_23 BrnT_toxin CopG_antitoxin Aminotran_3 OTCace_N OTCace ABC_tran TOBE_2 Cytochrom_C Cytochrom_C SH3_3 ODF2_C ILV_EDD_C ILVD_EDD_N

MCF6240429.1
JAKIUH010000035.1:1-18656
Bacteroidales bacterium

OmpA CarboxypepD_reg PD40 TPRs TPRs CoiA_N LuxE Peptidase_C1_2 Gly_kinase Aminotran_3 BrnT_toxin ? DUF6206 Polysacc_synt Glyco_transf_4 Glycos_transf_1 DUF4384 DCC1-like TPRs TPRs TPRs His-Kinase

MCF6251445.1
JAKIUK010000027.1:42952-50331
Methylococcaceae bacterium

RHH_1 RelB3_C ParE_toxin BrnT_toxin CopG_antitoxin ? CDP-OH_P_transf CTP_transf_like DHOR rve HTH_7

MCF6262726.1
JAKITK010000009.1:2084-15482
Lysobacterales bacterium

Transgly ? TPRs TPRs Thioredoxin ABC_membrane ABC_tran VTT_dom adh_short cNMPBD HTH_Crp_2 BrnT_toxin CopG_antitoxin ? HisKA_3 His-Kinase Receiver GerE DUF2730 AstE_AspA_cat ABC_tran

MCF6268170.1
JAKIUC010000001.1:118766-138591
Melioribacteraceae bacterium

His_kinase His-Kinase DUF559 Receiver GerE DUF4143 AAA_14 DUF4143 AAA_14 CBM96 PPC CarboxypepD_reg Por_Secre_tail ? BrnT_toxin CopG_antitoxin DUF2400 MerR_1 S1CSD-TOTE-2 GATase_7 CDP-OH_P_transf SRP54_N SRP54 DAGK_cat YegS_C RH3_dom

MCF6268533.1
JAKIUC010000005.1:3881-20551
Melioribacteraceae bacterium

OMP_b-brl ? AI-2E_transport cNMPBD Na_Ca_ex Na_Ca_ex ? PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I BrnT_toxin CopG_antitoxin SR1P Peptidase_S46 Peptidase_S46 Asn_synthase OMP_b-brl ?

MCF6274195.1
JAKIUA010000005.1:1-8963
Robiginitomaculum sp.

NMO Peptidase_M20 Peptidase_M28 PA PA Peptidase_M28 BrnT_toxin CopG_antitoxin ? ? ? HSP70

MCF6280590.1
JAKITT010000001.1:168-9476
Candidatus Polarisedimenticolaceae bacterium

ADH_zinc_N_2 RVT_1 BrnT_toxin CopG_antitoxin Cytochrome_CBB3 ? ? NAD_binding_2 6PGD G6PD_N G6PD_C Glucosamine_iso

MCF6319612.1
JAKISZ010000294.1:244-3220
Proteobacteria bacterium

? CopG_antitoxin ? MBOAT

MCF6320419.1
JAKISV010000010.1:1-9952
Rhizobiaceae bacterium

Aconitase Ribosomal_L19 tRNA_m1G_MT PRC_RimM RimM CM_2 SRP_SPB SRP54 SRP54_N ? ? CopG_antitoxin ?

MCF6334139.1
JAKISO010000005.1:235-7704
Spirochaetales bacterium

Condensation AAA_22 Mu-transpos_C rve_3 rve HTH_32 ? ? Polbeta HepT-like_2 BrnT_toxin CopG_antitoxin SWIM DUF8276

MCF6335194.1
JAKISO010000206.1:1-3558
Spirochaetales bacterium

? BPD_transp_2 BPD_transp_2 BrnT_toxin CopG_antitoxin PNPase

MCF6766653.1
JAKJJV010000062.1:134-5446
Thiotrichales bacterium 19S3-7

AbiEii DUF6088 ? HTH_3 ParE_toxin HTH_3 RelE BrnT_toxin ?

MCF6767714.1
JAKJJR010000006.1:64197-84627
Thiotrichales bacterium 19S11-10

TsaD Sugar_tr Amidohydro_1 SIS SIS BcrAD_BadFG MMR_HSR1 ? BrnT_toxin ? CCDC113_CCDC96_CC NAPRTase NAMPT_N Trp_Tyr_perm ? Peptidase_M23 LysM LysM GTP_EFTU MMR_HSR1 GTP1_OBG

MCF7813755.1
JAIPJH010000019.1:9965-25783
Candidatus Cloacimonadota bacterium

Sel1 Sel1 Peptidase_C39_2 ? FlgD_ig Peptidase_M28 HHH_3 HHH_3 ? Cons_hypoth95 BrnT_toxin CopG_antitoxin tRNA-synt_1e DALR_2 CysS_C AbiEii AbiEi_4 Peptidase_M48 TerB ANKs

MCF7873660.1
JAIPHU010000006.1:442-9484
Candidatus Omnitrophota bacterium

Polbeta HepT-like_2 DUF898 7TM_GPCR_Srh ? BrnT_toxin ? MFS_1 RnfC_N Complex1_51K Fer4_7 NQR2_RnfD_RnfE TRAP-gamma FMN_bind Rnf-Nqr Rnf-Nqr

MCF7942530.1
JAIPFY010000079.1:167-12075
Spirochaetia bacterium

PRiA4_ORF3 DUF6933 LacI Peripla_BP_3 PfkB ? ? NusG KOW ? BrnT_toxin CopG_antitoxin DruA Ig_mannosidase Mannosidase_ig Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N2

MCF7988699.1
JAIPFD010000125.1:588-7095
Methylovulum sp.

HsdM_N N6_Mtase N6_Mtase UPF0175 DUF3368 ? YafQ_toxin Methylase_S Methylase_S BrnT_toxin CopG_antitoxin REase

MCF8131202.1
JAIPDM010000357.1:1-6363
Deltaproteobacteria bacterium

PHB_acc_N PHB_depo_C BrnT_toxin CopG_antitoxin DUF4070 Radical_SAM B12-binding CheR_N CheR 23S_rRNA_IVP CheB_methylest

MCF8145083.1
JAIPDI010000027.1:57434-76271
Deltaproteobacteria bacterium

cNMPBD Rhodanese Pyr_redox_dim Pyr_redox_2 Calcineurin Radical_SAM LIAS_N LplA-B_cat Pyr_redox_dim Pyr_redox_2 Cu-oxidase_2 Cu-oxidase_3 BrnT_toxin CopG_antitoxin DUF1016_N GemA GcvP2_C GDC-P GDC-P GCV_H GCV_T GCV_T_C SHMT

MCF8145777.1
JAIPDI010000041.1:2098-20179
Deltaproteobacteria bacterium

Arnt_C PMT_2 DUF3786 DUF218 GARS_N GARS_A GARS_C TSCPD Ribonuc_red_lgC Ribonuc_red_lgN Peptidase_M16 Peptidase_M16_C Peptidase_M16 Peptidase_M16_C QueF BrnT_toxin CopG_antitoxin PNPOx_N MarR VapB_antitoxin PIN N6_Mtase Eco57I ?

MCF8240610.1
JAIPBP010000011.1:183-12150
Melioribacteraceae bacterium

ABC2_membrane_3 DrrA1-3_C ABC_tran ? Leu_Phe_trans CBS CBS EamA EamA Aldo_ket_red BrnT_toxin CopG_antitoxin Lactamase_B_2 TPM_phosphatase AMP_N Peptidase_M24 His_Phos_1 Zn_ribbon_ISL3 Por_Secre_tail

MCF8262341.1
JAIPBD010000022.1:35871-51741
Melioribacteraceae bacterium

DUF1684 Beta-lactamase Esterase Amidase MFS_1 Rhodanese NYN_YacP BrnT_toxin CopG_antitoxin Radical_SAM Zn_ribbon_HMPTM ? DUF7670 Peptidase_M1 SufE ARM_TT21_N HTH_18

MCF8316903.1
JAIPAS010000022.1:420-8189
Ignavibacteriales bacterium

NICE-3 DUF3671 Palm_thioest Peptidase_S9 MazE_antitoxin PemK_toxin Polbeta HepT-like_2 BrnT_toxin CopG_antitoxin ? Pyr_redox_2 GIY-YIG

MCF8359335.1
JAIOZY010000030.1:12088-32381
Prolixibacteraceae bacterium

Aldo_ket_red ADH_zinc_N ADH_N CBM_6 RicinB_lectin_2 Fucosidase_C Alpha_L_fucos PA14 Fn3_assoc Alpha_L_fucos MFS_1 RbsD_FucU Peripla_BP_1 LacI BrnT_toxin CopG_antitoxin DUF4143 AAA_14 DUF4143 AAA_14 FGGY_C FGGY_C_2 FGGY_N RhaA Bac_rhamnosid_C Bac_rhamnosid6H Bac_rhamnosid Bac_rhamnosid_N Ig_Rha78A_N rhaM

MCG2683816.1
JAKLRS010000195.1:187-3521
Planctomycetales bacterium

PhdYeFM_antitox ? LamininG BrnT_toxin CopG_antitoxin BrnA_antitoxin Uma2

MCG2711909.1
JAKLQT010000142.1:1-10685
Candidatus Omnitrophota bacterium

HepT-like Exosortase_EpsH DUF3485 ? CopG_antitoxin PEP-CTERM Bac_transf PEP-CTERM PEP-CTERM ? TPRs TPRs PLA2_B

MCG2712878.1
JAKLQT010000249.1:1-5447
Candidatus Omnitrophota bacterium

CopG_antitoxin DUF6941 ? HEPN DUF772 BrnT_toxin CopG_antitoxin ? ? DUF4160 DUF2442

MCG2725266.1
JAKLQJ010000074.1:5103-11556
Elusimicrobia bacterium

BrnT_toxin CopG_antitoxin ? ? ? FlgO CsgG ? Lactamase_B_2

MCG2726871.1
JAKLQJ010000280.1:4364-11440
Elusimicrobia bacterium

CopG_antitoxin BrnA_antitoxin BrnT_toxin ? PgpA CDP-OH_P_transf LolA FtsK_SpoIIIE FtsK_alpha FtsK_4TM

MCG3175184.1
JAJVIE010000001.1:134206-151496
Candidatus Omnitrophota bacterium

Calcineurin HsdM_N N6_Mtase Methylase_S Methylase_S REase SWI2_SNF2 T1R_D3-like T1RH-like_C ? HNH_2 ? ? CopG_antitoxin HTH_37 Radical_SAM ? Pterin_4a DUF6531 TEN_YD-shell TEN_YD-shell RHS_repeat RHS_repeat TEN_YD-shell TEN_YD-shell ?

MCG3769410.1
JAHCGY010000118.1:63-2322
Nitrosomonadaceae bacterium

BrnT_toxin CopG_antitoxin Abhydrolase_1 ParE_toxin

MCG5054081.1
JAGTJU010000003.1:607715-622669
Myxococcales bacterium

tRNA-synt_1g Anticodon_3 tRNA_bind CarboxypepD_reg GST_C GST_N_2 Phage_int_SAM_4 Phage_integrase BrnT_toxin BrnA_antitoxin Glyoxalase BrnT_toxin MqsA_antitoxin BrnT_toxin MqsA_antitoxin RHH_1 PemK_toxin BrnT_toxin BrnA_antitoxin AAA_33 Sirtuin

MCG5054101.1
JAGTJU010000003.1:624650-637242
Myxococcales bacterium

IstB_IS21 Mu-transpos_C_2 rve HTH_23 rve_3 rve HTH_32 ? ? BrnT_toxin BrnA_antitoxin ? MqsA_antitoxin PIN ? DDE_Tnp_1 ? AAA_22 ?

MCG5054504.1
JAGTJU010000004.1:281304-294033
Myxococcales bacterium

His-Kinase His-Kinase HAMP dCache_1 Trans_reg_C Receiver ? VWA_3 VIT DUF4440 Acetylase RsgA_GTPase BrnT_toxin MqsA_antitoxin BrnT_toxin BrnA_antitoxin ? ? BrnT_toxin MqsA_antitoxin

MCG6961748.1
JAJDNH010000008.1:54-5382
Acidobacteria bacterium

? BrnT_toxin CopG_antitoxin HTH_3 ?

MCG8338553.1
JAKSDL010000791.1:204-1275
Proteobacteria bacterium

BrnT_toxin CopG_antitoxin PIN

MCG8481664.1
JAKSCE010000869.1:86-4660
Spirochaetales bacterium

? ? ABC_membrane ABC_tran BrnT_toxin CopG_antitoxin

MCG8606835.1
JAKSBH010000953.1:1-4886
bacterium

TPRs TPRs TPRs HTH_77 AAA_22 STY-kinase BrnT_toxin CopG_antitoxin AIRC DUF4837

MCG8639339.1
JAKSAZ010000164.1:14042-21074
Desulfobacterales bacterium

MoaC MoaE NTP_transf_3 Pribosyltran MoCF_biosynth ? PUA_2 ATP-sulfurylase APS_kinase BrnT_toxin CopG_antitoxin NTase_sub_bind

MCG9129767.1
JAGFCA010000172.1:3413-13169
Candidatus Poribacteria bacterium

CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg ? ? ? Rhodanese MauE Sulfotransfer_5 BrnT_toxin ? DUF6242_C PSII_BNR Sortilin-Vps10 DUF6242_C

MCH1587538.1
JAKTAF010000030.1:1-4545
Staphylococcus epidermidis

MobA_MobL ? BrnT_toxin ? Rep3_C Rep3_N KfrB MobA_MobL

MCH2227790.1
JAKVAO010000045.1:12971-21720
Candidatus Caenarcanales bacterium

? BrnT_toxin CopG_antitoxin DnaGprimase_HBD NTP_transf_2 NTase_sub_bind ? N-OB tRNA-synt_2 ?

MCH7479176.1
JADFTB010000674.1:1-2449
SAR324 cluster bacterium

ACT_9 AA_kinase BrnT_toxin SpoU_methylase

MCH7516637.1
JADFSW010000259.1:132-3028
Bacteroidetes bacterium

BrnT_toxin CopG_antitoxin SelR PMSR ?

MCH7614113.1
JADFRD010000166.1:334-2092
Candidatus Neomarinimicrobiota bacterium

RecX_HTH1 RecX_HTH2 RecX_HTH3 BrnT_toxin CopG_antitoxin HicA_toxin HicB_lk_antitox

MCH7675627.1
JADFPY010000126.1:350-10559
candidate division KSB1 bacterium

MacB_PCD FtsX MacB_PCD FtsX ? BrnT_toxin CopG_antitoxin STY-kinase PD40 PD40 Tricorn_N Tricorn_N DUF8412 OmdA ? Sigma70_r4_2 Sigma70_r2 DUF3471 Beta-lactamase

MCH7730213.1
JADFPI010000006.1:22307-33705
Patescibacteria group bacterium

Zip Zip Peptidase_C39_2 TPRs GSDH SLT DUF411 ? YkoP DUF4258 CopG_antitoxin hAT-like_RNase-H DUF1045 ? ? NUDIX MazG-like

MCH7750265.1
JADFPE010000272.1:258-6109
Acidobacteria bacterium

PIN RHH_1 BrnT_toxin CopG_antitoxin DUF1800 DUF1501

MCH7764522.1
JADFOY010000118.1:1-8355
Candidatus Neomarinimicrobiota bacterium

PgpA MoCF_biosynth CinA_KH CinA 2H RecA_N RecA_C RecX_HTH1 RecX_HTH2 RecX_HTH3 BrnT_toxin ? HicA_toxin HicB_lk_antitox ? ? Dus

MCH7764756.1
JADFOY010000156.1:485-4529
Candidatus Neomarinimicrobiota bacterium

PhdYeFM_antitox PIN DUF3644 DUF4145 BrnT_toxin CopG_antitoxin ? ?

MCH7769476.1
JADFOR010000003.1:5209-22935
Bacteroidetes bacterium

DUF6161 ThiS MoaE ThiF Rhodanese F420_oxidored P5CR_dimer His_Phos_1 DUF1579 B12-binding Radical_SAM BrnT_toxin CopG_antitoxin SelR PMSR ? Por_Secre_tail Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10 ? TRCF Helicase_C DEAD CarD_TRCF_RID UvrB_inter GidB

MCH7771039.1
JADFOR010000115.1:1-9688
Bacteroidetes bacterium

DapB_C 23S_rRNA_IVP DapH_N Hexapep Hexapep Hexapep AA_kinase ACT_7 ? BrnT_toxin BrnA_antitoxin Peptidase_M20 ? NFACT-R_1 NFACT_N MFS_1 MFS_1 MFS_1

MCH7859401.1
JADFNF010000095.1:92-7639
Candidatus Neomarinimicrobiota bacterium

TIR N-OB tRNA-synt_2 BrnT_toxin DUF6364 PhoH ELFV_dehydrog_N ELFV_dehydrog ELFV_dehydrog NusB SecA_DEAD

MCH7882384.1
JADFMY010000115.1:5074-8004
Proteobacteria bacterium

BrnT_toxin CopG_antitoxin RH3_dom ParE_toxin ? BrnT_toxin BrnA_antitoxin CTP_transf_3

MCH7939551.1
JADFMC010000158.1:1-5361
Candidatus Neomarinimicrobiota bacterium

UvrA_inter nSTAND1 FabA GYD ? BrnT_toxin CopG_antitoxin RelB-like ParE_toxin S-methyl_trans

MCH8107627.1
JADFJS010000051.1:6808-12622
Chloroflexota bacterium

BrnT_toxin CopG_antitoxin HicA_toxin HicB_lk_antitox HGTP_anticodon tRNA-synt_2b tRNA_SAD G3P_acyltransf Polyketide_cyc2 PCRF RF-1

MCH8126481.1
JADFJB010000054.1:2280-17721
candidate division KSB1 bacterium

TPRs CytoC_RC DoxX_2 HTH_18 4HBT_2 ? RNase_Y_N AMP-binding AMP-binding_C_2 BrnT_toxin CopG_antitoxin RHH_8 PemK_toxin FtsX FtsX DUF2442 QueF 2-oxogl_dehyd_N E1_dh Transket_pyr OxoGdeHyase_C

MCH8311551.1
JADFGI010000011.1:155-12930
Nitrospinota bacterium

LAB_N Glycos_transf_2 Radical_SAM RlmN_N DnaJ DnaJ_C HBEGF_C Ham1p_like His-Kinase His-Kinase PAS CHASE BrnT_toxin Pox_I1 Lactamase_B_3 ? ?

MCH8538633.1
JAIUMG010000258.1:67-4912
Alkalimonas sp.

Aconitase Aconitase_C ? ? BrnT_toxin CopG_antitoxin

MCH8895328.1
JACZSR010000033.1:1-11410
Proteobacteria bacterium

Lyase_aromatic Amidohydro_1 HUTI_composite_bact Amidohydro_1 FGase FAD_binding_6 NAD_binding_1 ABC_tran ABC_tran_Xtn ABC_tran Hist_deacetyl BrnT_toxin CopG_antitoxin SHMT Zn_ribbon_NrdR ATP-cone dCMP_cyt_deam_1

MCH8989807.1
JACZUF010000052.1:1-9778
Chloroflexota bacterium

? ? Methyltransf_25 PRK Thioredox_DsbH Cupin_2 Cupin_2 Discoidin 2-Hacid_dh_C BrnT_toxin CopG_antitoxin ? DUF433

MCH9689601.1
JAHZJI010000087.1:182-5560
Gammaproteobacteria bacterium

FtsX ABC_tran MreB_Mbl MreC MreD TauE TauE ?

MCH9844321.1
JAHZLX010000080.1:1-7471
Alphaproteobacteria bacterium

3HCDH 3HCDH_N BKACE DinB_2 FGS Methyltransf_33 ? ? GmrSD_C GmrSD_C ? ?

MCH9844451.1
JAHZLX010000093.1:1-5918
Alphaproteobacteria bacterium

MatE MatE Fe-S_biosyn ? ? GmrSD_N GmrSD_C FeS_assembly_P

MCH9852218.1
JAHZMA010000027.1:1-7172
Alphaproteobacteria bacterium

AP_endonuc_2 ANKs ANKs ANKs ANKs ANKs ? ? Peripla_BP_3 LacI GFO_IDH_MocA GFO_IDH_MocA_C3 ABC_tran BPD_transp_2

MCH9852943.1
JAHZMA010000080.1:28064-39310
Alphaproteobacteria bacterium

Pribosyltran PRAI ? ? Phage_integrase Phage_int_SAM_1 DUF484 TAL_FSA PBECR3 ? ? PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C Polyketide_cyc2 peroxidase peroxidase

MCI0445310.1
JAKEHH010000770.1:460-4707
bacterium

UvrD_C UvrD-helicase BrnT_toxin CopG_antitoxin Xre_MbcA_ParS_C

MCI0449429.1
JAKEHE010000090.1:2246-18082
Chlorobiota bacterium

ABC_tran VapB_antitoxin PIN ? Radical_SAM ? ? ? CopG_antitoxin ? PAS PAS PAS PAS HisKA_2 His-Kinase ? Vsr ? His-Kinase HATPase_c_3

MCI0496305.1
JAKEGP010000657.1:350-5300
candidate division KSB1 bacterium

BrnT_toxin ? HpcH_HpaI ? ? Bac_rhamnosid_C Bac_rhamnosid6H

MCI5115855.1
QYLI01000163.1:1-5697
Candidatus Electrothrix sp. AX1

TEN_YD-shell DUF3885 ParD_like ParE-like_toxin BrnT_toxin CopG_antitoxin HigB-like_toxin HTH_3 dnstrm_HI1420 Gp49 ? PIN

MCI5137130.1
QYKV01000066.1:374-3409
Candidatus Electrothrix sp. AR1

? ? BrnT_toxin CopG_antitoxin PIN

MCI5144535.1
QYKU01000116.1:282-3897
Candidatus Electrothrix sp. AR3

BrnT_toxin CopG_antitoxin RimK Rimk_N Phage_int_SAM_4 Phage_integrase NAD_binding_1 ?

MCI5145221.1
QYKU01000308.1:1-2693
Candidatus Electrothrix sp. AR3

HEATs HEATs HEATs wHTH-HSP90_Na-assoc BrnT_toxin CopG_antitoxin

MCI5224291.1
QYKT01000819.1:86-1776
Candidatus Electrothrix sp. AR4

WD40_RLD BrnT_toxin CopG_antitoxin

MCI5569060.1
JALEQI010000030.1:7429-24132
Lachnospiraceae bacterium

DUF1275 Virulence_RhuM ? Patatin DUF6363 ? Thi4 FAD-depend_C Pribosyltran BrnT_toxin CopG_antitoxin Cna_B Cna_B Sortase STY-kinase ? ? Flg_new

MCI6086990.1
JALFCU010000114.1:1-16079
Selenomonas sp.

VitK2_biosynth FIST FIST_C GGDEF RelB YoeB_toxin MarR BSH_RND Beta-barrel_RND_2 YknX_C ACR_tran BrnT_toxin ? Mrr_N Mrr_cat PTS_EIIC PTS_EIIB Amidohydro_1 Glucosamine_iso PTS_EIIC PTS_EIIB LicD

MCI8676736.1
JAAWES010000001.1:33805-53604
Lawsonibacter sp.

Amidohydro_2 ABC_tran ABC_tran_Xtn ABC_tran ? SLH SLH SpoIIE SpoIIE_N Asp23 Dak2 FakA-like_M FakA-like_C BrnT_toxin BrnA_antitoxin TraB_PrgY_gumN ? HAD_2 SDH_alpha ParA Transpeptidase PBP_dimer

MCI9044730.1
JAAWKH010000010.1:33841-49918
Peptococcaceae bacterium

TP_methylase CbiG_C CbiG_mid CbiG_N PCMT TP_methylase CbiD CbiC CobD_Cbib GATase_3 CbiA BrnT_toxin ? ChlI AAA_25 Zn_ribbon_LapB ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N McsB_C ATP-gua_Ptrans UVR DUF7685 CtsR_C CtsR ?

MCJ7471351.1
JALHTA010000033.1:239-4488
Actinomycetota bacterium

? CopG_antitoxin MerR_1 HSP70 DnaJ DnaJ_C

MCJ7483940.1
JALHSV010000532.1:1-6015
Thermodesulfovibrionales bacterium

His-Kinase His-Kinase BrnT_toxin CopG_antitoxin Methyltransf_11 WYL PNPOx_N SHOCT FMN_red ? DUF6933

MCJ7524283.1
JALHTJ010000066.1:45-6489
Candidatus Aminicenantes bacterium

LytTR Receiver His-Kinase His_kinase ? Asparaginase_C ? CopG_antitoxin Asparaginase_C Asparaginase GatD_N GatB_Yqey GatB_N

MCJ7547574.1
JALHTM010000282.1:1-6688
Deltaproteobacteria bacterium

B12-binding ? BrnT_toxin CopG_antitoxin ParE_toxin PhdYeFM_antitox AMP-binding AMP-binding_C DUF362 Trep_dent_lipo DUF192

MCJ7666018.1
JALHVQ010000253.1:58-3345
Actinomycetota bacterium

DUF2807 BrnT_toxin CopG_antitoxin ABC_tran ABC2_membrane

MCJ7813668.1
JALHYD010000616.1:1-1220
bacterium

Peptidase_S9 Smr BrnT_toxin CopG_antitoxin

MCJ7826538.1
JALHYM010000164.1:1-2493

Patescibacteria group bacterium

EFG_C LepA_C ? BrnT_toxin CopG_antitoxin

MCK4249608.1
JAGMEQ010000477.1:1-7710
Candidatus Omnitrophota bacterium

Radical_SAM DUF362 Fer4_10 Epimerase Epimerase BrnT_toxin CopG_antitoxin S-methyl_trans MGDG_synth Glyco_tran_28_C Gp_dh_N Gp_dh_C

MCK4294202.1
JAGMDR010000385.1:1-16054
Planctomycetota bacterium

? DUF6677 ? ? DUF1015 NTP_transf_2 Aminotran_5 BrnT_toxin CopG_antitoxin tRNA_edit ? Trypsin PDZ_2 PDZ_6 Glyco_hydro_42M GAF_2 FHA FG-GAP_3 FG-GAP_3 FG-GAP_3

MCK4296437.1
JAGMDT010000522.1:1-3216
Candidatus Neomarinimicrobiota bacterium

Ribul_P_3_epim ? HepT-like BrnT_toxin ? BatA

MCK4402012.1
JAGMBN010000143.1:386-2439
bacterium

? BrnT_toxin CopG_antitoxin ?

MCK4505677.1
JAGLZM010000195.1:322-16624
Candidatus Aegiribacteria sp.

AbiEii ? CarboxypepD_reg Plug TonB_dep_Rec_b-barrel ? Peptidase_S41 Tricorn_C1 BrnT_toxin NikA-like Peptidase_M16_C Peptidase_M16_C Peptidase_M16 Beta-prop_THOC3 Beta-prop_THOC3 FlgD_ig ATPase_2 Methyltransf_31 Methyltransf_11 ?

MCK4506352.1
JAGLZM010000305.1:18-4267
Candidatus Aegiribacteria sp.

? MMR_HSR1 HydF_dimer HydF_tetramer ? ? ? CopG_antitoxin

MCK4514280.1
JAGLZP010000300.1:180-774
Spirochaetaceae bacterium

BrnT_toxin CopG_antitoxin

MCK4539396.1
JAGLYV010000143.1:383-14304
Candidatus Krumholzibacteriia bacterium

PKD_4 CarbopepD_reg_2 SdrD_B ? FlgD_ig PD40 BrnT_toxin ? Penicil_amidase ? FlgD_ig PEP-CTERM ?

MCK4540787.1
JAGLYY010000026.1:130-9203
Spirochaetales bacterium

Mg_chelatase_C Trypsin PDZ_2 ResIII UvrB_inter UvrB_3rd Helicase_C UvrB UVR BrnT_toxin CopG_antitoxin ? AbiEii ATC_hydrolase Caspase PhdYeFM_antitox ?

MCK4548825.1
JAGLYU010000046.1:1-5898
Candidatus Krumholzibacteriia bacterium

Big NtrC-AAA AAA_lid_14 HTH_8 BrnT_toxin CopG_antitoxin ? Adenosine_kin HTH_8 AAA_lid_14 NtrC-AAA GAF

MCK4561014.1
JAGLYF010000307.1:1-11120
Calditrichia bacterium

? CarboxypepD_reg CarboxypepD_reg ? ? Peptidase_C1_2 BrnT_toxin CopG_antitoxin STY-kinase TPRs TPRs TPRs TPRs Creatininase HicA_toxin UPF0150 HicB YhhN

MCK4639705.1
JAGLWX010000003.1:1-8298
Candidatus Neomarinimicrobiota bacterium

? fn3 fn3 fn3 Fib_succ_major DUF4365 ? BrnT_toxin CopG_antitoxin ? ? ? CsgG ? DDE_3

MCK4654944.1
JAGLWO010000581.1:278-1500
Candidatus Cloacimonadota bacterium

BrnT_toxin CopG_antitoxin RHH_1 NikR_C

MCK4707903.1
JAGLVT010000405.1:24-1088
Gammaproteobacteria bacterium

NMB0537_N BrnT_toxin CopG_antitoxin

MCK4761339.1
JAGLSW010000168.1:155-7855
Candidatus Aminicenantes bacterium

UPF0175 DUF3368 BrnT_toxin CopG_antitoxin ? ? HTH_24 Pribosyltran ? His-Kinase His-Kinase

MCK4764682.1
JAGLSW010000825.1:163-8361
Candidatus Aminicenantes bacterium

HEATs HEATs HEATs HEATs HEATs adh_short BrnT_toxin CopG_antitoxin TIR Pribosyltran AAA_23 ABC-ATPase SLFN_AlbA_2 HATPase_c_4

MCK4765728.1
JAGLSW010001028.1:183-7922
Candidatus Aminicenantes bacterium

BrnT_toxin CopG_antitoxin SpoIIE DUF6978 DUF1828 VWA Biotin_lipoyl E3_binding 2-oxoacid_dh Pyr_redox_2 Pyr_redox_dim

MCK4766620.1
JAGLSW010001230.1:331-8925
Candidatus Aminicenantes bacterium

BrnT_toxin CopG_antitoxin ATPase_2 TPRs TPRs TPRs TPRs ? TPRs TPRs TPRs TPR_NPHP3 ? HigB-like_toxin HTH_3

MCK4797988.1
JAGLTD010000948.1:1-5446
Spirochaetes bacterium

P-mevalo_kinase DbpA Methyltransf_4 PemK_toxin PemK_toxin AF2212-like BrnT_toxin DUF1564 DUF2229

MCK4829435.1
JAGLVY010008049.1:355-914
bacterium

BrnT_toxin CopG_antitoxin

MCK4866015.1
JAGLTQ010000172.1:29-7740
Gammaproteobacteria bacterium

Imm33 PhdYeFM_antitox YoeB_toxin ? DDE_Tnp_1_6 HigB-like_toxin HTH_3 BrnT_toxin CopG_antitoxin PhdYeFM_antitox ParE_toxin FKBP_N FKBP_C DUF2238 Methyltransf_25 Orn_DAP_Arg_deC Orn_Arg_deC_N

MCK4981791.1
JAGLQZ010000187.1:68-3213
Victivallaceae bacterium

FtsZ_C DUF6036 ? BrnT_toxin CopG_antitoxin

MCK4982469.1
JAGLQZ010000422.1:1-1935
Victivallaceae bacterium

GATase NAD_synthase GMP_synt_C ? BrnT_toxin

MCK4982765.1
JAGLQZ010000536.1:1-1571
Victivallaceae bacterium

GHMP_kinases_N BrnT_toxin CopG_antitoxin

MCK5082169.1
JAGLPI010000392.1:1-1643
Candidatus Omnitrophota bacterium

? CMD ?

MCK5105170.1
JAGLPF010002074.1:1-3462
Cyclobacteriaceae bacterium

TPRs LptC BrnT_toxin CopG_antitoxin ? ? ?

MCK5132229.1
JAGLOJ010000045.1:6-5650
Candidatus Sabulitectum sp.

? TrmB ? PPDK_N DUF5752 ?

MCK5133964.1
JAGLOJ010000159.1:170-1003
Candidatus Sabulitectum sp.

BrnT_toxin CopG_antitoxin ?

MCK5146933.1
JAGLOF010000058.1:38624-50660
bacterium

? CopG_antitoxin BrnT_toxin PhdYeFM_antitox PIN PhdYeFM_antitox ParE_toxin BrnT_toxin CopG_antitoxin ? PTE DUF4515 RNase_T DEAD DEAD_2 Helicase_C_2 IMPDH Ribosomal_S20p

MCK5152857.1
JAGLOB010000011.1:108-13557
Spirochaetales bacterium

AP_endonuc_2 EamA EamA Lactate_perm 2-Hacid_dh_C NUDIX TK BrnT_toxin CopG_antitoxin Zn_ribbon_LapB AAA_25 ChlI TraB_PrgY_gumN TPR_CcmH_CycH TPRs TPRs ?

MCK5154948.1
JAGLNY010000018.1:1-4904
Spirochaetales bacterium

NTP_transferase PTS_EIIA_2 Na_H_Exchanger LPG_synthase_C ? DUF4258 NikA-like

MCK5162118.1
JAGLNZ010000013.1:154-2867
Desulfobacula sp.

HigB-like_toxin HTH_3 DUF4258 CopG_antitoxin Acetylase dnstrm_HI1420

MCK5200628.1
JAGLNS010001816.1:228-4503
Spirochaetales bacterium

BrnT_toxin CopG_antitoxin ABC_tran ABC2_membrane_6 ABC2_membrane_6 PIN

MCK5243473.1
JAGLMY010000320.1:1-14252
bacterium

Nitroreductase YoyC CopG_antitoxin HTH_3 Receiver ?

MCK5250002.1
JAGLMU010000670.1:1-2016
Spirochaetaceae bacterium

AAA_14 BrnT_toxin CopG_antitoxin Glycos_trans_3N

MCK5287907.1
JAGLMF010000091.1:109-3696
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin VanZ AMP-binding AMP-binding_C PseudoU_synth_2

MCK5353103.1
JAGLKT010001192.1:1-1802
bacterium

? Cytidylate_kin IPPT BrnT_toxin

MCK5416287.1
JAGLJW010000029.1:16185-28997
Candidatus Parcubacteria bacterium

? ? ? ? Glucos_trans_II ? Wzy_C BrnT_toxin CopG_antitoxin ? Zn_ribbon_Top1 Zn_ribbon_Top1 Zn_ribbon_Top1 Topoisom_bac Toprim Ribosomal_S20p Radical_SAM Bro-N

MCK5454622.1
JAGLIT010000863.1:67-619
Calditrichia bacterium

BrnT_toxin CopG_antitoxin

MCK5492246.1
JAGLIJ010000353.1:76-4256
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin ? ? RsgA_GTPase ? ? PEP-CTERM

MCK5493386.1
JAGLIJ010000887.1:332-2099
Candidatus Omnitrophota bacterium

? CopG_antitoxin IF-2B

MCK5493729.1
JAGLIJ010001050.1:94-2956
Candidatus Omnitrophota bacterium

? DUF4145 ? ? BrnT_toxin

MCK5494173.1
JAGLIJ010001250.1:26-2154
Candidatus Omnitrophota bacterium

Csf1 BrnT_toxin DJC28_CD CopG_antitoxin ?

MCK5510327.1
JAGLIO010000020.1:43-9724
Candidatus Parcubacteria bacterium

Ribosomal_S5 Ribosomal_S5_C Ribosomal_L27A SecY AAA_14 DUF4143 ? ? LeuA_dimer BrnT_toxin CopG_antitoxin ? ABC_tran MacB_PCD FtsX

MCK5519882.1
JAGLHZ010000149.1:98-3191
Candidatus Neomarinimicrobiota bacterium

HTH_Tnp_1 ParD_like ParE-like_toxin ParB_N DUF1841 BrnT_toxin CopG_antitoxin PhdYeFM_antitox

MCK5520249.1
JAGLHZ010000287.1:1-8084
Candidatus Neomarinimicrobiota bacterium

BatA BrnT_toxin CopG_antitoxin Sigma70_r2 Sigma70_r4_2 zf-HC2 ? His-Kinase Receiver GAF_2 GAF_2

MCK5520850.1
JAGLHZ010000521.1:1-1586
Candidatus Neomarinimicrobiota bacterium

? BrnT_toxin CopG_antitoxin PEP-utilizers_C

MCK5521258.1
JAGLHZ010000688.1:1-4132
Candidatus Neomarinimicrobiota bacterium

ABC_membrane ABC_tran ? BrnT_toxin CopG_antitoxin ?

MCK5529037.1
JAGLIE010000252.1:1-3364
Kiritimatiellae bacterium

DUF3482 BrnT_toxin CopG_antitoxin Na_H_Exchanger Fe_hyd_SSU Fe_hyd_lg_C

MCK5569275.1
JAGLHD010000579.1:337-6588
Spirochaetes bacterium

BrnT_toxin CopG_antitoxin PemK_toxin MazE_antitoxin Ub-Mut7C Mut7-C BAR_3 Helicase_C DEAD DUF559

MCK5671876.1
JAGLFP010000066.1:1-1923
Spirochaetales bacterium

? BrnT_toxin CopG_antitoxin CsgG

MCK5673064.1
JAGLFP010000514.1:165-1358
Spirochaetales bacterium

Fer4_6 BrnT_toxin CopG_antitoxin

MCK5686406.1
JAGLFN010000960.1:86-3201
bacterium

HTH_AraC ? CopG_antitoxin Syntaxin MCPsignal ? PhdYeFM_antitox ParE_toxin

MCK5708883.1
JAGLFH010000074.1:117-3446
Candidatus Auribacterota bacterium

? ? Flavodoxin_2 AsnC_trans_reg BrnT_toxin CopG_antitoxin ?

MCK5803393.1
JAGLDW010000573.1:1-8257
Lentisphaeria bacterium

Sulfatase Spermine_synth BrnT_toxin CopG_antitoxin Acetyltransf_11 Hexapep Hexapep Hexapep SYY_C-terminal tRNA-synt_1b Ribosomal_L21p PhoH PIN BcrAD_BadFG

MCK5841423.1
JAGLDB010000059.1:2872-9944
Candidatus Sabulitectum sp.

? ? ? Radical_SAM ? Sacchrp_dh_NADP Sacchrp_dh_C Radical_SAM BrnT_toxin CopG_antitoxin

MCK5871774.1
JAGLCI010000109.1:104-1323
Methylococcales bacterium

BrnT_toxin ? MMR_HSR1

MCK6426915.1
JAKLLA010000022.1:20035-31516
Burkholderiaceae bacterium

CobA_CobO_BtuR CobA_CobO_BtuR Radical_SAM BATS CobD_Cbib Aminotran_1_2 Aminotran_1_2 CbiA GATase_3 Asp_decarbox DBI_PRT BrnT_toxin CopG_antitoxin CobS His_Phos_1 Peripla_BP_2 RHH_1 HhH-GPD ?

MCK6557104.1
JAKLJW010000057.1:345-11322
Candidatus Binatia bacterium

SDH_protease REase ResIII Helicase_C EcoEI_R_C HsdM_N N6_Mtase ? ? DUF4160 DUF2442 BrnT_toxin MqsA_antitoxin PemK_toxin RHH_1 RHH_5 PIN 23S_rRNA_IVP PIN

MCK9426333.1
JALNZG010000008.1:141523-154855

Ignavibacteriaceae bacterium

Lactamase_B DUF6588 Carboxyl_trans SLT LysM LysM LysM LysM LysM LysM CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel TPK_catalytic TPK_B1_binding SSF BrnT_toxin BrnA_antitoxin Por_Secre_tail

MCK9431189.1
JALNZI010000155.1:1-3386
Candidatus Omnitrophota bacterium

? ? CopG_antitoxin bAvd

MCL1883507.1
WRHB01000250.1:497-10437
Defluviitaleaceae bacterium

HAMP His-Kinase His-Kinase BrnT_toxin Rota_NSP3 Germane DUF4131 Competence GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C BipA_C Peptidase_U32_C Peptidase_U32 DUF624 Aldose_epim

MCL1946302.1
WRFX01000028.1:63-7754
Chitinivibrionia bacterium

? RUBY_RBDX Flavin_Reduct Flavodoxin_5 ODP FGS DUF4258 ? YggU Ribonuc_L-PSP DUF2442 DUF4160 Na_Ca_ex Na_Ca_ex BATS Radical_SAM

MCL2043639.1
WRLF01000009.1:63303-76168
Treponema sp.

Molybdopterin Molybdopterin Molybdop_Fe4S4 Fer4_Nqo3 Fer4_20 Pyr_redox_2 Fer4_20 Fer2_4 Fer4_21 NADH_4Fe-4S SLBB Complex1_51K 2Fe-2S_thioredx ? Flavodoxin_2 YggU DUF1801 BrnT_toxin CopG_antitoxin Fe_dep_repr_C Fe_dep_repress PTS_EIIA_2 ? ? ? CARDB

MCL2066012.1
WRLP01000002.1:1807-13839
Treponema sp.

MOSC MoCF_biosynth CsrA FliW Flagellin_C Flagellin_IN Flagellin_N Flg_bbr_C Flagellin_IN FlgK_D1 Flg_bb_rod ? ? CopG_antitoxin DUF433 DUF5615 MS_channel_3rd MS_channel_2nd MS_channel_1st RbsD_FucU Lip_prot_lig_C LplA-B_cat ADH_N ADH_zinc_N

MCL2087040.1
WRLV01000027.1:14333-24538
Oscillospiraceae bacterium

Glycos_transf_1 HipA_C Ribosomal_S6 SSB Ribosomal_S18 Sigma70_r4 Sigma70_r2 N_methyl BrnT_toxin ? RelB YafQ_toxin TSCPD PHP SpoVAC_SpoVAEB SpoVAD

MCL2205596.1
WQRS01000079.1:18-6393
Treponema sp.

BrnT_toxin CopG_antitoxin ADH_zinc_N ADH_N PfkB PTS_EIIA_2 PTS-HPr ? ?

MCL2209605.1
WQRZ01000013.1:39181-52845
Treponema sp.

? LPP20 ? SidD_C Methyltransf_21 CsgG DUF7724 BrnT_toxin CopG_antitoxin META Polbeta DUF3818 ABC_tran UvrA_inter UvrA_DNA-bind DnaJ_CXXCXGXG SMC_N Cellulase ? Hydrolase_3

MCL2224564.1
WQSG01000018.1:16050-28442
Defluviitaleaceae bacterium

? Cu_amine_oxidN1 LRRs LRRs LRRs LRRs BPD_transp_1 BPD_transp_1 ? gag-asp_proteas NTP_transf_2 BrnT_toxin ? NHL Yip1 DUF5696 HicA_toxin HicB_lk_antitox Big_14 Big_14

MCL2264920.1
WQTG01000011.1:34674-49303
Treponema sp.

HicA_toxin FGGY_N FGGY_C XylA2_C Glyco_hydro_43 CBM_6 DegT_DnrJ_EryC1 Aldolase ChlI Mg_chelatase Mg_chelatase_C BrnT_toxin BrnA_antitoxin PHP Amidohydro_1 cNMPBD YfiO cNMPBD HTH_Crp_2 ? RDD

MCL2293826.1
WQTU01000082.1:614-14961
Spirochaetes bacterium

PGAP1 RecR_C Toprim_4 RecR_ZnF RecR_HhH YbaB_DNA_bd DNA_pol3_gamma3 DNAX_ATPase_lid DNA_pol3_delta2 DUF4143 AAA_14 ParE_toxin ? BrnT_toxin CopG_antitoxin MethyltransfD12 Acetylase 4HBT tRNA-synt_2 N-OB GatB_Yqey GatB_N Amidase

MCL2480288.1
WRFK01000011.1:6634-16185
Spirochaetaceae bacterium

DUF2971 ? PIN PhdYeFM_antitox dnstrm_HI1420 HTH_3 HicB_lk_antitox BrnT_toxin CopG_antitoxin Radical_SAM RlmN_N ? IGPD NMT1_3 DUF1850 DctM

MCL2481333.1
WRFK01000148.1:1-4593
Spirochaetaceae bacterium

HH_YBHG His-Kinase His-Kinase HAMP Trans_reg_C Receiver BrnT_toxin BrnA_antitoxin His-Kinase DNA_gyraseB

MCL2519575.1
WRBN01000004.1:27436-40754
Spirochaetaceae bacterium

NTP_transferase Aminotran_5 TPRs TPR_BSK1_C RelA_SpoT ? ? EamA ? ? CopG_antitoxin Creatinase_N Creatinase_N_2 Peptidase_M24 Peptidase_M24_C Glycos_transf_2 Glyco_trans_2_3 ? cNMPBD YfiO YfiO cNMPBD ?

MCL2519737.1
WRBN01000008.1:22088-35015
Spirochaetaceae bacterium

ABC_tran PTPS EPSP_synthase Tautomerase ? RadC Glyco_hydro_77 ? CopG_antitoxin tRNA_SAD tRNA-synt_2c PAZ Peptidase_M78 Peptidase_M78 Phage_portal_3 Terminase_3C Terminase_3 ?

MCL2519954.1
WRBN01000014.1:15863-26286
Spirochaetaceae bacterium

Ala_racemase_N Peptidase_M15_4 SIMPL BE_C Glyco_hydro_57 DUF4912 LysM ? ? CopG_antitoxin LysM ? ABC_tran NMT1 BPD_transp_1

MCL2699467.1
WRCS01000120.1:1-4216
Defluviitaleaceae bacterium

? RelB BrnT_toxin BrnA_antitoxin Thia_YuaJ DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_gyraseA_C DNA_topoisoIV

MCL2755141.1
WQXP01000140.1:1-3903
Oscillospiraceae bacterium

? BrnT_toxin ? Trans_reg_C Receiver His-Kinase ? Lipoprotein_9 4HB_MCP_1

MCL2763397.1
WQXQ01000103.1:1-8925
Treponema sp.

BrnT_toxin CopG_antitoxin CsgG DUF5640 cobW ? Flg_new Flg_new Flg_new Flg_new Flg_new Flg_new Flg_new Flg_new Flg_new FGS RNase_H_2

MCL2765852.1
WQXU01000037.1:276-7368
Treponema sp.

ABC_tran ABC_tran BPD_transp_2 BPD_transp_2 Polysacc_deac_2 Polysacc_deac_1 Polysacc_deac_1 ? YafQ_toxin BrnT_toxin BrnA_antitoxin

MCL2804874.1
WQXG01000008.1:12861-31537
Treponema sp.

PP_kinase_C PP_kinase_C_1 PP_kinase PP_kinase_N Ppx-GppA_III Ppx-GppA MGH1-like_GH FGGY_N FGGY_C XylA2_C DegT_DnrJ_EryC1 ChlI Mg_chelatase Mg_chelatase_C BrnT_toxin BrnA_antitoxin PHP Amidohydro_1 cNMPBD TPRs cNMPBD Metallophos_2 Glyco_hydro_94 GH94_b-supersand

MCL2821232.1
WQXS01000056.1:1-12381
Oscillospiraceae bacterium

His-Kinase His-Kinase Receiver Hpt Receiver His-Kinase His-Kinase PAS PAS sCache_3_3 sCache_3_2 ? ABC_tran ABC_tran_Xtn ABC_tran DUF3887 Abhydrolase_6 ? RelB BrnT_toxin BrnA_antitoxin ?

MCL2844185.1
WRAZ01000002.1:124463-142952
Chitinivibrionia bacterium

? LPD3 YDG LRRs PemK_toxin ABC_tran ABC_membrane Por_Secre_tail Virulence_RhuM BrnT_toxin ? GyrI-like Abi_2 ABC_tran ABC_membrane Hydrolase_like Hydrolase_6 ? Anticodon_2 tRNA-synt_1c

MCL2845847.1
WRAZ01000051.1:1994-11943
Chitinivibrionia bacterium

SLFN_AlbA_2 HATPase_c_4 HTH_24 HEPN Polbeta BrnT_toxin ? Peptidase_U32 AIRC RS_preATP-grasp-like ATP-grasp PurK_C EAL HDOD PEGA ?

MCL4353390.1
JAMCTM010000027.1:88727-95170
Patescibacteria group bacterium

Pep_deformylase Formyl_trans_N PEMT Peptidase_M50 Peptidase_M50 ? AlkZ-like Ribosomal_L33 BrnT_toxin CopG_antitoxin

MCL4357449.1
JAMCTI010000013.1:1-5037
Patescibacteria group bacterium

? AAA_17 Ham1p_like ? BrnT_toxin ? PAP2 FMN_red

MCL4385159.1
JAMCSH010000023.1:39-5435
Cyanobacteriota bacterium

SHOCT Flg_new_2 RexA CopG_antitoxin Polysacc_deac_1 ? MazE_antitoxin HTH_3 HigB-like_toxin DUF6788

MCL4390272.1
JAMCSA010000017.1:12599-22850
Patescibacteria group bacterium

HTH_3 Gp49 DeoC LDcluster4 VKOR CopG_antitoxin Transgly Transpeptidase BrnT_toxin CopG_antitoxin ? PIN Transgly Transpeptidase CopG_antitoxin ? Methyltransf_11

MCL4475106.1
JAMCQB010000033.1:1-3530
Nitrospirae bacterium

? QRPTase_N QRPTase_C DUF4258 YokU BrnT_toxin FAD_SOX Lactamase_B_2

MCL4501575.1
JAMCQY010000127.1:1-11458
Deltaproteobacteria bacterium

Peptidase_U32 ARMT1-like_dom HMA DUF6600 Methyltransf_11 CMD Aminotran_1_2 BrnT_toxin ? ? NAD_binding_3 Homoserine_dh ACT ?

MCL4510348.1
JAMCQL010000030.1:30047-41946
Bacteroidetes bacterium

? YceG CM_2 Trp_syntA TsaD ? Por_Secre_tail DUF5110 Glyco_hydro_31_3rd Glyco_hydro_31_2nd Gal_mutarotas_2 TPRs TPRs Glyco_transf_9 BrnT_toxin CopG_antitoxin Por_Secre_tail

MCL4510958.1
JAMCQL010000044.1:32648-44153
Bacteroidetes bacterium

? SSF LptF_LptG LptF_LptG ? HicA_toxin DUF7387 BrnT_toxin CopG_antitoxin ? ? MPN635_N ? Por_Secre_tail

MCL4511755.1
JAMCQL010000067.1:5505-23101
Bacteroidetes bacterium

Rce1-like ? Beta_elim_lyase MFS_1 ? PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I OPT BrnT_toxin NikA-like N6_N4_Mtase HinfI ? PIN MazE_antitoxin Sortilin-Vps10 Sortilin-Vps10 Sortilin-Vps10

MCL4546390.1
JAMCTF010000058.1:611-2337
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin ? SEC-C UPF0149

MCL4702855.1
JAHDYC010000035.1:10444-18650
Burkholderiaceae bacterium

Bac_rhamnosid_C MazE_antitoxin MFS_1 RHH_1 RelB3_C RHH_1 ParE_toxin PIN Cupin_2 BrnT_toxin CopG_antitoxin Cupin_2 HigB-like_toxin HTH_3 ? GerE AAA_33

MCL4804201.1
JAHDWL010000015.1:604-15917
Anaerolineae bacterium

Emfourin Peptidase_M4_C Peptidase_M4 ? MRM3-like_sub_bind SpoU_methylase ? DUF6847 AAA AAA_assoc_2 MgsA_C BrnT_toxin CopG_antitoxin ADH_zinc_N ELFV_dehydrog ADH_N SH3_2 RNase_PH RNase_PH_C Peptidase_M20 Chromate_transp Chromate_transp CofD PG_binding_4 LysM YkuD

MCL4819166.1
JAHDVO010000005.1:241256-255956
Vicinamibacteria bacterium

ABC_tran FecCD Peripla_BP_2 Diphthami_syn_2_C Diphthami_syn_2_N TonB_dep_Rec_b-barrel Plug CarboxypepD_reg PIN RHH_1 BrnT_toxin CopG_antitoxin HNH DUF222 Methyltransf_25 Methyltransf_20 URO-D Amino_oxidase CBM9_1 DUF5916

MCL4860590.1
JAHDWG010000140.1:1056-15656
Caldilineaceae bacterium

TRAM Radical_SAM UPF0004 Sulfatase Cellulase PhyH Cupin_2 HTH_18 CarboxypepD_reg CarboxypepD_reg Amidohydro_3 BrnT_toxin ? MlrC_C DUF1485 Beta-lactamase2 PIN RHH_1 BPD_transp_1 OppC_N BPD_transp_1 BPD_transp_1_N

MCL4862729.1
JAHDWG010000374.1:122-7183
Caldilineaceae bacterium

BrnT_toxin BrnA_antitoxin Peptidase_C69 adh_short_C2 2-Hacid_dh_C TAT_signal SBP_bac_1 TAT_signal SBP_bac_1

MCL4867378.1
JAHDWJ010000001.1:60389-74202
Anaerolineae bacterium

BPD_transp_1 ABC_tran PhoU PhoU CopG_antitoxin ? PA26 ? BrnT_toxin ABC_tran ABC_membrane ABC_tran ABC_membrane ABC2_membrane_6 DUF2085 His-Kinase His-Kinase GAF Receiver His-Kinase His-Kinase RsgA_GTPase RsgA_N

MCL5011270.1
JAMCXQ010000035.1:9684-15395
Patescibacteria group bacterium

Synaptojanin ? Y1_Tnp Ribosom_S12_S23 Ribosomal_S7 GTP_EFTU EF-G_D2 EFG_III EFG_IV EFG_C ? BrnT_toxin

MCL5062199.1
JAMCVL010000053.1:16-7233
Nitrospirae bacterium

BrnT_toxin BrnA_antitoxin ? ? TsaD Cytochrom_C_asm ResB ResB His-Kinase His-Kinase HAMP dCache_1

MCL5408465.1
JAMCZX010000004.1:81-17242
Candidatus Omnitrophota bacterium

ATP-grasp_5 Ligase_CoA_2 Succ_CoA_lig CoA_binding_2 URO-D Glyco_hydro_42 Glyco_hydro_42 Peripla_BP_3 LacI tRNA-synt_1b S4 BrnT_toxin CopG_antitoxin MscL UPF0547 MazE_antitoxin PIN Aldolase_II Pterin_bind

MCL5675520.1
JAMDAL010000001.1:15590-30013
Patescibacteria group bacterium

CTP_transf_like VanW PG_binding_4 SBP_bac_8 AI-2E_transport ? CAP Colicin_V Transgly Transpeptidase BrnT_toxin CopG_antitoxin ? Sigma70_r2 Sigma70_r4_2 ? Yip1 HMG-CoA_red Thiolase_N ACP_syn_III_C

MCL5796867.1
JAMDBI010000123.1:1-13733
Gammaproteobacteria bacterium

ABC_tran ABC_tran_Xtn ABC_tran Impact_N DUF1949 CbiA GATase_3 REase ResIII Helicase_C EcoEI_R_C BrnT_toxin CopG_antitoxin Methylase_S Methylase_S NA37 RelA_SpoT SLFN_AlbA_2 HATPase_c_4 Fic-like_C HsdM_N N6_Mtase CobU

MCM0607111.1
JAGTJX010000004.1:167291-180824
Lysobacteraceae bacterium

UvrD-helicase UvrD_C PcrA_UvrD_tudor DUF6929 Peptidase_M28 HTH_3 AbiEii Phytochelatin ? ? CopG_antitoxin CTP_transf_like ? Dus PEMT Glycos_transf_2 MJ0711_N

MCM1025908.1
JAMPIG010000018.1:25040-39938
Roseburia sp.

RNase_T Peptidase_S11 Methyltransf_4 Thioredoxin YitT_membrane DUF2179 DUF1846_C DUF1846 FtsH_ext AAA AAA_lid_3 Peptidase_M41 BrnT_toxin ? YokU DUF4258 TipAS GIY-YIG UvrC_M UVR UvrC_RNaseH UvrC_RNaseH_dom HHH_2 Hpr_kinase_N Hpr_kinase_C ROK

MCM1184024.1
JAMPFV010000038.1:204-11302
Roseburia sp.

CCDC39 Sulfatase YfhO NFACT-R_1 NFACT_N Glycos_transf_1 Glyco_transf_5 ? ? BrnT_toxin ?

MCM1224054.1
JAMPGK010000797.1:362-3853
Lachnospiraceae bacterium

Flg_bb_rod LlgE_F_G_D1 Flg_bbr_C BrnT_toxin BrnA_antitoxin EamA EamA HTH_18 AraC_binding

MCM1316141.1
JAMPEF010000060.1:207-10462

Alistipes senegalensis

CBM_6 Glyco_hydro_43 Dockerin_1 CBM_4_9 Glyco_hydro_10 CBM_4_9 DUF7317 MmeI_C MmeI_TRD MmeI_Mtase MmeI_hel ? MmeI_Mtase MmeI_TRD MmeI_C Dockerin_1 Dockerin_1 BrnT_toxin BrnA_antitoxin

MCM2303146.1
JAMPYD010000003.1:9693-19352
Elusimicrobiota bacterium

CTP_transf_like TatA_B_E CopG_antitoxin ? OmdA 3-dmu-9_3-mt TauE ? CopG_antitoxin Cytochrom_C Cytochrom_C DUF1207 DUF1207 COX2 COX1 COX3

MCM8813546.1
JAMWCB010000066.1:1-10462
Candidatus Omnitrophota bacterium

ABC_tran ABC-3 Cupin_2 ? MG2 bMG5 A2M_BRD A2M TED_complement TED_complement bMG10 ? His_Phos_1 BrnT_toxin

MCO4094901.1
JAAVWB010000020.1:17537-37116
Acidovorax sp.

DDE_Tnp_ISL3 HTH_Tnp_ISL3 Zn_ribbon_ISL3 ? DDE_Tnp_1 ? Cupin_2 Transposase_20 DEDD_Tnp_IS110 Transposase_20 DEDD_Tnp_IS110 BrnT_toxin CopG_antitoxin ? DEDD_Tnp_IS110 Transposase_20 Tnp_DNA_bind DDE_Tnp_1 Dimer_Tnp_Tn5 DDE_Tnp_1 ? ?

MCO5044338.1
JAMLJG010000004.1:71222-87275
Kiritimatiellia bacterium

? GspH N_methyl ? N_methyl N_methyl PilW_C N_methyl T2SSG N_methyl BrnT_toxin CopG_antitoxin Cupin_2 polyprenyl_synt NQRA_N NQRA_2nd NQRA_SLBB NQR2_RnfD_RnfE FMN_bind Rnf-Nqr

MCO5163328.1
JAMLIG010000001.1:3870581-3883664
Mesorhizobium sp.

FA_desaturase HTH_3 S-methyl_trans PadR YfbK_C VWA vWF_A Sigma70_r4_2 Sigma70_r2 adh_short_C2 ? CopG_antitoxin ? ? ? DLH DLH LysE MTTB

MCP3680485.1
JAAGZJ010001163.1:138-380
Gammaproteobacteria bacterium

BrnT_toxin

MCP3681391.1
JAAGZO010000126.1:623-5549
bacterium

? BrnT_toxin CopG_antitoxin

MCP3890860.1
JAAELB010000738.1:15-9097
Desulfobulbaceae bacterium

PhdYeFM_antitox YoeB_toxin Resolvase PDZ_2 Trypsin LysR_substrate BrnT_toxin CopG_antitoxin ? PALP Hexapep MetW Abhydrolase_1 Cys_Met_Meta_PP

MCP3929866.1
JAAELF010000701.1:116-3719
Bacteroidetes bacterium

AbiEii DUF6088 RelB3_C ParE_toxin ? BrnT_toxin CopG_antitoxin

MCP3957245.1
JAAELT010000080.1:1-8134
bacterium

? BrnT_toxin CopG_antitoxin Cupin_2 Abhydrolase_11 ? Asp_protease_2 nSTAND3 AAA_11 AAA_12 FxsA BSH_YhbJ DctM

MCP4138891.1
JAAENC010000049.1:32152-44625
Chloroflexota bacterium

NUDIX Nudix_N CarbopepD_reg_2 ? ? OMPdecase OTCace OTCace_N ? BrnT_toxin CopG_antitoxin Abhydrolase_3 Amidohydro_1 Xan_ur_permease Iso_dh ? DUF1579

MCP4295736.1
JAAEOE010000546.1:7076-13029
Proteobacteria bacterium

Fer4_11 Fer4_11 Cytochrome_C7 Cytochrome_C554 OTCace OTCace_N YfdI_N ? ? ?

MCP4298722.1
JAAEOE010001442.1:1-14017
Proteobacteria bacterium

GAF AAA_16 STY-kinase tRNA_edit EamA EamA Methyltransf_23 Peptidase_M20 Amidohydro_3 Fic BrnT_toxin CopG_antitoxin ? RVT_1 ?

MCP4298800.1
JAAEOE010001466.1:13664-25142
Proteobacteria bacterium

Receiver His-Kinase Acetyltransf_5 His-Kinase His-Kinase HAMP Trans_reg_C Receiver CCP_MauG Cytochrom_C ? Oxidored_molyb BrnT_toxin CopG_antitoxin Peptidase_S8 ?

MCP4460542.1
JAAEPI010001294.1:20633-30228
Cytophagales bacterium

SmpB ExbD ExbD ? MotA_ExbB Asparaginase_C Asparaginase TatD_DNase BrnT_toxin CopG_antitoxin ? Polysacc_deac_1 SpoIIE MarR Glycos_transf_2 ?

MCP4593440.1
JAAEQG010001266.1:1-1017
bacterium

? BrnT_toxin

MCP4608287.1
JAAEQJ010000188.1:1-3555
Planctomycetota bacterium

rve Oxidored_FMN Pyr_redox_2 BrnT_toxin CopG_antitoxin

MCP4681362.1
JAAERB010000119.1:11229-21867
Desulfobacterales bacterium

PMI_typeI_cat PMI_typeI_hel MPI_cupin_dom P5CR_dimer F420_oxidored ATPase_2 FTHFS PGI Radical_SAM Spermine_synth BrnT_toxin

MCP4766108.1
JAAERM010000094.1:3216-17121
Gammaproteobacteria bacterium

SBP_bac_3 SBP_bac_3 SLT PPK2 Reductase_C Pyr_redox_2 Fer2 Pyridoxal_deC GATase Cytochrome_CBB3 BrnT_toxin CopG_antitoxin PseudoU_synth_1 PseudoU_synth_1 K319L-like_PKD K319L-like_PKD FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 HpcH_HpaI AIPR SduA_C LysR_substrate

MCP4896038.1
JAAESQ010000139.1:234-1763
bacterium

PIN Trimer_CC FitA-like_RHH BrnT_toxin CopG_antitoxin

MCP5061478.1
JAAETS010000075.1:1-9135
Ignavibacteriota bacterium

GCV_H Receiver His-Kinase His-Kinase HAMP Cytochrome_C7 Cytochrome_C7 Cytochrom_CIII NrfD Fer4_11 Fer4_6 Nitro_FeMo-Co BrnT_toxin BrnA_antitoxin HTH_8 AAA_lid_14 NtrC-AAA PAS Cytochrome_cB

MCP5062267.1
JAAETS010000197.1:1-7052
Ignavibacteriota bacterium

? Aconitase_C Aconitase ParE_toxin ? ? 23S_rRNA_IVP Iso_dh BrnT_toxin CopG_antitoxin NYN OST-HTH ?

MCP5131325.1
JACKNO010000005.1:160626-177171
Pseudomonadales bacterium

SGSH_C Sulfatase ? FMO-like NAD_binding_8 DUF2834 ? DSPc ? BrnT_toxin CopG_antitoxin DUF3309 Gp49 dnstrm_HI1420 DUF2442 ? ?

MCP5346017.1
JACKOH010000001.1:889905-908838
Pseudomonadales bacterium

DUF1289 ? Pro_isomerase tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2 tRNA-synt_1e DALR_2 CysS_C ? rve_3 rve HTH_21 HTH_Tnp_1 BrnT_toxin CopG_antitoxin MqsA_antitoxin Methyltransf_21 PNPase CsiV TRCF Helicase_C DEAD CarD_TRCF_RID MFD_D3 UvrB_inter Gp_dh_N Gp_dh_C

MCP5515468.1
JACKPD010000024.1:11085-27124
Spirochaetales bacterium

cNMPBD Ion_trans DUF2229 BcrAD_BadFG BcrAD_BadFG GGDEF GAF GIY-YIG SpoU_methylase AzlC AzlD BrnT_toxin CopG_antitoxin Cupin_2 Cupin_2 DUF3394 DctM NMT1_3 CitMHS TauE

MCQ2050224.1
JAKSIN010000078.1:115-4131
Candidatus Saccharibacteria bacterium

? DNA_methylase RE_HaeIII ? CopG_antitoxin

MCQ2092620.1
JAKSIE010000043.1:4-10788
Fibrobacter sp.

Omp85 MMPL MMPL Aminotran_1_2 FHA ABC_tran ABC-ATPase ABC2_membrane STY-kinase NADase_transloc UPF0102 Sua5_yciO_yrdC BrnT_toxin CopG_antitoxin

MCQ2097345.1
JAKSIH010000013.1:23243-34884
Fibrobacter sp.

ABC_tran MlaE MMR_HSR1 RecR_C Toprim_4 RecR_ZnF RecR_HhH HDOD EAL HisG_C HisG PRA-PH BrnT_toxin CopG_antitoxin ? DUF4423 ? ? ? ?

MCQ2099866.1
JAKSII010000013.1:77685-88404
Fibrobacter sp.

? Glyco_hydro_44 Discoidin UPF0175 ? ArAE_1_C ? ? BrnT_toxin CopG_antitoxin PIN RelB KilA-N ? ? ?

MCQ2121405.1
JAKSIJ010000007.1:39607-53335
Fibrobacter sp.

KilA-N KilA-N RelB DNA_methylase RE_NgoFVII RE_NgoFVII_C ResIII Z1 ? ? CopG_antitoxin Amidohydro_1 ABC_tran MlaE MMR_HSR1 RecR_C Toprim_4 RecR_ZnF RecR_HhH HDOD EAL

MCQ2121980.1
JAKSIJ010000014.1:29112-63569
Fibrobacter sp.

Sigma70_r1_2 Sigma70_r2 Sigma70_r4 ? ? Radical_SAM Flg_new MBG_3 MBG_2 MBG_2 MBG_2 MBG_2 MBG_2 MBG_2 Flg_new Flg_new Flg_new Flg_new Flg_new Flg_new HTH_Rpn_C PDDEXK_2 PTA_PTB Malic_M malic BrnT_toxin CopG_antitoxin HMGL-like Radical_SAM CofH_C ABC2_membrane_3 ABC2_membrane_3 BSH_RND OEP OEP

MCQ2122902.1
JAKSIB010000002.1:138248-164346
Fibrobacter sp.

? ? TraB_PrgY_gumN Pectate_lyase_4 DUF6383 Hydrolase PseudoU_synth_2 ? BrnT_toxin ? Fumerase_C Fumerase ? Glyco_hydro_9 Pur_ac_phosph_N CelD_N Glyco_hydro_9 Pur_ac_phosph_N Pur_ac_phosph_N PAS His-Kinase Receiver

MCQ2414526.1
JAKSPC010000001.1:107321-124705
Lachnospiraceae bacterium

Methyltrans_SAM PUA_3 TIR ? LRRs RyR SLATT Peptidase_C39_2 Glyco_hydro_25 BrnT_toxin BrnA_antitoxin HTH_3 Dockerin_1 Cohesin ? Flg_new Flg_new Flg_new_2 Flg_new Flg_new Big Dockerin_1 ?

MCQ2546554.1
JAKSSG010000013.1:1-11324
Clostridia bacterium

Lys-AminoMut_A OAM_dimer B12-binding CoA_trans CoA_trans RUBY_RBDX Rubrerythrin BrnT_toxin CopG_antitoxin Arginase Amidohydro_3 RMMBL Lactamase_B_2 Peptidase_M3 Peripla_BP_6 NifU

MCQ2609237.1
JAKSTX010000010.1:1-4686
Lachnospiraceae bacterium

RmuC ? SecD_1st SecDF_P1_head SecD_SecF_C SecD_SecF_C BrnT_toxin ? ZnR

MCQ4773976.1
JANFYK010000008.1:88760-103470
Lacrimispora saccharolytica

2Fe-2S_thioredx ? ABC_tran ABC_membrane HMA SpoIIE PAS FeS Fe_hyd_lg_C Fer4_16 Fer4_21 CCDC192 BrnT_toxin ? Ribosomal_TL5_C Ribosomal_L25p Gate FeoB_C Gate FeoB_Cyto FeoB_N FeoA Chromate_transp Chromate_transp Peptidase_S55 PDZ_2 SMC_N

MCR4277725.1
JANLIQ010000028.1:1327-8171
Candidatus Berkelbacteria bacterium

? AAA_18 Transgly_assoc CCB3_YggT ? GrpB ? BrnT_toxin CopG_antitoxin OmdA ? ? PTPlike_phytase Methyltransf_11 ?

MCR4293316.1
JANLIE010000514.1:212-5514
Candidatus Kuenenia sp.

Phage_int_SAM_4 TIR FtsH_ext ? ? PIN BrnT_toxin CopG_antitoxin

MCR4313686.1
JANLHT010000007.1:5482-15394
Candidatus Uhrbacteria bacterium

DUF4044 TspO_MBR Peptidase_C39_2 HigB-like_toxin HTH_3 DUF5654 ? BrnT_toxin CopG_antitoxin ? PDDEXK_1 YgjP-like DUF5775 ? DEAD Helicase_C RecQ_Zn_bind RQC HRDC

MCR4596800.1
JAIKXY010000122.1:1-3944
Lachnospiraceae bacterium

DUF4131 Competence Lactamase_B_2 BrnT_toxin ? HSP20 NAPRTase

MCR4764164.1
JAIKYU010000122.1:1412-15073
Lachnospiraceae bacterium

DHH DHHA1 TIR Beta-prop_THOC3 ? STY-kinase ? HTH_3 ? BrnT_toxin BrnA_antitoxin AAA_18 ? ? HAD YvzJ Acetylase

MCR4792079.1
JAIKZQ010000021.1:36656-55590
Lachnospiraceae bacterium

CoA_binding_3 Bac_transf LytR_cpsA_psr Glyco_tranf_2_3 GATase_6 SIS SIS PDZ_2 Trypsin ZnR Radical_SAM BrnT_toxin BrnA_antitoxin HMA E1-E2_ATPase Hydrolase AAA_2 ClpB_D2-small Acetylase EPSP_synthase GGDEF MatE MatE

MCR4962015.1
JAILCR010000091.1:6101-22224
Lachnospiraceae bacterium

SBP_bac_3 ? DUF4430 ? Acetate_kinase PTA_PTB PTA_PTB MCPsignal YkyA BrnT_toxin BrnA_antitoxin Big Big Flg_new Flg_new Flg_new Flg_new LRRs LRRs LRRs LRRs HTH_26 ETF_alpha ETF ETF Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N 3HCDH 3HCDH_N

MCR5211623.1
JAILGQ010000061.1:1-5776
Lachnospiraceae bacterium

T2SSF Flp1_like DUF5702 TadE ? ? FHA BrnT_toxin ?

MCR5309650.1
JAILIF010000102.1:224-4609
Bacilli bacterium

? BrnT_toxin BrnA_antitoxin DUF1016_N YhcG_C DUF3825 BrnA_antitoxin

MCR5487958.1
JAILLN010000054.1:6188-12602
Lachnospiraceae bacterium

Glyco_hydro_25 DBI_PRT CobU CobS CobU TrkA_C Ion_trans ? BrnT_toxin DUF6364

MCR5621425.1
JAILON010000071.1:26817-44078
Treponema sp.

Cas_Cas1 Cas_Cas1 Cas_Cas5d Cas_CT1975 CRISPR_assoc CRISPR_Cse2 CRISPR_Cse1 Cas3-like_C_2 DEAD Cas3_HD BrnT_toxin BrnA_antitoxin NikA-like ? ? ? Gp10_C NLPC_P60

MCR5669772.1
JAILOW010000031.1:1-2806
Butyrivibrio sp.

? BrnT_toxin BrnA_antitoxin Alpha-amylase

MCR5762558.1
JAILPY010000063.1:191-7397
Treponema sp.

YbjQ_1 dCache_1 HAMP CCDC171 MCPsignal ? HipA_C ? BrnT_toxin BrnA_antitoxin ? ? ? ?

MCR9121862.1
JANSXT010000006.1:61639-74442
Phyllobacteriaceae bacterium

ExoD Formyl_trans_N Virulence_fact MTHFR Pterin_bind DUF559 Fer2 RACo_linker Raco_middle RACo_C_ter ? ? GCS2 MFS_1 ? Hydrolase DJ-1_PfpI HTH_18 PBP_like_2

MCU0595219.1
JALOPA010000183.1:1-5224
Thermodesulfobacteriota bacterium

AFOR_C AFOR_N BrnT_toxin CopG_antitoxin Thiolase_N Thiolase_C EamA EamA Asparaginase_C

MCU0665577.1
JALOQK010000001.1:93828-107287
Candidatus Omnitrophota bacterium

TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C ACT_AHAS_ss ALS_ss_C KARI_N KARI_C HMGL-like HCS_D2 LeuA_dimer Shikimate_dh_N OCD_Mu_crystall SDH_C A24_N_bact Peptidase_A24 Glyco_hydro_77 ? CopG_antitoxin ? ? NUMOD3 NUMOD3 ATP_bind_3 RmuC Cu-oxidase_4

MCU0667581.1
JALOQO010000064.1:293-4073
Patescibacteria group bacterium

SIS bact-PGI_C PBECR3 CopG_antitoxin FtsX MacB_PCD ABC_tran

MCU0790343.1
JALORS010000176.1:171-3555
Nitratireductor sp.

Sacchrp_dh_C Sacchrp_dh_NADP Amidohydro_1 BrnT_toxin BrnA_antitoxin

MCU0846111.1
JALOTJ010000143.1:3780-11258
Spirochaetes bacterium

DUF6250 BrnT_toxin BrnA_antitoxin ? SMC_N ? TPRs OMP_b-brl Pribosyltran

MCU7489952.1
JAAXKT010000001.1:101432-120863

Ignavibacteria bacterium

Cytochrom_NNT Cytochrom_c3_2 ? OmpA Gly-zipper_YMGG OMP_b-brl CHASE PAS GAF His-Kinase His-Kinase ? NAPRTase_C NAPRTase NAPRTase_N BrnT_toxin CopG_antitoxin ? DUF6930 DUF7309 ? OxoGdeHyase_C Transket_pyr E1_dh 2-oxoacid_dh 2-oxogl_dehyd_N GATase_6 SIS SIS cobW

MCU7816228.1
JAHXIT010000140.1:5833-10987
Candidatus Thiodiazotropha sp. (ex Rostrolucina anterorostrata)

? ? Radical_SAM PP-binding rve_3 rve HTH_21 HTH_Tnp_1 BrnT_toxin CopG_antitoxin

MCU7958573.1
JAHXHF010000026.1:1-3012
gamma proteobacterium symbiont of Bathyaustriella thionipta

BrnT_toxin CopG_antitoxin ? Bflower_2 HTH_40 DUF4011

MCW8815726.1
JAPHUN010000259.1:358-1915
Chlorobium sp.

BrnT_toxin CopG_antitoxin ParE_toxin Unstab_antitox

MCX2983122.1
SHNN01000005.1:215384-226433

Candidatus Litorirhabdus singularis

? ? ? ? ? Glyoxalase LCM ? CopG_antitoxin DoxX ? ? Plug TonB_dep_Rec_b-barrel ? SnoaL_2

MCX5643805.1
JAPLMX010000054.1:30844-40992
Phycisphaerae bacterium

Ribosomal_L34 Peptidase_M90 TruD ? Polbeta HEPN Cellulase ? BrnA_antitoxin ? TnpB_IS66 DDE_Tnp_IS66_C SDH_protease ? ? Rrf2

MCX5697359.1
JAPLLV010000053.1:26122-33162
Candidatus Omnitrophota bacterium

? CopG_antitoxin ? Ion_trans_2 TIR DUF4145 GYD DPBB_1

MCX5699313.1
JAPLLY010000043.1:13675-26136
Candidatus Omnitrophota bacterium

CinA DHBP_synthase GTP_cyclohydro2 DMRL_synthase NusB PHP dCMP_cyt_deam_1 RibD_C Lum_binding Lum_binding BrnT_toxin CopG_antitoxin Acylphosphatase Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_2 BRCT ? Lactamase_B Phage_int_SAM_1 Phage_integrase PolyA_pol PolyA_pol_RNAbd

MCX5733064.1
JAPLLC010000017.1:31020-38167
Candidatus Methylomirabilota bacterium

TPRs TPRs DUF1810 ? Cupin_2 ARG ? ? ? BrnA_antitoxin Meth_synt_2 Meth_synt_2 ?

MCX5788538.1
JAPLKK010000116.1:3271-16797
Elusimicrobia bacterium

NifU_N FeS_assembly_P Rrf2 NifU PHP GDPD GDPD NadA BrnT_toxin CopG_antitoxin ? ? VWA_3 VIT Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 CsgG SDF

MCX5838539.1
JAPLJW010000232.1:1-8774
Deltaproteobacteria bacterium

Eco57I TaqI_C Mrr_cat Acetyltransf_7 BrnT_toxin CopG_antitoxin PhdYeFM_antitox ParE_toxin DUF3465 ? PfkB AMP-binding AMP-binding_C_2

MCX5898886.1
JAPLIZ010000696.1:163-3335
Proteobacteria bacterium

PIN RHH_1 PIN MazE_antitoxin BrnT_toxin CopG_antitoxin CpXC

MCX5946670.1
JAPLID010000008.1:210-5376
Cyanobacteriota bacterium

? CopG_antitoxin ? HicA_toxin HNH Unstab_antitox ? SLH OprB PepSY_TM

MCX6028757.1
JAPLHB010000054.1:69036-89178
Chloroflexota bacterium

PDT ACT DAHP_snth_FXD DAHP_synth_1 PDH_N PDH_C EPSP_synthase EPSP_synthase Shikimate_dh_N Shikimate_DH SDH_C Chorismate_synt SKI DHQ_synthase DHQS_C BrnT_toxin CopG_antitoxin DHquinase_II AzlD GATase_2 Glu_syn_central Glu_synthase GXGXG Citrate_synt Aldo_ket_red Acetyltransf_9

MCX6034340.1
JAPLGW010000291.1:525-2851
Chloroflexota bacterium

BrnT_toxin CopG_antitoxin DUF1178 Zn_ribbon_8 Guanylate_kin Rhomboid

MCX6037856.1
JAPLGV010000254.1:1-3954
Chloroflexota bacterium

DDE_5 BrnT_toxin CopG_antitoxin Amidohydro_1 Xan_ur_permease

MCX6047181.1
JAPLGZ010000371.1:309-8850
Chloroflexota bacterium

? Ala_racemase_N D-ser_dehydrat HTH_3 HigB-like_toxin ? MR_MLE_N MR_MLE_C Creatininase BrnT_toxin ? CobU

MCX6105706.1
JAPLFY010000040.1:134991-148737
Proteobacteria bacterium

ATPase_2 Peptidase_S74 PhdYeFM_antitox ParE_toxin HTH_3 Couple_hipA HipA_C Peptidase_S74 BrnT_toxin ? Peptidase_S74 ParE_toxin HTH_3 RNase_E_G_Thio RNase_E_G Glycos_transf_N Glyco_transf_9

MCX6135864.1
JAPLFQ010000036.1:20489-30236
Ignavibacteriales bacterium

SpoIIE DinB_2 DUF2200 ? RibD_C ? ? BrnT_toxin CopG_antitoxin ? Redoxin

MCX6145999.1
JAPLFM010000083.1:139-5664
Candidatus Kapaibacterium sp.

? YqeY Ubie_methyltran NTP_transf_4 Hexapep ? BrnT_toxin ? BrnT_toxin ?

MCX6154125.1
JAPLFL010000084.1:1-6458
Candidatus Kapaibacterium sp.

? Thiolase_N ACP_syn_III_C ? Thiolase_N Thiolase_C_1 OB_ChsH2_C BrnT_toxin ? DUF2283 ? ?

MCX6353753.1
JAPLCW010000035.1:79-2068
Candidatus Auribacterota bacterium

DUF8127 BrnT_toxin CopG_antitoxin ?

MCX6356218.1
JAPLCW010000204.1:8912-12982
Candidatus Auribacterota bacterium

BrnT_toxin CopG_antitoxin DUF433 ? LEM-3_GIY-YIG BrnT_toxin ? ?

MCX6567435.1
JAPKZF010000003.1:1-4339
Candidatus Aminicenantes bacterium

UvrA_inter AAA_16 BrnT_toxin CopG_antitoxin GntR ABC_tran ABC2_membrane_4 DEAD

MCX6791076.1
JAPLWD010000073.1:24-9622
Candidatus Gribaldobacteria bacterium

EamA DUF6938 Pept_tRNA_hydro Polbeta HepT-like DUF2341 BT_3657-like_N fn3 DUF4258 CopG_antitoxin Trp_repressor Radical_SAM ?

MCX6822883.1
JAPLUV010000021.1:408-13149
candidate division SR1 bacterium

? SIMPL ? CopG_antitoxin tRNA-synt_1c Anticodon_2 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 DUF5122 CLP_protease PG_binding_1 ? IPPT

MCX6958344.1
JAPLTJ010000144.1:47106-59463
Verrucomicrobiia bacterium

? ? CTP_transf_like ? E1-E2_ATPase Hydrolase His_Phos_2 His_Phos_2 AMMECR1 Memo BrnT_toxin ? BRCT HHH_2 HHH_5 DNA_ligase_ZBD OB_DNA_ligase DNA_ligase_aden Nlig-Ia

MCX6974525.1
JAPLTH010000269.1:4106-16993
Verrucomicrobia bacterium

Acyltransferase ? LysM ACCA LptF_LptG Abhydrolase_1 Peptidase_M16 Peptidase_M16_C ? CopG_antitoxin UPF0547 ? Porphobil_deamC Porphobil_deam Shikimate_DH GlutR_N Cytochrom_C_asm GreA_GreB GreA_GreB_N HTH_GRAD

MCX6975957.1
JAPLTH010000442.1:93-3054
Verrucomicrobia bacterium

? PemK_toxin BrnT_toxin CopG_antitoxin DUF6869 ?

MCX7096988.1
JAPLRR010000030.1:244-15081
Methylococcales bacterium

BrnA_antitoxin PEMT NifU_N SAM_MT Rieske Big_10 bMG3 MG2 bMG5 A2M_BRD A2M bMG10 BrnT_toxin CopG_antitoxin ? BrnT_toxin ? DO-GTPase1 DUF4131 YoeB_toxin

MCX7115428.1
JAPLRL010000048.1:60474-70976
Gammaproteobacteria bacterium

ASL_C2 Lyase_1 Arginosuc_syn_C Arginosuc_synth ABC_tran BPD_transp_1 SBP_bac_3 Arg_repressor Arg_repressor_C ? BrnT_toxin ? DnaG_DnaB_bind DnaB_bind Toprim_2 DNAG_N Zn_ribbon_DnaG YqeY Ribosomal_S21 TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N

MCX7121384.1
JAPLRF010000041.1:120-12453
Gammaproteobacteria bacterium

? ? CopG_antitoxin TK ATPase_2 DUF234 Thioredoxin Rho_N Rho_RNA_bind ATP-synt_ab TOP_N Peptidase_M3 Peptidase_S13

MCX7124012.1
JAPLRJ010000160.1:220-3422
Gammaproteobacteria bacterium

HepT-like Polbeta BrnT_toxin ? ? ? ?

MCX7125580.1
JAPLRG010000315.1:1-3916
Gammaproteobacteria bacterium

Aminotran_5 EPSP_synthase ? CopG_antitoxin Cytidylate_kin S1 S1

MCX7201819.1
JAPLQG010000250.1:1-10422
Burkholderiales bacterium

PutA_1st ? TctC Adhesin_E MlrC_C DUF1485 MCP-like_PDC_1 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 BrnT_toxin D-ser_dehydrat PMT_C3 Ala_racemase_N DUF4258 DUF2283 Hydrolase_3 S6PP MatE MatE

MCX7661217.1
JANZZX010000076.1:534-1711
Candidatus Omnitrophota bacterium

ParE_toxin DUF6290 ? CopG_antitoxin

MCX8098855.1
JAOAIF010000056.1:115191-128345
Casimicrobiaceae bacterium

Phosphonate-bd Acetylase Synaptojanin Gln-synt_N Gln-synt_C ? PAS His-Kinase His-Kinase Receiver NtrC-AAA AAA_lid_14 HTH_8 BrnT_toxin CopG_antitoxin FeS Fer4_21 HhH-GPD EndIII_4Fe-2S DUF1841 NAD_binding_11 NAD_binding_2 PQ-loop ASL_C2 Lyase_1

MCY3022501.1
JAPLOD010000640.1:4883-22063
Planctomycetota bacterium

MEKHLA LamininG NPCBM GFO_IDH_MocA GFO_IDH_MocA_C3 Cache_3-Cache_2 3keto-disac_hyd LamininG Pel9A-like_beta_helix Beta_helix Beta_helix HNH BrnT_toxin CopG_antitoxin PhdYeFM_antitox ParE-like_toxin ? ? ? Beta-prop_IFT122_1st Beta-prop_WDR3_1st

MCY3565199.1
JAPPBL010000163.1:1-7365
Gammaproteobacteria bacterium

PMSR Aminotran_3 EHN Abhydrolase_1 ABG_transport ? BrnT_toxin BrnA_antitoxin PEPcase

MCY4128829.1
JAPOVV010000054.1:395-14369
Gammaproteobacteria bacterium

Esterase ABC-3 ABC_tran OMP_b-brl_4 OMP_b-brl_4 Plug CarboxypepD_reg ZnuA ? ARM_ECM29 BrnT_toxin BrnA_antitoxin ? SF-assemblin MdcG ? ? FGS

MCY4185770.1
JAPOTQ010000044.1:28690-38979
Rhodobacteraceae bacterium

? ? Uma2 ? ? ? ? BrnT_toxin HTH_31 ? COX4_pro_2 B_lactamase-like_C Lactamase_B Acyl-CoA_dh_C Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N Amidohydro_2 RbsD_FucU

MCY4200865.1
JAPOTJ010000103.1:19566-32734
Gammaproteobacteria bacterium

AAA AAA ? HTH_3 Zeta_toxin BrnT_toxin CopG_antitoxin PhdYeFM_antitox M_Eco57I_C ? DUF1295 Anticodon_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 CN_hydrolase LNT_N

MCY4277949.1
JAPOSA010000084.1:12-13501
Gammaproteobacteria bacterium

PIN ? MoxR-AAA McrB_lid McrBC BrnT_toxin CopG_antitoxin AAA_14 DUF4143 DUF234 HTH_IclR ATPase_2 Methyltransf_33 FGS FGS DinB_2 NTP_transf_3 GMPPB_C Hexapep Hexapep GATase_6 SIS SIS

MCY4281707.1
JAPORZ010000037.1:179-9637
Gammaproteobacteria bacterium

DUF8426 Mrr_cat DUF433 BrnT_toxin CopG_antitoxin DUF4143 TIP39 AAA_14 GHMP_kinases_C GHMP_kinases_N LolB TPRs TPRs TPRs TPRs ARM_TT21_N GlutR_N Shikimate_DH GlutR_dimer PCRF RF-1

MCY4337907.1
JAPORE010000096.1:123-10868
Gammaproteobacteria bacterium

Acyltransferase DALR_1 tRNA_synt_2f tRNA-synt_2e Phasin_2 DUF6994 ? Type_ISP_C Eco57I N6_Mtase ISP_coupler Helicase_C ResIII Mrr_cat_2 BrnT_toxin

MCY4363688.1
JAPOQU010000100.1:1-9354
Gammaproteobacteria bacterium

NAD_synthase GATase IMPDH tRNA_anti_2 Exonuc_VII_L DUF4258 PhdYeFM_antitox ParE_toxin HTH_3 Radical_SAM LIAS_N LplA-B_cat ? ABC_tran ABC2_membrane

MCY4389210.1
JAPOQR010000048.1:239712-249343
Desulfurellaceae bacterium

? ? ? ? BrnA_antitoxin HTH_3 TPRs AP-ATPase Sirtuin BrnT_toxin BrnA_antitoxin MT-A70 Endonuc-BglII ? ? HigB-like_toxin HTH_31

MCY4415072.1
JAPOQH010000019.1:49634-65542
Chloroflexota bacterium

CinA Bac_luciferase Penicil_amidase MFS_1 MFS_1 MFS_1 Bac_luciferase Bac_luciferase DUF4258 ? GMC_oxred_N GMC_oxred_C Glutaredoxin DinB_2 PAC2 Glyoxalase MFS_1 Hydrolase

MCY4416143.1
JAPOQH010000059.1:10481-23960
Chloroflexota bacterium

Acetylase Lactamase_B Competence DUF4131 HHH_3 SLBB SGL ? Methyltransf_28 CoA_transf_3 BrnT_toxin ? STY-kinase Glutaredoxin Pyr_redox_2 ? ?

MCY4428205.1
JAPOQD010000180.1:28021-48910
Halieaceae bacterium

RNase_T TetR_N TetR_C_1 DUF1214 Amino_oxidase ? HipA_C Couple_hipA HTH_31 BrnT_toxin CopG_antitoxin DUF5131 DUF4143 AAA_14 AbiJ_NTD4 TerB AAA_19 UvrD_C Helicase_C MZB

MCY4557483.1
JAPOOV010000203.1:293103-309668
Chloroflexota bacterium

DUF7124 CoA_transf_3 CoA_transf_3 SGL ? ? Methyltransf_28 CoA_transf_3 ? ? Glutaredoxin Pyr_redox_2 ? ? MFS_3 PDH_N PDH_C EPSP_synthase

MCY4580079.1
JAPOOL010000461.1:1-7498
Chloroflexota bacterium

? ? BrnT_toxin TIR ? HMUDK_hel DCD MazG_C MazG

MCY4596024.1
JAPOOJ010000339.1:6259-23163
Bryobacterales bacterium

DUF1501 AMP-binding_C AMP-binding FAA_hydrolase HTH_3 Sulfatase SGSH_C OMP_b-brl_4 OMP_b-brl_4 OMP_b-brl_4 CarboxypepD_reg OMP_b-brl_4 OMP_b-brl_4 CarboxypepD_reg BrnT_toxin CopG_antitoxin UvrD_C_2 SLFN-g3_helicase NERD ? ARG

MCY4624936.1
JAPONS010000067.1:11609-26152
Chloroflexota bacterium

tRNA-synt_His tRNA-synt_His HisG HisG_C Histidinol_dh Aminotran_1_2 IGPD MFS_1 Bac_luciferase BrnT_toxin NikA-like T4_deiodinase GFO_IDH_MocA GFO_IDH_MocA_C3 LYTB DUF3800 IMS_C IMS_HHH IMS ?

MCY7367464.1
JAJAYU010000143.1:1-9011
Chamaesiphon sp.

? TPMT DUF8436 Pyr_redox_2 FAD-depend_C BrnT_toxin CopG_antitoxin ? ChpXY Proton_antipo_M Proton_antipo_M Proton_antipo_N ?

MCY7406435.1
JAJAYH010000110.1:442-5062

Alkalinema sp. CAN_BIN05

BrnT_toxin DUF2283 Uma2 TPRs TPRs ? IU_nuc_hydro Pribosyltran

MCZ2099366.1
JAIXLN010000424.1:152-6520
Anaerolineae bacterium

? TnpB_IS66 LZ_Tnp_IS66 zf-IS66 DDE_Tnp_IS66 DDE_Tnp_IS66_C HTH_29 rve Mu-transpos_C_2 IstB_IS21 ? BrnT_toxin CopG_antitoxin

MCZ2112431.1
JAIXKH010000007.1:1-7323
Anaerolineae bacterium

? SH3_2 SH3_2 BrnT_toxin CopG_antitoxin RNase_PH RNase_PH_C Esterase ? ? NEDD4_Bsd2 BPD_transp_1

MCZ6547395.1
JAPUGC010000134.1:153-5801
Deltaproteobacteria bacterium

Acetone_carb_G Hydantoinase_B PhyH Amidohydro_2 Glyoxalase BrnT_toxin

MCZ6802795.1
JAPUKP010000056.1:1-3787
Proteobacteria bacterium

? ? DUF6884 GIY_YIG_3 BrnT_toxin CopG_antitoxin ? ?

MCZ7609595.1
JAOSKD010000001.1:1253904-1265372
Ignavibacterium sp.

Epimerase ? NUDIX DUF4440 DUF8412 OmdA OmpA DUF6958 BrnT_toxin DUF6364 Polbeta HepT-like DNA_methylase RE_NgoFVII RE_NgoFVII_C Vsr DUF559 Transposase_20 DEDD_Tnp_IS110

MCZ8156841.1
JAPZRK010000062.1:8547-24381
Leptospira sp.

AhpC-TSA Peptidase_M17 TPRs SpoIIE 7TMR-DISM_7TM Ppx-GppA_III Ppx-GppA MreB_Mbl ? CsgG BrnT_toxin CopG_antitoxin DUF2442 DUF4160 ? ? ? DUF1232

MDA0243453.1
JAQCNZ010000012.1:54882-72934
Chloroflexota bacterium

? DUF2283 MgsA_C AAA_assoc_2 AAA Y1_Tnp TetR_N TetR_C_24 BrnT_toxin CopG_antitoxin Calx-beta Calx-beta Calx-beta Calx-beta Calx-beta Calx-beta LTD Ig_Rha78A_N CotH LTD Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 HTH_20 Beta-lactamase2 ? ?

MDA0246023.1
JAQCNZ010000056.1:142-14333
Chloroflexota bacterium

GntR FCD ? E1_dh Transket_pyr Transketolase_C ? E1_dh Transket_pyr Transketolase_C BrnT_toxin ? Biotin_lipoyl E3_binding 2-oxoacid_dh CitMHS Epimerase Acyltransferase SCP2 BrnT_toxin

MDA0263766.1
JAQCKM010000011.1:44813-59831
Chloroflexota bacterium

Abhydrolase_1 Peptidase_S9 NHL NHL NHL NHL NHL DUF1800 DUF1501 Thioredox_DsbH 2-Hacid_dh_C BrnT_toxin CopG_antitoxin HGTP_anticodon tRNA-synt_2b tRNA_SAD G3P_acyltransf Polyketide_cyc2 AXE1 AXE1 PCRF RF-1

MDA0348841.1
JAQCIW010000068.1:6022-25289
Verrucomicrobia bacterium

SGSH_C Sulfatase SGSH_C Sulfatase BNR_2 Sulfatase Phage_integrase Phage_int_SAM_4 ? ? BrnT_toxin CopG_antitoxin EVE NmeDI_C MacB_PCD FtsX MacB_PCD FtsX PQQ_2 PQQ_2 PQQ_2 PQQ_2 DUF1501

MDA0672486.1
JAQCHG010000044.1:1-8424
Cyanobacteriota bacterium

ADH_zinc_N Pept_tRNA_hydro ? Acetylase ? Uma2 BrnT_toxin Metal_resist ? NACHT

MDA0737804.1
JAQCFS010000006.1:19119-34283
Nitrospirae bacterium

MOSC Methylase_S Methylase_S AAA_13 HsdM_N N6_Mtase ? Gp49 dnstrm_HI1420 BrnT_toxin CopG_antitoxin ? DUF2835 VTT_dom NUDIX Zn_ribbon_Nudix DegT_DnrJ_EryC1 QueF

MDA0821418.1
JAQCEL010000014.1:24850-36154
Proteobacteria bacterium

MoCoBD_2 MoCoBD_1 Ald_Xan_dh_C Fer2_2 Fer2 CO_deh_flav_C FAD_binding_5 ECH_1 Abhydrolase_3 Glyoxalase Glyoxalase FAD_binding_3 BrnT_toxin CopG_antitoxin HigB-like_toxin HTH_3 DUF6082 Unstab_antitox ParE_toxin HicA_toxin

MDA1079848.1
JAQBYG010000027.1:4840-11155
bacterium

Acetylase ? ? Cyclase CopG_antitoxin BrnT_toxin Lysine_decarbox BrnT_toxin CopG_antitoxin ? ?

MDA1316694.1
JAQBTX010000006.1:15281-27016
bacterium

ABC1 UbiB_N DDE_Tnp_1 rve_3 ? RecA_N RecA_C RecX_HTH1 RecX_HTH2 RecX_HTH3 RNase_Y_N KH_1 HD BrnT_toxin CopG_antitoxin Bac_DNA_binding DUF5667 S1 S1 S1 DsbD_2 ? Pantoate_ligase

MDA1316709.1
JAQBTX010000007.1:179-6752
bacterium

BrnT_toxin CopG_antitoxin Glyoxalase Hydrolase_4 Glyco_hydro_18 FecR Thioredoxin_10 AhpC-TSA DsbD_TM Virul_fac_BrkB

MDA1337778.1
JAQBTZ010000002.1:8451-19383
bacterium

Ribosomal_S11 Ribosomal_S4 S4 RNA_pol_A_bac RNA_pol_A_CTD Ribosomal_L17 Ribosomal_L13 Ribosomal_S9 ? BrnT_toxin CopG_antitoxin Sua5_yciO_yrdC DUF4293 Cupredoxin_1 Trns_repr_metal HMA E1-E2_ATPase Hydrolase HMA DsbD_2 SurA_N_3 SurA_N_3

MDA1372068.1
JAQBSZ010000030.1:39888-47539
Proteobacteria bacterium

Gp49 HTH_26 AAA_14 DUF4143 HTH_3 HicB-like_2 HicA_toxin VanZ ? BrnT_toxin CopG_antitoxin ? DUF4160 ? DUF4143 Gp49 HTH_37

MDA3789269.1
JAQICW010000350.1:538-3549
Desulfobacula sp.

HipA_C ? ? BrnT_toxin ?

MDA3799238.1
JAQICX010000147.1:26803-40257
Kiritimatiellae bacterium

? Glyco_hydro_43 Glyco_hydro_43 Methyltransf_23 BrnT_toxin CopG_antitoxin ABC_tran UvrA_inter UvrA_DNA-bind AAA_29 ABC_tran ARM_TT21_C TPRs TPRs TPRs TPRs NAD_kinase NAD_kinase_C ? DUF4181 URO-D Thioredoxin_3

MDA3799480.1
JAQICX010000170.1:157-5886
Kiritimatiellae bacterium

KilA-N DUF6364 ParE_toxin BrnT_toxin CopG_antitoxin ? ? ?

MDA3807188.1
JAQIDE010000142.1:368-2553
Thiomicrorhabdus sp.

FG-GAP_3 FG-GAP_3 BrnT_toxin CopG_antitoxin

MDA3859983.1
JAQICK010000069.1:1-7132
Melioribacteraceae bacterium

? ? TPRs TPRs TPRs GAF_2 His-Kinase His-Kinase ? BrnT_toxin CopG_antitoxin ? DDE_3

MDA3922491.1
JAQIBT010000145.1:3512-17100
Salinisphaera sp.

BcrAD_BadFG Aminotran_3 AstA AstA Aldedh ? AstB BrnT_toxin CopG_antitoxin NIPSNAP HGLS SBP_bac_3 BPD_transp_1 BPD_transp_1 ABC_tran

MDA3924017.1
JAQIBS010000056.1:37548-58464
Kiritimatiellae bacterium

? ? zf-like ? DUF8255 ? AbiEi_4 BrnT_toxin CopG_antitoxin TetR_N CecR_C Discoidin Alpha_L_fucos Discoidin CHB_HEX_C_1 Bgal_small_N LacZ_4 Glyco_hydro_2_C Glyco_hydro_2 Glyco_hydro_2_N Discoidin Glyco_hydro_2 Glyco_hydro_2_N ? Bac_rhamnosid_C Bac_rhamnosid6H Bac_rhamnosid Bac_rhamnosid_N Ig_Rha78A_N

MDA3924667.1
JAQIBS010000090.1:68312-79867
Kiritimatiellae bacterium

? BrnT_toxin CopG_antitoxin ? ABC2_membrane_2 ABC_tran SpoU_methylase AXE1 GTP_EFTU EF-G_D2 EFG_III EFG_IV EFG_C

MDA3924861.1
JAQIBS010000106.1:228-5122
Kiritimatiellae bacterium

Competence ? ? Y1_Tnp ParE_toxin Unstab_antitox BrnT_toxin CopG_antitoxin

MDA3937951.1
JAQIBN010000019.1:4568-9679
Spirochaetia bacterium

ParE_toxin PhdYeFM_antitox FG-GAP_2 FG-GAP_2 TPRs Sec10_N ? ? BrnT_toxin

MDA3940077.1
JAQIBN010000179.1:6667-18189
Spirochaetia bacterium

Impact_N VWA_4 CsgG ? Oxidored_FMN Hydrolase_6 Hydrolase_like P-II NTPase_1 ? CopG_antitoxin Indigoidine_A PfkB ? ABC_tran FecCD Peripla_BP_2

MDA3940217.1
JAQIBN010000190.1:48-10383
Spirochaetia bacterium

tRNA-synt_1c Anticodon_2 tRNA-synt_2b HGTP_anticodon tRNA-synt_1b SYY_C-terminal DUF4143 AAA_14 PepX_C Peptidase_S15 HXXEE TetR_C_7 TetR_N BrnT_toxin CopG_antitoxin ?

MDA8273040.1
JAPTUU010000049.1:1-9183
Deltaproteobacteria bacterium

DUF234 HTH_45 ATPase_2 DUF4143 AAA_14 DUF4143 AAA_14 DUF4143 AAA_14 Glycos_transf_2 ? BrnT_toxin CopG_antitoxin HVO_A0114 ? AAA_14

MDA8273507.1
JAPTUU010000075.1:1-11638
Deltaproteobacteria bacterium

? ? HicA_toxin UPF0150 BrnT_toxin CopG_antitoxin PIN PhdYeFM_antitox MqsA_antitoxin YafQ_toxin RelB CopG_antitoxin BrnT_toxin HVO_A0114 MazE_antitoxin DUF433 Glycos_transf_2 Glyco_tran_28_C Glyco_trans_1_3 LPG_synthase_TM

MDA8306115.1
JAPTUL010000087.1:1-11616
Deltaproteobacteria bacterium

Prismane ydhR BrnT_toxin CopG_antitoxin PrkA AAA_PrkA SpoVR_C SpoVR DUF444 DUF444 PrkA AAA_PrkA YegS_C DAGK_cat UPF0150

MDA8401389.1
JAPTTJ010000019.1:1-2924
Deltaproteobacteria bacterium

FAM81B ? ? CopG_antitoxin Pyr_redox_2 ?

MDA8411249.1
JAPTTK010000295.1:250-7772
Treponema sp.

NTP_transf_2 NTase_sub_bind HTH_24 PfkB Indigoidine_A RibD_C Mrr_cat_2 Type_ISP_C ? BrnT_toxin CopG_antitoxin ABC_tran ABC-ATPase ABC_tran_Xtn ABC_tran

MDA8426490.1
JAPTSX010000255.1:1-10435
Treponema sp.

GGDEF ? ? NYN YgjP-like SocA_Panacea MqsA_antitoxin ? BrnA_antitoxin tRNA-synt_2b Seryl_tRNA_N PDDEXK_3 ? TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N ILV_EDD_C

MDA8426801.1
JAPTSX010000298.1:139-7545
Treponema sp.

Receiver Trans_reg_C Polbeta HepT-like_2 ? CopG_antitoxin NTP_transf_2 HEPN Trans_reg_C ? ? HepT-like_2

MDA9129103.1
JAOGXR010000005.1:671-11457
Candidatus Gracilibacteria bacterium

HypA RNase_T Sigma70_r4 ? N_methyl ? PseudoU_synth_2 ? CopG_antitoxin Abhydrolase_6 Lipase_GDSL_2 ? TsaD ? ?

MDA9129266.1
JAOGXR010000010.1:8935-21430
Candidatus Gracilibacteria bacterium

Ferritin His-Kinase His-Kinase Trans_reg_C Receiver ? NTF2_7 Peptidase_S8 TPRs HSP90 HATPase_c_3 BrnT_toxin CopG_antitoxin PrgI TraC-like T4SS_pilin DUF5673 Ser_hydrolase Ldh_2

MDA9129305.1
JAOGXR010000012.1:1-11567
Candidatus Gracilibacteria bacterium

AA_kinase Ribonuclease_P 60KD_IMP ? ? BipA_C EFG_C GTP_EFTU_D2 GTP_EFTU ? Gp44_lid CopG_antitoxin Ribosomal_S18 SSB Ribosomal_S6 MMR_HSR1 KH_2 SoFic-like_C Fic Fic_N DSBA

MDA9962793.1
JAOGFH010000041.1:1-14514
Opitutales bacterium

TPRs TPRs TPRs TPRs TPRs TPRs YfiO TPRs TPRs MotA_ExbB ExbD PD40 PD40 SQHop_cyclase_C ACDH_C Ig_3 FGS Bact_surface_Ig-like Bact_surface_Ig-like FGS FGS ? BrnT_toxin CopG_antitoxin

MDB5046986.1
JAQBPN010000001.1:56484-65166
Fibrobacteres bacterium

? Sulf_transp Sulf_transp ? CheW 4HB_MCP_1 Tweety MCPsignal ? BrnT_toxin HTH_3 ? SocA_Panacea ? DUF5683 ? Phage_CI_repr

MDB5850232.1
JAQGMQ010000662.1:1-2892
Rhodoferax sp.

? DUF3325 BrnT_toxin CopG_antitoxin Isochorismatase Acetyltransf_7 ?

MDC0547616.1
JAOMTU010000010.1:1-12032
Opitutales bacterium

DUF7133 HEATs Cytochrom_C 3keto-disac_hyd ? BrnT_toxin CopG_antitoxin PhoD PhoD DUF7800 BD-FAE ? Plug TonB_dep_Rec_b-barrel DUF6807 SnoaL_4

MDC0864640.1
JAOMYS010000018.1:80-11093
Rickettsiaceae bacterium

MFS_1 UPF0102 Coprogen_oxidas Ldh_1_N Ldh_1_C FOL3_C Mur_ligase_M BrnT_toxin CopG_antitoxin COX3 GatB_Yqey GatB_N Amidase GatC ? ?

MDC0920042.1
JAONBW010000002.1:55221-67330

Candidatus Thioglobus sp.

VIT VWA_3 Sortase Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Trypsin ? ? ? Phage_integrase BrnT_toxin CopG_antitoxin SocA_Panacea ? NTP_transferase Glyphos_transf IspD adh_short

MDC1205325.1
JAOEPB010000003.1:17908-34156
Candidatus Paceibacterota bacterium

adh_short dCMP_cyt_deam_1 NUDIX ? Nuc_deoxyri_tr2 ? ? BrnT_toxin CopG_antitoxin RelB ? Polysacc_deac_1 Hepar_II_III_C Hepar_II_III_N Glycos_transf_2 UDPG_MGDP_dh_C UDPG_MGDP_dh UDPG_MGDP_dh_N ?

MDC7231702.1
JAQQAM010000006.1:1-8775
Spirochaetales bacterium

? CTP_transf_like Glycos_transf_4 DUF6675 ? DHQ_synthase DHQS_C BrnT_toxin CopG_antitoxin PhdYeFM_antitox YoeB_toxin OB_NTP_bind HA2_C WHD_HA2

MDD1619936.1
JAJRDU010000010.1:7446-21944
Methylococcaceae bacterium

Band-7 Band-7 HflK_N MMR_HSR1 GTP-bdg_M GTP-bdg_N Hfq ABC_tran ABC_tran_Xtn ABC_tran Receiver Trans_reg_C dCache_1 HAMP His-Kinase His-Kinase BrnT_toxin CopG_antitoxin ? Maf Sulfotransfer_3 Nif11 Asp_Arg_Hydrox Acetylase

MDD2461910.1
JAQUEX010000270.1:142-4265
Kiritimatiellae bacterium

? BrnT_toxin CopG_antitoxin PQQ_2 PQQ_2 PQQ_2 Pterin_bind

MDD2679613.1
JAQUAP010000014.1:989-14508
Candidatus Omnitrophota bacterium

DUF4931_N GalP_UDP_tr_C Gp_dh_N Gp_dh_C Spermine_synth DUF3180 PGK TIM SecG OAD_gamma BT1 ? CopG_antitoxin ORF6N ORF6N ? MarC NTP_transferase Hexapep_GlmU Acetyltransf_7

MDD2806208.1
JAQTZB010000007.1:166396-178603
Elusimicrobiales bacterium

Glycos_transf_2 Glyco_trans_4_4 Glycos_transf_1 Bac_transf PglD_N Hexapep Hexapep DegT_DnrJ_EryC1 CoA_binding_3 Polysacc_synt_2 DUF1016_N BrnT_toxin CopG_antitoxin FecR dTDP_sugar_isom NTP_transferase dTDP_sugar_isom GDP_Man_Dehyd Lactamase_B_2

MDD3182095.1
JAQUYN010000018.1:7234-21542
Alphaproteobacteria bacterium

Peptidase_M48 LemA P5CR_dimer F420_oxidored Aldedh PUA AA_kinase Polyketide_cyc2 HSP70 BrnT_toxin CopG_antitoxin NUDIX SBP_bac_3 DnaJ DnaJ_C DnaJ_C_2 ? DNA_binding_1 HhH-GPD EndIII_4Fe-2S

MDD3262698.1
JAQUVI010000002.1:82798-102838
Candidatus Absconditabacteria bacterium

POTRA_1 dCMP_cyt_deam_1 Aminotran_5 ? VanY Peptidase_S24 LexA_DNA_bind RecA_N RecA_C ? CopG_antitoxin Acetate_kinase Bro-N ? RNase_Y_N HmpF HmpF

MDD3297085.1
JAQUWY010000035.1:592-7002
Candidatus Omnitrophota bacterium

? ? ? CopG_antitoxin HTH_3 Couple_hipA HipA_C Band-7 HMG_box N_methyl PilW_C

MDD4151288.1
JAQVPN010000018.1:1936-11913
Candidatus Gracilibacteria bacterium

? MatE MatE MarR Pyr_excise Mrr_cat Zn_ribbon_Top1 ? UPF0004 Radical_SAM ? CopG_antitoxin Phage_holin_4_2 SecG SBP_bac_5 SBP_bac_5 EI24

MDD4736703.1
JAQVYB010000111.1:5004-12994
Kiritimatiellae bacterium

? BrnT_toxin CopG_antitoxin DUF2283 ? Ferritin TauE ? tRNA-synt_2b HGTP_anticodon ProRS-C_1

MDD4887597.1
JAQTVS010000250.1:1-4768
Thiomonas sp.

Ala_racemase_C Ala_racemase_N MFS_1 MS_channel_1st_1 MS_channel_2nd LGT AhpC-TSA BrnT_toxin

MDD4893664.1
JAQTVI010000235.1:1-2566
Candidatus Rickettsiella isopodorum

? BrnT_toxin BrnA_antitoxin WWE_3 ? ?

MDD4905160.1
JAQTVD010000066.1:1-7171
Methylobacter tundripaludum

Anticodon_1 LptE DNA_pol3_delta DNA_pol3_delt_C His-Kinase His-Kinase TPRs TPRs Receiver HATPase_c_3 HSP90 BrnT_toxin

MDD4953839.1
JAQTUD010000026.1:1-8581
Candidatus Omnitrophota bacterium

P-loop_SecA SecA_SW SEC-C Cupin_2 ? CopG_antitoxin LNT_N CN_hydrolase Peptidase_M23 ZT_dimer Cation_efflux PIG-L Adenosine_kin Lip_A_acyltrans

MDD4979979.1
JAQTTO010000001.1:331971-341174
Candidatus Omnitrophota bacterium

DUF2062 NIL DUF3108 RF-1 B12-binding Radical_SAM ? zf_Tbcl_2 ? CopG_antitoxin Methyltransf_11 Vpu ? Radical_SAM DUF134 ?

MDD5031060.1
JAQTSL010000333.1:1-3862
Rhodoferax sp.

ABC-ATPase UvrA_DNA-bind UvrA_inter ABC_tran ParA-Soj-NTPase BrnT_toxin CopG_antitoxin YchJ_M-like

MDD5044056.1
JAQTTE010000006.1:1-12915
Candidatus Omnitrophota bacterium

DHH DHHA1 RecJ_OB Ribosomal_L28 ? CopG_antitoxin DUF488 ? HTH_3 Gp49 ? ?

MDD5085603.1
JAQTRP010000061.1:1-11443
Candidatus Omnitrophota bacterium

QRPTase_N QRPTase_C N6_Mtase REase REase Eco57I TaqI_C rve_2 rve HTH_21 HTH_Tnp_1 DUF5128 CopG_antitoxin ResIII UvrB_inter UvrB_3rd Helicase_C UvrB UVR Aldolase GDE_N

MDD5095519.1
JAQTRE010000156.1:1-5879
Dehalococcoidia bacterium

PIN FitA-like_RHH Patatin ? HTH_13 Fic Fic_N BrnT_toxin CopG_antitoxin ? ?

MDD5109569.1
JAQTQJ010000035.1:1-2834
Candidatus Omnitrophota bacterium

DUF5128 CopG_antitoxin ? BamE ? ? ?

MDD5129818.1
JAQTQG010000004.1:23950-39982
Candidatus Omnitrophota bacterium

Sua5_yciO_yrdC LMWPc LacAB_rpiB SHMT dCMP_cyt_deam_1 Zn_ribbon_NrdR ATP-cone DUF1016_N YhcG_C UPF0758_N RadC ? CopG_antitoxin HKD AAA_22 SNF2-rel_dom Helicase_C MmeI_Mtase Eco57I TaqI_C DUF4747 ? ? ?

MDD5197628.1
JAQTON010000001.1:479555-496421
Candidatus Gracilibacteria bacterium

ABC2_membrane_6 VF530 ? ? AAA AAA_lid_3 Peptidase_M41 DUF8237 ATP-cone ATP-cone Ribonuc_red_lgN Ribonuc_red_lgC ? CopG_antitoxin Trypsin ? ? Ribonuc_red_sm DHFR_1 ?

MDD5206787.1
JAQTOV010000554.1:1-3429
Desulfobacterales bacterium

? BrnT_toxin RHH_1 Glycos_transf_4

MDD5222379.1
JAQTNV010000001.1:37927-49624
bacterium

? ? DUF2283 ? DUF2283 Epimerase Mrx7 BrnT_toxin CopG_antitoxin DUF5670 ? PAS PAS HisKA_3 His-Kinase SNase Ada_Zn_binding Collagen ATP-synt_C

MDD5226316.1
JAQTQY010000096.1:1-5338
Candidatus Omnitrophota bacterium

Phage_integrase Phage_int_SAM_1 ? CopG_antitoxin DUF6804 SNase DUF1016_N ? BRCT HHH_2 DNA_ligase_ZBD OB_DNA_ligase DNA_ligase_aden

MDD5260073.1
JAQTPL010000051.1:4671-10771
bacterium

Thi4 VWA BrnT_toxin CopG_antitoxin ? BAR_3_WASP_bdg ? DUF8255 DUF4901 Y1_Tnp Bac_DnaA_C

MDD5299642.1
JAQTMZ010000003.1:137580-151946
Gallionella sp.

Dala_Dala_lig_N Dala_Dala_lig_N Dala_Dala_lig_C POTRA_1 FtsQ_DivIB_C SHS2_FTSA FtsA Tubulin FtsZ_C LpxC ABC_tran ABC_tran_Xtn ABC_tran ? BrnT_toxin ? ? ? DUF2157 Methyltransf_10 KTSC Peptidase_M20

MDD5306694.1
JAQTNF010000016.1:2855-16293
Deltaproteobacteria bacterium

Methyltransf_31 HTH_20 HD_5 Receiver ? DUF2229 HGD-D DUF2229 HGD-D ? BrnT_toxin CopG_antitoxin ExbD MotA_ExbB BacA HTH_25 ParA YqeY ?

MDD5432661.1
JAQURX010000013.1:36860-44157
Candidatus Omnitrophota bacterium

? DUF6291 HTH_5 ? Phr_regulator Receiver HTH_3 RadC UPF0758_N ? CopG_antitoxin ? Acetylase ? ? PLDc_N ?

MDD5457456.1
JAQURL010000078.1:1-10188
Candidatus Margulisiibacteriota bacterium

DUF3841 His_Phos_1 Memo ? GIY-YIG Methyltransf_25 His-Kinase His-Kinase ? CopG_antitoxin Receiver ?

MDD5465951.1
JAQURC010000007.1:81105-93873
Candidatus Omnitrophota bacterium

Ribosomal_L28 DUF488-N3i DUF6941 Peptidase_M48 Gp49 HTH_3 PDDEXK_1 ? CopG_antitoxin UvrD-helicase UvrD_C PDDEXK_1 MshEN T2SSE T2SSF_N T2SSF T2SSF N_methyl N_methyl N_methyl T2SSJ

MDD5552797.1
JAQUPI010000008.1:41428-49206
Candidatus Omnitrophota bacterium

DUF7937 ? ? Gp49 HTH_3 DUF748 DUF3568 ? BrnT_toxin CopG_antitoxin PDZ_2 Peptidase_M48

MDD5595673.1
JAQUNE010000092.1:1-6078
Candidatus Omnitrophota bacterium

? AF_1680 DUF6941 ORF6N BrnT_toxin CopG_antitoxin ? Smr DPBB_1 Fer4_7 Nitroreductase ?

MDD5657017.1
JAQUMZ010000144.1:1-6889
Elusimicrobia bacterium

LamininG FlgD_ig PorV Epimerase GTP_cyclohydroI RHH_3 BrnT_toxin CopG_antitoxin Methyltransf_25 DUF8026 Cupredoxin_1 Receiver

MDD5675715.1
JAQUMP010000434.1:114-3836
Chitinivibrionales bacterium

? Por_Secre_tail BrnT_toxin CopG_antitoxin AIRC Dus

MDD5731448.1
JAQULF010000001.1:157916-173111
Patescibacteria group bacterium

? ? VanY Phage_lysozyme Tricorn_2nd PEGA GGDEF NTP_transf_2 HepT-like BrnT_toxin CopG_antitoxin ? ? 23S_rRNA_IVP PrgI TraC-like CagE_TrbE_VirB P-loop_TraG P-loop_TraG

MDD5746608.1
JAQULA010000091.1:1-7011
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin ? Pyr_redox_2 Fer2_BFD DAO FGGY_C FGGY_N Hydrolase_4 ?

MDD7609693.1
JAQYPM010000037.1:1-5049
Spirochaetales bacterium

Toprim_2 VapE-like_dom BrnT_toxin BrnA_antitoxin Mob_Pre PEP-utilisers_N Zn_ribbon_DnaG

MDD8017240.1
JAQUOA010000018.1:9030-24124
Bacteroidota bacterium

AbiEii DegT_DnrJ_EryC1 DegT_DnrJ_EryC1 Poly_export SLBB_2 SLBB SLBB SLBB SLBB_2 SLBB ? ? Sigma70_r2 GerE DUF4258 CopG_antitoxin ? ? ? ? ? UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C

MDD9334466.1
JARBCI010000006.1:8161-20130
Rickettsiaceae bacterium

Thiolase_C Thiolase_N Abhydrolase_1 PGAP1 OMP_b-brl ? TLC CopG_antitoxin ? Peptidase_M17_N Peptidase_M17 MipZ LepB_N 120_Rick_ant MFS_1 MFS_1 Ribosomal_L36

MDD9989370.1
JAPPNY010000226.1:21253-40583
Spirochaetaceae bacterium

RecO_N RecO_C LigT_PEase LigT_PEase DHH DHHA1 RecJ_OB ? His-Kinase RNase_H ChlI Mg_chelatase Mg_chelatase_C ? Meth_synt_2 Pect-lyase_RHGA_epim DUF1349 ? BPD_transp_1 BPD_transp_1 DUF3502 ?

MDE0027800.1
JAPPNE010000002.1:12264-27862
Deltaproteobacteria bacterium

S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C AdoHcyase AdoHcyase_NAD FitA-like_RHH PIN N6_Mtase Eco57I Type_ISP_C ? HTH_3 BrnT_toxin CopG_antitoxin PhdYeFM_antitox PIN HemN_C Radical_SAM OCD_Mu_crystall Peptidase_M48 Peptidase_M48_N CBS CBS CBS CBS

MDE0031371.1
JAPPNE010000174.1:19934-30289
Deltaproteobacteria bacterium

ABC_membrane ABC_tran ABC_membrane ABC_tran Aminopep DUF983 Mut7-C ? GFA BrnT_toxin CopG_antitoxin NAD_binding_10 DUF2867

MDE0219846.1
JAPPKD010000142.1:503-11383
Spirochaetaceae bacterium

Transposase_mut Glyco_hydro_32C BrnT_toxin CopG_antitoxin Glyco_hydro_32C Glyco_hydro_32N PhyH Bac_rhamnosid_C Bac_rhamnosid6H Bac_rhamnosid Bac_rhamnosid_N p450 GFO_IDH_MocA GFO_IDH_MocA_C3 ABC_tran oligo_HPY

MDE0262627.1
JAPPIV010000096.1:1-12014
Bryobacterales bacterium

PrlF_antitoxin dnstrm_HI1420 ? UPF0150 DUF4143 ? Glyco_hydro_43 Glyco_hydro_43 BrnT_toxin CopG_antitoxin FtsX FtsX MacB_PCD Transposase_31 DUF4351 ? Cpn60_TCP1

MDE0769210.1
JAPKDW010000149.1:1-9195
Opitutaceae bacterium

UPF0758_N RadC Queuosine_synth TPR_CcmH_CycH ACCA TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C PhdYeFM_antitox YoeB_toxin BrnT_toxin CopG_antitoxin ? GFO_IDH_MocA

MDE0824610.1
JAPKDE010000196.1:65-4778
Dehalococcoidia bacterium

2-Hacid_dh_C BrnT_toxin CopG_antitoxin HGTP_anticodon tRNA-synt_2b tRNA_SAD G3P_acyltransf NHL NHL NHL

MDE1170885.1
JAPWJX010000003.1:1121334-1133819
Verrucomicrobium sp.

DUF3799 RecT ? ? ? HTH_17 DUF6290 Peptidase_S24 BrnT_toxin ? adh_short ACR_tran BSH_Fcp1 HH_MFP_RND TetR_N TetR_C_13 2-Hacid_dh_C ?

MDE1919833.1
JAGWIR010000001.1:15375-26880
Candidatus Omnitrophota bacterium

Ribosomal_L36 Ribosomal_S13 Ribosomal_S11 Ribosomal_S4 S4 RNA_pol_A_bac RNA_pol_A_CTD Ribosomal_L17 RES BrnT_toxin CopG_antitoxin UPF0093 RelE HTH_31 ? ? N-OB tRNA-synt_2

MDE1920066.1
JAGWIR010000001.1:234577-239352
Candidatus Omnitrophota bacterium

? CopG_antitoxin HxlR Aldo_ket_red CSD RNA_pol_Rpb6 eIF-1a LpxD Hexapep Hexapep Hexapep Hexapep

MDE2027266.1
JAGWKV010000027.1:815-7992
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin HHH_5 Helicase_dom4_arc HTH_61 Helicase_C DEAD Tyr_Deacylase TatD_DNase DUF3352 DUF3352 ? DHODB_Fe-S_bind

MDE2027879.1
JAGWKV010000122.1:1-4934
Candidatus Omnitrophota bacterium

His-Kinase Receiver NtrC-AAA AAA_lid_14 HTH_8 PocR GAF_2 BrnT_toxin CopG_antitoxin ? DUF1653 zf_Tbcl_2 ? ?

MDE2215195.1
JAGWOE010000027.1:101-8316
Candidatus Omnitrophota bacterium

DUF4131 BrnT_toxin CopG_antitoxin UPF0093 RelE HTH_31 ? ? N-OB tRNA-synt_2

MDE2215425.1
JAGWOE010000042.1:1-5920
Candidatus Omnitrophota bacterium

BrnT_toxin CopG_antitoxin ? TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs RH3_dom ParE_toxin RecG_wedge DEAD Helicase_C RecG_dom3_C ?

MDE2293674.1
JAGWPT010001147.1:57-5130
Elusimicrobia bacterium

Acetylase ThiS Sortilin-Vps10 ? ORF6N MazE_antitoxin PemK_toxin ? NikA-like Phage_int_SAM_4 Phage_integrase

MDE2320557.1
JAGWQH010000018.1:2119-19160
Candidatus Methylomirabilota bacterium

YgbB DUF5677 HsdM_N N6_Mtase Methylase_S Methylase_S 23S_rRNA_IVP HSDR_N_2 ResIII Helicase_C EcoEI_R_C DUF4440 BrnT_toxin ? HI0933_like ABC_tran ABC_tran_Xtn ABC_tran ABC_tran_CTD ABC_tran ABC_tran_Xtn ABC_tran Pyr_redox_2 Mito_fiss_Elm1 PEPcase_2

MDE2444555.1
JAGWUY010000001.1:20645-31871
Alphaproteobacteria bacterium

adh_short_C2 AHSA1 HTH_20 AHSA1 AHSA1 RibD_C Polyketide_cyc2 PNPOx_N PNP_phzG_C BrnT_toxin CopG_antitoxin Cys_Met_Meta_PP PRK Pyr_redox_2 Oxidored_FMN Ldh_2 MarR Flavin_Reduct

MDE2511543.1
JAGWRD010000550.1:57-1997
Elusimicrobia bacterium

? CopG_antitoxin CopG_antitoxin Phosphorylase

MDE2773390.1
JAPPGF010000022.1:21444-40443
Gemmatimonadota bacterium

SEC-C DUF6998 DUF6814 Acetylase Rhodanese ? ? BrnT_toxin BrnA_antitoxin Xan_ur_permease His-Kinase SusD_RagB TonB_dep_Rec_b-barrel Plug CarbopepD_reg_2 ? G_glu_transpept DUF1015

MDE2803882.1
JAPPGZ010000019.1:32551-46771
Gemmatimonadota bacterium

RimK TM2 DUF5777 CoaE ? Peptidase_M56 Penicillinase_R BrnT_toxin CopG_antitoxin Methyltransf_12 G_glu_transpept FG-GAP FG-GAP_3 FG-GAP_3 ? DUF5671 TetR_N TetR_C_33

MDE2893183.1
JAPPJZ010000071.1:1-2075
Chloroflexota bacterium

DUF4258 CopG_antitoxin PIN FitA-like_RHH AAA_14

MDE3175680.1
JAGWRL010000123.1:1-6697
Pseudomonadota bacterium

FA_synthesis ACP_syn_III ACP_syn_III_C 4HBT 4HBT DUF4258 CopG_antitoxin Ferric_reduct DUF697 DUF463

MDE3233871.1
JAGWUN010000229.1:1-4190
Pseudomonadota bacterium

tRNA-synt_1 BrnT_toxin CopG_antitoxin DUF2783 FAD_binding_3 Rubrerythrin

MDE5597812.1
JAIDMA010000744.1:43-1920
Lachnospiraceae bacterium

BrnT_toxin BrnA_antitoxin ? Acetylase

MDE6519359.1
JAIDTI010000023.1:95703-108178
Ruminococcus sp.

BPD_transp_1 BPD_transp_1 PDDEXK_2 HTH_Rpn_C DUF5906 NHL TPRs Yip1 BrnT_toxin BrnA_antitoxin DUF5696 BPD_transp_1 BPD_transp_1 ? Cytidylate_kin2 ?

MDF0667977.1
JARFXO010000006.1:18722-33697
Nitrospira sp.

? OEP OEP Rce1-like Lactamase_B_2 ? YbjQ_1 T6SS_Tle1-like_cat T6SS_Tle1-like_cat DUF4258 CopG_antitoxin FGS nSTAND1 DUF4062 HepT-like NTP_transf_2 ABC_tran UvrA_inter Zn_ribbon_8 UvrA_DNA-bind ABC_tran AAA_16 Rv0078B DUF3422

MDF1568607.1
JARGFR010000111.1:127-6112
Spirochaetaceae bacterium

RelE Cas_Cas4 DUF2779 BrnT_toxin CopG_antitoxin ? ? ? ? AbiEii AbiEi_4

MDF1827106.1
JARGOS010000001.1:475728-490031

Legionellaceae bacterium

Amino_oxidase MurB_C FAD_binding_4 FAD_binding_7 DNA_photolyase DUF4116 ATPase_2 Epimerase NAD_binding_8 Amino_oxidase BrnT_toxin CopG_antitoxin HTH_8 LpxB GFO_IDH_MocA_C3 GFO_IDH_MocA DnaB DnaB_C Ala_racemase_N Ala_racemase_C Rick_17kDa_Anti 17kDa_Anti_2

MDF2529797.1
JARBTI010000057.1:1-7116
Gammaproteobacteria bacterium

? He_PIG He_PIG ANKs ANKs ANKs ANKs BrnT_toxin CopG_antitoxin Thiolase_C Thiolase_N ?

MDF2939484.1
JAQSYQ010000001.1:114238-128091
Gammaproteobacteria bacterium

Enolase_N Enolase_C DivIC SurE LysM Peptidase_M23 NTP_transferase Glyco_transf_4 Glycos_transf_1 Glyco_trans_4_4 Glyco_trans_1_4 Bac_transf BrnT_toxin DUF1564 Acetylase ? TGS tRNA_SAD tRNA-synt_2b HGTP_anticodon IF3_N IF3_C Ribosomal_L35p Ribosomal_L20

MDF2939695.1
JAQSYQ010000003.1:75295-100727
Gammaproteobacteria bacterium

DAO DAO_C Fer4_2 Fer4 DUF3470 SesA DNA_pol_IIIA_C N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP ANKs ANKs DMP19 HTH_24 BrnT_toxin ? OTU NACHT DUF481 Lyase_1 ASL_C Arc AA_permease_2 MlaA

MDF2966057.1
JAQSWU010000043.1:1-10529
Rickettsiaceae bacterium

Resolvase PIN MazE_antitoxin BrnT_toxin CopG_antitoxin ? ? ? ? TPRs TPRs DDE_Tnp_IS240

MDF3047546.1
JAQSWL010000019.1:1-5241
Candidatus Midichloriaceae bacterium

AAA_14 BrnT_toxin CopG_antitoxin PhdYeFM_antitox ? DUF977 Fic HTH_3 HigB_toxin PH_30

MDF3130867.1
JARHXX010000045.1:10439-27522
Kiritimatiellaeota bacterium B1221

ChlI Mg_chelatase Mg_chelatase_C GH113 PEP-CTERM ? Peripla_BP_3 ? Phosphodiest TPRs TPRs BrnT_toxin CopG_antitoxin Transposase_mut ? zinc_ribbon_6 GYF Band_7_1 ? SBP_bac_3 SDF

MDG1122535.1
JAQWWM010000184.1:3492-16798
Glaciecola sp.

PqqD Radical_SAM SPASM Peptidase_M16 GerE Receiver His-Kinase HisKA_3 LapD_MoxY_N TonB_dep_Rec_b-barrel Plug Hydrolase_3 BrnT_toxin CopG_antitoxin TonB_C ? DsrE ? ? DUF3718

MDG1172757.1
JAQWVG010000031.1:1-2093
Opitutales bacterium

DEAD Helicase_C BrnT_toxin CopG_antitoxin ?

MDG1324133.1
JAQXBH010000003.1:1-5829
Opitutales bacterium

Aconitase_2_N Aconitase_B_N DUF4143 AAA_14 DoxX_2 DoxX_2 RF-1 BrnT_toxin CopG_antitoxin Fic

MDH2918172.1
JARCRQ010000030.1:1-4664
Sideroxydans sp.

PTPS Bax1-I ? PIN DUF4926 ParA-Soj-NTPase ? BrnT_toxin ?

MDH3402410.1
JAOTUA010000017.1:311-14849
Acidobacteriota bacterium

COXG MoCoBD_2 MoCoBD_1 Ald_Xan_dh_C AllG Ligase_CoA Succ_CoA_lig HMA_AllF_N Amidase Fer2_2 Fer2 CO_deh_flav_C FAD_binding_5 BrnT_toxin CopG_antitoxin ? Sigma70_r2 Sigma70_r4_2 ? ? ADH_zinc_N_2 AlaDh_PNT_C ADH_N AA_permease SLC12

MDH3601819.1
JAOTXM010000722.1:431-3716
Candidatus Tectomicrobia bacterium

BrnT_toxin CopG_antitoxin DUF952 ? ABC_tran BCA_ABC_TP_C ABC_tran ?

MDH3816370.1
JAOUAB010000473.1:1-3404
Acidobacteriota bacterium

Phage_integrase BrnT_toxin CopG_antitoxin OmdA TrmO_N ? ? ?

MDH4382431.1
JARXAA010000116.1:23116-36670
Gammaproteobacteria bacterium

adh_short_C2 Epimerase DEAD Helicase_C DeaD_dimer DbpA Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N ? DUF998 APH BrnT_toxin CopG_antitoxin CBS CBS PTP-NADK Abhydrolase_1 DUF6537 POR TPP_enzyme_C

MDH5201972.1
JAOUJR010000026.1:1-11477
Nitrospirota bacterium

DHH SecD_SecF_C Sec_GG SecD_SecF_C SecDF_P1_head SecD_1st Sec_GG YajC ? BrnT_toxin CopG_antitoxin Ion_trans_2 TrkA_N TrkA_C TrkA_N TrkA_C Inositol_P cNMPBD HTH_Crp_2 ? ?

MDH5444475.1
JAOULO010000004.1:67426-80991
Gammaproteobacteria bacterium

? DNA_pol3_beta DNA_pol3_beta_2 DNA_pol3_beta_3 ? ? ? ? ? ? CopG_antitoxin SMC_N ? ? ? His-Kinase DNA_gyraseB Toprim GyrB_hook GyrB_insert DNA_gyraseB_C ?

MDH5721545.1
JAOUOT010000033.1:264-10606
Spirochaetia bacterium

? Spermine_synth DEAD Helicase_C RecQ_Zn_bind FliD_N FliD_C Pyr_redox_dim Pyr_redox_2 2-oxoacid_dh E3_binding Biotin_lipoyl KARI_N KARI_C BrnT_toxin CopG_antitoxin DUF2196

MDH7514217.1
JARYMI010000115.1:149-1793
Clostridiales bacterium

BrnT_toxin CopG_antitoxin Polbeta HEPN

MDI9525521.1
JASGUA010000008.1:1-10064
Pseudomonadota bacterium

FliD_N Flagellin_IN FliD_C Flagellin_IN FliD_C DUF1285_N TetR_N TetR_C_10 BKACE LysM DUF4015 BrnT_toxin CopG_antitoxin TonB_C ? SurE Ribonucleas_3_3 dsrm

MDJ0597521.1
JASJDW010000009.1:7732-20758
Crocosphaera sp.

Asp_protease_2 RelB-like N6_Mtase N6_Mtase HsdM_N GHL10 FMN_dh DUF29 BrnT_toxin ? ? HicA_toxin HicB_lk_antitox CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS ? ?

MDJ0598206.1
JASJDW010000028.1:1-7527
Crocosphaera sp.

BrnT_toxin CopG_antitoxin UPF0367 TetR_N DUF5367 Lipoxygenase Catalase ? SSB

MDJ0728618.1
JASJDD010000006.1:24507-37851
Crocosphaera sp.

Uma2 ? ParE_like HicA_toxin HicB_lk_antitox ? CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS BrnT_toxin CopG_antitoxin DUF6887 STY-kinase STY-kinase TerB adh_short DUF5942 Peptidase_S8

MDJ0818252.1
JASJCE010000242.1:163-6658
Desulfobacterales bacterium

BrnT_toxin RHH_10 ? ? ? HSP20 AAA_14 DUF4143 MFS_1_like

MDJ0824945.1
JASJCA010000020.1:27019-37713
Rhodobacter sp.

GATase p450 p450 Gln-synt_C ? TraB_PrgY_gumN TIR Aminotran_3 Hydrolase TctC TctB TctA

MDJ0873018.1
JASJBY010000254.1:1-6935
Gammaproteobacteria bacterium

Helical_needle ? ELFV_dehydrog_N ELFV_dehydrog MMPL MMPL BrnT_toxin CopG_antitoxin

MDK1029566.1
JASJYH010000064.1:6421-15603
Anaerolineae bacterium

? Pribosyltran OMPdecase PDDEXK_3 OTCace OTCace_N PIN PhdYeFM_antitox BrnT_toxin CopG_antitoxin DHOase MutS_V MSSS Smr ?

MDK9694413.1
JAOTSA010000095.1:1-6435
Sulfurimonas sp.

FUR BrnT_toxin CopG_antitoxin PHP DNA_pol3_alpha DNA_pol3_finger HHH_6 Ala_racemase_N Peptidase_M50

MDM3858048.1
JAQSZA010000191.1:3439-22359
Aphanizomenon gracile PMC649.10

CNNM CBS CBS CorC_HlyC Caroten_synth ? Ribosomal_L27 Ribosomal_L21p RlaP STY-kinase GUN4 BrnT_toxin ? zf-HC2 Sigma70_r4_2 Sigma70_r2 YkuD 5_3_exonuc_N 5_3_exonuc DNA_pol_A_exo1 DNA_pol_A PsaA_PsaB PsaA_PsaB

MDM8537691.1
JAUCGD010000216.1:56-5724
Desulfobacterales bacterium HSG17

POTRA_2 ShlB BrnT_toxin ? ETFQO_UQ-bd Fer4_ETF_QO FUR Citrate_synt

MDM8564518.1
JAUCGL010000010.1:39041-57566

Candidatus Halobeggiatoa sp. HSG11

Mur_ligase_M Mur_ligase_C BacA HTH_1 LysR_substrate Beta_helix Beta_helix Thiolase_C Thiolase_N DrsE_2 TusA BrnT_toxin CopG_antitoxin Cys_Met_Meta_PP Calx-beta Calx-beta Flg_new_2 Flg_new_2 fn3 DUF8202 HTH_Rpn_C

MDM8568530.1
JAUCGN010000048.1:3276-10439
Thiotrichales bacterium HSG1

? ? ? ? ? BrnT_toxin ? DUF1375 ? ? BrnT_toxin ? HD

MDN5601157.1
JAKDIW010000433.1:1-2672
Brachybacterium sp.

? TetR_N TetR_C_6 BrnT_toxin CopG_antitoxin MFS_3

MDO4232895.1
JAUNHU010000141.1:19-6028
Lautropia sp.

? MarC Peripla_BP_4 BrnT_toxin CopG_antitoxin DUF2789 Fic PNTB

MDO4398849.1
JAUNEO010000007.1:9818-19658
Candidatus Saccharibacteria bacterium

PAP2 HIT TM2 ? tRNA-synt_1 Anticodon_1 Val_tRNA-synt_C BrnT_toxin BrnA_antitoxin ? YafQ_toxin ? ? FemAB FemAB FemAB FemAB

MDO5103828.1
JAUNKF010000033.1:75-12637
Lautropia sp.

Fer4_8 CCG CCG MarC ? ? ? BrnT_toxin CopG_antitoxin PNTB PNTB_4TM AlaDh_PNT_C AlaDh_PNT_N tRNA_Me_trans tRNA_Me_trans_M tRNA_Me_trans_C ? Lyase_1 ASL_C

MDO8474042.1
JAUSJT010000001.1:122361-137744
bacterium

Sec_GG SecD_SecF_C Sigma70_r4 HARE-HTH DUF8128 TrwB_AAD_bind zf_Tbcl_2 PGK PGK DHH DHHA1 RVT_3 DUF4258 CopG_antitoxin MurJ GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C DivIC PCRF RF-1 ABC_tran FtsX_ECD FtsX

MDO8492311.1
JAUSIV010000021.1:2738-16109
bacterium

? ? ? ? Bac_DNA_binding TrmB ? BrnT_toxin CopG_antitoxin HD KH_1 RNase_Y_N CLP_protease MazE_antitoxin Fic TrmB LacY_symp

MDO8497656.1
JAUSJK010000063.1:628-5509
bacterium

BrnT_toxin CopG_antitoxin ? RnlA-toxin_DBD DUF1514 Methyltransf_31 ? 23S_rRNA_IVP DUF5660

MDO8497977.1
JAUSJK010000079.1:2051-10733
bacterium

Pribosyltran ? Synaptojanin BrnT_toxin CopG_antitoxin PIN T6SS_Tle1-like_C PhdYeFM_antitox ? ? dCMP_cyt_deam_1 ?

MDO8523999.1
JAUSHZ010000001.1:335-10930
bacterium

? ? ? BrnT_toxin CopG_antitoxin PemK_toxin ? Peptidase_M48 MreC MreD PBP_dimer Transpeptidase

MDO8576710.1
JAUSHO010000001.1:1-5519
Candidatus Daviesbacteria bacterium

TRAM PIN LytR_C BrnT_toxin CopG_antitoxin ChlI DnaB_C Zn_ribbon_LapB S1 S1 S1 S1

MDO8599804.1
JAUSGC010000004.1:1-2463
bacterium

BacA BrnT_toxin Sortase ?

MDO8602700.1
JAUSEX010000025.1:30920-46742
Candidatus Omnitrophota bacterium

tRNA-synt_1c tRNA-synt_1c Anticodon_2 PilZ Zn_ribbon_recom Recombinase Resolvase DUF2924 OEP OEP BSH_CzcB Beta-barrel_RND_2 CzcB_C ACR_tran ? CopG_antitoxin VPg Receiver FAZ1_C PAS His-Kinase His-Kinase GGDEF His-Kinase His-Kinase ?

MDO8609980.1
JAUSFJ010000203.1:1-1756
bacterium

PaaX_M BrnT_toxin CopG_antitoxin MerR_1 ?

MDO8675067.1
JAUSEK010000011.1:35473-48249
Candidatus Omnitrophota bacterium

Epimerase Epimerase B12-binding Radical_SAM GHMP_kinases_N GHMP_kinases_C NTP_transferase ? ? BrnT_toxin CopG_antitoxin ? DegT_DnrJ_EryC1 Glyco_transf_4 Glycos_transf_1 Methyltransf_11 Glycos_transf_2 ?

MDO8737171.1
JAUSDE010000185.1:1-8407
Thermoleophilia bacterium

Topoisom_bac Zn_ribbon_Top1 Thymidylate_kin BrnT_toxin CopG_antitoxin DNA_pol3_delta2 PSP1 PSP1_3rd Asp_Glu_race QueT RNase_PH RNase_PH_C Ham1p_like Ham1p_like

MDO8954141.1
JAUSAW010000007.1:1-14372
Gammaproteobacteria bacterium

HTH_Tnp_1 ? MlaA AA_permease_2 Arc ASL_C Lyase_1 DUF481 BrnT_toxin ? Lipase_GDSL_2 Abhydrolase_3 Methyltransf_12 SnoaL_2 Methyltransf_31 Beta-lactamase

MDO8954313.1
JAUSAW010000009.1:32974-46541
Gammaproteobacteria bacterium

Peptidase_M16_C Peptidase_M16 ? Thioredoxin ADK RF3_C EF-G_D2 GTP_EFTU ANKs ANKs ANKs ? BrnT_toxin NikA-like Acetylase ? CDP-OH_P_transf DUF3579 FKBP_C Peptidase_A8

MDO9234009.1
JAURYE010000049.1:2857-11308
Methylotenera sp.

BrnT_toxin CopG_antitoxin Porin_4 HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase PAS DUF4124 Gln-synt_C Gln-synt_N Rhodanese

MDO9463742.1
JAUSPT010000005.1:6702-18154
bacterium

TauE Glycos_transf_N Epimerase_2 Glycos_transf_1 ? MpPF26 ? ? ? CopG_antitoxin Radical_SAM DUF4321 Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r1_2 Sigma70_r1_1 DnaG_DnaB_bind DnaB_bind Toprim_2 DNAG_N Zn_ribbon_DnaG HIT LysM DUF4015

MDO9541981.1
JAUSOX010000278.1:1-2687
Kiritimatiellia bacterium

Phage_int_SAM_4 BrnT_toxin CopG_antitoxin ? DUF3410 2-Hacid_dh_C

MDO9549048.1
JAUSPJ010000351.1:72-6213
Candidatus Neomarinimicrobiota bacterium

Fer4_5 Fer4_5 NosR Fer4_10 Fer4 DUF6599 Radical_SAM Memo AMMECR1 BrnT_toxin CopG_antitoxin DUF433

MDO9567729.1
JAUSPN010000107.1:736-18287
Candidatus Desulfaltia sp.

CCG CCG Fer4_8 FlpD Fer4_7 Fer4_20 Pyr_redox_2 Fer4_20 Fer4 POR TPP_enzyme_C PFOR_II POR_N Fer4_7 Fer4 TPP_enzyme_C POR_N BrnT_toxin ? HATPase_c_3 N6_N4_Mtase RE_EcoO109I VapB_antitoxin PIN SMC_N

MDP1597411.1
JAUXVM010000073.1:76709-89644
Methylotenera sp.

PilN PilO PilP AMIN AMIN STN Secretin_N Secretin SKI DHQ_synthase DHQS_C HD HD_assoc BrnT_toxin CopG_antitoxin CopG_antitoxin ParC-like ParE_toxin ? ? HTH_8 AAA_lid_14 NtrC-AAA Receiver

MDP1780188.1
JAUXSW010000770.1:1-1444
Anaerolineales bacterium

? BrnT_toxin ? ?

MDP2103478.1
JAUXOT010000075.1:1-1564
Candidatus Gracilibacteria bacterium

Trigger_C DUF4258 CopG_antitoxin SecA_DEAD

MDP2104163.1
JAUXOT010000335.1:1-1605
Candidatus Gracilibacteria bacterium

GatB_Yqey ? CopG_antitoxin Methyltransf_12

MDP2134171.1
JAUXOI010000461.1:1-8879
Sulfuritalea sp.

TetR_C_24 Hemerythrin BrnT_toxin CopG_antitoxin DUF1499 EAL GGDEF PAS NLPC_P60 ADH_N ADH_zinc_N S-AdoMet_synt_N S-AdoMet_synt_M S-AdoMet_synt_C Beta-lactamase

MDP2157870.1
JAUXNV010000538.1:37-7994
Nitrospirota bacterium

RNA_pol_Rpb1_1 RNA_pol_Rpb1_2 RNA_pol_Rpb1_2 RNA_pol_Rpb1_3 RNA_pol_Rpb1_4 RNA_pol_Rpb1_5 ? RAG1_RNase_H ABC-ATPase ABC-ATPase BrnT_toxin

MDP2283867.1
JAUXNB010000383.1:1572-5960

Pseudohongiella sp.

? DUF4160 DUF2442 PhdYeFM_antitox PIN ? NTP_transf_2 ? CopG_antitoxin HTH_3 ?

MDP2807855.1
JAUYFD010000530.1:1-1616
bacterium

? Oxidored_q2 BrnT_toxin ? Oxidored_q4

MDP2866761.1
JAUYEM010000787.1:1-5491
Elusimicrobiota bacterium

TPRs TPRs DUF2268 BrnT_toxin CopG_antitoxin ? THF_DHG_CYH THF_DHG_CYH_C

MDP2866767.1
JAUYEM010000789.1:1-1054
Elusimicrobiota bacterium

PNPase ? CopG_antitoxin

MDP2922475.1
JAUYCY010000030.1:1034-13186
Candidatus Omnitrophota bacterium

? ? OsmC YHS ArsP_1 HK ? TMP-TENI ? CopG_antitoxin ORF6N XhlA Radical_SAM HHH_3 ? CarboxypepD_reg CarboxypepD_reg PAN_4 ? DUF6788

MDP2938515.1
JAUYCX010000009.1:1-4769
Candidatus Omnitrophota bacterium

SecA_SW SEC-C LNT_N CN_hydrolase BrnT_toxin CopG_antitoxin TAT_signal ?

MDP3148766.1
JAUYAB010000041.1:194204-209403
Ignavibacteria bacterium

dCache_1 PAS GAF_2 PAS His-Kinase His-Kinase ? AbiEi_2 AbiEii Por_Secre_tail Peptidase_S8 LepA_C EFG_C GTP_EFTU_D2 GTP_EFTU ? BrnT_toxin CopG_antitoxin DUF4760 ? DUF4395 RsgA_GTPase DUF2180 TP_methylase

MDP3331764.1
JAUYUQ010000318.1:1-14728
Methylococcaceae bacterium

Neuraminidase ABC_toxin_N PA14 VRP1 PG_binding_1 PEP-CTERM PfkB UTRA GntR BrnT_toxin CopG_antitoxin His_Phos_1 ? PriA_C Helicase_C Zn_ribbon_PriA DEAD

MDP3465576.1
JAUYSQ010000149.1:1-2021
Sulfuricurvum sp.

GDP_Man_Dehyd GDP_Man_Dehyd BrnT_toxin CopG_antitoxin NTP_transf_2

MDP3515668.1
JAUYSM010000006.1:1-11358
Pseudohongiella sp.

HipA_C HTH_31 PAS PAS MCPsignal Peptidase_Mx Zn_ribbon_DZR_6 Alpha-E CP_ATPgrasp_2 Bact_transglu_N Transglutaminase DUF2126 BrnT_toxin CopG_antitoxin CP_ATPgrasp_2

MDP3541887.1
JAUYSH010000039.1:43815-54477
Elusimicrobiota bacterium

TsaE Carb_kinase HNH_5 CopG_antitoxin Phage_int_SAM_4 Phage_integrase MazE_antitoxin PemK_toxin ? CopG_antitoxin Transposase_20 DEDD_Tnp_IS110 ? ? ? DUF2703 Sigma70_r2 Sigma70_r4_2

MDP3661542.1
JAUYPY010000030.1:5899-26840
bacterium

? ? ? ? FTSW_RODA_SPOVE Glycos_transf_1 Glyco_transf_4 Bac_transf BrnT_toxin CopG_antitoxin SecD_1st SecDF_P1_head SecD_SecF_C Sec_GG SecD_SecF_C ? MurD-like_N Mur_ligase_M Mur_ligase_C Mur_ligase Mur_ligase_M Mur_ligase_C NUDIX

MDP3741348.1
JAUYPH010000075.1:3587-14007
bacterium

bPH_2 Acetylase DUF1428 3-dmu-9_3-mt 3-dmu-9_3-mt ? DUF1801 ? BrnT_toxin CopG_antitoxin SKI Thr_synth_N PALP THR4_C Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 Pterin_4a HicA_toxin

MDP3876377.1
JAUYMY010000022.1:78099-92148
Methylobacter sp.

Asp_Glu_race Gp_dh_N Gp_dh_C PK PK_C HicB Sel1 N_methyl Pilin Y1_Tnp BrnT_toxin CopG_antitoxin ? Secretin Secretin_N type_II_gspD_N0 AMIN AMIN PilP PilO PilN PilM_2

MDP3896950.1
JAUYMV010000346.1:1-5981
Mesorhizobium sp.

FA_hydroxylase adh_short YjdM TAT_signal YkuD ? HigB-like_toxin HTH_3 BrnT_toxin CopG_antitoxin HEPN_Apea

MDP3917569.1
JAUYMM010000003.1:2157-8933
Candidatus Woesebacteria bacterium

Cob_adeno_trans ? Aq_aa27 CopG_antitoxin ? DUF8128 TrwB_AAD_bind 2-Hacid_dh_C UDPG_MGDP_dh_N UDPG_MGDP_dh Erp_C Phage_integrase Phage_int_SAM_1 DivIC

MDP3918043.1
JAUYMM010000050.1:1-4290
Candidatus Woesebacteria bacterium

FtsH_ext AAA AAA_lid_3 Peptidase_M41 VTT_dom BrnT_toxin CopG_antitoxin 5_3_exonuc_N 5_3_exonuc Glyco_transf_4

MDP3920232.1
JAUYMS010000030.1:2101-54022
Candidatus Omnitrophota bacterium

ROK SNase T2SSF_N SelR ? ? SNase BrnT_toxin CopG_antitoxin ? ? TPRs TPR_CcmH_CycH Helicase_C SNF2-rel_dom PCMT HEATs His-Kinase Acetylase Rce1-like APH His-Kinase PHO4 Lactamase_B VanZ

MDP3948084.1
JAUYLH010000002.1:1-2692
bacterium

PCRF ? CopG_antitoxin Asparaginase Peptidase_M10

MDP3955852.1
JAUYLO010000010.1:3234-7952
bacterium

DUF4143 BrnT_toxin CopG_antitoxin Bact_surface_Ig-like ? MS_channel_1st_1 MS_channel_1st_1 YafQ_toxin RHH_1 ?

MDP3983270.1
JAUYLA010000111.1:141-7363
bacterium

? PsbP_2 ? FAD_binding_4 MurB_C DUF4258 CopG_antitoxin HTH_3 ? ATP-grasp_4 Mur_ligase_M ?

MDP3989039.1
JAUYLF010000025.1:1-9445
bacterium

DUF4143 AAA_14 Bact_surface_Ig-like Glycos_transf_2 ? ? ? TatA_B_E ? LamininG LamininG LamininG ?

MDP3995013.1
JAUYKJ010000003.1:273-9754
bacterium

Glycos_transf_1 Glyco_transf_4 BrnT_toxin CopG_antitoxin OTCace_N OTCace Hydrolase GST_N_3 HTH_5 ? HTH_23

MDP4011449.1
JAUYJP010000091.1:1-7545
Candidatus Roizmanbacteria bacterium

Transpeptidase Methyltransf_23 ? BrnT_toxin CopG_antitoxin FGase ? ? tRNA-synt_1f

MDP4610066.1
JANREU010000029.1:72-16584
Opitutales bacterium

? HipA_C Couple_hipA HTH_3 HTH_5 Methyltransf_11 ? ? Methyltransf_25 BrnT_toxin CopG_antitoxin RadC UPF0758_N RlmN_N Radical_SAM ? Aldedh Aldedh PfkB DUF3568

MDP4618669.1
JANREH010000116.1:106-3230
Burkholderiaceae bacterium

BrnT_toxin CopG_antitoxin NTP_transf_3 FeS_assembly_P ParA LpqE-like

MDP4623955.1
JANRFH010000006.1:3681-17454
Akkermansiaceae bacterium

? HSP20 MotA_ExbB ExbD TPRs TPRs TPRs TPRs TPRs LMWPc LepA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU BrnT_toxin CopG_antitoxin Peptidase_M24 SecY Ribosomal_L27A Ribosomal_L18p Ribosomal_L6 Ribosomal_L6 Ribosomal_S8

MDP8230078.1
JAVCCW010000017.1:1-4568

Candidatus Gorgyraea atricola

Cupin_2 ? HMG-CoA_red HisKA_7TM GAF His-Kinase ? CopG_antitoxin

MDP8246526.1
JAVCDC010000002.1:14126-27143
Candidatus Tritonobacter lacicola

HHH_3 Nucleos_tra2_N Nucleos_tra2_N Gate Nucleos_tra2_C ? ? ? RRM_1 Unstab_antitox HicB_lk_antitox BrnT_toxin ? MazE_antitoxin PemK_toxin CBS CBS Voltage_CLC STAS His-Kinase malic Malic_M PTA_PTB

MDP8248437.1
JAVCDC010000101.1:6696-26217
Candidatus Tritonobacter lacicola

ParB_N HTH_ParB BrnT_toxin CopG_antitoxin PhdYeFM_antitox Methyltransf_25 HicB_lk_antitox CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg CarboxypepD_reg PDZ ? CopG_antitoxin CAT Glyco_hydro_18 PMT_2 ? YfhO YfhO PMT_2

MDP8985172.1
JALYHP010000063.1:1-11064
Pseudomonadota bacterium

Band-7 DUF2065 Adenylsucc_synt MFS_1 DUF3096 ? MFS_1 BrnT_toxin CopG_antitoxin TPRs TPRs Sulfotransfer_3 OB_RNB CSD2 RNB S1

MDQ1333896.1
JAUPKI010000083.1:257-3650
Thermodesulfobacteriota bacterium

ECH_1 BrnT_toxin CopG_antitoxin ? HEPN ? ?

MDQ1350756.1
JAUPLE010000165.1:1-2436
Acidobacteriota bacterium

? ? BrnT_toxin CopG_antitoxin HTH_Rpn_C

MDQ3009620.1
JALYRZ010000045.1:352-6964
Acidobacteriota bacterium

DLH Abhydrolase_3 ? Abhydrolase_6 Peptidase_S15 BAAT_C DUF3089 RMD1-3 TPRs TPRs ? BrnT_toxin ?

MDQ3062184.1
JALYRE010000229.1:1-2402
Acidobacteriota bacterium

Pyr_redox_2 BrnT_toxin BrnA_antitoxin BrnT_toxin BrnA_antitoxin ? PIN

MDQ3564857.1
JALZJW010000075.1:8725-18118
Pseudomonadota bacterium

BrnT_toxin CopG_antitoxin GlnE GlnD_UR_UTase GlnE GlnD_UR_UTase Aminotran_4 zf-CHCC Mito_fiss_Elm1 Glyco_transf_4 Glycos_transf_1 Methyltransf_21

MDQ6638041.1
JALYXX010000002.1:1-2065
Pseudomonadota bacterium

ABC_tran BrnT_toxin CopG_antitoxin BPD_transp_1 BPD_transp_1

MDQ6785600.1
JALYZF010000035.1:1-9163
Acidobacteriota bacterium

Flg_new_2 Flg_new_2 Zn_ribbon_NOB1 PALP TonB_C ? Zn_ribbon_HMPTM Radical_SAM ? PGI BrnT_toxin

MDQ6955789.1
JAUZQR010000061.1:150-10652
Mariprofundaceae bacterium

ResIII UvrB_inter UvrB_3rd Helicase_C UvrB UVR SWIM SNF2-rel_dom Helicase_C ABC_tran oligo_HPY ABC_tran AAA_13 oligo_HPY TauE ? BrnT_toxin CopG_antitoxin CN_hydrolase

MDQ7069128.1
JAUZRZ010000090.1:428-5919
Sulfurimonas sp.

OEP OEP ABC_membrane ABC_tran HH_AprE Beta-barrel_AprE BrnT_toxin CopG_antitoxin

MDQ7796609.1
JAVHLQ010000237.1:1-3193
Spirochaetia bacterium

ZnuA ABC_tran ABC-3 CTXphi_pIII-N1 CopG_antitoxin FtsK_gamma ?

MDR0230663.1
JAIRVZ010000014.1:28475-43038
Dysgonamonadaceae bacterium

Biotin_lipoyl Gp_dh_C Gp_dh_N DUF3795 Glyoxalase GTP_EFTU GTP_EFTU_D2 EFG_C BipA_C DUF4831 Carb_kinase YjeF_N BrnT_toxin ? ? ? Flg_new ? ? CDP-OH_P_transf

MDR0311617.1
JAIRVC010000335.1:71-2998
Acidobacteriota bacterium

BrnT_toxin CopG_antitoxin ABC_membrane ABC_tran Receiver

MDR0320346.1
JAIRUQ010000016.1:10053-24596
Treponema sp.

Eno-Rase_FAD_bd Enoyl_reductase Eno-Rase_NADH_b FabA Ketoacyl-synt_C ketoacyl-synt adh_short_C2 Acyl_transf_1 ACP_syn_III_C ACP_syn_III HTH_3 BrnT_toxin CopG_antitoxin Radical_SAM ? ? Radical_SAM B12-binding NT5C Isoamylase_C Alpha-amylase Alpha-amylase CBM_48

MDR0322524.1
JAIRUQ010000118.1:1-16095
Treponema sp.

PBP_like_2 BD-FAE Peptidase_M3 Peptidase_M3_N Hexokinase_2 Hexokinase_1 PDDEXK_1 UvrD_C UvrD-helicase PDDEXK_1 ? DUF4258 CopG_antitoxin TRAM_2 Radical_SAM UPF0004 ?

MDR0458881.1
JAIRQI010000009.1:1-6292
Coriobacteriales bacterium

Zn_Ribbon_1 BrnT_toxin BrnA_antitoxin TRCF Helicase_C DEAD CarD_TRCF_RID UvrB_inter TSCPD

MDR0463644.1
JAIRQF010000013.1:1-11454
Treponema sp.

Gp_dh_N Gp_dh_C PGK TIM ? BrnT_toxin CopG_antitoxin BcrAD_BadFG BcrAD_BadFG DUF2229 SATase_N Hexapep tRNA-synt_1

MDR0597464.1
JAIROB010000095.1:1-4122
Treponema sp.

PIN RHH_5 ABC_tran BPD_transp_1 NMT1_2 BrnT_toxin

MDR0676265.1
JAIRLV010000161.1:1-3972
Elusimicrobiota bacterium

? ? BrnT_toxin CopG_antitoxin ? Terminase-T7_RNaseH-like ? ?

MDR0676302.1
JAIRLV010000173.1:1-6341
Elusimicrobiota bacterium

? ? ? ? ? ? ? CopG_antitoxin ? CheY-binding ? ?

MDR0774745.1
JAIRJV010000157.1:419-6620
Rickettsia sp.

BrnT_toxin CopG_antitoxin ? ? ? ProQ Fic Virulence_RhuM ?

MDR0829681.1
JAISKN010000145.1:115-15318
Prevotellaceae bacterium

Creatinase_N Creatinase_N_2 Peptidase_M24 Peptidase_M24_C Glyco_hydro_65C Glyco_hydro_65m Glyco_hydro_65N ? MMR_HSR1 DUF2400 ? Peptidase_M75 BrnT_toxin ? Unstab_antitox ParE_toxin Shr-like_HID DUF6722 ?

MDR0830751.1
JAISKN010000285.1:53-9847
Prevotellaceae bacterium

? AICARFT_IMPCHas MGS Amidase_3 MIP EamA EamA BrnT_toxin ? Redoxin Pantoate_ligase ? ?

MDR0859301.1
JAISLB010000174.1:1-9915
Oscillospiraceae bacterium

PmbA_TldD_3rd RuvB_N AAA_lid_4 WHD_RuvB BrnT_toxin ? PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV ? Pep_deformylase Glyco_hydro_8 HD_5 Receiver Hpt Receiver His-Kinase His-Kinase dCache_1

MDR0869644.1
JAISJZ010000074.1:1-8824
Planctomycetaceae bacterium

UDPG_MGDP_dh_N PAS PAS GAF_2 His-Kinase His-Kinase Receiver Receiver ThuA NTP_transf_2 HEPN SYY_C-terminal tRNA-synt_1b ? ? ?

MDR0873107.1
JAISJU010000108.1:113-10043
Prevotellaceae bacterium

iPGM_N Phosphodiest QueH ZT_dimer Cation_efflux OMP_b-brl BrnT_toxin ? ? Cpn10 Cpn60_TCP1 Ribosomal_L25p Ribosomal_TL5_C Pept_tRNA_hydro ?

MDR0910814.1
JAISIL010000206.1:4100-18559
Spirochaetaceae bacterium

TsaE TsaD GreA_GreB GreA_GreB_N GRAD_C HTH_GRAD TPRs Ribul_P_3_epim TPRs eIF-1a BrnT_toxin ? LRRs VanY Pribosyltran GARS_N GARS_A GARS_C MS_channel_3rd MS_channel_2nd TrpP

MDR1053643.1
JAISFE010000300.1:1-10371
Planctomycetaceae bacterium

? Trypsin PDZ_2 PTPS BrnT_toxin ? Lysine_decarbox ? SBP_bac_10 N_methyl SBP_bac_10 N_methyl ? ?

MDR1190071.1
JAISCS010000006.1:179-4063
Verrucomicrobiales bacterium

BrnT_toxin CopG_antitoxin HicB_lk_antitox HicA_toxin ? ? ? Phage_capsid_4

MDR1453107.1
JAIRWS010000064.1:318-8928
Candidatus Margulisiibacteriota bacterium

GSIII_N Gln-synt_C Gln-synt_C-ter BrnT_toxin DUF3016 CN_hydrolase LNT_N ? ? IBR_1 SmpB Sm_multidrug_ex

MDR1453784.1
JAIRWS010000127.1:2916-11280

Candidatus Margulisiibacteriota bacterium

Methyltransf_25 ? HTH_3 tRNA_anti_2 Exonuc_VII_L Exonuc_VII_L ? CopG_antitoxin Exonuc_VII_S polyprenyl_synt DUF212 DXP_synthase_N Transket_pyr Transketolase_C Mur_ligase_M Mur_ligase_C ?

MDR1785904.1
JAISTZ010000110.1:2807-7829
Spirochaetaceae bacterium

? BrnT_toxin ? ? PIN DUF3368 UPF0175 BPD_transp_2 ABC_tran ABC_tran

MDR1997587.1
JAISQA010000045.1:549-5619
Candidatus Margulisiibacteriota bacterium

BrnT_toxin CopG_antitoxin NTase_sub_bind Polbeta TGT SecD_1st SecDF_P1_head SecD_SecF_C Sec_GG SecD_SecF_C Ribosomal_L21p

MDR2049894.1
JAISOE010000404.1:82-5411
Treponema sp.

? HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase MFS_1 BrnT_toxin BrnA_antitoxin

MDR2147511.1
JAITMT010000348.1:3540-14709
Tannerella sp.

POR_N PFOR_II TPP_enzyme_C BrnT_toxin BrnA_antitoxin ? YccF SH3_3 YkuD POR Porin_10 SBP_bac_3

MDR2166760.1
JAITMQ010000044.1:9459-20004
Clostridiales bacterium

ParE_toxin ? ? Gate AAA_3 AAA_lid_2 vWA ? BrnT_toxin BrnA_antitoxin Sortase ABC_tran_2 Zeta_toxin ? Peptidase_M23 PIN

MDR2428799.1
JAITIE010000093.1:1-7924

Candidatus Margulisiibacteriota bacterium

BrnA_antitoxin Methyltransf_25 tRNA_anti_2 Exonuc_VII_L Exonuc_VII_L BrnT_toxin CopG_antitoxin Exonuc_VII_S polyprenyl_synt DUF212 DXP_synthase_N Transket_pyr Transketolase_C Mur_ligase_M Mur_ligase_C ?

MDR2431538.1
JAITIS010000080.1:1-7125

Candidatus Margulisiibacteriota bacterium

? ? ? CopG_antitoxin Polbeta HEPN LpxC Radical_SAM_N2 Radical_SAM Peptidase_S26 ?

MDR2555475.1
JAITFO010000083.1:5517-18141
Candidatus Fibrimonadaceae bacterium

? B12-binding B12-binding_2 Pterin_bind S-methyl_trans CelD_N Glyco_hydro_9 Discoidin Glyco_hydro_9 Por_Secre_tail HATPase_c_3 DNA_mis_repair MutL_C BrnT_toxin ? Glycos_transf_2 Histidinol_dh Glyco_hydro_8 Por_Secre_tail ? ? ?

MDR2756851.1
JAITBS010000498.1:181-2609
Planctomycetaceae bacterium

? Phage_integrase BrnT_toxin CopG_antitoxin

MDR2999947.1
JAITUM010000029.1:378-2666
Candidatus Fibrimonadaceae bacterium

BrnT_toxin BrnA_antitoxin Peptidase_M15_3 DUF1353 TorS_sensor_domain

MDR3334008.1
JAITNU010000076.1:119-7287
Treponema sp.

BrnT_toxin BrnT_toxin Epimerase ? GalKase_gal_bdg GHMP_kinases_N MurJ GDPD Cyano_EgtBD

MDR3335817.1
JAITNM010000030.1:501-2938
Treponema sp.

5_3_exonuc_N BrnT_toxin CopG_antitoxin FlgN FliS ?

MDR3491657.1
JARLJN010000089.1:9767-23064
Gammaproteobacteria bacterium

RelB Phage_int_SAM_1 Phage_integrase ? KdpA E1-E2_ATPase Hydrolase KdpC KdpD Usp DUF4118 GAF_3 BrnT_toxin CopG_antitoxin GGDEF GAF PTA_PTB ? CBS CBS ? ?

MDR3642783.1
JARLHZ010000025.1:58632-70217
Candidatus Doudnabacteria bacterium

? TsaE VanW RuvB_N AAA_lid_4 WHD_RuvB ? bPH_2 CTP_transf_like ? CopG_antitoxin ? Glycos_transf_2 VTT_dom Methyltrans_RNA PUA_4 DciA SMC_N

MDR4463294.1
JALDRE010000002.1:91709-109220
Nitrospira sp.

PagL Aldedh MFS_1 PRKG1_interact ? ? Nucleos_tra2_C Gate Nucleos_tra2_N BrnT_toxin CopG_antitoxin SDF Dala_Dala_lig_C Peptidase_Mx LMWPc ATP_bind_3 ACR_tran

MDR9404253.1
JAVLCL010000264.1:1-4963
Halothece sp. Uz-M2-17

Guanylate_cyc Guanylate_cyc BrnT_toxin ? DUF3038 DUF4335 Helicase_C_2

MDT3739822.1
JAVSMQ010000001.1:2483610-2495088
Candidatus Kapaibacterium sp.

Ribosomal_TL5_C Ribosomal_L25p Pribosyl_synth Pribosyltran_N ? Peptidase_S24 Peptidase_S24 ? Por_Secre_tail BrnT_toxin Portal_Mu ? ? ? ? ? Phage_Mu_Gp36

MDT3741174.1
JAVSMQ010000001.1:4118681-4137338
Candidatus Kapaibacterium sp.

Lon_C Lon_lid AAA LON_substr_bdg Por_Secre_tail Por_Secre_tail Por_Secre_tail PKD_4 PKD FAA_hydrolase Flavin_Reduct ? ? RepB-RCR_reg Phe_tRNA-synt_N tRNA-synt_2d PIF1 SH3_13 UvrD_C_2 DUF2157 GDYXXLXY HATPase_c_3 HSP90 HSP90 Queuosine_synth

MDT8438557.1
JAVVDV010000014.1:1-7755
Wenzhouxiangellaceae bacterium

DUF1508 DUF1508 Lactamase_B_2 DNA_ligase_A_N DNA_ligase_A_M DNA_ligase_A_C DEAD Helicase_C WHD_Lhr DEAD_assoc Calcineurin PIN DUF6364 BrnT_toxin CopG_antitoxin Acyl-CoA_dh_N

MDT8446499.1
JAVVEA010000004.1:55016-65980
bacterium

? Bac_export_3 FliP Pirin Pirin_C ? PhdYeFM_antitox ParE_toxin ? CopG_antitoxin TumE HVO_A0114 TPR_CcmH_CycH ? PGM_PMM_III PGM_PMM_I tRNA-synt_1b

MDT8446932.1
JAVVEA010000006.1:92148-105348
bacterium

NUDIX DinB A_deaminase HEM4 Porphobil_deamC Porphobil_deam GlutR_dimer Shikimate_DH GlutR_N HGTP_anticodon tRNA-synt_His BrnT_toxin CopG_antitoxin ? Radical_SAM B12-binding 2OG-FeII_Oxy_3 NUDIX_4 EndIII_4Fe-2S HhH-GPD Glycos_transf_2 ExbD

MDT8447386.1
JAVVEA010000010.1:13262-29179
bacterium

? MreC MreB_Mbl SNF2-rel_dom Helicase_C GreA_GreB_N GreA_GreB ? MerR-DNA-bind MerR_1 BrnT_toxin CopG_antitoxin APH Peptidase_M23 FtsX FtsX_ECD ABC_tran LepA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU ALS_ss_C ACT_AHAS_ss

MDT8448337.1
JAVVEA010000022.1:1-11937
bacterium

zf-CDGSH zf-CDGSH Lcl_C Lcl_C DEAD Helicase_C DsrC GST_N_2 GST_C_2 ? CopG_antitoxin ? TIR HigB-like_toxin HTH_3 Rubrerythrin Rubrerythrin Rhodanese Lactamase_B

MDT8451096.1
JAVVDX010000152.1:173-2890

Wenzhouxiangellaceae bacterium

HAMP_PhoQ His-Kinase His-Kinase BrnT_toxin CopG_antitoxin MazG MazG

MDW8215689.1
JAVQWA010000388.1:359-2443
Roseiflexaceae bacterium

PIN ? BrnT_toxin ? Amino_oxidase

MDX1900674.1
JANRLY010000011.1:8948-22419
Gammaproteobacteria bacterium

? ? IcmW ? ? DUF4143 AAA_14 ? BrnT_toxin CopG_antitoxin T2SSE T4SS_TraI DotD O_Spanin_T7 IcmT_2 T4bSS_IcmS ANKs ANKs

MDX1915611.1
JANRLR010000009.1:96406-110536
Methylophilus sp.

PTS-HPr PEP-utilisers_N PEP-utilizers PEP-utilizers_C MamF_MmsF TOP_N Peptidase_M3 ? SnoaL HxlR BrnT_toxin CopG_antitoxin HAD_2 HSP20 Synaptojanin HTH_8 AAA_lid_14 Sigma54_activ_2 Receiver His-Kinase His-Kinase PAS HAMP NtrY_N DUF4390

MDX1957853.1
JANRLP010000028.1:68-7269
Leptospiraceae bacterium

DUF4143 AAA_14 Pirin_C Pirin ABA4-like UDPGP ? BrnT_toxin CopG_antitoxin ? HepT-like Polbeta Uma2 DUF2442 DUF4160

MDX1959229.1
JANRLP010000066.1:93765-99694
Leptospiraceae bacterium

HTH_32 HTH_23 ? ? BrnT_toxin ? DUF2805 PIN ? VapB_antitoxin TPRs ECH_1

MDX1959507.1
JANRLP010000078.1:15803-25691
Leptospiraceae bacterium

SH3_3 HxlR OsmC YCII ? EVE HNH ? BrnT_toxin CopG_antitoxin ? ? HigB_toxin HTH_3 ? 4HBT_2

MDX1977762.1
JANRLH010000072.1:63291-84545
Pseudanabaenaceae cyanobacterium bins.68

Peptidase_S8 Creatininase NAD_binding_10 Cytochrome_CBB3 URO-D GUN4 STY-kinase Arm-DNA-bind_2 Phage_integrase BrnT_toxin CopG_antitoxin TPS RNaseH-CATRA+Caspase SQS_PSY Amino_oxidase PmbA_TldD_1st PmbA_TldD_2nd PmbA_TldD_3rd Uma2 Pro_CA

MDX2050630.1
JANRKH010000064.1:457-9443
Rickettsiaceae bacterium

Polbeta NTase_sub_bind AAA_18 Esterase DUF1016_N YhcG_C CPT CopG_antitoxin ? Acetylase PhdYeFM_antitox PIN APH Hydrolase TspO_MBR Asp_Glu_race

MDX2073830.1
JANRKF010000021.1:9510-27141
Alphaproteobacteria bacterium

Oxidored_q2 Proton_antipo_N Proton_antipo_M NADH5_C Oxidored_q5_N Proton_antipo_M Proton_antipo_M BPL_LplA_LipB BPL_C Pan_kinase Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C BrnT_toxin BrnA_antitoxin HMGL-like HCS_D2 LeuA_dimer ? PTA_PTB Malic_M malic Unstab_antitox ParE_toxin MutS_I MutS_II MutS_III MutS_V MutS_C

MDX2112881.1
JANRJW010000008.1:91335-108630
Alphaproteobacteria bacterium

? DUF475 ? Methyltransf_23 Sulfatase ? Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_C BrnT_toxin CopG_antitoxin A24_N_bact Peptidase_A24 NIR_SIR_ferr NIR_SIR NIR_SIR_ferr PAPS_reduct ? GTP_EFTU GTP-eEF1A_C APS_kinase RP005_N Pyr_redox_2

MDX2447636.1
JASJFZ010000397.1:1-2842
Desulfobacterales bacterium

? ? ? CopG_antitoxin

MDX2470789.1
JASJGK010000175.1:1-9408
SAR324 cluster bacterium

? ? BrnT_toxin CopG_antitoxin Beta-prop_EML_2 ? ? HTH_73 HsdM_N N6_Mtase

MDX5585553.1
JASJGX010000275.1:8-4722
Aureibaculum sp.

PIN ? BrnT_toxin CopG_antitoxin RHH_1 ? DUF4143 AAA_14 ORF6N 23S_rRNA_IVP

MDX9710864.1
JAVFAR010000227.1:1-2371
Trichloromonas sp.

DUF3089 BrnT_toxin CopG_antitoxin CBAH

MDX9796661.1
JAVEZJ010000140.1:322-4496
Arcobacteraceae bacterium

? AbiEii BrnT_toxin ? SapC TPRs TPRs TPRs TPRs TPRs Glyco_transf_9

MDX9800287.1
JAVEZE010000019.1:11431-19925
Spirochaetia bacterium

CopG_antitoxin ? Flavodoxin_5 PhdYeFM_antitox PIN PhdYeFM_antitox PIN BrnT_toxin CopG_antitoxin MliC Gpr1_Fun34_YaaH ? DUF1801 Pps Pps Glyoxalase

MDX9801799.1
JAVEZE010000141.1:1-7297
Spirochaetia bacterium

CBS CBS MgtE ? Acetate_kinase ? CopG_antitoxin His_Phos_1 Patatin

MDY2759090.1
JAXDWK010000015.1:67721-74771
Lachnospiraceae bacterium

Peptidase_A24 ? DUF6382 FHA ? HIT Glyco_tran_28_C Glyco_transf_28 PGI BrnT_toxin BrnA_antitoxin

MDY3740179.1
JAXELS010000071.1:9360-24639
Selenomonadaceae bacterium

Methylase_S Methylase_S HEPN Polbeta N6_Mtase HsdM_N Maf Form_Nir_trans Lactamase_B Competence DUF4131 BrnT_toxin BrnA_antitoxin MS_channel_3rd MS_channel_2nd MS_channel_1st HHH_3 SLBB Nitroreductase QueF EAL GGDEF PAS FxsA DctM

MDY6825212.1
JAXHVL010000177.1:9315-21168
Thermodesulfobacteriota bacterium

OmpA MotB_plug MotA_ExbB MotA_N ? RdgC Lactamase_B_3 ? MgsA_C AAA_assoc_2 AAA BrnT_toxin CopG_antitoxin KH_dom-like MMR_HSR1 MMR_HSR1 Orn_Arg_deC_N N_methyl GspH N_methyl PpdB N_methyl ?

MDY6905834.1
JAXHTY010000023.1:8266-24786
Thermodesulfobacteriota bacterium

? AAA_14 DUF4143 Caspase PEGA PEGA FGS Y1_Tnp Uma2 DUF4384 MobB ? BrnT_toxin CopG_antitoxin ? ? RHS_repeat 2-Hacid_dh_C CapK_C AMP-binding FG-GAP_3 FG-GAP FG-GAP_3 FG-GAP_3 FG-GAP_3

MDY6950645.1
JAXHTR010000043.1:5922-14762
Thermodesulfobacteriota bacterium

? Peptidase_M78 HTH_31 Unstab_antitox ? DUF2283 BrnT_toxin CopG_antitoxin AbiEii AbiEi_4 DUF4143 AAA_14 ? UPF0150 HicA_toxin

MDY7031084.1
JAXHSZ010000114.1:106-3658
Thermodesulfobacteriota bacterium

? DUF5711 Y1_Tnp BrnT_toxin CopG_antitoxin

MDY7033395.1
JAXHSZ010000441.1:237-4069
Thermodesulfobacteriota bacterium

BrnT_toxin CopG_antitoxin NTP_transf_2 HepT-like ? Calcineurin ZnR

MDZ7582405.1
JAXHLC010000035.1:182188-201582
Deltaproteobacteria bacterium

PBP_like_2 Beta-prop_WDR3_1st BPD_transp_1 BPD_transp_1 ABC_tran ? PhoU PhoU LMWPc PBP_like_2 BrnT_toxin CopG_antitoxin BrnT_toxin Helicase_C SNF2-rel_dom AAA_14 DUF4143 ELFV_dehydrog ELFV_dehydrog_N DUF3708 BPD_transp_1 DUF3333 BPD_transp_1

MDZ7644049.1
JAXHLH010000005.1:2006-19612
Woeseiaceae bacterium

? AbiEii HTH_13 Fic Fic_N DUF2283 ? RQC NTP_transf_8 BrnT_toxin CopG_antitoxin Caps_assemb_Wzi Caps_assemb_Wzi Poly_export SLBB SLBB SLBB Caps_syn_GfcC_C Wzz ? ?

MDZ7722296.1
JAXHMJ010000002.1:608915-625874
candidate division KSB1 bacterium

Sulfatase Sulfatase SGSH_C ? OMP_b-brl ? LVIVD Flg_new_2 Beta_helix NosD FlgD_ig BrnT_toxin CopG_antitoxin Gp49 HTH_26 HTH_3 ? PemK_toxin AAA_14 DUF4143

MDZ7725763.1
JAXHMJ010000005.1:2390378-2409822
candidate division KSB1 bacterium

Ala_racemase_N Ala_racemase_C Cupin_2 PfkB Aldo_ket_red Aldo_ket_red Fer4_10 DUF6599 Cep192_D4 PKD_4 Discoidin Big Discoidin FlgD_ig PKD_4 Cohesin PKD_4 Cohesin PKD FlgD_ig BrnT_toxin CopG_antitoxin Peptidase_C1 Peptidase_C1 PKD FlgD_ig GTP_EFTU EF-G_D2 EFG_C BipA_C ? PIN Peptidase_M20

MDZ7767045.1
JAXHMN010000019.1:66681-74307
Melioribacteraceae bacterium

ALIX_LYPXL_bnd DEDD_Tnp_IS110 DUF1801 ? ? YafQ_toxin RelB BrnT_toxin CopG_antitoxin ? Methyltransf_31 Methyltransf_23 SnoaL_3 DUF6597 HTH_18 OmpA

MDZ7792939.1
JAXHMY010000001.1:835205-846457
Spirochaetia bacterium

HTH_3 RH3_dom ParE_toxin MazE_antitoxin PIN HigB-like_toxin HTH_3 BrnT_toxin HsdM_N N6_Mtase DUF1016_N YhcG_C Methylase_S Methylase_S ? Virulence_RhuM HSDR_N_2 ResIII CagE_TrbE_VirB Helicase_C EcoEI_R_C DUF7686 DUF7713

MDZ7793178.1
JAXHMY010000001.1:1060378-1075200
Spirochaetia bacterium

? adh_short FapA FapA_N Receiver PAS PAS PAS His-Kinase His-Kinase Receiver Receiver HD_5 DUF4258 ? DapB_N DapB_C DCD HD ? EPSP_synthase DnaJ DjlA_TM

MDZ7832254.1
JAXHNP010000006.1:643082-655547
Desulfobacterales bacterium

Hexapep Methyltransf_11 DegT_DnrJ_EryC1 Glycos_transf_2 Epimerase TPRs TPRs ? ? BrnT_toxin CopG_antitoxin DUF2442 Lcl_C AAA_14 DUF4143 RHH_1 PIN NTP_transf_2

MDZ7832991.1
JAXHNP010000007.1:75745-98414
Desulfobacterales bacterium

Flg_new_2 Flg_new_2 Pel9A-like_beta_helix Beta_helix Flg_new_2 Flg_new_2 Flg_new_2 Flg_new_2 Lcl_C Lcl_C FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 ? ? Peptidase_S74 BrnT_toxin CopG_antitoxin Flg_new_2 Flg_new_2 Flg_new_2 Rax2 Rax2 RelE Mu-transpos_C_2 ? ? HTH_3

MEA1910879.1
JAYEFU010000174.1:46-2729
Spirochaetota bacterium

DUF1254 HTH_3 RelE BrnT_toxin CopG_antitoxin

MEA1920778.1
JAYEGP010000292.1:149-8627
Campylobacterota bacterium

? MerR_1 DnaJ_C DnaJ ? Lipase_GDSL_2 ABC_tran MacB_PCD FtsX FtsX BrnT_toxin CopG_antitoxin AICARFT_IMPCHas

MEA1923057.1
JAYEGS010000238.1:27045-37884
Pseudomonadota bacterium

? DUF2959 BrnT_toxin CopG_antitoxin Peptidase_M90 OEP OEP BSH_RND ACR_tran HhH-GPD Amidase_2

MEA1936681.1
JAYEHC010000023.1:825-15553

Patescibacteria group bacterium

CTD11 GIY-YIG TraC-like 23S_rRNA_IVP TraC_F_IV P-loop_TraG GIY-YIG ? BrnT_toxin CopG_antitoxin T4SS_pilin ? Peptidase_MA_2 Radical_SAM DHH T2SSE MshEN

MEA2017106.1
JAYEIM010000044.1:2370-15105
Campylobacterota bacterium

Peptidase_M23 GlutR_N Shikimate_DH GlutR_dimer VWA_2 ? DUF2461 ? SLT_superhelical SLT BrnT_toxin CopG_antitoxin Maf Acetate_kinase Acetate_kinase PTA_PTB DRTGG AAA_26 ? DUF2196 Methyltrans_SAM

MEA2017480.1
JAYEIM010000078.1:32061-40868
Campylobacterota bacterium

LpxK HD_5 HD GAF Receiver CBS CBS RraA-like B12-binding Met_asp_mut_E Zn_ribbon_21 BrnT_toxin CopG_antitoxin MutL AMP-binding

MEA2019586.1
JAYEIM010000418.1:1-5790
Campylobacterota bacterium

Aconitase_B_N Aldolase_II Methyltransf_5 FUN14 Peripla_BP_4 PAS His-Kinase His-Kinase BrnT_toxin CopG_antitoxin ?

MEA2028096.1
JAYEIB010000184.1:56-2560
Campylobacterota bacterium

TP_0789-like MacB_PCD FtsX BrnT_toxin CopG_antitoxin

MEA2029832.1
JAYEIB010000617.1:1-4541
Campylobacterota bacterium

OMP_b-brl BrnT_toxin CopG_antitoxin Thy1 ABC_tran_CTD ABC_tran ABC_tran_Xtn ABC_tran ?

MEA2050164.1
JAYEJC010000113.1:30394-45427
Campylobacterota bacterium

FtsX MacB_PCD ABC_tran PIF1 Pif1_2B_dom HTH_40 ? CarboxypepD_reg DUF2442 DUF4160 BrnT_toxin DUF6364 MFS_1 MFS_1 TACO1_YebC_N Transcrip_reg DHH DHHA1 ? Fer2_BFD DnaJ T6PP_N TerB

MEA2092579.1
JAYEKQ010000018.1:7163-14367
Patescibacteria group bacterium

Glutaredoxin Gp_dh_N Gp_dh_C ATP-grasp_5 CoA_binding_2 Succ_CoA_lig DHH DHHA2 zf-Di19 Receiver Ribosomal_L33 BrnT_toxin CopG_antitoxin DrHyd ?

MEA2098356.1
JAYEKL010000111.1:439-7186
Patescibacteria group bacterium

Glyco_transf_4 Glycos_transf_1 HTH_11 Pribosyltran BrnT_toxin CopG_antitoxin tRNA-synt_1 His_Phos_1 tRNA-synt_1 Anticodon_1 DUF5915

MEA3227569.1
JAYEGK010000017.1:269-8154
Campylobacterota bacterium

BrnT_toxin CopG_antitoxin FixG_C Fer4_18 Fer4_5 HTH_17 Resolvase DUF5723 ILVD_EDD_N ? His-Kinase His-Kinase dCache_3

MEA3231251.1
JAYEKI010000230.1:1-3073
Thermodesulfobacteriota bacterium

BrnT_toxin CopG_antitoxin Cupin_2 Acyltransferase FAA_hydrolase Plectin_PPL

MEA3241345.1
JAYEFN010000264.1:1-13834
Pseudomonadota bacterium

RuvA_N HHH_5 RuvA_C RuvB_N AAA_lid_4 WHD_RuvB AMP-binding AMP-binding_C_3 TetR_N TetR_C_4 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_C 3HCDH_N 3HCDH 3HCDH ECH_2 Thiolase_N Thiolase_C BrnT_toxin CopG_antitoxin BacA EAL GGDEF PAS

MEA3285941.1
JAYEKW010000014.1:11487-17896
Candidatus Neomarinimicrobiota bacterium

? CopG_antitoxin PhdYeFM_antitox PIN HTH_3 HicA_toxin HicB BrnT_toxin CopG_antitoxin PhdYeFM_antitox PIN PhdYeFM_antitox ParE_toxin ? PhdYeFM_antitox

MEA3307662.1
JAYEMA010000476.1:68-8180
Elusimicrobiota bacterium

PycTM DUF6788 DDE_Tnp_1_assoc DDE_Tnp_1_4 ? BrnT_toxin CopG_antitoxin NTase_sub_bind Polbeta PS_Dcarbxylase CDP-OH_P_transf ? ?

MEA3355275.1
JAYEMX010000047.1:5719-12449
Patescibacteria group bacterium

Y1_Tnp CarD_C ATP-grasp_6 RimK RimK CarD_C ? Amidohydro_2 VTT_dom BrnT_toxin

MEA3360260.1
JAYEMW010000326.1:1-11780
Thermodesulfobacteriota bacterium

HAMP PAS His-Kinase His-Kinase Receiver Y1_Tnp Bac_DnaA_C Beta-lactamase DUF3471 DUF4258 DUF2283 Pyr_redox_2 BrnT_toxin BrnA_antitoxin DUF5680 ? ? AAA_14 DUF4143 GATase_7

MEA3361543.1
JAYEMW010000516.1:102-4955
Thermodesulfobacteriota bacterium

MazE_antitoxin ZnR ? DUF2283 ? MazE_antitoxin ParE_toxin BrnT_toxin ?

MEA3361618.1
JAYEMW010000519.1:1-6433
Thermodesulfobacteriota bacterium

Hexapep Hexapep ? Acetyltransf_6 ? NikA-like ? DUF3368 DUF7718 DUF5647 HicA_toxin ?

MEA3363684.1
JAYEME010000182.1:4549-8307
Thermodesulfobacteriota bacterium

? CopG_antitoxin CopG_antitoxin RHH_1 PemK_toxin Y1_Tnp Bac_DnaA_C ? PIN

MEA3369535.1
JAYEMH010000039.1:61594-66703
Candidatus Ratteibacteria bacterium

? ? ? BrnT_toxin CopG_antitoxin Y1_Tnp ? DUF2283 ? ? ?

MEA3372886.1
JAYEMF010000039.1:6-8795
Campylobacterota bacterium

Glucosaminidase Radical_SAM AMMECR1 Na_H_Exchanger BATS Radical_SAM Metallophos_2 Zn_ribbon_Top1 Zn_ribbon_Top1 Zn_ribbon_Top1 Topoisom_bac Toprim BrnT_toxin CopG_antitoxin

MEA3415684.1
JAYENE010000203.1:177-13644
Thermodesulfobacteriota bacterium

Mob_synth_C Radical_SAM FdhE_C FdhE_central Fer4_11 NTP_transf_3 Abhydrolase_1 Hydrolase_4 DUF1302 BrnT_toxin CopG_antitoxin TP_0789-like ? MMPL MMPL TonB_dep_Rec_b-barrel Plug Plug

MEA3441848.1
JAYENV010000313.1:1-6193
Chloroflexota bacterium

AbiEi_4 AbiEii DUF433 DUF5615 ? ? BrnT_toxin CopG_antitoxin AAA_14 DUF4143 ?

MEA3450618.1
JAYENF010000099.1:84-3230
Bacteroidota bacterium

Eco57I TaqI_C BrnT_toxin ?

MEA3456481.1
JAYEOH010000143.1:349-1826
Campylobacterota bacterium

BrnT_toxin CopG_antitoxin Uma2

MEA3456779.1
JAYEOH010000179.1:132-1685
Campylobacterota bacterium

? BrnT_toxin CopG_antitoxin NTase_sub_bind Polbeta

MEA3466557.1
JAYEOA010000127.1:10095-27045
Thermodesulfobacteriota bacterium

YceG G3P_acyltransf Na_H_Exchanger TrkA_N TrkA_C PIN ? Peptidase_A8 zf-FPG_IleRS Anticodon_1 tRNA-synt_1 BrnT_toxin BrnA_antitoxin HhH-GPD LysR_substrate HTH_1 Cytochrome_C7 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 MtrB_PioB MtrB_PioB GIIM RVT_1 PhdYeFM_antitox

MEA3489170.1
JAYEOV010000023.1:23293-44094
Candidatus Omnitrophota bacterium

DUF134 STAS Cation_efflux ZT_dimer DUF2092 HTH_17 PilZ MoxR-AAA MoxR-AAA MoxR-AAA GAF_2 BrnT_toxin CopG_antitoxin

MEA3522298.1
JAYEPE010000119.1:384-16553
Campylobacterota bacterium

Pyr_redox_2 2-thiour_desulf Aldedh NIR_SIR_ferr NIR_SIR NIR_SIR_ferr NIR_SIR Aminotran_5 NifU_N Fer2_BFD NifU BrnT_toxin CopG_antitoxin His-Kinase His-Kinase PAS Tll0287-like Receiver His-Kinase His-Kinase Protoglobin Receiver HD_5 Receiver PhoH PIN ?

MEA4865260.1
JAYFQZ010000078.1:1-5565
Sphaerochaeta sp.

? ? RelB BrnT_toxin BrnA_antitoxin YjbQ Peptidase_M20 DcuC

MEB3232363.1
JAYKIC010000450.1:1-4235

Leptolyngbyaceae cyanobacterium

? BrnT_toxin ? BrnT_toxin RHH_1 HNH ? ? ParE_toxin GIT1_C ?

MEB3316389.1
JAYKHA010000122.1:146-12098
Candidatus Melainabacteria bacterium

PhoX ? BrnT_toxin CopG_antitoxin ? DHBP_synthase GTP_cyclohydro2 HATPase_c_3 DNA_mis_repair Spermine_synth TPRs Rib_mS39_PPR NUDIX

MEC5397348.1
JAYXHR010000001.1:1253797-1268640

Uliginosibacterium sp. H1

AstE_AspA_cat MarR_2 ROK ABC_tran ABC_tran BPD_transp_2 Peripla_BP_4 AP_endonuc_2 GFO_IDH_MocA GFO_IDH_MocA_C3 BrnT_toxin CopG_antitoxin SKI Glyoxalase ADH_N ADH_zinc_N_2 DUF2127 ? FAD_binding_2 Succ_DH_flav_C

MEC5410358.1
JAYXIK010000002.1:117702-131708
Aurantimonas sp. C2-4-R8

? MinE CbiA MinC_C MinC_N CoA_transf_3 NAD_binding_1 FAD_binding_6 Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans ? CopG_antitoxin Aminotran_3 LpqE-like ? Sigma70_r2 Sigma70_r4_2 NrsF ?

MEE4286879.1
JAZBNQ010000415.1:1-7305
Mariniphaga sp.

SIS Isochorismatase GATase_7 Lactamase_B TonB_C Acyl_transf_3 BaxI_1 BrnT_toxin CopG_antitoxin Peptidase_S9

MEE4357396.1
JAZBPH010000399.1:191-5543
Desulfococcaceae bacterium

FGS BrnT_toxin ? ? PIN ? DUF8371

MQY63385.1
WJOP01000193.1:305-4816
Calditrichaeota bacterium

BrnT_toxin CopG_antitoxin ? S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12

MRI58673.1
PUXX01000047.1:5156-16058
Epsilonproteobacteria bacterium

S-methyl_trans Pterin_bind B12-binding_2 B12-binding Met_synt_B12 TetR_N BrnT_toxin CopG_antitoxin Oxidored_q6 NiFe_hyd_SSU_C NiFeSe_Hases NiFeSe_Hases NiFeSe_Hases Uma2 TAT_signal Oxidored_q6 NiFe_hyd_SSU_C HoxH NiFeSe_Hases Ni_hydr_CYTB

MSP53793.1
SHZH01000062.1:239-9088
Gammaproteobacteria bacterium

Glyco_hydro_20b Glyco_hydro_20 UbiA ? ? ANKs ANKs ? MipA Beta-lactamase

MXY95753.1
VXRG01000169.1:40134-52570

Caldilineaceae bacterium SB0664_bin_27

Bmp BPD_transp_2 BPD_transp_2 Cupin_2 Methyltransf_11 ? DUF402 BrnT_toxin BrnA_antitoxin FAA_hydrolase Rv2993c-like_N adh_short ? 2-Hacid_dh_C UxuA UxuA ?

MYA27988.1
VXUK01000031.1:4687-19238

Nitrospira sp. SB0666_bin_27

CzcB_C Beta-barrel_RND_2 BSH_CzcB OEP OEP Zip Zip ? AnmK DUF2726 ? BrnT_toxin BrnA_antitoxin Synaptojanin PvlArgDC Aldedh Thioredoxin_4 DS ?

MYB94452.1
VXRH01000234.1:137-6109
Candidatus Poribacteria bacterium

ATP-sulfurylase GmrSD_N Uma2 ADH_N ADH_zinc_N BrnT_toxin ? DUF6785 DUF6784

MYC06205.1
VXNZ01000013.1:439-8610
Chloroflexi bacterium

ADH_N ADH_zinc_N ? ? Aldo_ket_red MFS_3 MFS_3 MR_MLE_C MR_MLE_N NAD_binding_2 NAD_binding_11 Uma2

MYC32161.1
VXOG01000027.1:375-9486
Chloroflexi bacterium

Uma2 Glug Glug Glug Glug SGL ? T7_gp14 Methyltransf_28 CoA_transf_3 ? ? ? ?

MYC34597.1
VXOG01000098.1:7039-21791
Chloroflexi bacterium

EamA EamA G3P_acyltransf Aminotran_4 MFS_1 MFS_1 MFS_1 MFS_1 Bac_luciferase Bac_luciferase ? ? GMC_oxred_N GMC_oxred_C DinB_2 PAC2 Glyoxalase MFS_1 Hydrolase

MYC35341.1
VXOI01000004.1:9923-21480
Chloroflexi bacterium

SecD_1st SecDF_P1_head SecD_SecF_C SecD_SecF_C ? ADH_zinc_N ADH_N AP_endonuc_2 Zincin_1 PgpA BrnT_toxin ? Ribosomal_S2 PaREP1 EF_TS EF_TS AA_kinase RRF Prenyltransf

MYD36592.1
VXMO01000045.1:123-5552
Dehalococcoidia bacterium

NinF PALP NIL BrnT_toxin CopG_antitoxin ThiS TGT UbiA SoxE

MYD63817.1
VXLK01000171.1:1-6401
Gemmatimonadetes bacterium

Sulfatase BrnT_toxin CopG_antitoxin Acetylase ChaC HTH_3 DUF2442 CopG_antitoxin PhyH PhyH

NBB91242.1
JAAAOR010000136.1:1-5385
Spirochaetes bacterium

HepT-like_2 Polbeta ART-PolyVal SLFN-g3_helicase BrnT_toxin CopG_antitoxin PhdYeFM_antitox PIN ? ?

NBD17175.1
JAAANX010000100.1:8397-13706
Cyanobacteria bacterium GSL.Bin1

? SNase BrnT_toxin ? ? PIN Acetylase ? Rv2175c_C ?

NBD73703.1
JAAAOO010000020.1:12412-26026

Patescibacteria group bacterium

T2SSE MshEN ? T2SSM_b ? His-Kinase His-Kinase HAMP Receiver Receiver RecG_wedge DEAD PDDEXK_3 Helicase_C RecG_dom3_C BrnT_toxin CopG_antitoxin ? Germane

NBQ02238.1
RGAS01000118.1:10-3537
Opitutia bacterium

RF-1 BrnT_toxin CopG_antitoxin DUF6036 ? DoxX_2 DoxX_2 ?

NBQ39308.1
RGAZ01000126.1:1-5430
Alphaproteobacteria bacterium

Ribonuc_red_2_N Ribonuc_red_lgC Ribonuc_red_lgC TSCPD GIY-YIG ? BrnT_toxin ?

NBS32888.1
RFVE01000126.1:289-7139
Planctomycetia bacterium

BrnT_toxin CopG_antitoxin ? ACR_tran ACR_tran ? ?

NBS53423.1
RFVR01000363.1:1-2009
Spartobacteria bacterium

Glyco_tran_WbsX BrnT_toxin CopG_antitoxin

NBV05696.1
RFSF01000002.1:102-10808
Proteobacteria bacterium

DUF3750 CopG_antitoxin BrnT_toxin Acetylase SEC-C SecA_SW P-loop_SecA SecA_DEAD EPSP_synthase Peptidase_M17 Peptidase_M17_N DsbD_2 COX1 FixO

NBX03718.1
RFOF01000064.1:1-9963
Alphaproteobacteria bacterium

Trypsin NAD_Gly3P_dh_C NAD_Gly3P_dh_N CagX VirB8 AnmK BrnT_toxin CopG_antitoxin tRNA-synt_1b TrbL TrbL ?

NBX57178.1
RFYH01000020.1:1-11349
Gammaproteobacteria bacterium

T2SSE ? N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP BrnT_toxin CopG_antitoxin Peripla_BP_2 MM_CoA_mutase MM_CoA_mutase MeaB B12-binding TetR_N

NBX61117.1
RFYK01000007.1:89-6596
Betaproteobacteria bacterium

GDP_Man_Dehyd ? Prenyltransf NusG Phasin TetR_N TetR Peptidase_M20 BrnT_toxin CopG_antitoxin

NBY27239.1
RFNA01000515.1:1-2139
Betaproteobacteria bacterium

RHH_1 PIN BrnT_toxin CopG_antitoxin ?

NCC34072.1
RZZQ01001298.1:197-1763
Chloroflexia bacterium

? ? BrnT_toxin ? DUF433

NCC62217.1
RZZE01000362.1:372-2060
Verrucomicrobiia bacterium

DUF3592 BrnT_toxin CopG_antitoxin

NCD42318.1
RZYD01000227.1:176-4285
Bacteroidia bacterium

Phage_int_SAM_4 Phage_integrase Phage_integrase Zn_Tnp_IS91 Y2_Tnp BrnT_toxin CopG_antitoxin HNH_4 ? DUF2442

NCN04738.1
JAACWS010000018.1:13126-27598
Spirochaetales bacterium

PhdYeFM_antitox HTH_5 HMA E1-E2_ATPase Hydrolase Hydrolase_3 Polysacc_lyase DNA_topoisoIV DNA_gyraseB_C Toprim DNA_gyraseB His-Kinase Polbeta ? CopG_antitoxin AAA_14 DUF4143 Polbeta HepT-like_2 EIIC-GAT DUF3450 HicA_toxin

NCN05772.1
JAACWS010000054.1:4040-18507
Spirochaetales bacterium

ROK HTH_24 PFK PIG-L SIS CBM9_2 NTP_transf_3 FKBP_C ? BrnA_antitoxin Phosphorylase DUF8084 DUF8084 His-Kinase STAS SpoIIE GAF FxsA

NCN82343.1
JAACWQ010000002.1:380-6355
Candidatus Paceibacterota bacterium

BrnT_toxin ? CopG_antitoxin BrnT_toxin Acetyltransf_10 Acetylase ? BrnT_toxin CopG_antitoxin ? Methyltransf_11

NCO01663.1
JAACVB010000048.1:148-17189
Epsilonproteobacteria bacterium

DsbD_2 TonB_dep_Rec_b-barrel Plug Acyltransferase Aldedh Helicase_C Methyltrans_SAM PUA_3 Glutaminase BrnT_toxin CopG_antitoxin Peptidase_M20 ? Trypsin TOP_N Peptidase_M3 Exo_endo_phos_3 UDG

NCO88672.1
JAACUC010000049.1:1-5035
Candidatus Roizmanbacteria bacterium

DUF2207_C DUF2207_C Radical_SAM B12-binding BrnT_toxin CopG_antitoxin Lyase_1 ASL_C

NCO88817.1
JAACUC010000080.1:140-3397
Candidatus Roizmanbacteria bacterium

Adenylsucc_synt IMPDH BrnT_toxin

NCO89202.1
JAACUC010000211.1:1-1391
Candidatus Roizmanbacteria bacterium

Bact_hydrolase BrnT_toxin CopG_antitoxin AlkZ-like

NCP86088.1
JAACSE010000005.1:6278-20704
bacterium

? Peptidase_M3_N Peptidase_M3 Pribosyltran OMPdecase OTCace OTCace_N Abhydrolase_3 Amidohydro_1 BrnT_toxin CopG_antitoxin Xan_ur_permease Isochorismatase ? Hydrolase Hpt Receiver Receiver His-Kinase His-Kinase PAS PAS

NCR41244.1
JAADBR010000088.1:1-16091
Microcystis aeruginosa W13-11

? ParE_like DUF7682 APH MgtE_N CBS CBS MgtE ? ? ? BrnA_antitoxin CobN-Mg_chel GAF His-Kinase His-Kinase Receiver PSI_8 PsaL Usp PGK

NCU31879.1
RZZP01000138.1:344-2595
Candidatus Moraniibacteriota bacterium

SBD_N DeoRC HTH_DeoR BrnT_toxin CopG_antitoxin

NDC53847.1
RGVT01000097.1:312-8468
Planctomycetia bacterium

? ? DUF6311 Glyco_transf_4 Glycos_transf_1 GDP_Man_Dehyd ? CopG_antitoxin

NDG99698.1
RGQS01000018.1:266-10687
Opitutia bacterium

? ? DEAD Helicase_C BrnT_toxin CopG_antitoxin HD_5 Receiver Aconitase Aconitase_2_N Aconitase_B_N DoxX_2 DoxX_2 HRDC RQC RecQ_Zn_bind Helicase_C DEAD ?

NEP13151.1
JAAHGL010000142.1:639-18347
Symploca sp. SIO2C1

DUF3288 ? ? Poly_export SLBB HTH_70 ParA-Soj-NTPase BrnT_toxin HTH_3 Peptidase_M78 Radical_SAM RlmN_N HAMP Guanylate_cyc CHASE2 PAS Guanylate_cyc TP_methylase

NEQ95313.1
JAAHGY010000014.1:1-9564
Cyanothece sp. SIO2G6

? PIN ? BrnT_toxin ? ? HUTI_composite_bact Amidohydro_3 AAA_29 ABC-ATPase ? GTP_CH_N GTP_cyclohydro2 ?

NER22589.1
JAAHFP010000018.1:48567-69125
Symploca sp. SIO1C2

Pro_isomerase CYP38_PsbQ-like EFP_N EFP Elong-fact-P_C Biotin_lipoyl HTH_5 CTP_transf_1 Amino_oxidase SQS_PSY BrnT_toxin Uma2 FOL3_C Mur_ligase_M ? His-Kinase His-Kinase PAS GAF AAA_16 STY-kinase ? Lactamase_B_3 GST_N_3 GST_C_2

NES17395.1
JAAHHL010000001.1:1-14366
Caldora sp. SIO3E6

Secretin Secretin_N STN AMIN SR_plectin_7 PilN PilM_2 PilM_2 Rieske HTH_31 RelE BrnT_toxin RHH_1 ? ParE_toxin PhdYeFM_antitox OKR_DC_1 OKR_DC_1 Glyoxalase

NET01416.1
JAAHFX010000006.1:172434-185629
Sphaerospermopsis sp. SIO1G1

SecG Metalloenzyme DUF4276 ABC-ATPase PIN ? ? BrnT_toxin ? ABC_tran ABC_tran_Xtn ABC_tran Sod_Ni Peptidase_S26 Peptidase_S26 Sod_Fe_C Sod_Fe_N Cytidylate_kin Pantoate_ligase DPBB_1

NET71469.1
JAAHFY010000083.1:5349-13892
Sphaerospermopsis sp. SIO1G2

2OG-FeII_Oxy_2 MazE_antitoxin PIN Y4jL_N ? Fic HTH_17 BrnT_toxin CopG_antitoxin ? PIN ? ? PhdYeFM_antitox ?

NET74314.1
JAAHFY010000876.1:1-1211
Sphaerospermopsis sp. SIO1G2

? BrnT_toxin ? ?

NGZ08832.1
PHGD01000014.1:16366-31975
Nitrospira sp. LK70

Citrate_synt AMP-binding AMP-binding_C ? ? ? GerE Receiver SNase BrnT_toxin CopG_antitoxin ? ? Fer4_7 DAO OmpA CRWN_N Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Sel1

NIM11851.1
WWBY01000150.1:24344-32538
Candidatus Aminicenantes bacterium

BrnT_toxin ? COR-B COR-A AP-GTPase LRRs LRRs LRRs DUF5615 DUF433 ? Lactamase_B_3 ABC2_membrane_2

NIM14344.1
WWBY01000373.1:1-7066
Candidatus Aminicenantes bacterium

YegS_C DUF4258 CopG_antitoxin TPRs TPRs nSTAND3 RNaseH-CATRA+Caspase RelB-like Lant_leader_dom Thymidylat_synt B12-binding_2 B12-binding

NIM16857.1
WWBY01000655.1:1-5405
Candidatus Aminicenantes bacterium

YhcG_C TIR BrnT_toxin CopG_antitoxin ? Methyltransf_11 ? ?

NIS38042.1
JAABZP010000112.1:1-2570
Candidatus Saccharibacteria bacterium

adh_short BrnT_toxin CopG_antitoxin HTH_3 BTRD1 BTRD1 DPBB_1

NJK55012.1
JAAUVB010000007.1:34258-45593

Pleurocapsa sp. SU_5_0

Sec_GG SecD_SecF_C ? ? XhlA PA14 HicB ? BrnT_toxin BrnA_antitoxin HicA_toxin ? DUF29 PNPOx_N PNP_phzG_C PatA_N LGT

NJL71776.1
JAAUTC010000197.1:1-4200
Candidatus Competibacteraceae bacterium

PIN RelA_SpoT ? ? BrnT_toxin CopG_antitoxin Mrr_cat

NJM30133.1
JAAURV010000106.1:1-9069
Hyphomicrobiales bacterium

rve HTH_29 BrnT_toxin Methyltransf_11 HlyIII PNPOx_N ThuA ? DUF559

NKB25496.1
WLWN01000001.1:2632012-2644097
Kiritimatiellae bacterium

? DDE_Tnp_1_4 Ribosomal_S2 EF_TS Sod_Fe_C Sod_Fe_N ? DDE_Tnp_1_4 DDE_Tnp_1_4 BrnT_toxin CopG_antitoxin Ald_Xan_dh_C MoCoBD_1 MoCoBD_2 YHS Sigma70_r2 Sigma70_r4_2 OSTMP1

NKB62895.1
WLWU01000003.1:138319-157036
Gammaproteobacteria bacterium

SLT LysM LysM LysM LysM Rhodanese Rhodanese TruD TruD DUF547 Phospholipase-like Phage_integrase ? DUF4258 CopG_antitoxin ? ? Cupin_7 Cupin_7 HTH_18 DJ-1_PfpI CoA_binding_2 Succ_CoA_lig ATP-grasp_5 ECH_1

NKB64909.1
WLWU01000014.1:461904-477323
Gammaproteobacteria bacterium

Thiolase_N ACP_syn_III ACP_syn_III_C HisKA_7TM HTH_Mga Aminotran_5 SIS_2 Phospholipase-like Phage_integrase ? DUF4258 CopG_antitoxin ? ? Cupin_7 Cupin_7 HTH_18 DJ-1_PfpI CoA_binding_2 Succ_CoA_lig ATP-grasp_5 ?

NKB65017.1
WLWU01000016.1:25942-40979
Gammaproteobacteria bacterium

GCV_T_C GCV_T FAO_M DAO MTTB Resolvase PRiA4_ORF3 PRiA4_ORF3 ? DDE_Tnp_Tn3 BrnT_toxin CopG_antitoxin Porin_8 ? ? ? ParA-Soj-NTPase HHH_5

NKC11228.1
WLWT01000001.1:635535-648506
Gammaproteobacteria bacterium

SBP_bac_5 SBP_bac_5 BPD_transp_1_N BPD_transp_1 OppC_N BPD_transp_1 ABC_tran oligo_HPY ABC_tran oligo_HPY ? DUF4372 BrnT_toxin CopG_antitoxin DUF6398 HTH_23 HTH_32 HTH_33 DDE_3 Methyltransf_12 DDE_3 HTH_33 HTH_29

NKC15185.1
WLWT01000002.1:178283-187187
Gammaproteobacteria bacterium

Lin0512_fam Lin0512_fam OmpH GreA_GreB GreA_GreB_N Phosphodiest HTH_29 HTH_33 DDE_3 SymE_toxin BrnT_toxin CopG_antitoxin DDE_Tnp_1 DDE_3 HTH_33 HTH_32 HTH_23 DUF2442 DUF4160 ?

NKQ37342.1
JAAXQA010000255.1:1-11675
Chloroflexota bacterium

TPR_MalT TPRs ? ? TerB BrnT_toxin CopG_antitoxin Penicil_amidase Peptidase_M23 NAGPA DUF11 GSDH Hydrolase PALP

NKQ40365.1
JAAXPZ010000015.1:22056-26761
Sulfurovum sp.

BrnT_toxin CopG_antitoxin BATS MIP LMWPc ? Sod_Ni Peptidase_S24

NLD15757.1
JAAZCG010000064.1:35476-44970
Gammaproteobacteria bacterium

His-Kinase His-Kinase PAS ? H2O2_YaaD ? HNH ydhR HTHP BrnT_toxin BrnA_antitoxin ? ? ? 5TM-5TMR_LYT ? ?

NLE59264.1
JAAZAJ010000848.1:236-2003
Planctomycetota bacterium

BrnT_toxin DUF4276

NND83079.1
JABDKW010000052.1:8018-18291
Gammaproteobacteria bacterium

PgpA Amidase_2 TauD PALP CBS CBS Cys_Met_Meta_PP ? Phage_integrase BrnT_toxin CopG_antitoxin ? Pterin_4a Thiolase_N

NNJ44779.1
JABDRP010000113.1:1-3202
Sulfurovum sp.

HTH_23 BrnT_toxin CopG_antitoxin PqqD GidB Metallothio_Pro ? Rhomboid

NNM58313.1
JABEVJ010000019.1:92-6467

Legionellales bacterium

Dala_Dala_lig_C Dala_Dala_lig_N Peptidase_M48 Inositol_P Aminotran_5 Fe-S_biosyn AAA_14 DUF4143 BrnT_toxin CopG_antitoxin

NOQ21342.1
WTBT01000006.1:123-9419

Candidatus Aegiribacteria sp.

QueF TPRs CorA Thioredoxin_3 ArsP_1 HTH_5 DEAD Helicase_C RecQ_Zn_bind RQC HRDC BrnT_toxin CopG_antitoxin HTH_5 DsrE TusA

NOQ22972.1
WTBT01000062.1:1-5119
Candidatus Aegiribacteria sp.

DDE_Tnp_ISL3 PIN BrnT_toxin CopG_antitoxin Cupin_2 CopG_antitoxin ? GIIM RVT_1

NOQ32362.1
WTBY01000074.1:58960-75209
Helicobacteraceae bacterium

Spermine_synth YgaP-like_TM HD_assoc HD DUF2779 ORF6N SbcC_Walker_B SMC_N Calcineurin BrnT_toxin CopG_antitoxin His-Kinase His-Kinase sCache_like Trans_reg_C Receiver Methylase_S Methylase_S N6_Mtase PhoU PhoU ABC_tran EAL

NOQ65499.1
WTCM01000263.1:534-6714
Methyloprofundus sp.

UPF0259 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 DUF5128 BrnT_toxin CopG_antitoxin Cytochrome_C7 ?

NOR13322.1
WTAW01000298.1:1-6002
Candidatus Aminicenantes bacterium

? Abhydrolase_1 STY-kinase Lactamase_B_3 BrnT_toxin CopG_antitoxin

NOR81639.1
WTAC01000621.1:1-3182
Methyloprofundus sp.

? ? CopG_antitoxin Calcineurin SbcD_C AAA_23

NOR82912.1
WSZX01000216.1:242-3838
Ardenticatenales bacterium

SpoIIE BrnT_toxin CopG_antitoxin Amidohydro_3 ACT CBS CBS

NOS73849.1
JABFSV010000145.1:1-4555

Methyloglobulus sp.

Glyoxalase Glyoxalase_5 ? PPC CarboxypepD_reg CarboxypepD_reg BrnT_toxin

NOU23898.1
JABFRD010000642.1:2458-14513
Methyloglobulus sp.

NAD_kinase DnaJ TerB DjlA_TM SAICAR_synt RadC UPF0758_N Flavoprotein DFP dUTPase dCache_1 BrnT_toxin ? dCache_1 PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV NUDIX GreA_GreB GreA_GreB_N

NOX16705.1
JAADIR010000006.1:1-8037
Chlorobiota bacterium

? ? Lactamase_B_2 Aminotran_1_2 CDP-OH_P_transf NTP_transferase TrmE_N MnmE_helical BrnT_toxin CopG_antitoxin HEPN

NOX16748.1
JAADIR010000009.1:19134-34867
Chlorobiota bacterium

GatB_Yqey GatB_N ? DA1-like DA1-like 4HBT_2 Por_Secre_tail PQQ_2 PQQ_2 DUF4136 Sortilin-Vps10 BrnT_toxin CopG_antitoxin PRAI Sirtuin 4HBT_2 GidB ABC-3 ZnuA DUF7951 ZnuA

NOX18708.1
JAADIR010000209.1:14076-23176
Chlorobiota bacterium

BrnT_toxin CopG_antitoxin DUF6029 Por_Secre_tail Omp28 CARDB Por_Secre_tail Omp28 Por_Secre_tail AhpC-TSA Por_Secre_tail Omp28

NOX19867.1
JAADIE010000015.1:22-9605
Nitrospirota bacterium

N_methyl UvrD_C UvrD-helicase Aconitase Aconitase_C DUF2274 BrnT_toxin CopG_antitoxin APH His_Phos_1 Isochorismatase Creatininase Methyltransf_28

NOX25692.1
JAADIV010000138.1:1-6128
Deltaproteobacteria bacterium

Bac_DnaA_C ? ? Acetyltransf_7 ParE_toxin RelB3_C RHH_1 SLFN_AlbA_2 BrnT_toxin CopG_antitoxin ABC_tran oligo_HPY ABC_tran BPD_transp_1

NOX62633.1
JAADII010000120.1:38408-44249
Chloroflexota bacterium

Sulfotransfer_1 BrnT_toxin ? Glycos_transf_2 Epimerase DegT_DnrJ_EryC1 UPF0150 HicA_toxin Methyltransf_11

NOY52650.1
JAADIO010000019.1:1226-16875
Deltaproteobacteria bacterium

Glycos_transf_2 DegT_DnrJ_EryC1 Methyltransf_9 ? Polysacc_synt_3 PEP-CTERM LamininG PMT_2 TPRs BrnT_toxin CopG_antitoxin Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 Receiver ? ? BPL_N CbiA GATase_3

NOZ11891.1
JAADGI010000034.1:75844-87796
Gammaproteobacteria bacterium

LplA-B_cat LIAS_N Radical_SAM DUF218 NTP_transferase APH LptD_N LptD_2 LptD SurA_N Rotamase Rotamase_3 BrnT_toxin CopG_antitoxin DUF4258 MqsA_antitoxin PIN FitA-like_RHH HicB_lk_antitox Monooxy_af470-like

NOZ41062.1
JAADGD010000275.1:67-5535
Planctomycetota bacterium

AMP-dom_DIP2-like PP-binding ABC_tran OEP OEP BrnT_toxin CopG_antitoxin FtsX MacB_PCD

NOZ60437.1
JAADFU010000022.1:184-9392
Calditrichota bacterium

GWxTD_dom TPRs TPRs TPRs Transgly DUF2207 DUF2207_C BrnT_toxin CopG_antitoxin DUF6599

NPA14257.1
JAADES010000028.1:5410-17323
Aquificota bacterium

RtcB Intein_splicing RtcB Guanylate_kin RNA_pol_Rpb6 DUF7132 polyprenyl_synt Lactamase_B_2 Anti-Pycsar_Apyc1 CDP-OH_P_transf ? BrnA_antitoxin HTH_3 ? N6_N4_Mtase N6_N4_Mtase Portal_Mu Portal_Mu Phage_P2_GpU RHH_1

NPA28282.1
JAADEQ010000119.1:59-3002
Campylobacterota bacterium

RibD_C dCMP_cyt_deam_1 DUF4258 CopG_antitoxin PUA_3 Methyltrans_SAM

NPV02214.1
JABLXV010000052.1:146-9777
Brevinematales bacterium

SH3_3 Beta_helix BrnT_toxin ? DUF4914 HEPN Polbeta PCRF RF-1 PrmC_N Methyltransf_25 SLT SLT_superhelical

NQT25161.1
JABMRT010000096.1:1-15922
candidate division KSB1 bacterium

PP2C_2 FG-GAP_3 PQQ_2 PQQ_2 FG-GAP FHA FHA Glycos_transf_2 GTP_EFTU GTP_EFTU_D2 EFG_III EFG_C LepA_C Peptidase_S26 Peptidase_S26 Peptidase_S26 Peptidase_S26 Radical_SAM BrnT_toxin CopG_antitoxin RMD1-3 UvrA_inter UvrA_DNA-bind ABC_tran AAA_29 ABC_tran HTH_3 Couple_hipA HipA_C ?

NQT25677.1
JABMRT010000121.1:1-4046
candidate division KSB1 bacterium

CoA_binding_3 Bac_transf 23S_rRNA_IVP BrnT_toxin CopG_antitoxin MqsA_antitoxin 23S_rRNA_IVP NTP_transf_2 ?

NQT58669.1
JABMQR010000125.1:523-7677
Bacteroidetes bacterium

Amidohydro_1 BsNagA_N Y1_Tnp DNA_processg_A Mg_chelatase_C Mg_chelatase ChlI DUF6290 ParE_toxin BrnT_toxin CopG_antitoxin

NQT58759.1
JABMQR010000135.1:197-7840
Bacteroidetes bacterium

HTH_24 PfkB Indigoidine_A URO-D HYDIN_VesB_CFA65-like_Ig MarR MatE MatE BrnT_toxin CopG_antitoxin

NQT72083.1
JABMQN010000053.1:18889-24963
Chloroflexota bacterium

HicA_toxin ? DUF2442 DUF7387 ? NTP_transf_2 ? ? CopG_antitoxin PemK_toxin MazE_antitoxin

NQT90873.1
JABMQS010000119.1:21671-34995
Candidatus Omnitrophota bacterium

AA_permease_2 PTS_EIIA_2 FUR FeoA FeoA FeoB_N FeoB_Cyto Gate FeoB_C Gate ? ? ? Peptidase_S24 CopG_antitoxin MafB19-deam GIY-YIG OprB ROK ROK MFS_1 SIS SIS

NQU63260.1
JABMPD010000159.1:1-2474
SAR324 cluster bacterium

HipA_C DUF4258 CopG_antitoxin HTH_40 HRDC

NQU63496.1
JABMPD010000226.1:150-2736
SAR324 cluster bacterium

BrnT_toxin CopG_antitoxin Anti-Pycsar_Apyc1 cNMPBD

NQU66239.1
JABMPD010000962.1:220-6287
SAR324 cluster bacterium

SoFic-like_C DUF4062 BrnT_toxin CopG_antitoxin POR Fer4 Pyr_redox_2 Fer4_20 TPP_enzyme_C POR_N

NQV49666.1
JABMOT010000103.1:1-6678
Candidatus Neomarinimicrobiota bacterium

Fic DUF4143 AAA_14 ? LTD FlgD_ig BrnT_toxin CopG_antitoxin HigB-like_toxin HTH_3 FlgD_ig ParE_toxin PhdYeFM_antitox

NQW24153.1
JABMNL010000053.1:3311-17367
SAR202 cluster bacterium

? Ferric_reduct ? PSII_BNR Sortilin-Vps10 Cytochrome_CBB3 Thioredox_DsbH 2-Hacid_dh_C BrnT_toxin CopG_antitoxin DUF3644 HGTP_anticodon tRNA-synt_2b tRNA_SAD G3P_acyltransf Polyketide_cyc2 AXE1 AXE1

NQW47197.1
JABMNA010000285.1:174-5693
Planctomycetota bacterium

Uma2 Catalase_C CopG_antitoxin DUF6797 Copper-bind AP_endonuc_2

NQW94003.1
JABMMI010000313.1:156-4331
Polaromonas sp.

Wzz Phasin TetR_N TetR RMD1-3 BrnT_toxin CopG_antitoxin GDP_Man_Dehyd YafQ_toxin RelB

NTU46272.1
JAAXWC010000003.1:43766-56654
Candidatus Roizmanbacteria bacterium

Ribosomal_S18 ? CtpB_N-like PDZ_6 Peptidase_S41 Trypsin PDZ_2 tRNA-synt_His HGTP_anticodon PCRF RF-1 ? BrnT_toxin CopG_antitoxin PRA-PH Ribosomal_L31 Ribosomal_L10 RL10_C Ribosomal_L12_N Ribosomal_L12 His-Kinase DNA_gyraseB Toprim DNA_gyraseB_C Pyrophosphatase

NTV30017.1
JAAXVU010000136.1:355-5709
Candidatus Omnitrophota bacterium

HD YmdB PilZ tRNA-synt_1c tRNA-synt_1c Anticodon_2 BrnT_toxin CopG_antitoxin HNH HNH

NTW49326.1
JAAXXH010000120.1:1-6956
Chlorobiales bacterium

Topoisom_bac Zn_ribbon_Top1 Zn_ribbon_Top1 Zn_ribbon_Top1 BrnT_toxin DUF2164 ? SH3_4 TPRs BatD BatD BatD P5CR_dimer F420_oxidored ?

NTW78355.1
JAAXXV010000574.1:1-1663
Syntrophaceae bacterium

BrnT_toxin CopG_antitoxin DEDD_Tnp_IS110

NUM65552.1
JABWBW010000049.1:243-7614
candidate division KSB1 bacterium

HigB-like_toxin HTH_3 ? Y2_Tnp ? Fic ? ? SMC_N ABC_tran ABC_membrane Hydrolase_4

NUN64296.1
JAATWA010000001.1:1255206-1271222
Pseudanabaena biceps

tRNA_NucTran2_2 PolyA_pol_RNAbd PolyA_pol DUF2288 UDG DUF433 DUF5615 DUF3226 ABC-ATPase BrnT_toxin Sin3_corepress ? FAD_oxidored FAD_oxidored FAD_oxidored DnaJ DXP_synthase_N Transket_pyr Transketolase_C FIST_C FIST Phycobilisome

NUO81186.1
JABWAN010000314.1:1-6689
candidate division KSB1 bacterium

TetR_N TetR_C_4 Thiolase_N Thiolase_C BrnT_toxin Tom5 ECH_1 3HCDH_N 3HCDH 3HCDH Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1

NUO82669.1
JABWAN010000601.1:258-4446
candidate division KSB1 bacterium

Tricorn_N Peptidase_S9 BrnT_toxin ? Aldedh

NUQ38032.1
JABWBG010000101.1:1-11822
Caldilineales bacterium

E3_binding Biotin_lipoyl ? UbiD_C UbiD UbiD_N PhdYeFM_antitox Amidohydro_1 Urease_alpha DeoC BrnT_toxin CopG_antitoxin FGGY_C FGGY_N HTH_24 HATPase_c_4 SLFN_AlbA_2 AP_endonuc_2 AAA_16 WHD_MalT TPR_MalT

NVO00125.1
JACABD010000043.1:16037-29175
Geobacteraceae bacterium

PRAI ? RVT_1 DUF4365 Crr6 CcdA BrnT_toxin CopG_antitoxin PALP SLT Trp_syntA Zn_ribbon_ACC Carboxyl_trans Mur_ligase_M Mur_ligase_C LptD_2 LptD

NWF88528.1
JACAFC010000055.1:15913-24428
Ignavibacteriaceae bacterium

Alpha-amylase AMPK1_CBM BrnT_toxin CopG_antitoxin ? ? AAA_13 DUF1801 Alpha-L-AF_C ASD1_dom Acetylase

NWG04290.1
JACAEV010000179.1:1-10177
Syntrophaceae bacterium

MM_CoA_mutase B12-binding Thiolase_C_1 Thiolase_N Thiolase_C_1 zf-ChsH2 OB_ChsH2_C zf-ChsH2 OB_ChsH2_C Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 BrnT_toxin ? Acetylase AMP-binding_C

NYT78345.1
JACCEP010000007.1:52587-74448

Alcaligenaceae bacterium

BPD_transp_2 BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran BCA_ABC_TP_C HTH_IclR IclR_C TauD TctC BrnT_toxin CopG_antitoxin GTP_EFTU GTP_EFTU_D2 EFG_III EFG_IV EFG_C ABC_membrane_2 ABC_tran ? AraC_binding_2 HTH_18 DUF2165 DUF1902

OAD19195.1
LUTY01002903.1:1-1365
Candidatus Thiomargarita nelsonii

BrnT_toxin ? ?

OAI46078.1
LSTC01000100.1:3309-13805
Nitrospira sp. SCGC AG-212-E16

? Amphi-Trp FitA-like_RHH PIN ? HicB_lk_antitox DUF899 BrnT_toxin CopG_antitoxin PilZ 3keto-disac_hyd ? TOBE_3 CysA_C_terminal ABC_tran BPD_transp_1 BPD_transp_1

OEU73428.1
MAXR01000198.1:6834-16463
Desulfuromonadales bacterium C00003068

Thioredoxin SIS_2 Cytochrom_C_asm GlutR_N Shikimate_DH GlutR_dimer HEM4 ALAD BrnT_toxin CopG_antitoxin DUF1499 HNH ? MraZ MraZ Methyltransf_5 DivIC

OFV95922.1
MEKX01000121.1:1-8102

Acidobacteria bacterium RIFCSPLOWO2_12_FULL_54_10

? ? Vgb_lyase HUTI_composite_bact Amidohydro_3 Amidohydro_3 BrnT_toxin BrnA_antitoxin

OFV96278.1
MEKX01000112.1:25487-38720

Acidobacteria bacterium RIFCSPLOWO2_12_FULL_54_10

RMNT_CmcI Fic MOSC Esterase FtsX MacB_PCD ABC_tran Methyltransf_11 BrnT_toxin CopG_antitoxin ? ? ? Aminotran_1_2 SpoIIE GAF DUF3389

OFW80284.1
MEMG01000032.1:26029-32034

Alphaproteobacteria bacterium RIFCSPHIGHO2_02_FULL_40_34

CopG_antitoxin DUF4258 Cpn60_TCP1 Cpn10 Glutaredoxin ? ? dCMP_cyt_deam_1 RibD_C Glycos_transf_1 Glyco_transf_4

OFX28229.1
MENA01000113.1:1-10551

Armatimonadetes bacterium RBG_16_67_12

DUF4258 PemK_toxin PTP-NADK Amino_oxidase Ferrochelatase SPASM Radical_SAM Chlor_dismutase

OFY33421.1
MEOG01000046.1:72892-85053

Bacteroidetes bacterium GWF2_35_48

S1_2 S1_2 WHD_CvfB Sacchrp_dh_C Sacchrp_dh_NADP WCX WYL ? Cas6b_N Cas6b_C DUF1016_N YhcG_C HD BrnT_toxin NikA-like ? ? DUF481 DUF1573 Aminotran_1_2 Flavoprotein DFP

OFY71564.1
MEOS01000006.1:239-8506

Bacteroidetes bacterium RBG_13_42_15

BrnT_toxin CopG_antitoxin PPK2 ? Rho_N DUF6326 DUF1579 PGI

OGA33292.1
MERH01000171.1:570-8651

Betaproteobacteria bacterium RIFCSPLOWO2_12_FULL_62_13b

Phosphonate-bd 2OG-FeII_Oxy DIOX_N BrnT_toxin CopG_antitoxin adh_short CoA_transf_3 ? HTH_11 Rieske NAD_binding_1

OGA55319.1
MERG01000001.1:8-7155

Betaproteobacteria bacterium RIFCSPLOWO2_12_FULL_62_13

Sacchrp_dh_NADP Sacchrp_dh_C ? BrnT_toxin CopG_antitoxin PrpF

OGB65988.1
MESS01000075.1:278-5169

Caldithrix sp. RBG_13_44_9

? BrnT_toxin BrnA_antitoxin HepT-like Polbeta

OGC56273.1
MEVN01000042.1:19531-28750

candidate division WWE3 bacterium RIFCSPLOWO2_12_FULL_36_10

Peptidase_M41 AAA_lid_3 AAA tRNA-synt_1b ? BrnT_toxin CopG_antitoxin DivIC Phage_int_SAM_1 Phage_integrase ? Transposase_mut PEGA HicB_lk_antitox

OGC57068.1
MEVN01000022.1:493-6850

candidate division WWE3 bacterium RIFCSPLOWO2_12_FULL_36_10

? PIN ? PIN ? ? CopG_antitoxin ? Ribosomal_L33 ? Transposase_mut

OGC97512.1
MEXG01000011.1:72388-82234

Candidatus Amesbacteria bacterium RIFCSPHIGHO2_01_FULL_48_32

TIM ? ? PsbP_2 ? ? ? BrnT_toxin CopG_antitoxin PIN MazE_antitoxin ? Peptidase_M73 Peptidase_S24 LTD Beta-Roll Beta-Roll

OGD05051.1
MEXT01000019.1:1-7191

Candidatus Amesbacteria bacterium RIFOXYB1_FULL_47_13

Phage_holin_4_2 BrnT_toxin CopG_antitoxin LTD Peptidase_S26 Peptidase_M73 ? PMT_2 ?

OGD19410.1
MEYB01000087.1:10228-16548

Candidatus Aminicenantes bacterium RBG_16_63_16

SYY_C-terminal tRNA-synt_1b BrnT_toxin CopG_antitoxin ? ? ? ? ? Transgly

OGE22101.1
MFCQ01000006.1:103483-115951
Candidatus Daviesbacteria bacterium RIFCSPHIGHO2_01_FULL_40_24

bPH_2 DUF5657 ? SmpB ? ? DZR_2 Pribosyltran BrnT_toxin CopG_antitoxin ? NUDIX FtsJ DUF5652 Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4

OGE26196.1
MFCY01000040.1:1-16429

Candidatus Daviesbacteria bacterium RIFCSPHIGHO2_02_FULL_39_41

? Pur_ac_phosph_N AbiEii AbiEi_4 AbiEi_4 AbiEii ? BrnT_toxin CopG_antitoxin Transpeptidase Transgly ? ? ?

OGE73598.1
MFEH01000006.1:5720-17183

Candidatus Doudnabacteria bacterium RIFCSPHIGHO2_01_FULL_41_86

Dockerin_1 RelB YafQ_toxin DUF6364 YafQ_toxin Dockerin_1 Dockerin_1 BrnT_toxin CopG_antitoxin ParE_toxin PhdYeFM_antitox Sod_Fe_C Sod_Fe_N RNase_Y_N KH_1 HD YmdB Methyltransf_31

OGE88324.1
MFEU01000010.1:60368-65872

Candidatus Doudnabacteria bacterium RIFCSPHIGHO2_02_FULL_49_24

TrmB DUF2798 BrnT_toxin CopG_antitoxin ? TrmB R3H KH_KhpA-B 60KD_IMP YidD Ribonuclease_P

OGF52778.1
MFGX01000128.1:16-5328

Candidatus Fraserbacteria bacterium RBG_16_55_9

? BrnT_toxin CopG_antitoxin

OGG02226.1
MFJA01000070.1:1897-19371

Candidatus Gottesmanbacteria bacterium RBG_16_37_8

Bac_transf ? G6PD_C G6PD_N Dockerin_1 Sulfatase SGSH_C LiaI-LiaF-like_TM1 ? Dockerin_1 Sulfatase SGSH_C DUF4258 CopG_antitoxin ? ParG LiaI-LiaF-like_TM1 DsbD_TM ArabinogalactanP DUF4832 ?

OGG12460.1
MFJJ01000062.1:4957-16581

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_01_FULL_46_14

? MazE_antitoxin PIN MJ0043_C Glyco_transf_7C Glycos_transf_2 Na_Ca_ex Na_Ca_ex Cation_ATPase_C Hydrolase E1-E2_ATPase Cation_ATPase_N BrnT_toxin CopG_antitoxin DNA_ligase_A_C DNA_ligase_A_M DNA_ligase_A_N ? Fic DUF5654 PIN RHH_1

OGG14133.1
MFJE01000024.1:14543-27278

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_01_FULL_39_10

Lyase_1 EamA EamA EamA Cyclase Arg_tRNA_synt_N tRNA-synt_1d DALR_1 ? zf-C2H2_AcuF DUF4258 CopG_antitoxin Malic_M Malic_M malic TrmB ? ? Rhodanese Maf

OGG14294.1
MFJE01000022.1:76937-81404

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_01_FULL_39_10

? ? BrnT_toxin CopG_antitoxin ? ? ? YafQ_toxin Pterin_4a ?

OGG15514.1
MFJL01000024.1:38062-51202

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_02_FULL_39_11

PhdYeFM_antitox YceG Dockerin_1 ? ? Ribosomal_L33 ? BrnT_toxin CopG_antitoxin ? ? ? ? ? ?

OGG15544.1
MFJL01000023.1:11396-22755

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_02_FULL_39_11

Zn_ribbon_IS1595 ? ? ? AbiEi_4 AbiEii ? BrnT_toxin CopG_antitoxin ? AbiEii ? AbiEi_4 AbiEii ?

OGG18461.1
MFJK01000014.1:32231-44744

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_01_FULL_47_48

Transketolase_N Transket_pyr Transketolase_C ? ? ? ? PfkB ? ? DUF11 ? ? F_bP_aldolase ? BrnT_toxin

OGG18487.1
MFJK01000014.1:51585-64442

Candidatus Gottesmanbacteria bacterium RIFCSPHIGHO2_01_FULL_47_48

BrnT_toxin ? Pribosyltran Fic MazE_antitoxin CobA_CobO_BtuR ? DUF4258 BrnA_antitoxin TSCPD Ribonuc_red_lgC Ribonuc_red_lgN ATP-cone Zn_ribbon_NrdR DUF5667 ? ? PIN

OGG26026.1
MFJR01000014.1:19387-30302

Candidatus Gottesmanbacteria bacterium RIFCSPLOWO2_01_FULL_39_12b

Thioredoxin Pyr_redox_2 ? ? ? ? Seryl_tRNA_N tRNA-synt_2b BrnT_toxin CopG_antitoxin ? RH3_dom PIN ? ? ?

OGG53390.1
MFKT01000013.1:30342-39358

Candidatus Kaiserbacteria bacterium RIFCSPHIGHO2_01_FULL_53_29

? ? BrnT_toxin BrnA_antitoxin CopG_antitoxin BrnT_toxin PKD PG_binding_1 ParE_toxin ? EFG_C EFG_IV EFG_III EF-G_D2 GTP_EFTU

OGG93235.1
MFNE01000052.1:10854-26028

Candidatus Lambdaproteobacteria bacterium RIFOXYD2_FULL_50_16

? Mem_trans Fer2 Glycos_transf_2 PMT_2 Citrate_synt ? DUF4258 CopG_antitoxin TPRs TPRs TPRs TPRs TPR_CcmH_CycH TPRs TPRs Beta-lactamase ABC_tran oligo_HPY ABC_tran oligo_HPY Phr_regulator PilZ

OGG94725.1
MFNE01000035.1:51546-66806

Candidatus Lambdaproteobacteria bacterium RIFOXYD2_FULL_50_16

? Nitroreductase GDPD DAO_C DAO FGGY_C FGGY_N SRAP Hemerythrin BrnT_toxin CopG_antitoxin ? APS_kinase GTP-eEF1A_C GTP_EFTU_D2 GTP_EFTU PAPS_reduct PAPS_reduct TusA NIR_SIR NIR_SIR_ferr NIR_SIR NIR_SIR_ferr Prok-JAB

OGH06391.1
MFNG01000038.1:9-4717

Candidatus Levybacteria bacterium RBG_13_35_9

NUDIX BrnT_toxin CopG_antitoxin DUF5673 VIT1 VIT1 PIN PrlF_antitoxin His_Phos_1 DZR_2 Pribosyltran

OGH14538.1
MFNM01000023.1:1-7542

Candidatus Levybacteria bacterium RIFCSPHIGHO2_01_FULL_37_33

BrnT_toxin CopG_antitoxin Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r1_2 Sigma70_r1_1 DnaB_bind Toprim_2 DNAG_N Zn_ribbon_DnaG LAGLIDADG_3 Zn_ribbon_DnaG ? ?

OGH18938.1
MFNR01000015.1:1-7738

Candidatus Levybacteria bacterium RIFCSPHIGHO2_01_FULL_40_15b

GIY-YIG ? ? ? ? ? Cupin_2 ? CopG_antitoxin ? Methyltransf_23 Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r1_2 Sigma70_r1_1

OGH37431.1
MFOM01000065.1:2675-13504

Candidatus Levybacteria bacterium RIFCSPLOWO2_01_FULL_36_10

? Fic ? PadR DsrE ? Beta_helix hAT-like_RNase-H CopG_antitoxin ? Methyltransf_25 ? ? 4TM_phosphoesterase GSDH

OGI07720.1
MFRO01000035.1:1-8074

Candidatus Melainabacteria bacterium RIFCSPLOWO2_12_FULL_35_11

? AAA_14 DUF4143 PHA02142 RNA_ligase BrnT_toxin BrnA_antitoxin ? MnmE_helical TrmE_N GGDEF GAF GAF

OGI20649.1
MFRM01000003.1:32839-48130

Candidatus Melainabacteria bacterium RIFCSPHIGHO2_02_FULL_34_12

? GMP_synt_C NAD_synthase GATase ? ? CBS CBS ? Bro-N DUF5128 CopG_antitoxin ? adh_short_C2 CheR PP-binding ketoacyl-synt Ketoacyl-synt_C S4 FtsJ

OGJ15620.1
MFWR01000009.1:29307-43435

Candidatus Pacebacteria bacterium RIFCSPHIGHO2_01_FULL_46_16

? ? NTP_transferase Fic_N Fic NYN Ribosomal_L33 DUF6938 ? CopG_antitoxin AbiEi_4 AbiEii ? ? CPDase EamA EamA

OGJ16277.1
MFWR01000005.1:39-12819

Candidatus Pacebacteria bacterium RIFCSPHIGHO2_01_FULL_46_16

? AbiEi_4 AbiEii AbiEi_4 AbiEii BrnT_toxin CopG_antitoxin DUF4143 AAA_14 ? AbiEii N_methyl T2SSG N_methyl ?

OGJ43096.1
MFXB01000034.1:1-9394

Candidatus Peregrinibacteria bacterium RIFCSPLOWO2_01_FULL_48_20

STY-kinase HSP20 NTP_transferase TPRs TPRs TPRs CoA_binding_3 Bac_transf BrnT_toxin CopG_antitoxin SecA_DEAD P-loop_SecA P-loop_SecA SecA_SW SEC-C HTH_3

OGK10196.1
MFZA01000021.1:47-11698

Candidatus Roizmanbacteria bacterium RIFCSPHIGHO2_01_FULL_35_10

? AbiEii ? tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 ? ? ? BrnT_toxin CopG_antitoxin ? Calcineurin CBM96 DDE_Tnp_1 ? ? ?

OGK13539.1
MFZD01000016.1:51623-62780

Candidatus Roizmanbacteria bacterium RIFCSPHIGHO2_01_FULL_38_15

? Pterin_4a MGDG_synth SMAGP Receiver G6PD_N G6PD_C NAD_binding_2 6PGD DUF4258 CopG_antitoxin Cyclase Pribosyltran Pribosyltran Pribosyltran LDcluster4 PIN

OGK19991.1
MFZI01000041.1:7502-22988

Candidatus Roizmanbacteria bacterium RIFCSPHIGHO2_01_FULL_39_8

N_methyl T2SSG ? Peptidase_M16 Peptidase_M16_C A24_N_bact Peptidase_A24 NDK NDK ? Methyltransf_13 Methyltransf_23 Methyltransf_14 BrnT_toxin CopG_antitoxin Glycos_transf_2 Glycos_transf_1 Glycos_transf_2 Hexapep_2 LbH_EIF2B TPRs TPRs TPRs TPRs 23S_rRNA_IVP

OGK23569.1
MFZN01000065.1:10818-17010

Candidatus Roizmanbacteria bacterium RIFCSPHIGHO2_02_FULL_37_9b

? tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 ? ? PIN ? ? DUF4258 CopG_antitoxin

OGK24179.1
MFZO01000038.1:9620-18087

Candidatus Roizmanbacteria bacterium RIFCSPHIGHO2_02_FULL_38_11

TetR_C_13 TetR_N Mrr_cat Aldo_ket_red PNPOx_N BrnT_toxin CopG_antitoxin Na_Ca_ex Na_Ca_ex DUF2000 Glyoxalase PhdYeFM_antitox ParE_toxin ?

OGK39142.1
MGAE01000025.1:1-10383

Candidatus Roizmanbacteria bacterium RIFCSPHIGHO2_12_FULL_44_10

? CopG_antitoxin ? ? Pep_deformylase GT-C_AftD Glycos_transf_2

OGK66569.1
MGBG01000006.1:67762-82074

Candidatus Roizmanbacteria bacterium RIFOXYA1_FULL_41_12

Glyco_transf_4 Glycos_transf_1 BSP Carbam_trans_C Carbam_trans_N ? ? ? DUF4143 AAA_14 BrnT_toxin CopG_antitoxin SNase Radical_SAM APH AP_endonuc_2 Peptidase_M20 Radical_SAM SPASM Wzy_C

OGL60564.1
MGDQ01000010.1:2704-10795
Candidatus Uhrbacteria bacterium RIFCSPHIGHO2_01_FULL_46_23

BrnT_toxin BrnA_antitoxin PD40 Peptidase_S9_N ClpB_D2-small AAA_2 AAA_lid_9 AAA Clp_N ? ? Apolipoprotein

OGM04571.1
MGFK01000010.1:11548-19707

Candidatus Woesebacteria bacterium GWA1_42_12

? ? ? CBM96 ? UbiA New_glue BrnT_toxin CopG_antitoxin Gp49 HTH_3 VTT_dom HD HTH_23 ?

OGM18969.1
MGGD01000084.1:1-6448

Candidatus Woesebacteria bacterium RIFCSPHIGHO2_01_FULL_38_26b

PIN PhdYeFM_antitox ? ? ? ? ? DUF4258 CopG_antitoxin CTP_transf_like NUDIX ? ?

OGM28184.1
MGGM01000033.1:172-5274

Candidatus Woesebacteria bacterium RIFCSPHIGHO2_01_FULL_41_10

DUF4258 CopG_antitoxin ? HIT Ldh_2 RibD_C ? ?

OGM59330.1
MGHC01000023.1:1-5726

Candidatus Woesebacteria bacterium RIFCSPLOWO2_01_FULL_39_10

? ? CopG_antitoxin ? Sulfotransfer_1 Lysine_decarbox

OGM64487.1
MGHH01000009.1:1-5086

Candidatus Woesebacteria bacterium RIFCSPLOWO2_01_FULL_39_25

HTH_29 rve rve_3 BrnT_toxin CopG_antitoxin PsbP_2 DUF7387 HicA_toxin ?

OGO22587.1
MGNL01000125.1:1-10511

Chloroflexi bacterium RBG_16_51_16

? DHO_dh DinB OmdA DUF1905 OMPdecase OTCace OTCace_N Amidohydro_1 BrnT_toxin CopG_antitoxin DUF2867 NmrA DUF2867 NAD_binding_10

OGO92793.1
MGOZ01000026.1:27088-42805

Coxiella sp. RIFCSPHIGHO2_12_FULL_42_15

TraW-like ? IcmW Phage_AlpA ? ? ArsP_1 BrnT_toxin CopG_antitoxin MFS_1 BCCT T2SSE T4SS_TraI DotD Trp_Tyr_perm

OGP04925.1
MGPE01000006.1:84246-97762

Deltaproteobacteria bacterium GWA2_38_16

Band-7 ? HTH_3 HicB-like_2 HicA_toxin DUF4143 AAA_14 ? ? BrnT_toxin ? ? SLT ? AMMECR1 Competence Lactamase_B Peptidase_S41

OGP07281.1
MGPH01000066.1:40-15837

Deltaproteobacteria bacterium GWA2_45_12

? DUF4258 MqsA_antitoxin BrnT_toxin BrnA_antitoxin CopG_antitoxin BrnT_toxin BrnT_toxin CopG_antitoxin ? ? PutA_N Pro_dh Aldedh Methyltransf_25

OGP87012.1
MGQP01000192.1:476-5228
Deltaproteobacteria bacterium RBG_16_48_10

BrnT_toxin CopG_antitoxin LysM ? TIR Receiver His-Kinase His-Kinase ?

OGP93414.1
MGQP01000010.1:9074-21012
Deltaproteobacteria bacterium RBG_16_48_10

GDC-P Aminotran_5 GcvP2_C LplA-B_cat Pyr_redox_2 Pyr_redox_dim LIAS_N Radical_SAM Transketolase_N Transket_pyr Transketolase_C_1 DUF4019 BrnT_toxin CopG_antitoxin ? ?

OGQ06074.1
MGRG01000005.1:9102-20080

Deltaproteobacteria bacterium RIFCSPHIGHO2_01_FULL_43_49

Aldo_ket_red FolB PTPS adh_short Radical_SAM GTP_cyclohydroI Methyltransf_33 BrnT_toxin ? AhpC-TSA GCV_T_C GCV_T GCV_T MFS_1_like 2Fe-2S_thioredx FGGY_N FGGY_C ?

OGQ36172.1
MGRR01000040.1:7021-19468

Deltaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_43_9

PBP E1_dh Transket_pyr Transketolase_C LMBR1 CorA Glyco_transf_9 BrnT_toxin ? NTP_transf_5 RtcB Archease CopG_antitoxin BrnT_toxin HTH_3

OGQ44517.1
MGSD01000120.1:120-10196

Deltaproteobacteria bacterium RIFCSPLOWO2_02_FULL_46_8

Acyltransferase GPAT_C ? TPRs TPRs TPRs ? FGS DUF4160 ? BrnT_toxin BrnA_antitoxin Asp_protease_2 ? ?

OGR45946.1
MGTX01000034.1:60692-70470

Elusimicrobia bacterium GWA2_66_18

BrnT_toxin CopG_antitoxin RHH_1 BrnA_antitoxin BrnT_toxin Peptidase_M48 ? FIST_C FIST ? NadA GDPD PHP

OGR46636.1
MGTX01000028.1:5355-19244

Elusimicrobia bacterium GWA2_66_18

Aldedh Aminotran_3 PNPOx_N HTH_18 CoA_trans PhdYeFM_antitox YoeB_toxin CopG_antitoxin PhdYeFM_antitox ParE_toxin CoA_trans ABC_tran ABC_tran_Xtn ABC_tran Acyl-CoA_dh_1 Acyl-CoA_dh_M Acyl-CoA_dh_N 3HCDH 3HCDH_N ?

OGR52092.1
MGTR01000115.1:1-16938

Elusimicrobia bacterium GWA2_38_7

BrnT_toxin CopG_antitoxin H_PPase BCA_ABC_TP_C ABC_tran BCA_ABC_TP_C ABC_tran BPD_transp_2 GGDEF Receiver PDZ_2 PDZ_2 Trypsin

OGR53299.1
MGTV01000007.1:87661-94758

Elusimicrobia bacterium GWA2_62_23

ORF6N Calcineurin ORF6N PHM7_cyt PEMT BrnT_toxin CopG_antitoxin 2-thiour_desulf Sigma70_r2 Sigma70_r4_2 ? ? Tll0287-like Phage_holin_3_3

OGR82415.1
MGUM01000070.1:94-7140

Elusimicrobia bacterium RIFCSPHIGHO2_01_FULL_64_10

? HTH_31 ? SelR UDG BrnT_toxin CopG_antitoxin GIY-YIG DUF2283

OGS27674.1
MGVM01000050.1:2733-11574

Elusimicrobia bacterium RIFOXYB2_FULL_48_7

DUF1846 DUF1846_C NADase_transloc Intein_splicing BrnT_toxin CopG_antitoxin Amidohydro_2 POR Symport_access SSF SSF Alginate_exp AMP-binding AMP-binding_C_2

OGS37376.1
MGVN01000028.1:1-15454

Elusimicrobia bacterium RIFOXYB2_FULL_49_7

Glyco_transf_4 Glycos_transf_1 Wzz FPP GNVR Wzy_C Glycos_transf_1 Glyco_transf_4 Glyco_transf_4 Glycos_transf_1 Methyltransf_25 BrnT_toxin CopG_antitoxin DUF4143 AAA_14 AAA_14 DUF4143 DUF4143 AAA_14 YfhO

OGS95593.1
MGWZ01000079.1:100-9060

Gallionellales bacterium RIFCSPLOWO2_02_FULL_57_47

? Phage_int_SAM_4 Phage_integrase BrnT_toxin CopG_antitoxin NTP_transf_3 GMPPB_C GMPPB_C Hexapep GATase_6 SIS SIS UbiB_N ABC1 Tim44

OGT08932.1
MGXG01000005.1:2978-21280

Gammaproteobacteria bacterium GWE2_37_16

ABC-ATPase RloB RloB Colicin_V CysS_C DALR_2 tRNA-synt_1e Calcineurin FAD_syn Flavokinase BrnT_toxin CopG_antitoxin tRNA-synt_1 Anticodon_1 zf-FPG_IleRS Gp49 HTH_3 Peptidase_A8 MraY_sig1 Glycos_transf_4 MurD-like_N Mur_ligase_M Mur_ligase_C

OGT09393.1
MGXH01000079.1:1-6145

Gammaproteobacteria bacterium GWE2_42_36

DUF1634 TauE DNA_gyraseB_C Toprim DNA_gyraseB His-Kinase BrnT_toxin CopG_antitoxin CopG_antitoxin BrnT_toxin

OGT32089.1
MGXK01000096.1:195-7355

Gammaproteobacteria bacterium RBG_16_51_14

? Fer4_5 Fer4_18 FixG_C ? SBP_bac_11 BPD_transp_1 ABC_tran HTH_Crp_2 cNMPBD BrnT_toxin CopG_antitoxin

OGT37876.1
MGXY01000037.1:13477-20177

Gammaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_38_14

? CopG_antitoxin Glucokinase Sugar_tr FMN_red Zn_ribbon_8 DUF502 N-OB tRNA-synt_2

OGT39309.1
MGXU01000010.1:438193-452225

Gammaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_36_30

CDP-OH_P_transf ? ? ? Tc-R-P Arginase DS BrnT_toxin DUF6364 MarR GCS2 MacB_PCD FtsX ABC_tran GTP-bdg_N GTP-bdg_M MMR_HSR1 MMR_HSR1_Xtn ?

OGT46056.1
MGXX01000059.1:83920-111497

Gammaproteobacteria bacterium RIFCSPHIGHO2_12_FULL_38_11

PBP CSD Ribosomal_S30AE ? MFS_1 MFS_1 MFS_1 MFS_1 ? ? ? CopG_antitoxin BrnT_toxin CopG_antitoxin TK Thioredoxin DUF6444 DDE_Tnp_IS66

OGU37842.1
MGZV01000103.1:5186-16983

Ignavibacteria bacterium GWB2_35_12

SfNPV_sf27 ? PIN ? RHH_1 PIN Beta-prop_THOC3 Por_Secre_tail BrnT_toxin ? ? ? ? Adenine_deam_C Amidohydro_1 PIN ?

OGU72046.1
MHAK01000103.1:464-8879

Ignavibacteria bacterium RIFCSPLOWO2_12_FULL_56_21

DUF6242_C DUF6242_C ? ? ? BrnT_toxin CopG_antitoxin Guanylate_cyc HNOBA DUF4238

OGV19018.1
MHBE01000040.1:132107-147354

Lentisphaerae bacterium GWF2_38_69

Cas_Csd1 Cas_Cas7 HD DEAD Cas3-like_C_2 Cnd1 Cas_Cas4 Cas_Cas1 CRISPR_Cas2 DUF4258 CopG_antitoxin Flavodoxin_2 Na_H_Exchanger TrkA_N KefB_C Peptidase_S9 Abhydrolase_1 FAD-oxidase_C FAD_binding_4 PIN FitA-like_RHH

OGV27165.1
MHBC01000020.1:245-8800

Legionellales bacterium RIFCSPHIGHO2_12_FULL_37_14

DUF4143 AAA_14 BrnT_toxin CopG_antitoxin AAA_14 DUF4143 ? NMT1 Pyr_redox_2 DciA IMPDH

OGV37756.1
MHBH01000067.1:68225-87998

Lentisphaerae bacterium GWF2_49_21

GDPD CBM9_1 SBP_bac_10 N_methyl Cellulase SBP_bac_10 N_methyl SBP_bac_8 GntR DUF5123 Beta_helix Beta_helix GH141_M BrnT_toxin CopG_antitoxin ? LacI Peripla_BP_3 ? Glyco_hydro_16 Glyco_hydro_130 ?

OGV60773.1
MHBM01000255.1:6391-20995

Lentisphaerae bacterium RIFOXYA12_FULL_48_11

GFO_IDH_MocA GFO_IDH_MocA_C3 TAT_signal TPRs TPRs TPRs DUF4034 ? ? ? DUF4019 BrnT_toxin CopG_antitoxin C_GCAxxG_C_C Unstab_antitox ParE_toxin PcRGLX_1st PcRGLX_2nd PcRGLX_2nd PcRGLX_3rd Calcineurin AP_endonuc_2

OGV89376.1
MHBP01000181.1:160-5953

Lentisphaerae bacterium RIFOXYB12_FULL_65_16

BrnT_toxin CopG_antitoxin VapB_antitoxin PIN HTH_18 ASL_C2 ? Ig_mannosidase Mannosidase_ig Glyco_hydro_2 Glyco_hydro_2_N2

OGV95251.1
MHBP01000012.1:270-6745

Lentisphaerae bacterium RIFOXYB12_FULL_65_16

N_NLPC_P60 BrnT_toxin CopG_antitoxin VapB_antitoxin PIN HTH_18 ASL_C2 ? Ig_mannosidase Mannosidase_ig Glyco_hydro_2 Glyco_hydro_2_N2

OGW60490.1
MHEK01000072.1:18515-31680

Nitrospirae bacterium RBG_16_64_22

Polbeta HepT-like Phenol_MetA_deg FeoA Fe_dep_repr_C Gate FeoB_C Gate FeoB_Cyto FeoB_N FTR1 FUR BrnT_toxin CopG_antitoxin MFS_1 MFS_1 Chromate_transp Chromate_transp Transglutaminase ? KhtT_N TrkA_C Na_H_Exchanger

OGW76331.1
MHFH01000083.1:1-7371

Omnitrophica bacterium RIFCSPHIGHO2_02_FULL_49_9

LNT_N CN_hydrolase ? BrnT_toxin CopG_antitoxin

OGW84600.1
MHFM01000073.1:1-7625

Omnitrophica bacterium RIFCSPLOWO2_01_FULL_45_10b

DUF5128 CopG_antitoxin ? ? Complex1_30kDa NiFeSe_Hases Complex1_49kDa Fer4_7 Fer4_16 NADHdh

OGW90472.1
MHFK01000097.1:832-5947

Omnitrophica bacterium RIFCSPHIGHO2_12_FULL_44_12

? DUF2934 DUF1016_N YhcG_C BrnT_toxin CopG_antitoxin ?

OGX05186.1
MHFT01000055.1:6058-19516

Omnitrophica bacterium RIFCSPLOWO2_12_FULL_50_11

B12-binding Radical_SAM DUF4070 ? ? DoxX LTXXQ CopG_antitoxin QueH ? CopG_antitoxin ? Yip1 Peptidase_M20 AstA Aldedh DDAH_eukar

OGX05657.1
MHFY01000150.1:18592-31675

Omnitrophica WOR_2 bacterium GWA2_47_8

HD_4 ORF6N ? DUF6036 ? DUF6691 CSD ? CopG_antitoxin MBOAT ? ECH_1 UbiA MR_MLE_C AMP-binding AMP-binding AMP-binding_C

OGX08373.1
MHFY01000030.1:1-15924

Omnitrophica WOR_2 bacterium GWA2_47_8

HHH_3 Anticodon_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 AdoMet_dc ThiS ThiG Radical_SAM BATS DUF4145 ? VTT_dom SAM_GIDA_C GIDA_C_1st GIDA ? Peripla_BP_3 LacI FlgT_C HAUS1

OGX11005.1
MHFU01000159.1:6743-16230

Omnitrophica bacterium RIFOXYB12_FULL_50_7

Receiver tRNA-synt_1c tRNA-synt_1c Anticodon_2 Receiver ? CopG_antitoxin tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2 ? YhcG_C DUF1016_N ? ? ?

OGX16052.1
MHGH01000010.1:33228-45549

Omnitrophica WOR_2 bacterium RBG_13_41_10

HAMP ? RRM_1 Radical_SAM B12-binding ? GIY-YIG ? ? CopG_antitoxin FMN_red ? Zn_Ribbon_1 SpoIIIAH E1-E2_ATPase Hydrolase Hydrolase_3 CSD

OGX27370.1
MHGM01000099.1:1-9126

Omnitrophica WOR_2 bacterium RIFCSPHIGHO2_01_FULL_52_10

PseudoU_synth_1 PseudoU_synth_1 ? ? ? CopG_antitoxin Ribosomal_L13 Ribosomal_S9 Semialdhyde_dh Semialdhyde_dhC_1 ArgJ ArgJ AA_kinase Aminotran_3

OGX28372.1
MHGM01000066.1:1-5115

Omnitrophica WOR_2 bacterium RIFCSPHIGHO2_01_FULL_52_10

? Unstab_antitox BrnT_toxin CopG_antitoxin Pribosyltran OTCace_N OTCace

OGX37979.1
MHGQ01000044.1:189-7855
Omnitrophica WOR_2 bacterium RIFCSPHIGHO2_02_FULL_50_17

BrnT_toxin BrnA_antitoxin DUF2442 BrnT_toxin CopG_antitoxin BrnT_toxin CopG_antitoxin MazE_antitoxin ParE_toxin TMEM164 Glyco_hydro_1 Glyco_hydro_1 ? YfiO

OGY13047.1
MHCA01000003.1:1-6121

Candidatus Blackburnbacteria bacterium RIFCSPHIGHO2_12_FULL_41_13b

? RelB YafQ_toxin DUF4258 CopG_antitoxin ? Acetylase RHH_1 PIN ? DEAD Helicase_C

OGY88036.1
MHKI01000005.1:29861-41103

Candidatus Kerfeldbacteria bacterium RIFOXYB2_FULL_38_14

Fer2 Acyl_transf_3 POR POR_N PFOR_II TPP_enzyme_C PFO_beta_C T2SSE Acetylase ? BrnT_toxin CopG_antitoxin 60KD_IMP YidD Ribonuclease_P Ribosomal_L34 DnaA_N Bac_DnaA Bac_DnaA_C DNA_pol3_beta DNA_pol3_beta_2 DNA_pol3_beta_3

OGY91417.1
MHKO01000047.1:14491-25852

Candidatus Komeilibacteria bacterium RIFCSPLOWO2_02_FULL_48_11

? ParE_toxin UPF0020 YchF-GTPase_C MMR_HSR1 CopG_antitoxin BrnT_toxin ? BrnT_toxin CopG_antitoxin Translin ? 5_3_exonuc_N 5_3_exonuc DNA_pol_A_exo1 DNA_pol_A PDDEXK_3 DNA_pol3_delta_C Ribosomal_S20p

OGZ08567.1
MHLL01000032.1:24912-40257

Candidatus Lloydbacteria bacterium RIFCSPHIGHO2_02_FULL_50_13

AAA_14 DUF4143 Polbeta HepT-like DUF6922 AbiEii GDP_Man_Dehyd NTP_transferase BrnT_toxin CopG_antitoxin PIN Wzy_C TPRs TPRs FAD_binding_4 MurB_C SecA_DEAD P-loop_SecA P-loop_SecA SecA_SW SecA_SW SEC-C ? Cu-oxidase_4

OHA65871.1
MHTV01000040.1:640-5384

Candidatus Wildermuthbacteria bacterium RIFCSPHIGHO2_02_FULL_45_25

BrnT_toxin CopG_antitoxin Ribosomal_S20p HepT-like Polbeta UVR UvrB Helicase_C UvrB_3rd UvrB_inter ResIII RuvA_N HHH_5 RuvA_C ?

OHA82998.1
MHUW01000021.1:612-10764

Candidatus Yonathbacteria bacterium RIFCSPLOWO2_01_FULL_47_33b

DUF3825 ? S1 RNB CSD2 OB_RNB CBM96 Peptidase_M23 PsbP_2 Ribosomal_S16 KH_KhpA-B BrnT_toxin CopG_antitoxin tRNA_m1G_MT DUF5678 Asp_protease_2 DUF1664 ? SurA_N_3 Trypsin

OHD08215.1
MIAO01000125.1:6768-20130

Spirochaetes bacterium GWB1_36_13

Receiver His-Kinase HisKA_2 SBP_bac_3 TfoX_N DUF2196 Abhydrolase_6 ? DUF3795 BrnT_toxin ? Nitroreductase TPR_BSK1_C TPRs TPRs Band-7 Guanylate_cyc ? ?

OHD44747.1
MIBA01000110.1:1-12902

Spirochaetes bacterium GWE2_31_10

? ? His-Kinase His-Kinase SpoIIE 7TMR-DISM_7TM ? AAA_lid_2 AAA_3 Acetylase BrnT_toxin CopG_antitoxin ? PIN

OHD67145.1
MIBJ01000083.1:20705-26867

Spirochaetes bacterium RBG_16_49_21

? DUF4258 BrnT_toxin CopG_antitoxin ? HTH_3 HSP90 HATPase_c_3 ? DUF1178 ?

OHD76498.1
MIBL01000456.1:1-875

Spirochaetes bacterium RIFOXYB1_FULL_32_8

BrnT_toxin

OHD76702.1
MIBK01000017.1:15902-25812

Spirochaetes bacterium RBG_16_67_19

? ? HicB_lk_antitox Plug TonB_dep_Rec_b-barrel ? CopG_antitoxin TonB_C ExbD MotA_ExbB ? ? SSB

OHD83243.1
MIBL01000025.1:1-3319

Spirochaetes bacterium RIFOXYB1_FULL_32_8

Acetylase BrnT_toxin CopG_antitoxin ? PIN

OHE23387.1
MICN01000068.1:1-6976

Syntrophus sp. GWC2_56_31

HHH_3 Tex_YqgF Tex_central_region Tex_N BrnT_toxin CopG_antitoxin P-II Ammonium_transp T2SSF_N T2SSF T2SSF N_methyl T2SSG N_methyl

OHE59157.1
MIDI01000078.1:1-9832

Thermodesulfovibrio sp. RBG_19FT_COMBO_42_12

BrnT_toxin CopG_antitoxin ? LysM FecR CHASE2 TRAP-gamma HD_5 ? Zn_ribbon_TFIIB Zn_ribbon_TFIIB Peptidase_M48 LemA

OHE73874.1
MIDP01000015.1:19385-34960

Verrucomicrobia bacterium GWF2_51_19

DUF374 AAA_14 DUF4143 PEP-utilizers_C PEP-utilizers PEP-utilisers_N MARVEL Biotin_lipoyl OAD_beta AcetylCoA_hydro AcetylCoA_hyd_C BrnT_toxin CopG_antitoxin Choline_kinase Zn_ribbon_9 CT398_CC Epimerase YjeF_N Carb_kinase RrnaAD ANKs

OIO08198.1
MNUU01000020.1:11120-14639

Candidatus Falkowbacteria bacterium CG1_02_37_44

Polbeta DUF3467 ? HicA_toxin ? ? ? BrnT_toxin CopG_antitoxin

OIO47510.1
MNWF01000023.1:23792-39198

Parcubacteria group bacterium CG1_02_40_25

HGTP_anticodon tRNA-synt_2b tRNA_SAD ? N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP OmpH GMP_synt_C NAD_synthase GATase ? RNase_HII RNase_HII RNase_HII BrnT_toxin CopG_antitoxin DNA_pol_B_exo2 CDP-OH_P_transf DUF498 YgjP-like ? Ribosomal_L32p

OIO58945.1
MNWS01000230.1:17164-26181

Candidatus Marinimicrobia bacterium CG1_02_48_14

? ? ? BrnT_toxin CopG_antitoxin ? AF_1327 RibD_C Synaptojanin Por_Secre_tail UbiA Metallophos_2

OIO78807.1
MNXF01000059.1:407-5441

Candidatus Omnitrophica bacterium CG1_02_43_210

? CopG_antitoxin ? MarC NTP_transferase Hexapep_GlmU Acetylase Aconitase Aconitase Aconitase Aconitase_C

OIP03985.1
MNXQ01000018.1:157-5326
Candidatus Beckwithbacteria bacterium CG2_30_44_31

DHO_dh ? Pribosyltran OMPdecase ? DUF4258 CopG_antitoxin ?

OIP37768.1
MNYI01000198.1:14145-26759

Candidatus Desantisbacteria bacterium CG2_30_40_21

Aminotran_4 AICARFT_IMPCHas MGS Glyco_transf_4 Lip_A_acyltrans PmbA_TldD_3rd PmbA_TldD_2nd PmbA_TldD_1st His-Kinase His-Kinase PAS 6TM_PilS Zn_ribbon_TFIIB BrnT_toxin ? GIDA GIDA_C_1st SAM_GIDA_C gag-asp_proteas GidB

OIP91314.1
MNZR01000268.1:393-6503

Syntrophobacteraceae bacterium CG2_30_61_12

POTRA Omp85 TamB_N TamB_N TamB_N TamB BrnT_toxin CopG_antitoxin

OJZ17500.1
MKWN01000044.1:3622-24973
Thiobacillus sp. 65-29

DUF494 Toprim Topoisom_bac Zn_ribbon_Top1 Toprim_C_rpt Toprim_C_rpt Toprim_C_rpt ? Phosphonate-bd Rhodanese EAL GGDEF PAS CMD ? BrnT_toxin TauE EVE HsdM_N N6_Mtase Methylase_S Methylase_S Fic_N Fic SoFic-like_C ResIII Helicase_C EcoEI_R_C AAA_16 BTAD

OPZ75870.1
MWAB01000004.1:33-10083
Verrucomicrobia bacterium ADurb.Bin474

? ARG Mrr_N Mrr_cat ? ? ? BrnT_toxin CopG_antitoxin DUF4214 Pyr_redox_2 NDH2_C Cupin_5 ? Zn_ribbon_8 ?

OQC15119.1
MWEA01000067.1:687-6774
Alphaproteobacteria bacterium ADurb.Bin100

PIN ? ? HIRAN ? ? ? BrnT_toxin CopG_antitoxin EamA EamA

OQC30226.1
MWEM01000055.1:515-10317
Thermotogae bacterium ADurb.Bin062

BrnT_toxin CopG_antitoxin DEAD Helicase_C DeaD_dimer DbpA Epimerase_2 ? Epimerase Epimerase Epimerase

OQW89759.1
MTEI01000001.1:80152-93641
Rhodoferax ferrireducens

N-OB tRNA-synt_2 YgjP-like Hpt Receiver His-Kinase His-Kinase PAS PAS CHASE Receiver CBS QSregVF_b SUI1 ? CopG_antitoxin ? Glyco_hydro_3 ACPS PdxJ RecO_C RecO_N KH_2 MMR_HSR1

OQX13444.1
MTEO01000147.1:1-11855
Desulfobulbaceae bacterium A2

Hydrolase KdpC KdpD Usp DUF4118 GAF_3 His-Kinase His-Kinase Receiver Trans_reg_C BrnT_toxin CopG_antitoxin AAA_14 DUF4143 TauE PadR PAP2 Receiver Trans_reg_C HAMP His-Kinase His-Kinase

OQX57973.1
MZGN01000088.1:22-5950

Helicobacteraceae bacterium 4484_230

DCD DUF465 CBS CBS zf-CDGSH zf-CDGSH BrnT_toxin CopG_antitoxin CopG_antitoxin ? ?

OQX94164.1
NBLK01000022.1:762-13986

Rickettsiaceae bacterium 4572_127

? SocA_Panacea Exonuc_VII_L tRNA_anti_2 Virulence_RhuM TsaD ? ? DUF4920 CopG_antitoxin AAA_14 DUF4143 TrmE_N MnmE_helical FAD_binding_6 NAD_binding_1 FtsJ Dus PIN

OQX95023.1
NBLI01000186.1:327-5247

candidate division KSB1 bacterium 4572_119

BrnT_toxin CopG_antitoxin ? Acetylase PNPase PFK Tricorn_N

OQY13114.1
NBLS01000028.1:512-6646

Desulfobacteraceae bacterium 4572_19

Polbeta ? DUF4172 Fic DIP2311-like_C ? ? Acetylase FMN_red BrnT_toxin BrnA_antitoxin

OQY65514.1
MWSR01000072.1:9176-20614
Microgenomates bacterium UTCPR1

? tRNA-synt_1b RRM_1 CTP_transf_like CTP_transf_like DUF359 DUF1599 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 tRNA-synt_1 Anticodon_1 ? ? CopG_antitoxin ? BrnT_toxin CopG_antitoxin HD EamA EamA EamA

OQY68538.1
MWSR01000018.1:114-7643

Microgenomates bacterium UTCPR1

Polbeta HD FtsK_gamma SIMPL ? adh_short_C2 ? BrnT_toxin CopG_antitoxin APH ?

OTE95834.1
MBQZ01000053.1:1-10722
Crenothrix sp. D3

HsdM_N N6_Mtase Methylase_S Methylase_S BrnT_toxin CopG_antitoxin ResIII Helicase_C EcoEI_R_C ABC_tran ABC_tran_Xtn ABC_tran ABC_tran Lipase_GDSL_2 HbpS-like ?

OUR96565.1
MAAO01000006.1:486455-501256
Halobacteriovorax marinus

DHO_dh TPR_NPHP3 QueC PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I Fer4_13 Na_H_antiport_3 ORF6N BrnT_toxin BrnA_antitoxin ORF6N Aminotran_4 ? ? DUF4160 ?

OWV00779.1
MWQE01000003.1:230162-245310
Fibrobacter sp. UWR2

Glycos_transf_1 ? ? Methyltransf_21 Wzt_C ABC_tran AAA_15 ? BrnT_toxin CopG_antitoxin ? ? HTH_3 Peptidase_M78 ABC2_membrane Glycos_transf_2 GFO_IDH_MocA_C3 GFO_IDH_MocA

OWY64260.1
KZ263035.1:73-11109
cyanobacterium TDX16

GIIM RVT_1 RVT_N DDE_Tnp_1_5 ? BrnT_toxin CopG_antitoxin DDE_Tnp_IS1 DUF7838 Zn_ribbon_ISL3 HTH_Tnp_ISL3 DDE_Tnp_ISL3

OYU20017.1
NKIT01000007.1:5573-23250
Rhodobacteraceae bacterium PARR1

CoA_transf_3 Synaptojanin DAP_epimerase DAP_epimerase UPF0004 Radical_SAM ? ABhydrolase AAA_2 AAA BrnT_toxin BrnA_antitoxin Proteasome Thioredoxin PDDEXK_1 UvrD_C UvrD-helicase PDDEXK_1 NTP_transf_3 APH

OYW00949.1
NCBU01000058.1:1337-15626
Acidiphilium sp. 37-64-53

DUF6536 ? HicB_lk_antitox HicA_toxin HNH_2 HTH_7 Resolvase MmeI_N MmeI_hel MmeI_Mtase MmeI_TRD MmeI_C BrnT_toxin BrnA_antitoxin NicO TPRs Resolvase HTH_7 ParE_toxin PEP-CTERM PEP-CTERM

OYZ34150.1
NCGU01000026.1:2992-16852
Campylobacterales bacterium 16-40-21

Hpt H-kinase_dim His-Kinase CheW Receiver CheW AMP_N Peptidase_M24 DUF445 DUF445 Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small ? ? BrnT_toxin CopG_antitoxin CiaI Ribosomal_L34 Ribonuclease_P YidD YidC_periplas 60KD_IMP Jag_N KH_KhpA-B HP1451_C

PCH56887.1
NVQV01000028.1:1-6353
Legionellales bacterium

AMP-binding_C BrnT_toxin CopG_antitoxin ? ATP-synt_A ATP-synt_C OSCP ATP-synt_ab_N ATP-synt_ab ATP-synt_ab_C ATP-synt

PCI28024.1
NVSR01000042.1:1-5980
SAR324 cluster bacterium

Couple_hipA HipA_C PhdYeFM_antitox ParE_toxin PhdYeFM_antitox YoeB_toxin Wzz GNVR BrnT_toxin CopG_antitoxin ? LepA_C EFG_C EFG_III

PCI39210.1
NVTF01000021.1:4471-21027
Elusimicrobia bacterium

Phage_int_SAM_4 Phage_integrase Zn_Tnp_IS91 Y2_Tnp IstB_IS21 ? ? Gp49 HTH_3 ? CopG_antitoxin TPRs TPRs HEATs HEATs BrnA_antitoxin Amidase_3 Peptidase_S8 HEATs HEATs Peptidase_S8

PCJ44656.1
NVVG01000047.1:17857-23793
Moraxellaceae bacterium

tRNA-synt_2d Phe_tRNA-synt_N Ribosomal_L20 Ribosomal_L35p IF3_C IF3_N Epimerase ParE_toxin RelB3_C RHH_1 BrnT_toxin BrnA_antitoxin UVR UvrB Helicase_C

PCJ50484.1
NVWF01000002.1:570-8363
Gammaproteobacteria bacterium

MmgE_PrpD_C MmgE_PrpD_N RHH_1 PIN HTH_18 Cupin_2 Cys_Met_Meta_PP ? ? BrnT_toxin CopG_antitoxin DUF6090

PHR53380.1
NVXJ01000048.1:204-8181
Arcobacter sp.

PhdYeFM_antitox ParE_toxin Polbeta HepT-like PhdYeFM_antitox BrnT_toxin CopG_antitoxin DegT_DnrJ_EryC1 Polysacc_synt_2 HHH_3 DNA_ligase_OB_2

PHS15905.1
NVVT01000013.1:84653-101363
Kangiella sp.

Ni_hydr_CYTB HTH_1 LysR_substrate G_glu_transpept Sigma70_r2 Sigma70_r4_2 FecR FecR_C Plug TonB_dep_Rec_b-barrel Pepdidase_M14_N Peptidase_M14 ? CopG_antitoxin Acetylase UPF0014 Arylesterase FA_hydroxylase GGDEF ?

PHS24795.1
NVVW01000014.1:767-8630
Methylophaga sp.

BrnT_toxin CopG_antitoxin GlnD_5th HD GlnD_3rd GlnD_UR_UTase NTP_transf_2 Peptidase_M24 Ribosomal_S2 EF-Ts_N EF_TS AA_kinase RRF

PHS37474.1
NVUY01000021.1:38286-47517
Sulfurovum sp.

SocA_Panacea ? BrnT_toxin CopG_antitoxin ? Thioredoxin MFS_1 MFS_1 NIR_SIR_ferr NIR_SIR NIR_SIR_ferr NIR_SIR Fer4_11 DmsC

PHS42280.1
NVUY01000001.1:117701-134499
Sulfurovum sp.

EF-Ts_N EF_TS EF_TS ABC_tran RF-1 PCRF Pantoate_ligase Transpeptidase PBP_dimer FTSW_RODA_SPOVE Glyco_transf_28 Glyco_tran_28_C ? CopG_antitoxin ? tRNA-synt_2c tRNA_SAD DHHA1 ? Maf Aminotran_1_2 Transgly Transpeptidase

PID34901.1
PDOX01000016.1:1-7544
candidate division SR1 bacterium

Glycos_trans_3N PYNP_C tRNA-synt_1c Anticodon_2 ? Abi_2 ? CopG_antitoxin ?

PID34937.1
PDOX01000012.1:1-4450
candidate division SR1 bacterium

DUF1902 HicA_toxin bPH_2 SHOCT DUF2759 ? DUF4258 CopG_antitoxin ? PhageMin_Tail

PID35034.1
PDOX01000004.1:1-4891
candidate division SR1 bacterium

? DUF4258 CopG_antitoxin NUDIX ResIII UvrB_inter UvrB_3rd Enolase_N Enolase_C ?

PID84358.1
PDRA01000010.1:2205-9863
Candidatus Gracilibacteria bacterium

Ribosomal_S2 MS_channel_1st MS_channel_2nd ? ? HD_4 RelA_SpoT TGS RelA_RIS ACT ? ? BrnT_toxin CopG_antitoxin

PIE70024.1
PDTL01000032.1:40219-49785
Deltaproteobacteria bacterium

? CopG_antitoxin HTH_18 Hydrolase E1-E2_ATPase Hydrolase E1-E2_ATPase HMA_2 FeoA Hydrolase E1-E2_ATPase HMA_2 ?

PIE97368.1
PDOP01000056.1:7091-12678
Treponema sp.

LytR_C LytR_cpsA_psr HD CTP_transf_like MMR_HSR1 GTP1_OBG Ribosomal_L27 Peptidase_Prp Ribosomal_L21p BrnT_toxin CopG_antitoxin

PIP04548.1
PCQY01000024.1:168-13674
candidate division WWE3 bacterium CG23_combo_of_CG06-09_8_20_14_all_40_14

Q_salvage HSBP1 HTH_3 Radical_SAM SPASM ? DUF4143 AAA_14 ? AbiEii BrnT_toxin CopG_antitoxin Radical_SAM GlyRS_ins_1 tRNA-synt_2b HGTP_anticodon ? ? ? AbiEii

PIP62774.1
PCTB01000042.1:4-4661

Candidatus Roizmanbacteria bacterium CG22_combo_CG10-13_8_21_14_all_35_9

GT87 Polbeta DUF3467 ? HicA_toxin ? ? BrnT_toxin

PIP63122.1
PCTB01000011.1:114-3986

Candidatus Roizmanbacteria bacterium CG22_combo_CG10-13_8_21_14_all_35_9

PIN ? CopG_antitoxin ? ?

PIP75572.1
PCTN01000132.1:32-1656

Candidatus Kuenenbacteria bacterium CG22_combo_CG10-13_8_21_14_all_39_9

BrnT_toxin CopG_antitoxin ? ? Lysis_col

PIQ72048.1
PCVL01000084.1:29-6922

Candidatus Roizmanbacteria bacterium CG11_big_fil_rev_8_21_14_0_20_35_14

? DUF5663 ? ? CopG_antitoxin

PIQ80106.1
PCVT01000084.1:1-4463

Parcubacteria group bacterium CG11_big_fil_rev_8_21_14_0_20_41_14

Band-7 DNA_alkylation AAA AAA_assoc_2 MgsA_C BrnT_toxin CopG_antitoxin ?

PIQ91549.1
PCWC01000083.1:6388-15164

Parcubacteria group bacterium CG11_big_fil_rev_8_21_14_0_20_39_22

HGTP_anticodon tRNA-synt_2b tRNA_SAD Cob_adeno_trans BrnT_toxin CopG_antitoxin ? YafQ_toxin ? DUF5659 ? HTH_17

PIR42544.1
PCXT01000015.1:448-8411

candidate division WWE3 bacterium CG10_big_fil_rev_8_21_14_0_10_35_32

? Mg_chelatase_C Mg_chelatase ChlI YceG MTS DUF11 PBECR3 CopG_antitoxin Y1_Tnp ? ? AbiEii MazE_antitoxin PIN

PIR43193.1
PCXU01000035.1:19118-32318

candidate division WWE3 bacterium CG10_big_fil_rev_8_21_14_0_10_32_10

? SurA_N_3 FtsX MacB_PCD ABC_tran YknX_C Beta-barrel_YknX BSH_RND His-Kinase His-Kinase Trans_reg_C Receiver ? CopG_antitoxin DUF4349 Y1_Tnp AGOG UvrD_C UvrD-helicase Peptidase_S11 Thioredoxin

PIR45236.1
PCYI01000002.1:225-7471

Candidatus Vogelbacteria bacterium CG10_big_fil_rev_8_21_14_0_10_51_16

PycTM ? ? ? TrmB RGS ? ? BrnT_toxin CopG_antitoxin TrmB NT5C His_Phos_1 His_Phos_1

PIR58716.1
PFDK01000024.1:3044-14339

Candidatus Pacebacteria bacterium CG10_big_fil_rev_8_21_14_0_10_56_10

PIN AbiEii AbiEi_4 ? Ribosomal_L33 YafQ_toxin RelB ? CopG_antitoxin DUF4200 NTP_transferase ? ? SecE NusG KOW

PIR62982.1
PFDG01000005.1:4597-12539

Candidatus Pacebacteria bacterium CG10_big_fil_rev_8_21_14_0_10_42_12

Thioredoxin_9 ABC-3 ABC_tran FUR NUDIX Methyltransf_11 DUF1189 BrnT_toxin CopG_antitoxin ? DUF5674 HIT PIN PhdYeFM_antitox Peptidase_S51

PIR86887.1
PFBD01000023.1:54331-66225

Candidatus Harrisonbacteria bacterium CG10_big_fil_rev_8_21_14_0_10_49_15

HTH_25 SLT_2 APG6_N Ribosomal_TL5_C Ribosomal_L25p dCMP_cyt_deam_1 DCD Thymidylat_synt DHFR_1 BrnT_toxin CopG_antitoxin TrmB ? F_bP_aldolase S1 S1 S1 Ribosomal_S15 NYN

PIS21802.1
PEYV01000016.1:710-9092

candidate division WWE3 bacterium CG08_land_8_20_14_0_20_41_15

? BrnT_toxin CopG_antitoxin ? ? ? ? ? ?

PIS31118.1
PEYN01000005.1:7895-14130

Candidatus Marinimicrobia bacterium CG08_land_8_20_14_0_20_45_22

BrnT_toxin CopG_antitoxin ? ? Por_Secre_tail PHP Acetylase Peptidase_S9_N Peptidase_S9

PIU00877.1
PEZS01000002.1:150642-166748

Bdellovibrionales bacterium CG10_big_fil_rev_8_21_14_0_10_45_34

HNH His_Phos_1 ? Synaptojanin Exo_endo_phos_3 Peptidase_M3 ? DUF4423 HTH_31 BrnT_toxin BrnA_antitoxin ? DUF4423 HTH_3 NTase_sub_bind Epimerase NTP_transferase GHMP_kinases_N T2SSM GHMP_kinases_C

PIU02244.1
PEZK01000020.1:19034-32535

Candidatus Shapirobacteria bacterium CG09_land_8_20_14_0_10_49_15

Mur_ligase_C Mur_ligase_M VTT_dom Ala_racemase_C Ala_racemase_N EPSP_synthase MurB_C FAD_binding_4 Mur_ligase_C Mur_ligase_M Mur_ligase Mur_ligase_C Mur_ligase_M DUF4258 CopG_antitoxin ? Transpeptidase PBP_dimer ? Methyltransf_5 MraZ MraZ ?

PIU36589.1
PEWY01000147.1:62-2799

Candidatus Roizmanbacteria bacterium CG07_land_8_20_14_0_80_34_15

? ? CopG_antitoxin ? ? CopG_antitoxin PhdYeFM_antitox PIN

PIU37122.1
PEWY01000068.1:1-8251

Candidatus Roizmanbacteria bacterium CG07_land_8_20_14_0_80_34_15

DUF2207_C HTH_23 PEMT ? Lactamase_B_2 BrnT_toxin CopG_antitoxin Pox_EPC_I2-L1 PP2C Glutaredoxin 23S_rRNA_IVP IpaB_EvcA ?

PIV08135.1
PEVA01000182.1:1-8439

Candidatus Roizmanbacteria bacterium CG03_land_8_20_14_0_80_39_12

ABC_tran FOL3_C Mur_ligase_M ? PhdYeFM_antitox BrnT_toxin CopG_antitoxin BipA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU ? Anticodon_1 tRNA-synt_1 tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1

PIV18473.1
PEUP01000057.1:248-7938

Elusimicrobia bacterium CG03_land_8_20_14_0_80_50_18

DEAD Helicase_C BrnT_toxin CopG_antitoxin Ribosomal_L20 Phe_tRNA-synt_N tRNA-synt_2d tRNA_bind B3_4 B5 tRNA_synthFbeta FDX-ACB PMT_2 Geminin ZapA

PIV70944.1
PFEV01000111.1:1-5069

Candidatus Roizmanbacteria bacterium CG17_big_fil_post_rev_8_21_14_2_50_39_7

FtsK_4TM ? ? BrnT_toxin CopG_antitoxin PaaX_M

PIV71045.1
PFEV01000084.1:1-9634

Candidatus Roizmanbacteria bacterium CG17_big_fil_post_rev_8_21_14_2_50_39_7

Peptidase_M16 ABC2_membrane_6 ? ABC_tran FOL3_C Mur_ligase_M ? PhdYeFM_antitox BrnT_toxin CopG_antitoxin ? Anticodon_1 tRNA-synt_1 tRNA-synt_1

PIY29098.1
PFKM01000096.1:1-6693

Comamonadaceae bacterium CG_4_10_14_3_um_filter_60_42

? GtrA_DPMS_TM ? BrnT_toxin CopG_antitoxin ? ? Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 AhpC-TSA TPRs Paired_CXXCH_1 Paired_CXXCH_1 Paired_CXXCH_1 NTP_transf_8

PIY79229.1
PFLO01000033.1:28114-38218

Candidatus Pacebacteria bacterium CG_4_10_14_0_8_um_filter_43_12

? ? BrnT_toxin CopG_antitoxin ? Transgly Transpeptidase Big Wzy_C FDX-ACB tRNA_synthFbeta B5 B3_4 ? tRNA-synt_2d Phe_tRNA-synt_N

PIY80362.1
PFLP01000041.1:4607-12475

Candidatus Pacebacteria bacterium CG_4_10_14_0_8_um_filter_42_14

PhdYeFM_antitox PIN CopG_antitoxin Methyltransf_11 NUDIX ? DUF1653 BrnT_toxin ? HIT HTH_Tnp_1 rve rve_3 ? Acyltransferase DUF998

PIZ65481.1
PFOC01000055.1:74-7476

Candidatus Roizmanbacteria bacterium CG_4_10_14_0_2_um_filter_39_12

BrnT_toxin CopG_antitoxin MazE_antitoxin PemK_toxin tRNA_edit Amino_oxidase GIY-YIG PBECR3 CopG_antitoxin Glyoxalase ? His-Kinase His-Kinase Trans_reg_C Receiver PDDEXK_1

PIZ65868.1
PFOD01000032.1:1-7406

Candidatus Roizmanbacteria bacterium CG_4_10_14_0_2_um_filter_36_9

DDE_Tnp_IS1595 ? CopG_antitoxin AbiEii F420_ligase Asn_synthase GATase_7 Peptidase_M48 DUF5993 LemA LysM AAA_33

PIZ66156.1
PFOD01000022.1:219-8877

Candidatus Roizmanbacteria bacterium CG_4_10_14_0_2_um_filter_36_9

? Caroten_synth ? CopG_antitoxin Glyoxalase EamA EamA Cupredoxin_1 PMT_2 Glyco_hydro_18 ?

PJA21575.1
PFQG01000152.1:1-2058
Candidatus Beckwithbacteria bacterium CG_4_10_14_0_2_um_filter_47_25

? BrnT_toxin CopG_antitoxin ? HicA_toxin ?

PJA37338.1
PFXB01000105.1:1-3691

candidate division WWE3 bacterium CG_4_9_14_3_um_filter_43_9

rve BrnT_toxin CopG_antitoxin ? ?

PJA55365.1
PFWL01000160.1:1-6404

Candidatus Roizmanbacteria bacterium CG_4_9_14_3_um_filter_33_18

? ? ? ? ZT_dimer Cation_efflux Hydrolase_3 DAO Lycopene_cycl EamA

PJC56123.1
PFRD01000078.1:1-9008

Candidatus Kaiserbacteria bacterium CG_4_9_14_0_2_um_filter_41_32

? DUF6364 ? BrnT_toxin CopG_antitoxin Hydrolase_2 ? Y1_Tnp DUF3892 T6SS_Tse3_cat ? LCAT

PJC74587.1
PFQQ01000112.1:5570-10577
Syntrophobacterales bacterium CG_4_8_14_3_um_filter_58_8

DUF4258 MqsA_antitoxin CCG CCG Fer4_8 LUD_dom ? LUD_dom Unstab_antitox ? BrnT_toxin

PKB72696.1
MUCU01000068.1:3544-19190
SAR202 cluster bacterium Io17-Chloro-G7

CCG CCG Aminotran_3 HIT Ldh_1_C Ldh_1_N DUF6932 IDH Thioredox_DsbH BrnT_toxin CopG_antitoxin NucS_C HicB_lk_antitox SnoaL HGTP_anticodon tRNA-synt_2b tRNA_SAD G3P_acyltransf ?

PKL09547.1
PGXR01000030.1:1-8272
Spirochaetae bacterium HGW-Spirochaetae-7

APH DEAD Helicase_C HisKA_2 His-Kinase RNase_T ExoI_SH3 ExoI_C PseudoU_synth_2 PIN RHH_1 BrnT_toxin

PKL14032.1
PGXS01000161.1:1-2189
Spirochaetae bacterium HGW-Spirochaetae-6

Succ_DH_flav_C MS_channel_1st MS_channel_2nd MS_channel_3rd ? NikA-like

PKM16705.1
PGZE01000005.1:52110-64137
Gammaproteobacteria bacterium HGW-Gammaproteobacteria-2

? LysR_substrate HTH_1 MS_N MS_TIM-barrel MS_C BrnT_toxin BrnA_antitoxin Fer4_20 Pyr_redox_2 DHO_dh Fer4_21 BrnT_toxin CN_hydrolase CN_hydrolase DNA_methylase

PKN76671.1
PHBM01000048.1:1-8147
Deltaproteobacteria bacterium HGW-Deltaproteobacteria-10

HEPN_AbiU2 ? BrnT_toxin CopG_antitoxin GPI2 ? MoxR-AAA VWA_CoxE Acyltransferase Ketoacyl-synt_C ketoacyl-synt

PPD31120.1
PERU01000031.1:34437-42413
Methylomonas sp.

ABC_tran MlaE MlaD STAS EPSP_synthase HisG Histidinol_dh BrnT_toxin CopG_antitoxin ? ? ?

PSP16499.1
PXQH01000055.1:672-8414
Cyanobacteria bacterium QS_8_64_29

OEP OEP Fer2 PsbM Usp Usp BT1 RPE65 SapC BrnT_toxin CopG_antitoxin

PUB74774.1
QBVC01000019.1:76817-91823
gamma proteobacterium symbiont of Ctena orbiculata

LMWPc SBF Mtrc-MtrF_II-IV_dom Cytochrom_C Cytochrom_C Pyr_redox_2 FCSD_central FCSD-flav_bind DUF1697 ? ? Peptidase_C39_2 ABC_tran ABC_tran BPD_transp_1 BPD_transp_1 LMWPc PhoU PhoU PBP_like_2

PWB76020.1
PQAR01000057.1:24259-35602
Anaerolineales bacterium

OmdA DUF1905 RelE MqsA_antitoxin Pribosyltran OMPdecase OTCace OTCace_N Acetylase BrnT_toxin CopG_antitoxin Amidohydro_1 Iso_dh Cytochrome_CBB3 Cytochrom_B_C ? Rieske Sdh_cyt

PYQ15468.1
QHVM01000053.1:40156-53587
Acidobacteria bacterium

Ribonuc_L-PSP BrnT_toxin UnbV_ASPIC FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 Cytochrom_C TrAA12 GH97_C Glyco_hydro_97 GH97_N MFS_1 MFS_1 Oxidored_FMN ? SMC_N

PYR72541.1
QHWR01000283.1:1-8587
Acidobacteria bacterium

MacB_PCD SSF Glyco_hydro_109_C GFO_IDH_MocA Phage_integrase Phage_int_SAM_1 Phage_integrase Phage_integrase Phage_int_SAM_1 BrnT_toxin ?

PYU25669.1
QHYS01000025.1:1-3866
Acidobacteria bacterium

TPRs ? Sigma70_r4_2 AhpC-TSA ? ? CopG_antitoxin

PZM83827.1
QKOG01000055.1:158-2439
Candidatus Gracilibacteria bacterium

? ? ?

PZM86995.1
QKOH01000040.1:102-10579
candidate division SR1 bacterium

DUF2304 Glycos_transf_2 DNA_gyraseB_C Toprim DNA_gyraseB His-Kinase Bps ? BrnT_toxin CopG_antitoxin ? H2O2_YaaD Pyr_excise ? DNA_gyraseA_C

PZM87285.1
QKOH01000017.1:1-14708
candidate division SR1 bacterium

Methyltransf_33 TrmB ? ? ? ? BrnA_antitoxin ? DUF4143 AAA_14 AAA_14 DUF4143 ? ? tRNA-synt_2b HGTP_anticodon ProRS-C_1

PZO44157.1
QBML01000003.1:78420-92880
Pseudanabaena frigida

NAPRTase_N NAPRTase NAPRTase_C CTP_transf_like NUDIX WHD_AraR DUF7136 MMR_HSR1 MMR_HSR1 KH_dom-like ? DXP_reductoisom DXP_redisom_C DXPR_C BrnT_toxin FANCL_d2 FA_desaturase MerR_1 Aldedh TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C DUF1643 Trypsin

PZO61197.1
QBMP01000003.1:1-11666
Phormidesmis priestleyi

ABC_tran oligo_HPY ABC_tran BPD_transp_1 BPD_transp_1 BPD_transp_1_N SBP_bac_5 AI-2E_transport ResIII RE_endonuc BrnT_toxin ?

PZV25679.1
QBMS01000067.1:1128-14539
Snowella sp.

PspA LRAT GntR PilZ Nuf2_DHR10-like ? ? GATase_2 Glu_syn_central Glu_synthase GXGXG BrnT_toxin ? DUF2283 NTP_transf_2 HepT-like ? ? FAD_binding_2 Succ_DH_flav_C

QCW84895.1
CP035467.1:3885057-3897546
Methylotuvimicrobium buryatense

Hexapep Hexapep SATase_N HCS_D2 HMGL-like ? ? ? ? HNH BrnT_toxin CopG_antitoxin ? DUF2281 ? ? NifU Fer2_BFD NifU_N Fe-S_biosyn

QKE42258.1
CP053676.1:889219-909040
Ferrovum myxofaciens

? GST_N GST_C LysR_substrate HTH_1 Biotin_lipoyl HH_EMRA Beta-barrel_AAEA MFS_1 ? MFS_1 BrnT_toxin CopG_antitoxin DUF3293 ? HTH_20 LMWPc ArsD ArsA_ATPase ArsA_ATPase ArsA_ATPase

QTA78963.1
CP061799.1:1337200-1345752
Desulfonema limicola

Aldedh ? PIN ? HEPN NTP_transf_2 ? BrnT_toxin NikA-like ? Fic ? ? ? ?

QTA83781.1
CP061799.1:6622480-6633544
Desulfonema limicola

Methyltransf_11 ? Peptidase_S8 EamA EamA UPF0270 Aminotran_1_2 IMPDH BrnT_toxin ? ? ? Phage_int_M Phage_integrase AbiEi_3_N YdjM AAA_28

RIK53686.1
QEUY01000160.1:183-10447
Chloroflexi bacterium

Hydrolase PHP BPD_transp_1 ABC_tran NMT1 META META IU_nuc_hydro BrnT_toxin BrnA_antitoxin MoCoBD_2 MoCoBD_1 Ald_Xan_dh_C Fer2_2

RJP65921.1
QZKI01000121.1:51121-63506
Candidatus Abyssobacteria bacterium SURF_17

His-Kinase PHP_C PHP DRTGG FeS Fe_hyd_lg_C Fer4_9 His-Kinase ? HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase PAS BrnT_toxin CopG_antitoxin ? YafQ_toxin HNH_5 ? ABC2_membrane_3 ABC2_membrane_3

RJX32804.1
QZKY01000007.1:45395-58128
Oxalobacter sp.

DUF898 DUF7092 DUF7977 Peptidase_M48 SET ? SET DUF202 GGDEF PAS PilJ BrnT_toxin BibA_N CopG_antitoxin HTH_3 ? HSP90 HATPase_c_3 GDT1 GDT1 ? YjgF_endoribonc

RKX43628.1
QNAK01000230.1:1-5212
Verrucomicrobia bacterium

SEC-C EamA EamA ? ? BrnT_toxin CopG_antitoxin Asp-Al_Ex TrkA_C

RKX77914.1
QNBF01000158.1:1-4871
Spirochaetes bacterium

BrnT_toxin CopG_antitoxin DRTGG DUF6036 DUF4140 NFACT_N DUF4139 YgaP-like_TM AAA_14 DUF4143

RKX91086.1
QNBM01000403.1:111-3401
Spirochaetes bacterium

DinB_2 ? CopG_antitoxin Acetyltransf_9 URO-D GIDA

RKX91298.1
QNBM01000390.1:1-3478
Spirochaetes bacterium

PfkB BrnT_toxin CopG_antitoxin DinB_2 ? ? NodA PhdYeFM_antitox

RKY15028.1
QNBX01000365.1:121-2918
Planctomycetota bacterium

BrnT_toxin CopG_antitoxin HTH_38 rve Mu-transpos_C_2 IstB_IS21

RKY80196.1
QNDG01000162.1:1-5231
candidate division KSB1 bacterium

Tricorn_C1 Tricorn_2nd Tricorn_N BrnT_toxin CopG_antitoxin BPD_transp_1 ABC_tran

RKZ93153.1
QNER01000020.1:59-21331
Candidatus Parabeggiatoa sp. nov. 1

YafQ_toxin ? Na_H_Exchanger DUF4382 Peptidase_M23 SLH Amidase_2 ? XisI BrnT_toxin ? XisI VF530 SpvB ? AAA_15 ?

RLA06639.1
QNFE01000047.1:207-3719
Gammaproteobacteria bacterium

? LPG_synthase_TM ? ParE_toxin ? DUF2442 ? BrnT_toxin DUF2777

RLA17444.1
QNFH01000304.1:2-3065
Gammaproteobacteria bacterium

TetR_N Acyl-CoA_dh_M Acyl-CoA_dh_1 HTH_37 BrnT_toxin CopG_antitoxin

RLA19935.1
QNFJ01000233.1:62-3705
Gammaproteobacteria bacterium

MS_C MS_TIM-barrel MS_N HTH_1 LysR_substrate ? CopG_antitoxin

RLA23966.1
QNFJ01000017.1:1-6851
Gammaproteobacteria bacterium

DDE_Tnp_1_6 DUF772 ? CopG_antitoxin PIN PhdYeFM_antitox YafQ_toxin PhdYeFM_antitox HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase PAS 6TM_PilS

RLA41364.1
QNFQ01000636.1:1-3241
Gammaproteobacteria bacterium

DUF1254 DUF1214 Kazal_1 BrnT_toxin CopG_antitoxin GDPD

RLA58409.1
QNFV01000312.1:69-2886
Epsilonproteobacteria bacterium

? YoeB_toxin PhdYeFM_antitox BrnT_toxin CopG_antitoxin CoiA_N

RLA69414.1
QMKV01000176.1:190-5764
Epsilonproteobacteria bacterium

BrnT_toxin CopG_antitoxin ? ABC_tran Transgly Transpeptidase

RLA73564.1
QMKX01000040.1:1-4766
Epsilonproteobacteria bacterium

DUF637 ? RelE POTRA_2 POTRA_3 ShlB BrnT_toxin Crinivirus_P26

RLB14069.1
QMLR01000246.1:471-2838
Deltaproteobacteria bacterium

BrnT_toxin CopG_antitoxin PIN

RLC10752.1
QMMV01000120.1:1-6404
Deltaproteobacteria bacterium

Y1_Tnp DUF3368 RHH_1 UPF0175 BrnT_toxin CopG_antitoxin ? DUF4143 AAA_14 HKD Helicase_C ResIII AEP-TOTE

RLC35470.1
QMND01000014.1:32-9929
Candidatus Shapirobacteria bacterium

? TPR_CcmH_CycH K1-lyase_C BrnT_toxin CopG_antitoxin TPRs TPRs AbiEii Peptidase_M23 LysM LysM

RLD06176.1
QMOK01000106.1:444-9157
Chloroflexota bacterium

BrnA_antitoxin BrnT_toxin CopG_antitoxin NTP_transf_2 MFS_1 GAF GAF His-Kinase His-Kinase ? His-Kinase His-Kinase GAF Receiver

RLD11985.1
QMOL01000023.1:11-10508
Chlamydiae bacterium

HTH_5 Pyr_redox_2 Pyr_redox_dim Rhodanese TusA DrsE_2 Radical_SAM Fumerase_C Glyco_hydro_20 Glyco_hydro_20b Phage_integrase Phage_int_SAM_4 Peptidase_S51 BrnT_toxin CopG_antitoxin ? BrnT_toxin

RLD37340.1
QMOZ01000089.1:34-8885
Bacteroidetes bacterium

? CbiA ? RepA_C Phage_integrase BrnT_toxin CopG_antitoxin ? ? ? ResIII

RLD60080.1
QMPH01000416.1:110-5035
Bacteroidetes bacterium

? Toluene_X OMP_b-brl TPR_SusR BrnT_toxin ? cNMPBD HTH_Crp_2 YgaP-like_TM

RLE23140.1
QMQF01000271.1:1-7398
Acidobacteria bacterium

MOFRL DUF4147 Thioredoxin_4 SurA_N_3 Peptidase_M20 Glyco_trans_1_4 Glyco_trans_4_4 DUF2293 BrnT_toxin

RMD59416.1
RFLN01000123.1:1-2720
Nitrospirae bacterium

? BrnT_toxin CopG_antitoxin Zn_ribbon_DnaG Toprim_2 DUF927

RNC66683.1
RHLS01000043.1:289-9327
Desulfuromonadales bacterium

SBP_bac_11 HicB HicA_toxin BrnT_toxin CopG_antitoxin Y1_Tnp CsoSCA_cat MCPsignal HAMP dCache_2 ? LpxC

RNJ48451.1
QWDD01000001.1:241282-255782
Methylocystis hirsuta

Polysacc_synt Rhomboid Rhomboid ETF ETF ETF_alpha 3HCDH_N 3HCDH Ferritin tRNA_m1G_MT BrnT_toxin CopG_antitoxin PRC_RimM RimM Ribosomal_S16 SRP_SPB SRP54 SRP54_N SocA_Panacea ? DEAD Helicase_C RecQ_Zn_bind HRDC

RPH38180.1
RPRP01000013.1:1-7457
bacterium

? ? Peptidase_M28 Zn_ribbon_TFIIB ORF6N BrnT_toxin CopG_antitoxin DUF2779

RPJ13503.1
RPPM01000410.1:1-2416
Deltaproteobacteria bacterium

Phage_integrase Phage_integrase Phage_int_SAM_4 Gp49 dnstrm_HI1420 BrnT_toxin CopG_antitoxin PhdYeFM_antitox

RPJ52015.1
RPOY01001186.1:1-1840
Acidobacteria bacterium

? BrnT_toxin RHH_1 DUF6298

RQD58787.1
QYZX01000104.1:22471-35626
Desulfonatronovibrio sp. MSAO_Bac4

ThiC_Rad_SAM EamA EamA Thioredoxin OB_YrrC HHH_5 HHH_RecD2 AAA_19 SH3_13 UvrD_C_2 GAF_2 PAS His-Kinase His-Kinase Receiver tRNA-synt_1c tRNA-synt_1c_C tRNA-synt_1c_C2 Phos_pyr_kin BrnT_toxin CopG_antitoxin REase-ARP

RUM41618.1
QNYH01000100.1:1-4069
Desulfurobacterium sp.

Usp BrnT_toxin ? IF-2B T2SSK_1st T2SSK PilN ?

RUM43589.1
QNYI01000093.1:1-8910
Desulfurobacterium sp.

Fer4_7 PAPS_reduct Zn_ribbon_9 Rod_CreS DUF34_NIF3 UbiD_N UbiD UbiD_C BrnT_toxin ? NMN_transporter SurE VTT_dom Fer4_7 Fer4_21 DNA_pol3_delta_C Peptidase_M50

RUM58548.1
QNZC01000025.1:720-11786
Persephonella sp.

Cas10-Cmr2_palm2 Csm1_B HD Cas_Cas6_C Cas6_I-A BrnT_toxin CopG_antitoxin Endonuclease_5 PLA1 NADH_4Fe-4S SLBB_2 Complex1_51K Pro_CA T2SSF T2SSF T2SSF_N T2SSE

RVT87391.1
RZWH01000001.1:968527-983490
Rhodobacteraceae bacterium CCMM004

? Cas_CT1975 Cas_Cas5d CRISPR_assoc Cas_Cas1 Cas_Cas2CT1978 CbiA ? CopG_antitoxin rve_2 FtsX FtsX MacB_PCD ABC_tran Lipase_GDSL_2 FAD_binding_4 FAD-oxidase_C Porphobil_deamC Porphobil_deam

RWX44229.1
MTKO01000099.1:4078-21803
Candidatus Electrothrix aarhusensis

? DUF4382 Golvesin_C Lcl_C RHS_repeat RHS_repeat TEN_YD-shell CDI_toxin_EC869_like CdiI PIN ? BrnT_toxin CopG_antitoxin CTP_transf_3 Ldh_1_C CofD ? Receiver ABC_membrane ABC_tran

RYE12838.1
SEEX01000582.1:106-1079
Rickettsiales bacterium

BrnT_toxin HTH_26

RZD14805.1
SGBD01000001.1:90357-105670
Candidatus Acidulodesulfobacterium ferriphilum

? ? Epimerase_2 ? ? Receiver His-Kinase His-Kinase PAS Receiver ? CopG_antitoxin Receiver His-Kinase His-Kinase PAS PAS HTH_8 AAA_lid_14 NtrC-AAA GAF_2 CCDC90-like Bac_DNA_binding ClpB_D2-small AAA_2 IstB_IS21 zf-C4_ClpX CLP_protease

RZD15675.1
SGBC01000004.1:105310-126134

Candidatus Acididesulfobacter guangdongensis

Fungal_TACC ZapA 5-FTHF_cyc-lig RNase_Y_N KH_1 HD YmdB tRNA-synt_1b SYY_C-terminal ? BrnT_toxin CopG_antitoxin Pro_CA Aldedh DUF779 DHQ_synthase DHQS_C TPRs Creatinase_N Peptidase_M24

RZD17560.1
SGBB01000032.1:140-13308
Candidatus Acididesulfobacter diazotrophicus

CHAD DsbD_TM Redoxin Sigma_reg_N DUF3108 TPRs TPRs TPR_CcmH_CycH Gpr1_Fun34_YaaH Aldedh BrnT_toxin CopG_antitoxin ALAD N6_N4_Mtase ? TPRs

RZD17640.1
SGBB01000028.1:15106-22906
Candidatus Acididesulfobacter diazotrophicus

Polbeta NTase_sub_bind BrnT_toxin CopG_antitoxin ? Glyco_transf_4 Glycos_transf_1 ZT_dimer Cation_efflux Acylphosphatase Ribosomal_L19 tRNA_m1G_MT tRNA_m1G_MT

RZD17745.1
SGBB01000023.1:211-8078
Candidatus Acididesulfobacter diazotrophicus

? MatE Polysacc_synt Inositol_P APS_kinase Epimerase Epimerase GDP_Man_Dehyd GNVR Wzz BrnT_toxin ?

SJM92419.1
FUKJ01000190.1:17483-22397
Crenothrix polyspora

DUF475 BrnT_toxin CopG_antitoxin ? Polbeta NTase_sub_bind ? ? ?

SMF69854.1
FXAF01000011.1:472899-481789
Xaviernesmea oryzae

Acetylase ADH_N ADH_zinc_N ParD_antitoxin ParE_toxin DUF188 DUF1345 Esterase ? CopG_antitoxin ? ATP12 ADH_zinc_N HAD_2 PseudoU_synth_2 S4 CRCB ?

SMN00694.1
FXLF01000040.1:5680-17944

uncultured Candidatus Thioglobus sp.

KARI_C KARI_N ALS_ss_C ACT_AHAS_ss TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N LpxC FtsA PilM_2 Dus HicB_lk_antitox BrnT_toxin CopG_antitoxin ? McrBC McrB_lid MoxR-AAA Glycos_transf_2 ?

SMN00938.1
FXLF01000047.1:2457-13082

uncultured Candidatus Thioglobus sp.

NUDIX Leu_Phe_trans ? Orn_Arg_deC_N TPRs TPRs TPRs TPRs TPRs TPRs TPRs TPRs DUF2207_C NmeDI_C ? ? CopG_antitoxin CRS1_YhbY LpxB QRPTase_N QRPTase_C SurE tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c ?

SMN01491.1
FXLF01000076.1:24-6980
uncultured Candidatus Thioglobus sp.

Rhomboid ? CopG_antitoxin Phage_integrase Phage_int_SAM_1 Ribosomal_L19 Fer4_20 Pyr_redox_2 Pyr_redox_2 NIR_SIR NIR_SIR_ferr NIR_SIR DsrE

SMN15944.1
FXLX01000008.1:11862-22960

uncultured Candidatus Thioglobus sp.

IF3_C IF3_N HGTP_anticodon tRNA-synt_2b tRNA_SAD TGS ParE_toxin Unstab_antitox ? ? IAT_beta ? CopG_antitoxin RecJ_OB DHHA1 DHH Lcl_C DAO Peptidase_A8 BT1 adh_short_C2

SMP76740.1
FXUJ01000025.1:7342-23330
Desulfonatronum zhilinae

Mltc_N SLT ? ? Rotamase_2 T2SSF T2SSF T2SSF_N T2SSE MshEN T2SSE DUF4258 CopG_antitoxin Secretin Secretin_N ? ? PilN N_methyl MazE_antitoxin

SPD76048.1
LT984854.1:144535-162811
uncultured Desulfobacterium sp.

HhH-GPD GTP_EFTU EF-G_D2 EFG_III EFG_IV EFG_C Molydop_binding Molybdopterin Molybdop_Fe4S4 DUF488-N3a OEP OEP ACR_tran RND-MFP_C Beta-barrel_RND HH_MFP_RND BrnT_toxin CopG_antitoxin ? ? LarA_C Lar_N FAD-oxidase_C FAD_binding_4 ? Pyr_redox_2

TAE61752.1
RDXV01000002.1:25707-45806
Nostocales cyanobacterium

4HBT ? BrxL_ATPase BrxL_N ? ? ? DUF4007 BrnT_toxin ? DXPR_C DXP_redisom_C DXP_reductoisom Beta-barrel_AprE BSH_CzcB ABC_tran ABC_membrane Peptidase_C39 cNMPBD Rotamase Beta-Roll Beta-Roll Beta-Roll Beta-Roll Beta-Roll FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 COP23

TAJ91627.1
SCUJ01000011.1:279915-290734
Gammaproteobacteria bacterium

TatC TatA_B_E TatA_B_E PRA-PH PRA-CH His_biosynth ? BrnT_toxin CopG_antitoxin DNA_pol_A DNA_pol_A_exo1 5_3_exonuc 5_3_exonuc_N Lysine_decarbox DUF2782 PTS-HPr PapZ_C RapZ-like_N Hpr_kinase_C Hpr_kinase_N

TAK65921.1
SCTX01000155.1:18075-33359
Betaproteobacteria bacterium

NusB RsmF-B_ferredox Methyltr_RsmB-F DUF4390 HAMP PAS His-Kinase His-Kinase Receiver Sigma54_activ_2 AAA_lid_14 HTH_8 TrkA_N TrkA_C TrkA_N TrkA_C TrkH Phage_int_SAM_4 BrnT_toxin CopG_antitoxin BrnT_toxin BrnA_antitoxin Ribonuc_L-PSP Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans HVO_A0114 TumE

TAL32190.1
SCSX01000075.1:1-10120
Spirochaetes bacterium

Fer4_7 Fer4_7 Oxidored_q6 Receiver TPRs CoA_binding_2 Succ_CoA_lig ORF6N ? BrnT_toxin CopG_antitoxin ? FKBP15 SpoIIE

TAN22672.1
SCQP01000008.1:121282-143766
Acidobacteria bacterium

? AAA_2 ClpB_D2-small BetaGal_gal-bd BetaGal_ABD_1 Glyco_hydro_35 TAT_signal Peptidase_S41 PDZ_Tricorn Tricorn_C1 Tricorn_2nd Tricorn_N Tricorn_N CPSase_sm_chain GATase CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS GmrSD_N BrnT_toxin CopG_antitoxin RHH_1 FtsX MacB_PCD FtsX MacB_PCD FtsX MacB_PCD FtsX MacB_PCD FtsX MacB_PCD FtsX MacB_PCD PadR YacG

TAN38778.1
SCQH01000093.1:34211-51455
Nitrospirae bacterium

HhH-GPD EndIII_4Fe-2S ? SRP54 SRP54_N Glyco_hydro_3 DUF748 DUF748 DUF748 MscL MFS_4 DUF4258 CopG_antitoxin DUF7737 DUF4132 ? DUF4276 ABC-ATPase ? DAHP_synth_1

TAN45023.1
SCQE01000021.1:106-7069
Nitrospirae bacterium

BrnT_toxin CopG_antitoxin AlaDh_PNT_C AlaDh_PNT_N tRNA-synt_2b Seryl_tRNA_N TPRs TPRs TPRs Flg_hook Bac_export_2 Fer4_7

TFH16022.1
SPBP01000117.1:261-7616
Lentisphaerales bacterium

? KH_dom-like MMR_HSR1 MMR_HSR1 BrnT_toxin CopG_antitoxin tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c Anticodon_2 tRNA-synt_1c tRNA-synt_1c

TLU85658.1
SZXV01000004.1:82-5214
Chlorobium sp.

adh_short PIN PhdYeFM_antitox BrnT_toxin CopG_antitoxin ? DUF2283 ? HTH_3 Gp49 ?

TLY28884.1
VBOC01000184.1:1-4431
Ignavibacteria bacterium

TPRs RPN7 TPRs TPRs ? BrnT_toxin CopG_antitoxin

TNE74018.1
SBGQ01000035.1:506-5361
bacterium

ALAD FitA-like_RHH PIN Aminotran_3 BrnT_toxin CopG_antitoxin MFS_1

TNF31512.1
SBGO01000409.1:1-7623
Deltaproteobacteria bacterium

Abhydrolase_2 BrnT_toxin BrnA_antitoxin DUF7107 EB Receiver NtrC-AAA AAA_lid_14 HTH_8 EMYY_lipop UbiA ? LPG_synthase_TM

TNF66423.1
SBFH01000037.1:187583-230855
Gammaproteobacteria bacterium

MFS_1 MFS_1 DUF805 Peptidase_M20 Pyr_redox_3 Peripla_BP_2 PhdYeFM_antitox PIN BrnT_toxin BrnA_antitoxin DUF4347 FG-GAP_2 Beta-prop_THOC3 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2 FG-GAP_2

TRZ53082.1
QYOX01000039.1:58-5572
bacterium

ParE-like_toxin RPA Phage_integrase BrnT_toxin CopG_antitoxin ? ? T4SS_LegC2C7 ? Toxin_YhaV

TSC58959.1
VMES01000001.1:1-7692

Parcubacteria group bacterium Greene0416_79

Amidase_3 PG_binding_1 BrnT_toxin CopG_antitoxin Mur_ligase_C Mur_ligase_M ? RuvA_C HHH_5 RuvA_N Ribosomal_S20p ? tRNA-synt_2

TVP97527.1
RECI01000323.1:303-4999
Balneolaceae bacterium

? F420_oxidored HTH_Crp_2 cNMPBD DUF4440 BrnT_toxin CopG_antitoxin HTH_3 Gp49

TVQ01471.1
RECI01000219.1:1-10302
Balneolaceae bacterium

ArsC ? HTH_IclR CMD CMD CarbopepD_reg_2 Plug TonB_dep_Rec_b-barrel MFS_1 DcuC BrnT_toxin CopG_antitoxin GIY-YIG ?

TVQ11421.1
RECM01000175.1:1-10953
Balneolaceae bacterium

Por_Secre_tail adh_short adh_short BrnT_toxin CopG_antitoxin MannoseP_isomer AP_endonuc_2 MFS_2 GH5_mannosidase ? Ig_mannosidase Mannosidase_ig Glyco_hydro_2 Glyco_hydro_2_N2

TVQ28000.1
RECN01000025.1:19431-23628
Spirochaetaceae bacterium

PhdYeFM_antitox BrnT_toxin CopG_antitoxin RHH_1 ParE_toxin ? SSB ? ?

TVQ36474.1
RECP01000100.1:235-8263
Spirochaetaceae bacterium

HTH_32 HTH_23 DUF5372 Zn_Tnp_IS91 MazE_antitoxin PemK_toxin BrnT_toxin CopG_antitoxin Gp49 HTH_3 Cupin_2 Lactamase_B_2 Hydrolase BPD_transp_1

TVQ49402.1
RECR01000017.1:221-4173
Gloeocapsa sp. DLM2.Bin57

NTP_transf_2 BrnT_toxin RHH_1 ?

TVQ97675.1
REDM01000240.1:66-8230
Spirochaetaceae bacterium

DHDPS FCD GntR DctP DctQ DctM His-Kinase HisKA_2 PAS Receiver ? CopG_antitoxin PrsW-protease

TVR30820.1
REDT01000185.1:1-10557
Balneolaceae bacterium

? TPRs Tricorn_N Tricorn_N Tricorn_N Tricorn_N Amidohydro_1 BrnT_toxin CopG_antitoxin ? ? ? ? GerE Amidohydro_1

TVR85923.1
REEM01000213.1:158-7153
Spirochaetaceae bacterium

HTH_18 ARG RHH_1 PIN BrnT_toxin CopG_antitoxin Receiver rve_3 rve HTH_32 HTH_23 HTH_Tnp_1

TVR90333.1
REEM01000123.1:2519-19651
Spirochaetaceae bacterium

SBP_bac_1 TOBE_2 ABC_tran BPD_transp_1 BPD_transp_1 SBP_bac_11 ? Abhydrolase_1 Glyco_hydro_79n BrnT_toxin CopG_antitoxin Formyl_trans_N ACT GcvP2_C GDC-P GDC-P GCV_T_C GCV_T AAA_14 DUF4143 Ank ANKs PIN PhdYeFM_antitox

TVS01243.1
REEV01000137.1:71-4688
Cyanobium sp. PLM2.Bin73

HTH_3 HNH BrnT_toxin CopG_antitoxin HicB_lk_antitox HicA_toxin FA_desaturase

TWW12703.1
SRHE01000003.1:1459-12757
Planctomyces bekefii

? Phenol_MetA_deg Peptidase_M14 Beta_helix PNPase Pantoate_transf AAA BrnT_toxin CopG_antitoxin ? CTP_transf_3 PNPase Ribosomal_S6 SSB Asp_Glu_race

TXH69760.1
SSEY01000203.1:1827-8847
Thiothrix sp.

CMD ? ? BrnT_toxin CopG_antitoxin VapB_antitoxin Monooxy_af470-like Nitroreductase CNNM CBS CBS CorC_HlyC DUF308 CCB3_YggT CCB3_YggT

TXT46399.1
VKGY01000058.1:7-9971
Spirochaetes bacterium

Prismane Rrf2 ? CopG_antitoxin SNF2-rel_dom ResIII RE_endonuc ABC_tran BPD_transp_1 SBP_bac_6

TZG33904.1
VTPU01000016.1:609-17139
Halomonas eurihalina

T6SS_HCP LplA-B_cat Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Carboxyl_trans ECH_1 Biotin_carb_N CPSase_L_D2 Biotin_carb_C BT_MCC_alpha Biotin_lipoyl HMGL-like BrnT_toxin CopG_antitoxin YcaO_C YcaO OsmC ? DUF4747 DUF4062 AadA_C Polbeta ?

UCC41267.1
CP070825.1:1688918-1700453
Candidatus Aminicenantes bacterium

? HipA_2 NDK PIN ? Zn_Ribbon_1 STY-kinase AAA_16 Zn_Ribbon_1 BrnT_toxin CopG_antitoxin DUF5128 DUF6364 ParE_toxin Abhydrolase_1 Hydrolase_4 PIN VapB_antitoxin

UCD55914.1
CP070780.1:428489-441599
Candidatus Omnitrophota bacterium

Zn_ribbon_DnaG DNAG_N Toprim_2 DnaB_bind DnaB Sigma70_r1_1 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 ? ? ? HTH_3 HTH_3 ? CopG_antitoxin ? MafB19-deam Gate FeoB_C Gate FeoB_N FeoA Pyr_redox_2 Radical_SAM

UCD77545.1
CP070771.1:4349125-4366330
Desulfobacterales bacterium

Amidase ? GT87 HATPase_c_3 DNA_mis_repair MutL_C MazE_antitoxin PIN ? BrnT_toxin CopG_antitoxin FG-GAP_3 LpoB GatB_Yqey GatB_N PhdYeFM_antitox PIN TPRs TPRs BPD_transp_2

UCG64761.1
CP070673.1:4200448-4217265
Deltaproteobacteria bacterium

? IclR_C HTH_IclR ? ? ? HTH_1 ? BrnT_toxin CopG_antitoxin LysM LysM SLT bPH_3 Zn_ribbon_4 PilZ ? ? PlsB_N Acyltransferase GPAT_C

UCH96898.1
CP070627.1:2372601-2394290
Candidatus Aminicenantes bacterium

AF_1457 ATP-synt_8 TcpS TPRs FecR_C Secretin_N DHODB_Fe-S_bind NAD_binding_1 FAD_binding_6 Peptidase_S41 PDZ_Tricorn Tricorn_C1 Tricorn_2nd Tricorn_N ? BrnT_toxin CopG_antitoxin PDDEXK_9 AAA-ATPase_like SPASM Radical_SAM AMP-binding AMP-binding_C PP-binding MacB_PCD FtsX MacB_PCD FtsX Condensation PP-binding AMP-binding_C AMP-binding Condensation TumE

UCM85258.1
CP084573.1:928223-941274
Rickettsia endosymbiont of Culicoides impunctatus

Fe-S_assembly RickCE_N RickCE_cat Dus ? YbeY CBS CBS CorC_HlyC ABC_tran ABC_tran_Xtn ABC_tran BrnT_toxin CopG_antitoxin Acetyltransf_8 Acetyltransf_8 Pyr_redox_2 Pyr_redox_dim DUF2628 Proteasome AAA AAA_2 ClpB_D2-small

USN57772.1
CP060226.1:736917-744803
Candidatus Peribacteria bacterium

Mnd1 ? PG_binding_1 ? ? ? ? BrnT_toxin SHOCT-like_DUF2089-C ? ? SLT ? ? DCD

UXE59581.1
CP073041.1:5535419-5545700

Woronichinia naegeliana WA131

? DUF6887 CopG_antitoxin BrnT_toxin ? BrnT_toxin ? BrnT_toxin HicB_lk_antitox HicA_toxin DUF2442 ? Caspase Peripla_BP_6 XisH XisI

VEN74313.1
CAACVI010000023.1:239115-252272

Candidatus Desulfarcum epimagneticum

CueP CueP HTH_3 TaqI_C Eco57I UPF0020 MmeI_hel Mrr_cat DUF8426 ? NatA_aux_su BrnT_toxin CopG_antitoxin ? ? DUF726 CCG CCG Cation_efflux Zn_ribbon_IS1595 Methyltransf_23

VEN74721.1
CAACVI010000036.1:5-777
Candidatus Desulfarcum epimagneticum

BrnT_toxin CopG_antitoxin ?

VFJ47105.1
CAADEZ010000044.1:1-7762
Candidatus Kentrum sp. FM

? DUF4926 DUF433 NTP_transf_2 HepT-like ? ? BrnT_toxin ? Resolvase HTH_7 ? DUF4351 Yae1_N

VFJ51173.1
CAADEX010000032.1:15882-28412
Candidatus Kentrum sp. DK

? ? Glycos_transf_2 DUF4258 MqsA_antitoxin Fer2_4 NADH-G_4Fe-4S_3 Fer4_Nqo3 Fer4_NDSU1 Molybdopterin Molydop_binding ? BrnT_toxin ? ? NADHdh ? ? ? polyprenyl_synt

VFJ52376.1
CAADEW010000036.1:1-13933
Candidatus Kentrum sp. FW

HTH_Rpn_C HTH_Rpn_C FeS Fer4_21 BPD_transp_1 NMT1 ? Peripla_BP_5 DUF4258 CopG_antitoxin BPD_transp_2 BPD_transp_2 ABC_tran BCA_ABC_TP_C ABC_tran LGT ATP-synt_ab Rho_RNA_bind Rho_N

VFJ52895.1
CAADEZ010000107.1:148-8782
Candidatus Kentrum sp. FM

TIR AAA_22 DUF7779 TPRs TPRs TPRs TPRs BrnT_toxin ? DUF4258 ? Histidinol_dh Epimerase ? Sel1 Sel1 Sel1 Zn_ribbon_FGT1_1 C19orf12

VFJ55343.1
CAADEW010000053.1:12189-21684
Candidatus Kentrum sp. FW

Apolipoprotein PfkB PfkB DAHP_synth_1 Sulf_transp Sulf_transp RelB ? CobA_CobO_BtuR BrnT_toxin CopG_antitoxin Sulf_transp HicA_toxin HicB_lk_antitox ? Beta-prop_THOC3 Beta-prop_IFT122_1st PEGA

VFJ65351.1
CAADFE010000007.1:45760-63969
Candidatus Kentrum sp. FW

Ribosomal_L28 Ribosomal_L33 TPRs HGTP_anticodon tRNA-synt_His FGS PEGA Caspase S1 RNase_E_G RNase_E_G_Thio Mur_ligase Mur_ligase_M Mur_ligase_C BrnT_toxin CopG_antitoxin Sua5_yciO_yrdC Peptidase_M50 Peptidase_M50 tRNA-synt_1b CCDC172 tRNA-synt_1b SMC_ScpA ThiG Amidase

VFJ88581.1
CAADFG010000010.1:257-10450
Candidatus Kentrum eta

? ? DUF3782 DUF8196 NTP_transf_2 CBU_0592 ? DIOX_N 2OG-FeII_Oxy ? CopG_antitoxin FMO-like FMO-like Fumerase_C Fumerase PRA-PH AAA_26

VFJ90410.1
CAADFH010000011.1:26590-46511
Candidatus Kentrum sp. LFY

T2SSM DUF3616 ? ? GATase_2 Glu_syn_central Glu_synthase GXGXG GXGXG NA-iREase2 RelA_SpoT BrnT_toxin CopG_antitoxin GXGXG GXGXG Receiver His-Kinase Receiver His-Kinase Receiver ThiS DUF8196 DUF948

VFJ95786.1
CAADFG010000092.1:277-5335
Candidatus Kentrum eta

? CopG_antitoxin BrnT_toxin Metal_resist LepA_C EFG_C EFG_III GTP_EFTU_D2 GTP_EFTU ? PIN ? Cupin_3 TM_HPP

VFK01104.1
CAADFH010000135.1:330-6409
Candidatus Kentrum sp. LFY

BrnT_toxin CopG_antitoxin B12-binding Radical_SAM DUF4070 HhH-GPD NUDIX_4 N6_N4_Mtase N6_N4_Mtase ?

VFK38506.1
CAADFR010000025.1:6031-22138
Candidatus Kentrum sp. SD

WHD_MCM7 RNase_Y_N CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS GreA_GreB_N GreA_GreB FtsJ FtsH_ext AAA AAA_lid_3 Peptidase_M41 ? BrnT_toxin ? MMPL MMPL Queuosine_synth ? DUF2442 DUF2283 MAM

VFK67433.1
CAADFZ010000142.1:265-9456
Candidatus Kentrum sp. UNK

Methyltransf_25 Mob_synth_C Radical_SAM ? ZnuA ? Orn_DAP_Arg_deC BrnT_toxin ? PqqF-like_C_4 Peptidase_M16_M Peptidase_M16_C Peptidase_M16 ?

VGO11688.1
CAAHFG010000001.1:278014-295313
Pontiella desulfatans

? MotA_ExbB MotA_ExbB ExbD TonB_C TPRs TPRs TPRs TPRs VWA_3_C TPRs ? BrnT_toxin CopG_antitoxin ? ? ? ? ? DUF2334

VUZ86586.1
CABIKM010000070.1:53-5760
Candidatus Methylomirabilis lanthanidiphila

DHODB_Fe-S_bind Anth_synt_I_N Chorismate_bind Aminotran_4 DHO_dh HhH-GPD BrnT_toxin CopG_antitoxin ?

VVA43820.1
CABIKN020000021.1:9681-14015
Candidatus Roizmanbacteria bacterium

PEMT BrnT_toxin CopG_antitoxin ? Glutaredoxin ? ParE_toxin ? Glycos_transf_1 Glyco_transf_5

WGK68693.1
CP123443.1:2506428-2519343
Candidatus Haliotispira prima

? ? ? ? ? ? ? ? CopG_antitoxin ? ? nSTAND3 ? ? ?

WPX99877.1
CP113871.1:26220-41881
Candidatus Megaera polyxenophila

DUF6071 SLT ? ? Sel1 Sel1 TraD_N TrwB_AAD_bind LRR_TMOD-LMOD LRR_TMOD-LMOD LRR_TMOD-LMOD BrnT_toxin CopG_antitoxin ? MFS_1 HD_4 ACR_tran YknX_C BSH_CzcB ?

WP_029746160.1
NZ_AKWL02000054.1:6022-16744
Leptospira kirschneri str. 200801774

Collar DUF4376 ? ? ? ? ParB_N BrnT_toxin CopG_antitoxin RtcB RtcB ? HH_YknX ParB_N ParA-Soj-NTPase DMPK_coil

WP_002672173.1
NZ_CAUUNX010000008.1:2032-19739
Treponema lecithinolyticum

Iso_dh ? ThrE ThrE_2 FMN_bind Rhodanese Flg_new DUF4998 BrnT_toxin BrnA_antitoxin Lipoprotein_17 FGS ? AA_kinase ACT_9 ACT_9 NAD_binding_3 Homoserine_dh HMGL-like HCS_D2 LeuA_dimer ABC_tran OB_MalK TOBE_2

WP_253698167.1
NZ_CP051302.1:1238058-1248462
Treponema denticola

? Transgly Hydrolase_3 DUF3969 ? ? NERD Zn_ribbon_Top1 ? ? PhdYeFM_antitox ParE_toxin ? ? Glyoxalase TfoX_N

WP_242034569.1
NZ_JACJSW010000096.1:85-8199
Microcystis flos-aquae FACHB-1344

DUF1517 Flavoprotein DFP Fer2 Sigma70_r1_2 Sigma70_r2 Sigma70_r3 Sigma70_r4 PNPase Urease_beta Urease_gamma ? CopG_antitoxin

WP_323289587.1
NZ_JAYGHQ010000001.1:191566-201460
Crocosphaera sp. XPORK-15E

Uma2 NTP_transf_2 Sulfatase DUF6883 DUF4926 BrnA_antitoxin BrnT_toxin BrnT_toxin ? ? Sulfatase Sulfatase_C SBF DUF2808 ? DUF433

WP_272081673.1
NZ_JADBSW010000069.1:1238-18273
Microcystis sp. M074S1

Rieske PaO TPRs TPRs TPRs TPRs TPRs TPRs STY-kinase ? Trypsin PDZ_2 WCX DEAD Cas3-like_C_2 DUF4258 ? Polbeta HepT-like Polbeta HepT-like ? Csc2

WP_004077634.1
NZ_VIRB01000098.1:317-6979
Schaedlerella arabinosiphila

? FtsX FtsX ABC_tran HTH_3 BrnT_toxin ? ? ? ?

WP_040917434.1
NZ_AKXE01000005.1:23456-42162
Leptospira meyeri serovar Hardjo str. Went 5

TIR ? ? ABC-ATPase ? ? SMC_N ? CopG_antitoxin ? TPRs TPRs ? ? OMP_Lepto PhdYeFM_antitox

WP_118253934.1
NZ_FR886662.1:26-7743
Blautia sp. CAG:237

Na_H_antiporter Na_H_antiporter ? MerR_1 YbjQ_1 MatE MatE BrnT_toxin ? Synaptojanin His-Kinase Receiver Calcineurin

WP_279457315.1
NZ_CP103834.1:2043068-2059399
Actinobacillus genomosp. 1

Proteasome SfsA SfsA_N Glyoxalase HhH-GPD EndIII_4Fe-2S NUDIX_4 Iron_traffic Mltc_N SLT SRP54_N SRP54 SRP_SPB BrnT_toxin CopG_antitoxin 5_nucleotid_C Calcineurin FAD_binding_3 ChlI Mg_chelatase Mg_chelatase_C ArabinogalactanP Abhydrolase_1 TPP_enzyme_C TPP_enzyme_M_2 TPP_enzyme_N

WP_244885015.1
NZ_LZXY01000010.1:92948-115607
Glaciecola punicea

MipA JmjC_2 WHD_ROXA Alpha-amylase Alpha-amylase Alpha-amylase_C Trp_halogenase TonB_dep_Rec_b-barrel Plug Peptidase_M14 HTH_6 SIS BrnT_toxin AAA_14 DUF4143 CcdA MR_MLE_C MR_MLE_N MFS_1 MFS_1 Aldose_epim ESPR TPS MBG_2 YDG YDG YDG

WP_006190337.1
NZ_ACYH01000073.1:11310-25295
Treponema vincentii ATCC 35580

? TetR_C_34 TetR_N ? ARPC4 ABC_tran ABC_tran CbiQ Trep_Strep BrnT_toxin BrnA_antitoxin HTH_18 MazE_antitoxin PemK_toxin IMS_C IMS_HHH IMS Na_H_antiporter Phage_int_SAM_4 Phage_integrase

WP_093446168.1
NZ_FOYY01000003.1:171186-189037
Thiomicrospira sp. ALE5

Ribosomal_S15 RNase_PH RNase_PH_C PNPase RNase_PH RNase_PH_C KH_1 S1 Peptidase_M16 Peptidase_M16_C M16C_assoc PreP_C adh_short_C2 SBP_bac_5 BPD_transp_1_N BPD_transp_1 BPD_transp_1 BrnT_toxin CopG_antitoxin ? ABC_tran oligo_HPY ABC_tran oligo_HPY ComFB ? DsrE FemAB_like

WP_034937683.1
NZ_ALVY01000223.1:37792-54758
Gloeocapsa sp. PCC 73106

? Calcineurin Glrx-like Beta_helix Beta-Roll Mur_ligase Mur_ligase_M Mur_ligase_C CTP_transf_like NUDIX WHD_AraR BrnT_toxin CRR7 SelO Aldo_ket_red AI-2E_transport DXPR_C DXP_redisom_C DXP_reductoisom Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r3 Sigma70_r1_2 HerA-FtsK HAS-barrel

WP_006745909.1
NC_019902.2:3386411-3397033
Thioalkalivibrio nitratireducens DSM 14787

? DUF2237 AbrB ? ? Dioxygenase_C SLT_4 BrnT_toxin CopG_antitoxin HTH_23 HTH_32 rve rve_3 AAA_22 ? DUF4351 NTP_transf_2

WP_027150903.1
NZ_JMLA01000001.1:4630226-4645299
Methylobacter tundripaludum 21/22

N_methyl N_methyl N_methyl GspH N_methyl N_methyl PilX_N DUF7305 DUF6701_3rd DUF6701_4th BrnT_toxin CopG_antitoxin ? CPSase_sm_chain GATase CPSase_L_D1 CPSase_L_D2 CPSase_L_D3 CPSase_L_D1 CPSase_L_D2 MGS GreA_GreB_N GreA_GreB CRS1_YhbY FtsJ

WP_300510852.1
NZ_JASJEB010000119.1:176-12267
Crocosphaera sp.

PAP2 Chloroa_b-bind ? Pps HAD_2 PEP-CTERM ? BrnT_toxin ? tRNA-synt_His HGTP_anticodon ? MS_channel_1st MS_channel_2nd MS_channel_3rd DUF2079 SurE ?

WP_272030250.1
NZ_JAQMUA010000053.1:96-11109
Kamptonema animale CS-326

? ? ? NAPRTase_N NAPRTase NAPRTase_C CTP_transf_like NUDIX WHD_AraR CN_hydrolase NAD_synthase BrnT_toxin CopG_antitoxin Beta-Roll Beta-Roll Peptidase_M10_C DUF192 dCache_1 HAMP HNOBA Guanylate_cyc TPRs

WP_008273333.1
NZ_AAXW01000002.1:152392-163289
Crocosphaera chwakensis CCY0110

HEATs HEATs HEATs ? ? ? ? Peptidase_M3 TOP_N Rv1322 BrnT_toxin CopG_antitoxin HEATs HEATs HEATs CpcD CpcD PBS_linker_poly CpcD PBS_linker_poly Phycobilisome Phycobilisome

WP_008274495.1
NZ_AAXW01000007.1:2496-22232
Crocosphaera chwakensis CCY0110

Peptidase_M50 Uma2 Methyltransf_11 UvrA_inter UvrA_DNA-bind ABC_tran ABC_tran HisG Clp_N AAA AAA_lid_9 AAA_2 ClpB_D2-small TPRs BrnT_toxin BrnA_antitoxin DUF3479 CobN-Mg_chel CobN-Mg_chel PIN MazE_antitoxin Glyco_transf_4 Glycos_transf_1 DUF772 DDE_Tnp_1_2 Glycos_transf_1

WP_008580036.1
NZ_CM000770.1:822545-834141
Rickettsia endosymbiont of Ixodes scapularis

DUF637 ? HTH_Tnp_IS630 DDE_3 Y1_Tnp PRC RimM ? Exonuc_VII_S ? CopG_antitoxin ? DDE_Tnp_1 ? Pur_DNA_glyco YidD 2Fe-2S_thioredx

WP_008871280.1
NZ_ACJN02000003.1:860491-874347
Desulfonatronospira thiodismutans ASO3-1

? MGS AICARFT_IMPCHas ? GTP-bdg_N GTP-bdg_M MMR_HSR1 Aldo_ket_red ? Transposase_31 BrnT_toxin CopG_antitoxin Campylo_MOMP ? ? ? SocA_Panacea HTH_3 ?

WP_009535856.1
NZ_AJZT01000010.1:13014-32405
Oribacterium parvum ACB8

? Sortase DUF5979 ? ? DUF3791 SIS_2 SIS BrnT_toxin BrnA_antitoxin Hydrolase_like PNK3P YoeB_toxin RelB NTP_transferase HTH_24 HATPase_c_4 SLFN_AlbA_2 Glyco_trans_1_4 Glycos_transf_2

WP_010420149.1
NZ_MCRM02000015.1:46697-59616
Leptospira inadai serovar Lyme

DNA_pol_B DUF8156 ? ? HTH_3 AcrVA2-like ? ? CopG_antitoxin DUF2335 ? ? ? ? ?

WP_147142503.1
NZ_AP019563.1:655312-671431
Rickettsia asiatica

Sgo0707_N2 ? Ldh_1_C Ldh_1_N Seryl_tRNA_N TLC GATase CTP_synth_N CTP_transf_3 BrnT_toxin CopG_antitoxin NUDIX GTP_cyclohydroI tRNA-synt_2b HGTP_anticodon DUF6608 RuvA_N HHH_5 RuvA_C ?

WP_010598277.1
NZ_AJGC01000002.1:166590-179381
Rickettsiella massiliensis 20B

? ? ? Sua5_yciO_yrdC ? AMP-binding NAD_binding_4 BrnT_toxin ? RHH_10 ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ?

WP_011243418.1
NC_006576.1:1212521-1223482
Synechococcus elongatus PCC 6301

AAA_lid_3 AAA CoiA_N Pps Pps Pps HcyBio NdhL DUF3007 Trp_syntA BrnT_toxin RHH_1 PRAI PSI_PSAK DUF4168 DNA_pol3_delta_C DNA_pol3_delta PCRF RF-1 DUF3285

WP_256211914.1
NZ_AP040997.1:53939-66727
Nitrosomonas sp. TNE100

Epimerase Glyco_transf_7C Glycos_transf_2 ParE-like_toxin Transposase_31 DUF4351 Polbeta NTase_sub_bind Transposase_mut BrnT_toxin CopG_antitoxin ? HigB-like_toxin HTH_3 Glycos_transf_2 Inositol_P GTP_EFTU GTP-eEF1A_C APS_kinase

WP_276670521.1
NZ_JAAXUX010000002.1:115633-124650
Chlorobium limicola

LysE ParE_toxin PhdYeFM_antitox ? HpcH_HpaI DUF2442 DUF4160 BrnT_toxin CopG_antitoxin Y1_Tnp HicB

WP_011980415.1
NC_009663.1:734919-749696
Sulfurovum sp. NBC37-1

Asparaginase adh_short_C2 23S_rRNA_IVP SpoU_sub_bind SpoU_methylase CAP SPATA24 Tex_N Tex_central_region Tex_YqgF HHH_3 HHH_9 S1 BrnT_toxin CopG_antitoxin Calcineurin AAA_23 SbcC_Walker_B Lactamase_B DDE_Tnp_1_4 Lactamase_B Trypsin CTD4

WP_012176167.1
NC_009943.1:3312836-3327135
Desulfosudis oleivorans Hxd3

SMC_ScpA tRNA-synt_1b Peptidase_M50 ? tRNA_NucTran2_2 PolyA_pol_RNAbd PolyA_pol CBS CBS DHHA1 DHH ? adh_short BrnT_toxin CopG_antitoxin HicB Dala_Dala_lig_C Rce1-like TPRs TPRs TPRs TPRs PALP tRNA-synt_2b Seryl_tRNA_N

WP_012280256.1
NZ_CP009343.1:1039938-1055454
Francisella philomiragia

? Pyr_redox_2 Lactamase_B_2 Ureide_permease Ureide_permease PfkB ? ? BrnT_toxin CopG_antitoxin Glyco_hydro_64 DUF3458_C DUF3458 Peptidase_M1 Peptidase_M1_N PRAI IGPS Cass2 PMSR Glycos_transf_3 Glycos_trans_3N

WP_012470273.1
NC_010814.1:2363633-2379171
Trichlorobacter lovleyi SZ

NAD_synthase DAC 4HBT Receiver CheW Radical_SAM UPF0004 DUF748 DUF748 DUF748 Ferritin BrnT_toxin BrnA_antitoxin ? DUF3373 Hydrolase Pyr_redox_3 Bac_DNA_binding Sdh_cyt Sdh_cyt

WP_012502109.1
NC_011027.1:932626-953194
Chlorobaculum parvum NCIB 8327

Glycos_trans_3N Glycos_transf_3 Pyr_redox_2 Thioredoxin N-OB HHH_6 DNA_pol3_finger DNA_pol3_alpha PHP Glycos_transf_1 Glyco_transf_4 Glyco_hydro_65C Glyco_hydro_65m Glyco_hydro_65N Zf_IFT144_1st LRRs LRRs LRRs BrnT_toxin CopG_antitoxin SPASM Radical_SAM Biotin_lipoyl PYC_OADA HMGL-like Fumerase_C Fumerase Usp APH AAA_33 VIT1 Rubrerythrin

WP_012508629.1
NC_011060.1:2012242-2023278
Pelodictyon phaeoclathratiforme BU-1

DUF2442 DUF4160 ? HTH_3 Gp49 ? ? BrnT_toxin CopG_antitoxin REase ? Aldo_ket_red DDE_Tnp_1 MreB_Mbl HTH_8 AAA_lid_14 NtrC-AAA GAF

WP_012842521.1
NC_013500.1:1-4602
Moraxella bovis

? ? Rep3_C Rep3_N KfrB MobA_MobL ? BrnT_toxin BrnT_toxin

WP_012856349.1
NC_013512.1:523814-537924
Sulfurospirillum deleyianum DSM 6946

Synaptojanin ? Cytochrome_CBB3 Cyt_bd_oxida_I MarR Cupin_2 PNPOx_N ? BrnT_toxin CopG_antitoxin ? ThiC_Rad_SAM ThiC-associated CheX AA_permease TauE ABC_tran UvrA_inter UvrA_DNA-bind AAA_29 ABC_tran

WP_013031237.1
NC_013960.1:129994-136224
Nitrosococcus halophilus Nc 4

? PEP-CTERM DUF4160 ? DUF4160 DUF2442 ? BrnT_toxin CopG_antitoxin PhdYeFM_antitox YoeB_toxin ? HicA_toxin HicB_lk_antitox ?

WP_013221768.1
NC_014315.1:3151662-3163500
Nitrosococcus watsonii C-113

MshEN T2SSE T2SSF_N T2SSF T2SSF N_methyl T2SSG N_methyl GspH N_methyl Exonuc_V_gamma Glyco_trans_4_4 Glycos_transf_1 BrnT_toxin CopG_antitoxin ? RVT_1 DDE_3 HTH_Tnp_IS630 Glyoxalase ?

WP_013681705.1
NC_015318.1:687923-699240
Hippea maritima DSM 10411

tRNA-synt_1b His-Kinase His-Kinase 6TM_PilS Filament GGDEF Creatinase_N Peptidase_M24 ECH_1 ApbA ApbA_C Zn_Ribbon_1 Phage_int_M Phage_integrase BrnT_toxin CopG_antitoxin DUF6364 Peptidase_M78 HTH_3 HTH_35 ? ?

WP_041263675.1
NC_015696.1:919003-928324
Francisella salina

RNA_pol_L RNA_pol_A_CTD Arm-DNA-bind_3 Phage_integrase ? DUF1441 ? Halo_RE BrnT_toxin CopG_antitoxin ? Phage_AlpA ? Arm-DNA-bind_3 Phage_int_M Phage_integrase HTH_39 ?

WP_215894458.1
NZ_JAAZUD010000028.1:185-7166
Acidithiobacillus ferrivorans

BrnT_toxin CopG_antitoxin Cpn60_TCP1 Cpn10 Hydrolase_4 DsbD_N DsbD_TM Thioredoxin_2 zf-CHCC Aminotran_4

WP_014803810.1
NC_018020.1:2732318-2754199
Turneriella parva DSM 21527

5_3_exonuc_N 5_3_exonuc DNA_pol_A_exo1 DNA_pol_A DNA_pol_A ? DDE_Tnp_IS1595 Phage_integrase AAA AAA ? HKD ResIII Helicase_C BrnT_toxin CopG_antitoxin B-block_TFIIIC DDE_Tnp_IS1595 HsdM_N N6_Mtase Methylase_S Methylase_S DUF6161 ? ?

WP_014804453.1
NC_018020.1:3473206-3485257
Turneriella parva DSM 21527

? ? DUF4184 Flagellin_N ? Lcl_C ? BrnT_toxin BrnA_antitoxin ? DUF7273 ? ? ? ?

WP_015219171.1
NC_019776.1:1587422-1606382
Cyanobacterium aponinum PCC 10605

DUF1810 Mo-nitro_C Hydrolase E1-E2_ATPase FOL3_C Mur_ligase_M Proton_antipo_N Proton_antipo_M Proton_antipo_M Cpn60_TCP1 BrnT_toxin ? UvrD-helicase UvrD_C_2 cNMPBD MS_channel_3rd MS_channel_2nd MS_channel_1st Slr1339 ECH_1 DUF8015 ?

WP_015521876.1
NC_021012.1:3247745-3257607
Roseburia intestinalis XB6B4

? ? ? Flg_new Ig_NUP210_5th Big Big ? ? BrnT_toxin BrnA_antitoxin HTH_26 CorC_HlyC ? ? ? ?

WP_015653788.1
NC_020505.1:1360542-1368538
Candidatus Sulfuricurvum sp. RIFRC-1

cwf21 PBP Acetylase ? Fer4_7 Nitro_FeMo-Co Fer4_7 BrnT_toxin BrnA_antitoxin Peptidase_S41 TOBE NifU Sel1 Sel1 HicB-like_2 Nitro_FeMo-Co ? ?

WP_016212424.1
NZ_CP012413.1:847896-865983
Piscirickettsia salmonis

? HTH_Tnp_1 HTH_21 rve rve_3 DCD DrHyd Abi_2 Lactamase_B Lactamase_B Phage_integrase BrnT_toxin CopG_antitoxin rve_3 rve HTH_29 Mrr_cat MTES_1575 AAA_12 AAA_11 AAA_11 DUF4011 Bac_DNA_binding HTH_Tnp_IS630 DDE_3 DDE_5 DDE_Tnp_1

WP_016211674.1
NZ_AMFF02000091.1:247-11595
Piscirickettsia salmonis LF-89 = ATCC VR-1361

? ParA ? BrnT_toxin CopG_antitoxin ? HTH_26 MFS_1 MFS_1 ?

WP_103239999.1
NZ_JANJZD010000011.1:159-10091
Acetatifactor muris

Receiver Trans_reg_C His-Kinase His-Kinase ? PDDEXK_2 Phage_integrase TnpV Acetyltransf_7 BrnT_toxin ? ? DUF5720

WP_310822531.1
NZ_JACDIO010000001.1:63216-75555
Treponema socranskii

tRNA_synthFbeta B5 PhetRS_B1 Ribosomal_L34 Ribonuclease_P YidD YidC_periplas 60KD_IMP Jag_N KH_KhpA-B R3H Pro_isomerase FKBP_C ? ? Acetylase AA_kinase ? ? GDPD DctP DctQ

WP_016523312.1
NZ_AP041040.1:780276-801696
Treponema medium

ABC_tran_CTD ABC_tran ABC_tran_Xtn ABC_tran SH3_3 SBP_bac_1 ? ? NTP_transferase LbH_GLGC ? BrnT_toxin BrnA_antitoxin Peptidase_S9 GCE_fung HTH_Rpn_C PDDEXK_2 HTH_28 ? ? Radical_SAM

WP_017713440.1
NZ_KB235941.1:62101-75287
Prochlorothrix hollandica PCC 9006 = CALU 1027

? GFO_IDH_MocA GFO_IDH_MocA_C3 ? GvpG ATP-synt_DE ATP-synt_DE_N ATP-synt_VA_C ATP-synt_ab ATP-synt_ab_N Proton_antipo_N Proton_antipo_M Proton_antipo_C BrnT_toxin ? DUF441 ? CTD6 MC2 SMC_N T4SS_LegC2C7 NTP_transf_2

WP_237741252.1
NZ_CM001775.1:669141-679948
Geminocystis herdmanii PCC 6308

CENP-F_leu_zip Peptidase_M15 Rce1-like A24_N_bact Peptidase_A24 BPD_transp_1_N BPD_transp_1 Aldedh DNA_methylase BrnT_toxin CopG_antitoxin ParE-like_toxin PhdYeFM_antitox ? PIN ? ?

WP_018867907.1
NZ_KB897941.1:1-10137
Thioalkalivibrio sp. ALE25

PepX_C Peptidase_S15 ATP-grasp_4 Fer4_22 cNMPBD NAD_binding_1 DHODB_Fe-S_bind Oxidored_q6 NiFeSe_Hases NiFeSe_Hases BrnT_toxin CopG_antitoxin Unstab_antitox HycI RES Transposase_mut

WP_019499482.1
NZ_KB235914.1:513168-531407
Portococcus stanieri PCC 6802

DUF2779 DUF4351 NMT1_2 BPD_transp_1 ABC_tran CYNS_N Cyanate_lyase ? BrnT_toxin ? DUF3685 Receiver Na_H_Exchanger DUF565 GNVR ParA-Soj-NTPase Glyco_trans_4_4 Glycos_transf_1 Pect-lyase_RHGA_epim Beta_helix

WP_245546794.1
NZ_KB913033.1:1314374-1325991
Hydrogenovibrio halophilus DSM 15072

Ham1p_like DUF1538 DUF1538 P-II ? NAD_binding_1 FAD_binding_6 Synaptojanin ACT Formyl_trans_N BrnT_toxin CopG_antitoxin DUF3573 TonB_C ExbD MotA_ExbB GNAT_acetyltr_2 GNAT_acetyltr_2 NAT10_TcmA_helicase TmcA_N MASE10 Thioredoxin_2

WP_020886031.1
NZ_ATHI01000004.1:107668-122462
Alkalidesulfovibrio alkalitolerans DSM 16529

S1 S1 S1 S1 S1 HD HTH_5 ArsP_1 HDOD EAL Semialdhyde_dhC_1 Semialdhyde_dh DUF1844 BrnT_toxin CopG_antitoxin MscL DNA_pol_A 5_3_exonuc 5_3_exonuc_N DHH CobU AP_endonuc_2 ?

WP_021330327.1
NZ_AUZJ01000034.1:96871-114822
Treponema socranskii subsp. socranskii VPI DR56BR1116 = ATCC 35536

Tyr_Deacylase Guanylate_cyc AAA_16 PMI_typeI_cat MPI_cupin_dom ThiF Bmp Alpha-amylase CBM_48 PUD ACT_6 BrnT_toxin BrnA_antitoxin BPD_transp_1 BPD_transp_1 SBP_bac_8 HD_assoc HD_assoc HD CitMHS MscL NikA-like

WP_021330682.1
NZ_AUZJ01000043.1:270322-287222
Treponema socranskii subsp. socranskii VPI DR56BR1116 = ATCC 35536

AP_endonuc_2 FGGY_C FGGY_N Peripla_BP_4 BPD_transp_2 ABC_tran ABC_tran RhaA 2H BrnT_toxin BrnA_antitoxin PfkB FliG_N FliG_M FliG_C Peptidase_M16 Peptidase_M16_C M16C_assoc PreP_C ? ? ?

WP_038081509.1
NZ_AUZJ01000050.1:1-9593
Treponema socranskii subsp. socranskii VPI DR56BR1116 = ATCC 35536

Peripla_BP_3 LacI Glyco_hydro_4C Glyco_hydro_4 BPD_transp_1 BPD_transp_1 SBP_bac_8 RecQ_Zn_bind Helicase_C DEAD BrnT_toxin BrnA_antitoxin MazE_antitoxin ?

WP_022267830.1
NZ_DBGCRE010000012.1:31865-58073
Blautia sp.

MMPL MMPL T4SS_LegC2C7 Na_Ala_symp tRNA-synt_1g tRNA-synt_1g Anticodon_3 tRNA_bind Cation_ATPase_N E1-E2_ATPase Cation_ATPase Hydrolase_3 Cation_ATPase_C AI-2E_transport TatD_DNase BrnT_toxin ? ? ? PP2C_2 STY-kinase ? YDG Big_13

WP_022732678.1
NZ_ATWN01000006.1:96019-109572
Thalassospira lucentensis MCCC 1A00383 = DSM 14000

Pirin Pirin_C LigT_PEase LigT_PEase F420_oxidored P5CR_dimer DUF3800 EamA EamA MarR_2 ? BrnT_toxin BrnA_antitoxin ? ADH_N ADH_zinc_N Methyltransf_12 YjiS-like HTH_1 LysR_substrate SBP_bac_5

WP_248403187.1
NZ_CP065800.1:2368823-2384460
Butyrivibrio fibrisolvens

Glycos_transf_2 Acetylase MazE_antitoxin HTH_3 Acyl_transf_3 PfkB zf-HC2 AMP-binding_C AMP-binding BrnT_toxin CopG_antitoxin Glyco_tranf_2_3 Glycos_transf_2 Glyco_trans_1_4 Glycos_transf_2 SH3_3 ?

WP_024538360.1
NZ_AXVM01000003.1:276085-289401
Comamonas badia DSM 17552

AMP-binding_C_3 AMP-binding ECH_1 PNPase AMP-binding_C AMP-binding PNTB PNTB_4TM AlaDh_PNT_C AlaDh_PNT_N GATase_4 BrnT_toxin CopG_antitoxin Phage_integrase Phage_int_SAM_4 ? DUF1010 ? ? HigB-like_toxin

WP_024547176.1
NZ_BAUB01000036.1:1-9021
Picosynechococcus sp. NKBG15041c

Abi_C ? ? DUF4258 HTH_3 Peptidase_M78 MJ1433_N Phage_holin_2_1 HKD SNF2-rel_dom SNF2-rel_dom Helicase_C DMPK_coil

WP_238624618.1
NZ_JAJNQR010000008.1:17299-26592
Ectothiorhodospira haloalkaliphila

FumaraseC_C Lyase_1 Lyase_1 ASL_C DUF2442 Dus NTP_transf_2 ? TauE ? CopG_antitoxin NAD_binding_10 HTH_3 FliS PIN PhdYeFM_antitox ?

WP_198015426.1
NZ_AYXM01000006.1:550-2053
Psychrobacter sp. AC24

Rep3_N Rep3_C BrnT_toxin ?

WP_026789156.1
NZ_KE734718.1:2121398-2133603
Planktothrix rubescens NIVA-CYA 407

CLP_protease CLP_protease DnaJ TPRs COP23 Trypsin Trypsin ? BrnT_toxin BrnA_antitoxin DUF4926 HicA_toxin HicB_lk_antitox PemK_toxin ? Beta-Roll Beta-Roll Beta-Roll Beta-Roll

WP_027353635.1
NZ_AXAM01000025.1:12092-31215
Desulfosarcina sp. BuS5

HD TraI_2 N-OB SurE Radical_SAM_C Radical_SAM HhH-GPD DnaA_N Bac_DnaA Bac_DnaA_C FAD_binding_4 FAD-oxidase_C HSP90 HATPase_c_3 BrnT_toxin CopG_antitoxin DHHA1 tRNA_SAD tRNA-synt_2c RecA_C RecA_N VCA0040-like DUF4301 ? LNT_N CN_hydrolase

WP_027371405.1
NZ_JIAK01000045.1:1-8247
Desulfovermiculus halophilus DSM 18834

DDE_Tnp_IS66 ? PDDEXK_3 HsdM_N N6_Mtase BrnT_toxin ? Eco57I MipA TrkH ?

WP_027848205.1
NZ_AULO01000010.1:110188-129609
Marinospirillum minutulum DSM 6287

? MoeA_N MoCF_biosynth MoeA_C Tudor_1_RapA Tudor_RapA SNF2-rel_dom Helicase_C RapA_C PLA1 DUF1731 Epimerase NAD_binding_8 DNA_photolyase FAD_binding_7 BrnT_toxin CopG_antitoxin YjdM Zn_Ribbon_YjdM S-AdoMet_synt_C S-AdoMet_synt_M S-AdoMet_synt_N MethyltransfD12 AAA_15 AAA_15 Transketolase_N Transket_pyr Transketolase_C_1 Yip1

WP_029461513.1
NZ_BAWN01000004.1:40423-57042
Serpentinimonas barnesii

Peptidase_S49 ODP TacA1 Acetyltransf_7 RdgC Lactamase_B DFP Flavoprotein BrnT_toxin CopG_antitoxin Amidase FtsX FtsX AMP-binding_C AMP-binding Abhydrolase_6 HTH_37 PCRF RF-1

WP_029891679.1
NZ_JOMH01000001.1:3607213-3616408
Polycyclovorans algicola TG408

N_methyl Pilin ? DUF4258 DUF2283 Y1_Tnp BrnA_antitoxin BrnT_toxin BrnT_toxin CopG_antitoxin MICAH GAF_2 GerE CHASE3 His-Kinase His-Kinase Receiver Receiver ? ?

WP_032026933.1
NZ_JMNZ01000294.1:1-273
Acinetobacter sp. 72431

BrnT_toxin

WP_034950452.1
NZ_JDST02000063.1:275-8482
Candidatus Accumulibacter cognatus

HTH_Tnp_1 HTH_32 rve rve_3 BrnT_toxin CopG_antitoxin ? ? PhdYeFM_antitox Cache_3-Cache_2 HAMP MCPsignal MS_channel_2nd cNMPBD ?

WP_262171100.1
NZ_CP104965.1:1490299-1508158
Devosia neptuniae

FAD_binding_2 Succ_DH_flav_C ? Fer2_3 Fer4_17 ? EamA EamA Phage_integrase Phage_int_SAM_1 ? BrnT_toxin BrnA_antitoxin OSCP ATP-synt_ab_N ATP-synt_ab ATP-synt_ab_C ATP-synt ATP-synt_ab_N ATP-synt_ab ATP-synt_VA_C ATP-synt_DE_N sCache_3_3 HAMP PAS MCPsignal

WP_035515709.1
NZ_KN234757.1:27388-35991
Pseudohaliea rubra DSM 19751

ParE_toxin HTH_24 RHH_1 ? ? Fic Phage_integrase BrnT_toxin CopG_antitoxin HTH_3 HTH_24 Polysacc_synt Methyltransf_24 Methyltransf_21 ?

WP_207667055.1
NZ_CDRTDA010000005.1:5126-20047
Lachnoanaerobaculum sp.

Aminotran_5 2-Hacid_dh_C ACT_AHAS_ss ? ? YgjP-like HAAS SH3_3 NLPC_P60 BrnT_toxin ? GNAT_acetyltran PRTase_2 TRSP StiP PELOTA_1 ? YceG_bac Trypsin YceG_bac YceG_bac YceG_bac

WP_045217753.1
NZ_KN882194.1:4152-11376
Desulfonatronovibrio magnus

NTP_transf_2 HicB_lk_antitox HicA_toxin HTH_23 HTH_32 ? GIY-YIG ? BrnT_toxin ? 23S_rRNA_IVP ? GATase_7 Asn_synthase ? ?

WP_046630889.1
NZ_CCRK01000001.1:449918-465210
Neorhizobium galegae bv. officinalis

Ppx-GppA FtsJ MFS_1 MFS_1 Lactamase_B IMPDH MAPEG YqhI DLH BrnT_toxin BrnA_antitoxin LysR_substrate HTH_1 YjiS-like DUF937 GCS2 Methyltrans_RNA PUA_4 PHO4

WP_047031853.1
NZ_CP011479.1:4528255-4543044
Hoeflea sp. IMCC20628

HemY_N Mitofilin HEM4 Porphobil_deamC Porphobil_deam TsaD YCII EVE BrnT_toxin BrnA_antitoxin YkuD Receiver Trans_reg_C cNMPBD Synaptojanin LolA FtsK_gamma FtsK_SpoIIIE FtsK_alpha FtsK_4TM

WP_048425627.1
NZ_AP024145.1:4977276-4992765
Methylobacterium indicum

MFS_1 GD_AH_C GD_AH_second SAF ABC_tran BCA_ABC_TP_C ABC_tran BPD_transp_2 BPD_transp_2 Peripla_BP_6 Ldh_2 BrnT_toxin HTH_3 ? SGL DUF6894 Bact_transglu_N_2 Transglutaminase EutB EutC

WP_053173426.1
NZ_LFYT02000007.1:20018-59011
Limnohabitans planktonicus II-D5

Big_12 Big_12 Big_12 Big_12 Big_12 Big_13 Big_13 Big_12 Big_13 Big_13 Big_12 Big_13 Big_13 Big_13 Big_12 Big_13 Big_13 Big_13 Big_13 Big Big Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big_13 Big Big_13 Big_13 Big_13 FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP Beta-Roll FG-GAP FG-GAP_3 FG-GAP Beta-Roll FG-GAP_3 FG-GAP Beta-Roll FG-GAP_3 FG-GAP_3 FG-GAP Beta-Roll RelE HTH_3 Gp49 dnstrm_HI1420 NTP_transf_2 HepT-like BrnT_toxin CopG_antitoxin HTH_3 DUF6088 AbiEii Glycos_transf_1 Methyltransf_21 TPRs TPRs TPRs TPRs Glyco_transf_41 Glyco_transf_41

WP_056189117.1
NZ_LMMV01000026.1:176577-190580
Pseudorhodoferax sp. Leaf267

Phage_sheath_1C Phage_sheath_1 ? HTH_8 AAA_lid_14 NtrC-AAA ParE_toxin CopD ? Beta-lactamase2 BrnT_toxin CopG_antitoxin His-Kinase His-Kinase HAMP_PhoQ 2CSK_N Trans_reg_C Receiver TctC ? MFS_1 Nramp

WP_056334477.1
NZ_LMFA01000020.1:326963-338781
Rhizobium sp. Root482

Cu-oxidase_4 Methyltransf_28 LGT BMFP YbjN F420_oxidored P5CR_dimer tRNA_bind ? CopG_antitoxin His-Kinase His-Kinase HAMP Trans_reg_C Receiver MarR_2 Aminotran_4 DUF930 AHSA1

WP_291822368.1
NZ_JACIZI010000011.1:42140-52143
Marinobacter sp.

PhdYeFM_antitox YoeB_toxin Y2_Tnp ? PIN VapB_antitoxin ? DUF4258 CopG_antitoxin EndIII_4Fe-2S HhH-GPD Rnf-Nqr FMN_bind NQR2_RnfD_RnfE Fer4_7 SLBB Complex1_51K RnfC_N Fer4_21 FeS

WP_058453365.1
NZ_CAAAIB010000002.1:339222-352606
Legionella maceachernii

HTH_Tnp_1 Phage_integrase Phage_int_SAM_1 ? CopG_antitoxin BrnT_toxin PIP49_N ? T4SS_LegC2C7 Hydro_N_hd ? ? RAP-1

WP_062295654.1
NZ_CP012036.1:5219107-5234032
Nostoc piscinale CENA21

GST_C_2 GST_N_3 Fasciclin GAF GAF GAF Tr-sialidase_C AAA_lid_2 Mg_chelatase Uma2 UPF0004 Radical_SAM TRAM BrnT_toxin ? ? ? Cupin_2 Arabinose_bd HTH_18 RNA_ligase ?

WP_064010101.1
NZ_LUUG01000105.1:55664-64430
Methylomonas methanica

Polyketide_cyc2 AHSA1 3-dmu-9_3-mt Transpeptidase FMN_red ArsC PfkB CTP_transf_like ? CopG_antitoxin ? UbiA dCMP_cyt_deam_1 ? GST_N_2 GST_C_2 Lipocalin_4

WP_319961856.1
NZ_JAXARY010000011.1:27746-37906
Methylomonas defluvii

HoxH NiFeSe_Hases ? Ni_hydr_CYTB HycI HupF_HypC HyaE Unstab_antitox BrnT_toxin CopG_antitoxin ? HupH_C HupH_C Rubredoxin NiFe-hyd_HybE NiFeSe_Hases HypA

WP_115007109.1
NZ_JBPAGO010000002.1:13153-22392
Moraxella lacunata

Lactamase_B sCache_3_2 His-Kinase His-Kinase ? CreD CreD ? DDE_Tnp_1 DUF4096 ? BrnT_toxin ? NqrM ? ? HTH_28 HTH_28 HTH_21 rve rve_2 MFS_1

WP_320194757.1
NZ_CP192797.1:2030250-2049003
Agrobacterium rosae

SIMPL ? YjbQ PAS GAF PHY HWE_HK Receiver DUF930 Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 DnaB_bind Toprim_2 DNAG_N Zn_ribbon_DnaG BrnT_toxin BrnA_antitoxin YqeY CPSase_sm_chain GATase MatE MatE dCache_1 HAMP MCPsignal DUF2938 Zn_peptidase

WP_069470703.1
NZ_MCBV01000012.1:23653-37351
Candidatus Marithrix sp. Canyon 246

YbjQ_1 YbjQ_1 Flavoprotein DUF346 DUF346 DUF346 DUF346 Bac_globin ? ? BrnT_toxin CopG_antitoxin Glyco_transf_5 Glycos_transf_1 Receiver Guanylate_cyc Receiver Receiver His-Kinase His-Kinase dCache_1 His-Kinase His-Kinase Receiver H_PPase

WP_069471235.1
NZ_MCBV01000007.1:31553-43337
Candidatus Marithrix sp. Canyon 246

ABC-ATPase AAA_23 NQR2_RnfD_RnfE ? DUF6717 ? ? Peptidase_M10 DUF346 BrnT_toxin CopG_antitoxin Methyltransf_5 ? ? ? HSP15_C S4 GTP_EFTU EF-G_D2 RF3_C

WP_069477892.1
NZ_CP017111.1:1313049-1324756
Sulfurospirillum halorespirans DSM 13726

Complex1_49kDa Complex1_49kDa NiFeSe_Hases Complex1_30kDa Fer4_7 Fer4_9 Oxidored_q6 HD OST-HTH NYN MFS_3 BrnT_toxin CopG_antitoxin LUD_dom LUD_dom Fer4_8 CCG CCG FKBP_C Aminotran_1_2 CobW_C cobW Sel_put

WP_071453917.1
NZ_CP017675.1:1017485-1032161
Gloeomargarita lithophora Alchichica-D10

Fer2 STY-kinase DUF4189 Glyco_transf_5 Glycos_transf_1 Phage_integrase XisAC_2nd XisAC_N HTH_31 ? CENP-B_N BrnT_toxin HTH_3 ? HicA_toxin DHDPS Lactamase_B RNase_J_b_CASP RMMBL RNase_J_C FAD_binding_7 ?

WP_290740563.1
NZ_DFHR01000024.1:3337-14695
Fibrobacter sp. UBA3718

ABC2_membrane ? BrnA_antitoxin ? ? FloA ? ? CopG_antitoxin TM2 ? ? ? ABC_tran ABC_tran_Xtn ABC_tran Por_Secre_tail Glyco_hydro_16

WP_073304476.1
NZ_FRAW01000015.1:63513-68946
Fibrobacter intestinalis

? ? HTH_3 ? ? CopG_antitoxin PepSY_2 CopG_antitoxin DUF2213 Phage_cement Major_capside

WP_073611059.1
NZ_JACJQN010000010.1:14973-30930
Phormidium tenue FACHB-1052

PFK DUF4070 Radical_SAM B12-binding ? DUF5671 DUF2281 DUF4926 DUF6883 Peptidase_S9 Peptidase_S9_N BrnT_toxin CopG_antitoxin Nlig-Ia DNA_ligase_aden DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_5 HHH_2 BRCT GAF EAL DUF559 GATase_6 SIS SIS XisI GYF

WP_074933827.1
NZ_FORI01000015.1:5128-16252
Treponema bryantii

N6_N4_Mtase SBP_bac_1 CBM9_1 Glyco_hydro_2_C Glyco_hydro_16 ? Big BrnT_toxin BrnA_antitoxin rve_2 rve

WP_075628565.1
NZ_FOAM01000004.1:75727-89645
Xaviernesmea oryzae

ABC_tran BCA_ABC_TP_C Sigma54_AID Sigma54_CBD Sigma54_DBD Ribosomal_S30AE Ribosom_S30AE_C PTS_EIIA_2 EamA EamA GntR Aminotran_1_2 GrpE BrnT_toxin BrnA_antitoxin HrcA RNase_PH RNase_PH_C Glyoxalase Ham1p_like Radical_SAM HemN_C Amidohydro_3 Amidohydro_3

WP_077732030.1
NZ_CM132196.1:4894612-4908419
Methylocaldum gracile

HerA-FtsK DndE SMC_N AAA_23 Sp-DndD PAPS_reduct PIN ? Molybdop_Fe4S4 Molybdopterin Molydop_binding BrnT_toxin Ribosomal_L26 Uma2 Fer4_7 Fer4_11 ? Ni_hydr_CYTB Dodecin DsbB

WP_078276909.1
NZ_CAACXO010000041.1:1-15621
Moraxella caviae

Acetate_kinase PTA_PTB H2O2_YaaD ECH_2 Catalase-rel Catalase Fe-S_assembly ABG_transport BrnT_toxin ? Peptidase_M20 2OG-FeII_Oxy_3 RsmJ CDP-OH_P_transf

WP_078777528.1
NZ_FUWU01000097.1:4-5357
Fibrobacter intestinalis

? ? HTH_3 DUF4258 CopG_antitoxin HicA_toxin HicB_lk_antitox HTH_3 DUF2213 Phage_cement Major_capside

WP_276990684.1
NZ_JAQYFD010000048.1:1-2430
Fibrobacter intestinalis

? ? HTH_31 ? BrnA_antitoxin CopG_antitoxin DUF4160

WP_080804912.1
NZ_LT828549.1:82544-100328
Desulfamplus magnetovallimortis

4HBT DUF3786 HTH_8 AAA_lid_14 NtrC-AAA GAF_2 Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 Acyl-CoA_dh_C AMP-binding AMP-binding_C 3HCDH_N 3HCDH Thiolase_N Thiolase_N Thiolase_C BrnT_toxin RHH_1 MNS-STAND SBP_bac_3 PAS His-Kinase His-Kinase Receiver QueF QueF_N QueC Radical_SAM PTPS

WP_083048650.1
NZ_CAXXQO010000003.1:2062954-2078698
Marispirochaeta aestuarii

DUF6933 Mrr_N Mrr_cat ? rve HTH_32 HTH_38 Polbeta HEPN Fic_N Fic SoFic-like_C BrnT_toxin CopG_antitoxin ? ? AF_1680 AAA_15 GmrSD_N GmrSD_C DUF2188

WP_083592613.1
NZ_AQQS01000021.1:342-6858
Aurantimonas sp. 22II-16-19i

? CopG_antitoxin ? SPASM Radical_SAM PqqD TENA_THI-4 Lactamase_B_2 OmpA

WP_083593570.1
NZ_AQQS01000031.1:5786-20899
Aurantimonas sp. 22II-16-19i

PhoD_2 Hydrolase_2 NMT1 BPD_transp_1 ABC_tran Arginase ? ? RHH_10 DUF1731 Epimerase ? HEATs Fer4_16 QueG_DUF1730 GMC_oxred_N GMC_oxred_C MFS_1 DeoRC HTH_DeoR

WP_084275469.1
NZ_AP026671.1:1066512-1082358
Nitratiruptor tergarcus DSM 16512

NTP_transferase ? TonB_dep_Rec_b-barrel Plug CobA_CobO_BtuR ? DUF6962 NUDIX_4 HhH-GPD BrnT_toxin CopG_antitoxin Met_synt_B12 B12-binding B12-binding_2 Pterin_bind S-methyl_trans ? CCP_MauG RoxA-like_Cyt-c Sua5_yciO_yrdC MGS CPSase_L_D2 CPSase_L_D1 CPSase_L_D3 CPSase_L_D2 CPSase_L_D1 MreC

WP_244556775.1
NZ_FWXR01000002.1:179297-195723
Fulvimarina manganoxydans

ACT_9 AA_kinase Ectoine_synth Aminotran_3 Acetylase MarR_2 CorC_HlyC CBS CBS CNNM PhoX TAT_signal ? CopG_antitoxin ? ? UVR UvrB Helicase_C UvrB_3rd UvrB_inter ResIII PRC MS_channel_2nd DUF302

WP_086445620.1
NZ_JBAMXX010000030.1:379-1896
Leptospira venezuelensis

BrnT_toxin CopG_antitoxin ?

WP_087144499.1
NZ_FUKI01000136.1:103-4640
Crenothrix polyspora

YfiO Radical_SAM QueC ? ? BrnT_toxin ?

WP_087438431.1
NZ_CP021416.1:1262044-1274328
Sulfurospirillum diekertiae

EAL dCache_1 OmpA MotB_plug TPR_AP5Z1_C MotA_ExbB CHAD YgjP-like BrnT_toxin CopG_antitoxin ? PhzC-PhzF Phage_holin_8 ? PBP_like_2 AA_permease_2

WP_192866743.1
NZ_CP021416.1:2157936-2170065
Sulfurospirillum diekertiae

? Polyketide_cyc Sel1 Sel1 Sel1 Sel1 AraC_binding HTH_18 Glug Glug Glug TPS LysE Hemerythrin BrnT_toxin CopG_antitoxin ? ? 6PGD HTH_28 ? TrbC Tom5

WP_174664571.1
NZ_BDMN01000137.1:119-2348
Bathymodiolus platifrons methanotrophic gill symbiont

N6_Mtase BrnT_toxin CopG_antitoxin Phage_int_SAM_1 Phage_integrase

WP_089355458.1
NZ_FZPD01000001.1:752756-772467
Ekhidna lutea

Methyltransf_11 TonB_dep_Rec_b-barrel Plug FIST FIST_C His-Kinase His-Kinase Receiver ABC_tran_CTD ABC_tran ABC_tran_Xtn ABC_tran ? BrnT_toxin CopG_antitoxin NDPSase_N NUDIX MatE MatE MFS_1 FtsX MacB_PCD FtsX MacB_PCD PadR Por_Secre_tail

WP_133512985.1
NZ_JAAEFD010000058.1:17825-24639
Candidatus Thiosymbion oneisti

ParE_toxin ? RuBisCO_large_N RuBisCO_large RuBisCO_small Tll0287-like HAMP TIR DUF8082 DUF8082 DUF2442 BrnT_toxin

WP_089728779.1
NZ_FLUY01001595.1:254-14409
Candidatus Thiosymbion oneisti

SprT-like Zn_ribbon_SprT TatA_B_E TatA_B_E Xre_MbcA_ParS_C ACT_4 RelA_RIS RelA_AH TGS RelA_SpoT HD_4 RNA_pol_Rpb6 Guanylate_kin ? ? YicC-like_C YicC-like_N PRAI PEP-utilizers_C PEP-utilizers PPDK_N PPDK_N PPDK_N ? ?

WP_090143051.1
NZ_CXOO01000063.1:1-9649
Limnohabitans sp. DM1

FdhD-NarQ ECH_1 ECH_1 4HBT AMP-binding AMP-binding_C_2 Oxidored_FMN PseudoU_synth_2 BrnT_toxin CopG_antitoxin H2O2_YaaD QueF_N QueF Zn_ribbon_ISL3 HTH_Tnp_ISL3

WP_306374545.1
NZ_JASAYK010000006.1:39625-50763
Phocoenobacter skyensis

DAGK_prokar MaAIMP_sms SNF SNF Rho_N Rho_RNA_bind ATP-synt_ab Epimerase Pep_deformylase SAM_SMF DNA_processg_A WHD_DprA BrnT_toxin CopG_antitoxin EbsA ? SecE NusG KOW Ribosomal_L11_N Ribosomal_L11 Ribosomal_L1

WP_092119956.1
NZ_FMXO01000008.1:144136-161028
Desulfonatronum thiosulfatophilum

Peptidase_C93 LapD_MoxY_N GGDEF EAL ? Peptidase_C39 ABC_membrane ABC_tran HH_AprE Beta-barrel_AprE GerE LapA BrnT_toxin CopG_antitoxin PDDEXK_2 HTH_Rpn_C dCMP_cyt_deam_1 TspO_MBR Receiver His-Kinase His-Kinase PAS PAS PAS PAS PAS PAS ? Lcl_C

WP_094561189.1
NZ_NQKZ01000013.1:6514-16941
Synechococcus sp. 8F6

LPG_synthase_TM HAD UbiA Sigma70_r4_2 Sigma70_r2 DUF4278 p450 ? Nif11 BrnT_toxin CopG_antitoxin ? Pyridox_oxase_2 DUF2256 SnoaL_2 ? DrHyd

WP_097035400.1
NZ_OCSS01000001.1:1340337-1357777
Fibrobacter sp. UWT3

MBOAT MBOAT Lipase_GDSL_2 Lipase_GDSL_2 tRNA-synt_His Beta_helix CBM_35 ? ? BrnT_toxin ? NTP_transferase SpoU_methylase ? Proton_antipo_M Proton_antipo_M Proton_antipo_M Proton_antipo_N

WP_233203634.1
NZ_NESI01000007.1:173078-190277
Limnohabitans sp. B9-3

TPP_enzyme_C HTH_1 LysR_substrate Acyl-CoA_dh_N Acyl-CoA_dh_M Acyl-CoA_dh_1 TctC EamA EamA PhoD_N PhoD PurT_C ATP-grasp RS_preATP-grasp-like BrnT_toxin CopG_antitoxin ABC_tran BCA_ABC_TP_C ABC_tran BPD_transp_2 BPD_transp_2 Peripla_BP_6 DUF4118 GAF_3 His-Kinase His-Kinase Receiver Trans_reg_C

WP_199244206.1
NZ_PGGB02000007.1:75071-86717
Caedibacter taeniospiralis

AAA AAA_assoc_2 MgsA_C adh_short_C2 Acetyltransf_8 Acetylase Acetyltransf_7 DUF4440 DUF1016_N YhcG_C Acetyltransf_8 Acetylase CopG_antitoxin ? Acetylase ? APH HTH_18 GyrI-like ? ? DDE_Tnp_IS1595

WP_100708281.1
NZ_JBBBVK010000079.1:124-630
Leptospira licerasiae

BrnT_toxin ?

WP_100727343.1
NZ_NPDM01000007.1:76847-92663
Leptospira levettii

DUF2071 DUF1554 ? PIN FitA-like_RHH HepT-like DUF1554 ? CopG_antitoxin Abhydrolase_6 TMEM234 Sulfatase SGSH_C Calx-beta CoA_transf_3 Calcineurin

WP_135568636.1
NZ_JANLJQ010000021.1:296-3621
Leptospira sp. 85282-16

UPF0175 DUF3368 ? PIN BrnT_toxin CopG_antitoxin

WP_101745773.1
NZ_JBQGQA010000001.1:4959183-4971058
Endozoicomonas acroporae

HTH_26 Vut_1 Vut_1 PRiA4_ORF3 SEC-C ? DUF7986 DUF7986 BrnT_toxin CopG_antitoxin ? Transposase_20 DEDD_Tnp_IS110 CpsB_CapC ? HTH_Tnp_1 HTH_Tnp_1 HTH_23 HTH_32 rve rve_3 Wzy_C

WP_106455152.1
NZ_PXOH01000001.1:358157-371763
Aphanothece hegewaldii CCALA 016

CNNM CBS CorC_HlyC PALP ? UbiA GCV_H Collagen Aminotran_5 BrnT_toxin ? Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_r1_2 PriA_3primeBD DEAD Zn_ribbon_PriA Helicase_C PriA_C ? PIN DUF4359 ?

WP_106458343.1
NZ_PXOH01000024.1:20060-45704
Aphanothece hegewaldii CCALA 016

CARDB PPC PPC PPC PPC PPC PPC PPC PPC PPC PPC Peptidase_C11 Beta-Roll Beta-Roll Beta-Roll DUF4351 AAA_25 Phage_pRha HTH_26 ? ? BrnT_toxin DUF2884 ? ? DUF5895 TPRs TPRs TPRs TPRs TPR_NPHP3 TPR_NPHP3 AP-ATPase ? nSTAND1

WP_106459606.1
NZ_PXOH01000088.1:125-2299
Aphanothece hegewaldii CCALA 016

? ? BrnT_toxin ? RepB-RCR_reg

WP_109735791.1
NZ_QGDU01000019.1:46885-61111
Fibrobacter sp. UWR4

DNA_pol_A_exo1 HRDC 5-FTHF_cyc-lig SpoU_sub_bind SpoU_methylase DegT_DnrJ_EryC1 GFO_IDH_MocA_C3 GFO_IDH_MocA ATP-grasp_3 ATPgrasp_N ? BrnT_toxin CopG_antitoxin Glycos_transf_2 ? ABC2_membrane HipA_C Couple_hipA HTH_3 PIN

WP_203428454.1
NZ_CP044125.1:3242637-3253660
Rhizobium sp. BG4

Cu-oxidase_4 Methyltransf_28 LGT BMFP YbjN F420_oxidored P5CR_dimer tRNA_bind ? CopG_antitoxin DUF1349 His-Kinase His-Kinase HAMP Trans_reg_C Receiver MarR_2 DUF1236 SH3_3 Lactamase_B

WP_114705618.1
NZ_QDKL01000001.1:517180-531312
Halobacteriovorax vibrionivorans

SLT LysM ? ? OMP_b-brl OMP_b-brl ? ? ? BrnT_toxin CopG_antitoxin ? ? SpoU_methylase Peptidase_M17 Peptidase_M17_N LptF_LptG RodZ_C HTH_25

WP_115341179.1
NZ_UGQB01000005.1:1-4495
Moraxella caprae

? BrnT_toxin ? ? Rep3_C Rep3_N Rep3_N KfrB MobA_MobL

WP_116037062.1
NZ_JADDJG010000005.1:354902-372060
Rhodohalobacter sp. SW132

DUF5128 DUF5128 TolB_like Por_Secre_tail Bps ? TPRs Tricorn_N Tricorn_N Tricorn_N Tricorn_N Amidohydro_1 BrnT_toxin CopG_antitoxin ? ? ? Peptidase_M23 DUF5128 DUF5128

WP_283157430.1
NZ_CP103983.1:1191741-1214095
Cysteiniphilum sp. QT6929

DUF1664 ? ? IglE_N ? DUF4280 ? BrnT_toxin CopG_antitoxin Thioredoxin DsbD_TM DsbD_N ? Exonuc_VII_L tRNA_anti_2 ? Peptidase_S24 LexA_DNA_bind Acid_phosphat_B

WP_117004157.1
NZ_BMJS01000099.1:194-2652
Cysteiniphilum litorale

Phage_portal_2 ? BrnT_toxin BrnA_antitoxin HicB GpA_nuclease

WP_117004157.1
NZ_CP103983.1:2141621-2152623
Cysteiniphilum sp. QT6929

HTH_Tnp_1 HTH_21 rve rve_3 DDE_Tnp_1_assoc HTH_3 DDE_Tnp_IS1595 Zn_ribbon_IS1595 Phage_portal_2 ? BrnT_toxin BrnA_antitoxin HicB GpA_nuclease DDE_Tnp_1_3 Fic DDE_Tnp_1 Sugar_tr

WP_118849876.1
NZ_CP025512.1:1566580-1576379
Neorhizobium sp. SOG26

EamA EamA ? Acetylase ADH_N ADH_zinc_N Acetyltransf_10 DUF188 Esterase ? CopG_antitoxin ? ATP12 HAD_2 PseudoU_synth_2 CRCB ?

WP_287791902.1
NZ_JAQVCD010000015.1:49897-62956
Arcobacter sp.

ACR_tran zf_DPOE_2 ? MMR_HSR1 MMR_HSR1 DLP_helical DUF726 Zn_ribbon_Elf1 ? CopG_antitoxin ? EAL GGDEF HAMP LapD_MoxY_N LapD_MoxY_N Peptidase_C93 MFS_1 Ham1p_like ?

WP_119696427.1
NZ_KZ984051.1:909933-929165
Microbacterium halotolerans

Pyr_redox_2 Flavin_Reduct Sod_Fe_N Sod_Fe_C Big Oxidored_molyb DAO FAO_M GCV_T GCV_T_C GCV_T GCV_T_C FTHFS BrnT_toxin CopG_antitoxin Aminotran_3 Bac_luciferase CofC Bac_luciferase Pyr_redox_2 CofH_C Radical_SAM Radical_SAM

WP_245962587.1
NZ_REFJ01000001.1:607652-619075
Umboniibacter marinipuniceus

PIN VapB_antitoxin Glyoxalase Glyoxalase_3 RelE MqsA_antitoxin TonB_C BrnT_toxin CopG_antitoxin Beta-lactamase adh_short Spermine_synth HTH_3 ? T6SS_Tle1-like_cat

WP_124948364.1
NZ_BHGL01000001.1:520-6387
Sulfuriferula thiophila

? NERD Zn_ribbon_Top1 BrnT_toxin CopG_antitoxin DUF4160 DUF2442 NrsF Sigma70_r4_2 Sigma70_r2 PMSR Ni_hydr_CYTB

WP_124973389.1
NZ_BDQK01000017.1:189975-203628
Aphanothece sacrum FPU1

DciA MFS_1 MFS_1 RecO_C RecO_N Peptidase_M50 HhH-GPD EndIII_4Fe-2S ABC2_membrane HTH_OrfB_IS605 OrfB_IS605 Cas12f1-like_TNB BrnT_toxin BrnA_antitoxin Pps STY-kinase Lipase_GDSL_2 RNA_pol_Rpb6 ? DUF1818

WP_169362493.1
NZ_JAAVJL010000001.1:1116227-1129911
Pseudanabaena yagii GIHE-NHR1

? Epimerase_2 NMT1_2 TY-Chap1 TY-Chap1 ? Aconitase_B_N Aconitase_2_N Aconitase BrnT_toxin ? ? ? XisH XisI ? BolA

WP_197725109.1
NZ_AP017560.1:2761674-2775123
Pseudanabaena sp. ABRG5-3

Peptidase_M23 FitA-like_RHH PIN ? ? DUF938 AAA_22 AAA_lid_2 DUF4258 ? AIRS_C AIRS Semialdhyde_dhC_1 Semialdhyde_dh AbrB-like TatD_DNase Guanylate_cyc GAF FHA Aminotran_3

WP_131617037.1
NZ_CM147841.1:2580793-2594346
Roseitalea porphyridii

DUF1638 ExoD Formyl_trans_N MTHFR Pterin_bind GIY-YIG Fer2 RACo_linker Raco_middle RACo_C_ter ? ? GCS2 MFS_1 ? Hydrolase DJ-1_PfpI HTH_18 PBP_like_2

WP_134761855.1
NZ_SOZD01000002.1:614443-627389
Jiella endophytica

Trans_reg_C Receiver YkuD MgtE Molybdopterin Molydop_binding DUF6460 Epimerase ? ? CopG_antitoxin DUF1731 Epimerase ? HEATs Fer4_16 QueG_DUF1730 GFA Aldo_ket_red GST_C GST_N_3

WP_134763994.1
NZ_SOZD01000009.1:63591-78766
Jiella endophytica

TctB TctA 2-Hacid_dh_C ? BenE MscL NMT1_3 ? CopG_antitoxin Thioredox_DsbH LysE TrmE_N MnmE_helical Acetylase ChaC Uma2

WP_135626729.1
NZ_RQGU01000086.1:236-12352
Leptospira selangorensis

? HATPase_c_3 SduA_C ? BrnT_toxin BrnA_antitoxin Peptidase_C92 GT87 BrnT_toxin CopG_antitoxin ? DUF2007 3-dmu-9_3-mt ? DUF6790 ?

WP_136059779.1
NZ_CAAHFH010000001.1:388659-401263
Pontiella sulfatireligans

PIN PhdYeFM_antitox CoA_binding_3 Bac_transf Hexapep Hexapep Hexapep PDDEXK_3 FdtA FdtA GDP_Man_Dehyd BrnT_toxin CopG_antitoxin Glycos_transf_1 Glyco_transf_4 Glyco_trans_1_4 Glyco_transf_4 PMT_2 GtrA_DPMS_TM Glycos_transf_2 Hexapep

WP_241664455.1
NZ_RAHT01000003.1:174878-191654
Ningiella ruwaisensis

Sacchrp_dh_NADP ? DUF885 gag-asp_proteas Peptidase_M16 Peptidase_M16_C Peptidase_M16 Peptidase_M16_C ParE_toxin ParD_antitoxin BrnT_toxin CopG_antitoxin ? ? DUF6702 Peptidase_M1 AA_kinase PUA Aldedh

WP_142935520.1
NZ_FXTM01000012.1:13304-26443
Balnearium lithotrophicum

DUF8042 A24_N_bact Peptidase_A24 ? TsaD Acetylase DUF465 ILVD_EDD_N ILV_EDD_C BrnT_toxin ? POR Fer4_21 TPP_enzyme_C Transketolase_C POR_N DHO_dh tRNA_U5-meth_tr PabTrmU54_TRAM_dom HAMP MCPsignal PilZ

WP_147917812.1
NZ_CP042828.1:3226233-3244708
Ruania zhangjianzhongii

Peripla_BP_3 HTH_DeoR BPD_transp_1 BPD_transp_1 ? ? ? ? BrnT_toxin CopG_antitoxin BPD_transp_1 BPD_transp_1 SBP_bac_1 TAT_signal Cellulase-like TPRs DUF5107 AraC_binding HTH_18

WP_148066972.1
NZ_VRZA01000001.1:959587-971896
Parahaliea maris

RimB-like UDPG_MGDP_dh_C UDPG_MGDP_dh UDPG_MGDP_dh_N SHMT HTH_3 ? Glycos_transf_2 G_glu_transpept BrnT_toxin CopG_antitoxin Zn_ribbon_NrdR ATP-cone dCMP_cyt_deam_1 RibD_C Lum_binding Lum_binding DHBP_synthase GTP_cyclohydro2 DMRL_synthase NusB

WP_149308782.1
NZ_SRSD01000009.1:141292-156301
Oryzomonas rubra

4HBT MCPsignal sCache_2 ? DcuA_DcuB Radical_SAM ? YCII BrnT_toxin CopG_antitoxin DAO FUSC_2 DUF1634 TauE HTH_8 AAA_lid_14 NtrC-AAA Receiver His-Kinase His-Kinase SBP_bac_3

WP_149569289.1
NZ_CP035807.1:3357056-3377794
Thiospirochaeta perfilievii

Methyltransf_4 Citrate_synt ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs ANKs Tex_N Tex_central_region Tex_YqgF HHH_3 HHH_9 S1 ? ? Phage_integrase ? CopG_antitoxin ? ? MatE MatE NAD_Gly3P_dh_C NAD_Gly3P_dh_N NDK ?

WP_150464709.1
NZ_VXKP01000009.1:4347-11409
Francisella sp. XLW-1

HTH_3 Fic Methylase_S Methylase_S N6_Mtase ? ? MipZ BrnT_toxin CopG_antitoxin ?

WP_150864128.1
NZ_VYXP01000005.1:265388-284209
Marinihelvus fidelis

? Arabinose_bd HTH_18 DUF456 THR4_C PALP Thr_synth_N GHMP_kinases_C GHMP_kinases_N Homoserine_dh NAD_binding_3 ACT_9 ACT_9 AA_kinase Asn_synthase Asn_synthase GATase_7 BrnT_toxin CopG_antitoxin PFO_beta_C TPP_enzyme_C PFOR_II POR_N POR Fer4_7 Pyr_redox_2 Fer4_20 DEAD Helicase_C GSDH Cytochrome_CBB3 Aldedh

WP_150968198.1
NZ_VZDO01000002.1:239452-254445
Plantimonas leprariae

MFS_1 MFS_1 ? SCO1-SenC PrmA ? BA14K Na_Ca_ex Na_Ca_ex ? CopG_antitoxin PcrA_UvrD_tudor UvrD_C UvrD-helicase YqhI DLH Kdo AsmA RHH_4 DUF4169

WP_152586853.1
NZ_CP045423.1:2928766-2945336
Microvirga thermotolerans

SDF Catalase-rel Catalase HTH_1 LysR_substrate COesterase MgtC DUF4010 Sigma70_r4 Sigma70_r3 Sigma70_r2 Sigma70_ner Sigma70_r1_2 Sigma70_r1_1 DnaB_bind Toprim_4 DNAG_N Zn_ribbon_DnaG BrnT_toxin BrnA_antitoxin AsnC_trans_reg HTH_AsnC-type Isochorismatase YqeY CPSase_sm_chain GATase ? Methyltransf_25

WP_188503796.1
NZ_BMFX01000028.1:12125-24255
Nesterenkonia alkaliphila

Pyridox_ox_2 Glyco_transf_4 Glycos_transf_1 APP1_cat MFS_1 ? DUF7120 Hydrolase_3 ? CopG_antitoxin ? DUF7120 PLDc_N VIT1 Inositol_P Ligase_CoA ATP-grasp_2

WP_162029459.1
NZ_CACVAI010000017.1:77974-88073
Lentimonas sp. CC10

? ? BrnT_toxin CopG_antitoxin MotA_ExbB ? ? SGSH_C Sulfatase Sialidase-like_CBM Beta_helix GH141_M SGSH_C Sulfatase

WP_162027975.1
NZ_CACVAG010000026.1:47092-59563
Lentimonas sp. CC4

? Cytochrom_C_asm ResB ? YchF-GTPase_C MMR_HSR1 Receiver His-Kinase His-Kinase FAD_binding_4 FAD-oxidase_C BrnT_toxin CopG_antitoxin CutA1 PhdYeFM_antitox YoeB_toxin ? ? ?

WP_163329441.1
NZ_CP176814.1:1-11360
Desulfurobacterium thermolithotrophum

HTH_3 ? Rep3_C DUF8404 Myb_DNA-binding CENP-H ? Phage_int_SAM_2 Phage_integrase ? BrnT_toxin ? DUF2283 Toprim_2 ? DNA_methylase DNA_methylase

WP_163464746.1
NZ_JAAAMG010000015.1:42384-58041
Jiella pacifica

TctA 2-Hacid_dh_C DNA_binding_1 MFS_1 AraC_binding HTH_18 MscL NMT1_3 ? CopG_antitoxin Thioredox_DsbH HxlR Oxidored_FMN Oxidored_FMN LysE TrmE_N MnmE_helical ChaC

WP_164212490.1
NZ_JAAGSC010000044.1:408682-417616
Wenzhouxiangella limi

tPilZ HepT-like NTP_transf_2 MazE_antitoxin PemK_toxin MazE_antitoxin PIN BrnT_toxin CopG_antitoxin FliT FliS FliD_C Flagellin_IN FliD_N FlaG Flagellin_C Flagellin_N ?

WP_164663374.1
NZ_JAACKB010000069.1:223-6364
Sulfurovum sp. bin170

BrnT_toxin CopG_antitoxin ? ? YgjP-like GXGXG Glu_synthase Glu_syn_central GATase_2

WP_165282315.1
NZ_JAAMFG010000035.1:134425-148877
Wenzhouxiangella sp. XN24

MrpF_PhaF MNHE UDPG_MGDP_dh_N UDPG_MGDP_dh UDPG_MGDP_dh_C Epimerase PGI MvaT_DBD Pyr_redox_2 BrnT_toxin CopG_antitoxin HTH_3 HigB-like_toxin HicB Caps_assemb_Wzi Poly_export SLBB_2 SLBB SLBB SLBB SLBB Wzz GNVR

WP_166224187.1
NZ_CP049989.1:561848-573369
Hydrogenophaga crocea

His_Phos_1 Thiolase_N Thiolase_C Meth_synt_2 Beta-lactamase Cytochrome_CBB3 Cytochrome_CBB3 DUF2889 Cupin_2 BrnT_toxin CopG_antitoxin DUF1232 2OG-Fe_Oxy_2 DUF2189 ? ? AMP-binding_C AMP-binding

WP_167082246.1
NZ_VVIW01000055.1:362-5288
Massilia aquatica

? BrnT_toxin ? NikA-like Mobilization_B RepB_primase RepB_primase_C ?

WP_168621482.1
NZ_JAAZQQ010000001.1:116126-131044
Roseicyclus persicicus

FMN_red TPP_enzyme_N TPP_enzyme_M TPP_enzyme_C MS_channel_3rd MS_channel_2nd MS_channel_1st Aconitase MFS_1 MFS_1 HupE_UreJ_2 DUF4198 ? CopG_antitoxin Aconitase_C LeuD LeuD_C ? Synaptojanin Iso_dh adh_short_C2 EamA EamA

WP_171696947.1
NZ_CP180476.1:161131-177209
Methylomonas sp. TEB

Anth_synt_I_N Chorismate_bind GATase DUF4160 DUF2442 HI0933_like ? Glycos_trans_3N Glycos_transf_3 BrnT_toxin CopG_antitoxin ? IGPS MTHFR OEP OEP BSH_CzcB RND-MFP_C ACR_tran

WP_176226380.1
NZ_BLRU01000106.1:255-4842
Candidatus Hakubella thermalkaliphila

? TumE BrnT_toxin CopG_antitoxin ? Cas12f1-like_TNB ARG QueH

WP_187972129.1
NZ_CP061080.1:2075378-2088499
Aquibium microcysteis

DDE_Tnp_1 PaaX_C PaaX TetR_C_24 TetR_N AMP-binding_C_2 AMP-binding 4HBT p450 p450 Aldedh MaoC_dehydratas BrnT_toxin CopG_antitoxin MFS_1 Lyase_1 Dioxygenase_C ParE_toxin RHH_1 Dioxygenase_C PCDO_beta_N

WP_188363827.1
NZ_BAABJF010000011.1:182376-194305
Marinicella pacifica

FtsH_ext AAA AAA_lid_3 Peptidase_M41 Pterin_bind PGM_PMM_I PGM_PMM_II PGM_PMM_III PGM_PMM_IV LpqE-like Queuosine_synth ? ? BrnT_toxin CopG_antitoxin Peptidase_M23 YjdM Acyl-CoA_dh_1 Acyl-CoA_dh_M AidB_N SlyX DUF938 PALP

WP_188915111.1
NZ_BMMF01000014.1:65855-83744
Salinarimonas ramus

Usp HKD HKD VTT_dom ? BCCT ? sll0997_C MASE1 PAS PAS PAS PAS His-Kinase His-Kinase Receiver BrnT_toxin HTH_31 Aminotran_1_2 GntR Pyridox_ox_2 DUF1622 APS_kinase GTP-eEF1A_C GTP_EFTU_D2 GTP_EFTU PAPS_reduct MFS_1

WP_190763064.1
NZ_JACXLD010000002.1:165043-182620
Spongiibacter pelagi

ECH_1 NMO NMO ? CoA_trans AsnC_trans_reg_3 HTH_AsnC-type AsnC_trans_reg HTH_AsnC-type AMP-binding_C AMP-binding BrnT_toxin CopG_antitoxin Helicase_C DEAD WHD_BLACT Lactamase_B Plug TonB_dep_Rec_b-barrel PepSY_TM MatE MatE TonB_dep_Rec_b-barrel Plug

WP_197210437.1
NZ_CP060397.1:2296933-2309796
Synechococcus sp. CBW1004

DUF2555 PCP_red ? DUF565 OTCace OTCace_N Pur_DNA_glyco DUF4143 AAA_14 BrnT_toxin CopG_antitoxin YoeB_toxin PhdYeFM_antitox N_methyl Methyltransf_11 Methyltransf_12 MethyTransf_Reg ? Glyco_transf_41 Glyco_transf_41 TPRs

WP_198262543.1
NZ_JAEHGE010000001.1:130868-143896
sulfur-oxidizing endosymbiont of Gigantopelta aegis

RNA_pol_A_bac RNA_pol_A_CTD Ribosomal_L17 ? rve_2 rve HTH_21 DegS HTH_Tnp_1 MFS_1 MFS_1 YajR_YAM SSB BrnT_toxin BrnA_antitoxin ? rve_2 rve HTH_21 DegS HTH_Tnp_1 ? ?

WP_200154008.1
NZ_NRRQ01000013.1:28502-58416
Chromatium okenii

? SapC ABC2_membrane ABC_tran Wzt_C Methyltransf_23 Glycos_transf_2 Glyco_transf_4 Glyco_trans_1_4 Glycos_transf_2 Glyco_trans_1_4 DUF4879 ? BrnT_toxin CopG_antitoxin ? Mur_ligase_M Mur_ligase_C Peptidase_S24 Glyco_hydro_94 GH94_b-supersand GH94_b-supersand MASE3 PAS PAS PAS PAS GGDEF PAS GAF PHY PAS PAS PAS

WP_200348199.1
NZ_NRSJ01000049.1:12981-28479
Halochromatium glycolicum

GATase_3 CbiA ? CbiM Big HI_1622_C CbiQ ABC_tran CbiX CbiX BrnT_toxin CopG_antitoxin FitA-like_RHH PIN Methylase_S Methylase_S TP_methylase CbiG_N CbiG_mid CbiG_C Uma2 Methyltransf_31

WP_200594439.1
NZ_JAEPBG010000008.1:244899-257940
Noviherbaspirillum pedocola

Receiver Receiver His-Kinase His-Kinase MAD HAMP CHASE8 DUF4154 TonB_dep_Rec_b-barrel Plug DUF2721 Cupin_2 ? adh_short BrnT_toxin CopG_antitoxin Pterin_4a adh_short AsnC_trans_reg HTH_24 Glyoxalase_5 Glyoxalase Glyco_transf_4 Glycos_transf_1 ?

WP_207350615.1
NZ_JAFMPY010000008.1:133010-145205
Jiella sonneratiae

NMT1_2 DIOX_N 2OG-FeII_Oxy ? BPD_transp_1 ABC_tran Arginase ? ? ? DUF1731 Epimerase ? HEAT_EZ Fer4_16 QueG_DUF1730 GFA Aldo_ket_red GST_C GST_N_2

WP_207351405.1
NZ_JAFMPY010000014.1:63124-79183
Jiella sonneratiae

DNA_ligase_A_M DNA_ligase_A_C DUF6460 Molydop_binding Molybdopterin YkuD Receiver Trans_reg_C cNMPBD Synaptojanin ? CopG_antitoxin Abhydrolase_1 LolA Cys_Met_Meta_PP SBP_bac_3 BPD_transp_1 BPD_transp_1

WP_207561194.1
NZ_CP046072.1:2051204-2067780
Sulfurimonas aquatica

? ? ? HTH_Tnp_1 ? RelE DUF4258 BrnT_toxin CopG_antitoxin Y1_Tnp Glycos_transf_4 8xMP YjbH Poly_export SLBB SLBB SLBB SLBB SLBB Wzz SelO

WP_207690767.1
NZ_CP061799.1:1336802-1346251
Desulfonema limicola

Receiver Aldedh PIN ? HEPN NTP_transf_2 ? BrnT_toxin NikA-like ? Fic ? ? ? ?

WP_213445606.1
NZ_JAHCDN010000002.1:439097-458018
Variovorax sp. PCZ-1

GXGXG Glu_synthase Glu_syn_central GATase_2 Y1_Tnp ? HD_assoc HD ParE_toxin ParD_antitoxin DHQS_C DHQ_synthase BrnT_toxin CopG_antitoxin Glyoxalase HATPase_c_3 Ferric_reduct Oxidored_molyb Cytochrom_C_asm ResB

WP_226067042.1
NZ_CP084578.1:103247-114868
Sulfurimonas sp. SWIR-19

Isocitrate_DH_C_bact Iso_dh GyrI-like HTH_18 CobD_Cbib GATase_3 CbiA Thioredoxin_2 Aminotran_1_2 ? BrnT_toxin CopG_antitoxin BrnT_toxin CopG_antitoxin HTH_3 Excalibur APH ?

WP_233721465.1
NZ_JAJUWU010000024.1:29847-44089
Jiella avicenniae

TctA 2-Hacid_dh_C Astacin MscL NMT1_3 BrnT_toxin HTH_31 ? CopG_antitoxin Thioredox_DsbH LysE TrmE_N MnmE_helical Acetylase ChaC BPD_transp_2

WP_320188612.1
NZ_CP192802.1:2398-11883
Agrobacterium rosae

? ? ? ? ? SieB rve_3 rve HTH_21 HTH_Tnp_1 BrnT_toxin BrnA_antitoxin ? DUF982 CN_hydrolase LNT_N DUF3617 Peptidase_M23 ?

WP_320188612.1
NZ_CP192770.1:2398-11221
Agrobacterium rosae

? ? ? ? ? SieB rve_3 rve HTH_21 HTH_Tnp_1 BrnT_toxin BrnA_antitoxin ? DUF982 ? CN_hydrolase LNT_N DUF3617 Peptidase_M23

WP_251812548.1
NZ_CP101527.1:3610192-3617237
Alkalimarinus sediminis

UPF0150 HicB RelB3_C ParE_toxin ParE-like_toxin ParD_like Acetyltransf_7 Imp-YgjV BrnT_toxin CopG_antitoxin PAAR_motif PAAR_motif YoeB_toxin PhdYeFM_antitox ? YdjM ?

WP_257263113.1
NZ_JAHHPO010000051.1:104-2643
Endozoicomonas sp. ONNA2

HicB ? PIN FitA-like_RHH BrnT_toxin CopG_antitoxin ?

WP_252179526.1
NZ_CP082909.1:1648937-1664659
Endozoicomonas sp. 4G

TPRs TPRs ATG16 FimV_N Semialdhyde_dhC Semialdhyde_dh 4HBT 4HBT Glt_symporter AAA_23 ABC-ATPase ? BrnT_toxin BrnA_antitoxin KH_dom-like MMR_HSR1 MMR_HSR1 PQQ_2 PQQ_2 PQQ TPRs HGTP_anticodon tRNA-synt_His HTH_25 TPR_CcmH_CycH TPRs TPRs TPRs

WP_253307527.1
NZ_OZ032162.1:1109338-1121842
Rickettsia endosymbiont of Ceutorhynchus obstrictus

DUF2310 DUF5915 Anticodon_1 tRNA-synt_1 COQ9 Ribosomal_S21 ? ABC-3 Methyltransf_23 BrnT_toxin CopG_antitoxin tRNA-synt_1c Anticodon_2 ? ? DUF6314 ? ?

WP_253448952.1
NZ_JALJYF010000002.1:485213-507163
Natronospira proteinivora

Collagen Beta-prop_ACSF4 Beta-prop_ACSF4 Beta-prop_ACSF4 Peptidase_S8 CHB_HEX_C_1 Glycos_transf_2 ? Epimerase Glycos_transf_4 CoA_binding_3 Polysacc_synt_2 BrnT_toxin CopG_antitoxin HTH_31 HipA_C NTP_transferase PGI ? WEMBL UPF0020

WP_253687318.1
NZ_CP051200.1:663345-680626
Treponema sp. OMZ 789

EVI2A Ig_CFAP65_4th HYDIN_VesB_CFA65-like_Ig LamininG DEAD Helicase_C DbpA CarboxypepD_reg Amidohydro_1 NYN LPAM_1 TdpA_C BrnT_toxin BrnA_antitoxin ? Transposase_31 Transposase_31 PDDEXK_2 HTH_Rpn_C NamZ_C NamZ_N Beta-lactamase

WP_255106058.1
NZ_JAGQBZ010000018.1:1-9001
Synechococcus sp. Edmonson 11F2

HsdM_N N6_Mtase ? Methylase_S dCMP_cyt_deam_1 HATPase_c_4 DUF4357 PIN MazE_antitoxin BrnT_toxin CopG_antitoxin ? HTH_38 Virulence_RhuM DUF772

WP_255824134.1
NZ_CP038803.1:937649-944049
Treponema putidum

RelB YoeB_toxin ? Gp49 HTH_19 PhdYeFM_antitox ParE_toxin BrnT_toxin BrnA_antitoxin Acetylase DUF3781 ? ? PhdYeFM_antitox ParE_toxin

WP_262601291.1
NZ_CP103300.1:2987803-2994128
Endozoicomonas euniceicola

NTP_transferase PhdYeFM_antitox MazE-like PemK_toxin PemK_toxin DUF433 DUF5615 BrnT_toxin CopG_antitoxin MazE-like PemK_toxin DUF2442 DUF4160 BrnA_antitoxin HVO_A0114

WP_267184253.1
NZ_JAPMLN010000015.1:62535-72842
Marinicella gelatinilytica

MFS_1 SSF ? TspO_MBR CopG_antitoxin BrnT_toxin UDG BrnT_toxin CopG_antitoxin ? ? DUF1287 DUF7379 MFS_1 ASCH

WP_269281364.1
NZ_CP098247.1:1680572-1689145
Oxalobacter aliiformigenes

HicB-like_2 Phage_antitermQ ? VapE-like_dom ? ? ? BrnT_toxin BrnA_antitoxin ? ? YdaT_toxin YdaS_toxin HTH_3 ?

WP_269540184.1
NZ_JAPWDU010000004.1:121-10449
Cerasicoccus fimbriatus

Esterase AXE1 BrnT_toxin CopG_antitoxin Peptidase_C15 DUF294 GlnD_UR_UTase GlnD_3rd HD GAF_2 GAF_2 His-Kinase His-Kinase Pantoate_transf IU_nuc_hydro ?

WP_272120334.1
NZ_JAQMTY010000137.1:394-7984
Spirulina subsalsa CS-330

MannoseP_isomer Peptidase_M48 BrnT_toxin ? ? ? Beta-lactamase2 SH3_16 NLPC_P60 Maf PMT_2

WP_272743951.1
NZ_JAQQKV010000001.1:1226204-1243077
Asticcacaulis machinosus

MgtE CBS MgtE_N FliMN_C LplA-B_cat DUF805 VWA Tad VWA VWA_2 Tad PseudoU_synth_2 BrnT_toxin BrnA_antitoxin DUF805 DUF805 MgsA_C AAA_assoc_2 AAA ACR_tran ACR_tran RND-MFP_C Beta-barrel_RND_2 HH_MFP_RND TetR_N TetR_C_11

WP_281009509.1
NZ_CP112875.1:179553-199058
Pseudanabaena galeata CCNP1313

DUF7219 UDPGT Glyco_transf_28 ? Abhydrolase_1 AAA Helicase_C MZB BrnT_toxin BrnA_antitoxin HKD PIN RHH_1 Uma2 ? MmeI_TRD Eco57I

WP_286393913.1
NZ_CP097329.1:286890-302876
Pseudanabaena mucicola str. Chao 1806

DUF5615 DUF433 PIN DUF2281 PhdYeFM_antitox DUF7008 DDE_Tnp_1 Eco57I BrnT_toxin ? DUF4258 DUF2283 DUF29 RNA_pol_A_CTD STY-kinase STY-kinase NERD ? RNaseH-CATRA+Caspase TPRs TPRs TPRs TPRs TPRs

WP_289153315.1
NZ_JATAAF010000007.1:98146-111697
Porticoccus sp. W117

Thiolase_C Thiolase_N 3HCDH 3HCDH 3HCDH_N ECH_1 Methyltransf_23 Tex_N Tex_central_region Tex_YqgF HHH_3 HHH_9 S1 DUF6249 Sigma70_r2 Sigma70_r4_2 ? BrnT_toxin CopG_antitoxin ? NAD_binding_1 FAD_binding_6 NnrS ? Rrf2 ATPgrasp_ST

WP_289223644.1
NZ_CP121121.1:1511325-1525831
Thiomicrospira sp. R3

HTH_7 Resolvase HNH_5 dnstrm_HI1420 Gp49 ? ? rve_3 rve BrnT_toxin CopG_antitoxin Helicase_C DEAD PhdYeFM_antitox ParE_toxin HTH_3 PAS PAS MCPsignal PGM_PMM_I PGM_PMM_II PGM_PMM_III

WP_289995587.1
NZ_JAOAOQ010000031.1:10621-29891
Aquibium sp. ELW1220

DDE_Tnp_1 DDE_Tnp_1_assoc N-OB tRNA-synt_2 Peptidase_C15 ? Abi_2 DNA_topoisoIV DNA_gyraseA_C DNA_gyraseA_C Abhydrolase_6 ? CopG_antitoxin Ribonuc_L-PSP DPRP TetR_N TetR_C_17 BSH_CzcB Beta-barrel_RND_2 YknX_C ACR_tran ?

WP_290543718.1
NZ_JADCDA010000076.1:2640-15420
Aestuariivirga sp.

? ? YkuD DUF995 ? PliI DUF3011 ? CopG_antitoxin UVR UvrB Helicase_C UvrB_3rd UvrB_inter ResIII RsmJ TPRs FAD_binding_3 Aminotran_1_2 DUF1993

WP_290661871.1
NZ_JACMNS010000122.1:55-3991
Aquabacterium sp.

ParE_toxin ParD_antitoxin ? Gp49 dnstrm_HI1420 DUF6788 ? BrnT_toxin CopG_antitoxin

WP_290734065.1
NZ_NJER01000002.1:432451-445118
Halobacteriovorax sp. JY17

NTP_transferase ? PMT_2 Glycos_transf_1 Glyco_transf_4 Glycos_transf_2 ? ? ? CopG_antitoxin AbiEi_4 AbiEii Glyco_transf_9 Peptidase_C69 Fic Y1_Tnp

WP_295684807.1
NZ_CAMHPO010000053.1:6084-13566
uncultured Fibrobacter sp.

? ? ? ? STB2_C ? ? BrnT_toxin CopG_antitoxin ? ? ? Terminase_3 ? Extensin_1

WP_291018800.1
NZ_WVWM01000060.1:43907-54843
Hydrogenophaga sp.

Cytochrome_CBB3 Cytochrome_CBB3 AMP-binding AMP-binding_C DoxX_2 HTH_1 DUF2889 Cupin_2 MOSC BrnT_toxin CopG_antitoxin 2OG-Fe_Oxy_2 DUF2189 ? ? AMP-binding_C AMP-binding SBP_bac_3

WP_291322590.1
NZ_PWSL01000305.1:274-3627
Desulfonatronospira sp.

BrnT_toxin CopG_antitoxin Virulence_RhuM Fic_N

WP_291347939.1
NZ_JABZFL010000078.1:24128-33209
Desulfobacula sp.

Peptidase_M78 HTH_3 BrnT_toxin CopG_antitoxin ELFV_dehydrog ELFV_dehydrog_N TrkA_N TrkA_C TrkA_N TrkA_C TrkH cNMPBD Proton_antipo_M ?

WP_292570648.1
NZ_JALOBS010000018.1:92684-107218
Methylomonas sp.

? Alpha-amylase Sigma70_r4_2 Sigma70_r2 ? Y1_Tnp PepSY_TM TonB_dep_Rec_b-barrel Plug STN BrnT_toxin CopG_antitoxin FecR DUF4880 Sigma70_r4_2 Sigma70_r2 PEP_mutase MFS_3 Fic-like_C Fic ?

WP_292655743.1
NZ_WFQN01000062.1:21989-33061
Nitratifractor sp.

adh_short_C2 Hydrolase SMC_N NAD_kinase_C NAD_kinase Synaptojanin Methyltransf_25 BrnT_toxin CopG_antitoxin DPBB_1 Arginosuc_syn_C Arginosuc_synth PdxJ ? PdxA CCDC39 DUF2730

WP_293037260.1
NZ_JAAHFV010000020.1:20608-40013
Moorena sp. SIO1F2

DDE_Tnp_1_2 NOMO1-like_2nd DEAD Helicase_C DSHCT ? TPRs Beta-prop_WDR5 Beta-prop_WDR5 Beta-prop_WDR5 Beta-prop_IFT122_1st MNS-STAND MNS-STAND STY-kinase BrnT_toxin BrnA_antitoxin GATase GHL10 HTH_17 PPC CobS ? TGT

WP_293621790.1
NZ_NZYJ01000095.1:4823-21206
Salinisphaera sp.

ABC_tran ABC_membrane SMC_N ABC_tran ABC_membrane OmpW cNMPBD HTH_Crp_2 Synaptojanin UPF0004 Radical_SAM TRAM PhoH BrnT_toxin CopG_antitoxin YbeY NMB0537_N CBS CBS CorC_HlyC LNT_N CN_hydrolase tRNA-synt_1 tRNA-synt_1_2 tRNA-synt_1 Anticodon_1 LptE Nitroreductase

WP_294882275.1
NZ_MICC01000126.1:4766-19773
Sulfurimonas sp. RIFOXYB12_FULL_35_9

? Receiver Trans_reg_C His-Kinase His-Kinase Cytochrome_CBB3 nos_propeller Cupredoxin_1 Cytochrom_C DNA_pol3_gamma3 DNAX_ATPase_lid DNA_pol3_delta2 BrnT_toxin ? ? PemK_toxin YokU DUF155 ? PhoH PIN

WP_294956048.1
NZ_WFRE01000010.1:1-3810
Sulfurovum sp.

HHH_5 HHH_8 DEAD Helicase_C BrnT_toxin CopG_antitoxin tRNA-synt_1c

WP_295002639.1
NZ_CAXBUV010000005.1:17685-32207
uncultured Sulfurimonas sp.

? MoeA_N MoCF_biosynth MoeA_C ? PTPS Fer4_12 Mob_synth_C Radical_SAM RDD BrnT_toxin CopG_antitoxin IF3_C IF3_N HGTP_anticodon tRNA-synt_2b tRNA_SAD ILV_EDD_C ILVD_EDD_N Phage_integrase ? ?

WP_295022906.1
NZ_JAKFWK010000004.1:345721-355282
Sulfurimonas sp.

Pribosyltran LPP20 ? MMR_HSR1 MMR_HSR1 KH_dom-like UPF0093 Ribosomal_S30AE tRNA-synt_1b BrnT_toxin CopG_antitoxin ? ParE_toxin Hemerythrin OTCace OTCace_N ALAD GTP_cyclohydro2

WP_295066176.1
NZ_JALNYS010000009.1:28006-37848
Sulfuricurvum sp.

Aminotran_4 ? LysR_substrate HTH_1 UvrD-helicase UvrD_C TruB_N CsrA GHMP_kinases_N BrnT_toxin CopG_antitoxin SmpB CorA Ribosomal_L19 tRNA_m1G_MT PRC_RimM RimM KH_KhpA-B

WP_295066722.1
NZ_JALNYS010000012.1:39598-54495
Sulfuricurvum sp.

? Hydrolase MTHFR ? ? PhdYeFM_antitox LprI BrnT_toxin CopG_antitoxin Seryl_tRNA_N tRNA-synt_2b TPRs ARM_TT21_5th TPRs ARM_TT21 CDP-OH_P_transf PS_Dcarbxylase Peptidase_M41 AAA_lid_3 AAA FtsH_ext PrmA

WP_295067407.1
NZ_JALNYS010000015.1:62437-74333
Sulfuricurvum sp.

Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_5 HHH_2 BRCT Pyr_redox_2 S4 FtsJ FAD_syn FAD_syn Flavokinase Cytochrom_C ATP-synt_A ? BrnT_toxin CopG_antitoxin ? ParE_toxin TMP-TENI GatB_N GatB_Yqey Ion_trans TrkA_N TrkA_N TrkA_C

WP_297432295.1
NZ_JALXBK010000047.1:144302-155401
Sulfurimonas sp.

TP_0481_C MCPsignal NifU Flavin_Reduct TauE PEMT MS_channel_1st MS_channel_2nd MS_channel_3rd Na_Ca_ex Na_Ca_ex BrnT_toxin CopG_antitoxin AhpC-TSA 1-cysPrx_C Rhodanese Cytochrome_C554 DUF134 Nitro_FeMo-Co Cytochrom_C Cytochrom_C

WP_299973007.1
NZ_JAQTUH010000002.1:179333-194398
Sulfuricurvum sp.

Polysacc_deac_1 ? Aminotran_1_2 ? Mur_ligase Mur_ligase_M Mur_ligase_C MutS_V MSSS Smr ? BrnT_toxin BrnA_antitoxin DUF1771 Sel1 ? Sel1 Sel1 Sel1 Sel1 Sel1 Sel1 Nlig-Ia DNA_ligase_aden OB_DNA_ligase DNA_ligase_ZBD HHH_5 HHH_2 BRCT Pyr_redox_2 S4 FtsJ

WP_304433307.1
NZ_CAPVCI010000054.1:2556-12039
Acutalibacter muris

ParA-Soj-NTPase ? HTH_3 ? ? ? BrnT_toxin ? ? ? RelB YafQ_toxin ? ?

WP_305314592.1
NZ_JAUXUS010000147.1:10431-26453
Hydrogenophaga sp.

Phage_sheath_1C Phage_T4_gp19 Phage_sheath_1C Phage_sheath_1 ? Pvc16_N TctC FAD_binding_3 BrnT_toxin CopG_antitoxin VWA VWA_2 BatA vWA AAA_lid_2 AAA_3 Trypsin Aconitase_C Aconitase DUF3309

WP_306119343.1
NZ_JASHJO010000011.1:51140-67120
Roseitalea sp. MMSF_3504

? DeoRC Rib_5-P_isom_A DUF2059 Pyr_redox_2 Pyr_redox_dim ? DAHP_synth_2 CN_hydrolase NAD_synthase DUF4258 CopG_antitoxin VIT1 MFS_1 tRNA-synt_1c Anticodon_2 DUF2865 PTA_PTB Malic_M malic adh_short_C2

WP_306728608.1
NZ_JAVDDT010000006.1:364-9169
Natronospira bacteriovora

BrnT_toxin CopG_antitoxin Epimerase Methyltransf_11 Glycos_transf_4 Wzy_C Wzy_C TRAM Radical_SAM UPF0004

WP_308056729.1
NZ_JAVFVV010000006.1:201940-217161
Mesorhizobium sp. LHD-90

SBP_bac_3 MerR-DNA-bind MerR_1 ? MgtE CBS CBS MgtE_N LplA-B_cat Phytochelatin ? ? CopG_antitoxin ABC_tran FtsX MacB_PCD HGTP_anticodon tRNA-synt_2b DUF1467 Glyoxalase_4 RNase_J_C RMMBL RNase_J_b_CASP Lactamase_B_2

WP_308253403.1
NZ_VRZC01000003.1:25519-40017
Geminocystis sp. GBBB08

tRNA_NucTran2_2 PolyA_pol_RNAbd PolyA_pol Thioredoxin Lysine_decarbox Usp Usp DUF1517 ? OprB SLH BrnT_toxin ? HcyBio DUF2499 HTH_Crp_2 PNP_phzG_C PNPOx_N ? POTRA_1

WP_309361386.1
NZ_JARLHN010000037.1:1-6786
Oryzomonas sp.

POR TPP_enzyme_C PFOR_II POR_N Fer4_7 CbiM PDGLE CbiQ ABC_tran BrnT_toxin CopG_antitoxin ?

WP_310447590.1
NZ_JASJPD010000047.1:5103-22724
Thiobacillus sp.

EamA EamA STAS Sulfate_transp Acetate_kinase PGM_PMM_IV PGM_PMM_III PGM_PMM_II PGM_PMM_I XFP_C XFP XFP_N Sulfate_transp STAS TonB_C BrnT_toxin CopG_antitoxin ? ? ? HNH AP2 tRNA-synt_1c_C2 tRNA-synt_1c_C tRNA-synt_1c Aldolase

WP_310780102.1
NZ_CANPWG010000060.1:102548-114604
Candidatus Palauibacter scopulicola

PIN ? adh_short ? HTH_3 HTH_15 DUF5128 ? BrnT_toxin CopG_antitoxin ? AdoHcyase Methyltransf_11 HTH_5 PIN ? DUF1499

WP_311363941.1
NZ_JAVRIC010000004.1:69516-87876
Banduia mediterranea

RecA_C RecA_N 2H CinA MutS_I MutS_II MutS_III MutS_V MutS_C Hydantoinase_B Hydant_A_C Hydantoinase_A Hydant_A_N PIN VapB_antitoxin BrnT_toxin CopG_antitoxin DUF885 HGLS ParA FeS_assembly_P MFS_1 Beta-barrel_AAEA BSH_RND OEP OEP

WP_321990719.1
NZ_OY821668.1:2117293-2133484
Marispirochaeta aestuarii

Peripla_BP_1 LacI ? ? Beta_helix Glyco_trans_4_2 Glycos_transf_1 Glycos_transf_4 GDP_Man_Dehyd BrnT_toxin CopG_antitoxin AAA_18 Glycos_transf_2 ? CTP_transf_like ? Hydrolase CTP_transf_3

WP_322124895.1
NZ_CP133777.1:1600581-1623771
Actimicrobium sp. CCC2.4

PAS PAS GGDEF EAL ? CbiX LptF_LptG LptF_LptG Peptidase_M17_N Peptidase_M17 TPRs BrnT_toxin CopG_antitoxin Glyco_trans_2_3 Glycoamylase GH94_b-supersand GH94_b-supersand Glyco_hydro_94 Glyco_hydro_65C LRRs ? MazE_antitoxin PemK_toxin ?

WP_323104167.1
NZ_JAYESJ010000001.1:1166826-1177011
Rhizobium sp. CC-YZS058

? ? GreA_GreB_N GreA_GreB Glyco_trans_1_4 YbjN F420_oxidored P5CR_dimer tRNA_bind ? CopG_antitoxin His-Kinase His-Kinase HAMP Trans_reg_C Receiver MarR_2 ? Lactamase_B BA14K

WP_323291536.1
NZ_JAYGHQ010000009.1:46240-64304
Crocosphaera sp. XPORK-15E

Pps Pps Pps Pps AAA_26 DRTGG PTA_PTB Uma2 CAF1A_acidic HR1_TOCA MlaD ABC_tran Pps Pps Pps Pps Pps HD_5 CHASE6_C DICT BrnT_toxin ? Amidohydro_3 Amidohydro_3 OprB SLH OprB SLH PhoU PhoU His-Kinase His-Kinase PAS Lipase_GDSL

WP_323300148.1
NZ_JAYGHP010000002.1:452152-464117
Crocosphaera sp. UHCC 0190

PHP DNA_pol3_alpha DNA_pol3_finger Intein_splicing Psb34 P-II YegS_C DAGK_cat HigB_toxin HTH_3 Epimerase_2 BrnT_toxin CopG_antitoxin PEP-CTERM DUF29 tRNA-synt_1e DALR_2 CysS_C DUF29 DUF29 ?

WP_323843489.1
NZ_JAXCJK010000003.1:68111-81471
Moraxella sp. Pampa

RNase_E_G_Thio RNase_E_G S1 Maf MreD MreC MreB_Mbl GatC Amidase BrnT_toxin ? Acetylase ? PIN GatB_N GatB_Yqey N6_Mtase Methylase_S Methylase_S

WP_325586600.1
NZ_DATMGA010000028.1:173857-183508
Eoetvoesiella sp.

DEAD Helicase_C DUF1697 3-dmu-9_3-mt Glyoxalase ? BrnT_toxin CopG_antitoxin HTH_3 LysR_substrate HTH_1 Pirin Pirin_C_2 FMN_red ? Methyltransf_28